Chapter 1 Introduction

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Key Points

Basic Concepts	Concepts, Components, Functions	Proficient
	Classification and Topology	Proficient
	Development	Understand
	Internet Standardization	Master
Network Architecture	Layered Architecture / Related Concepts	Proficient
	ISO/OSI Reference Model	Proficient
	TCP/IP Reference Model	Proficient
	LAN Reference Model	Proficient

Chapter 1: Roadmap

1.1 Computer network and the Internet

1.2 Network edge and core

1.3 Network access and physical media

1.4 Transmission Technology

1.5 Type and Topology

1.6 Network Performance

1.7 Protocol layers, service models

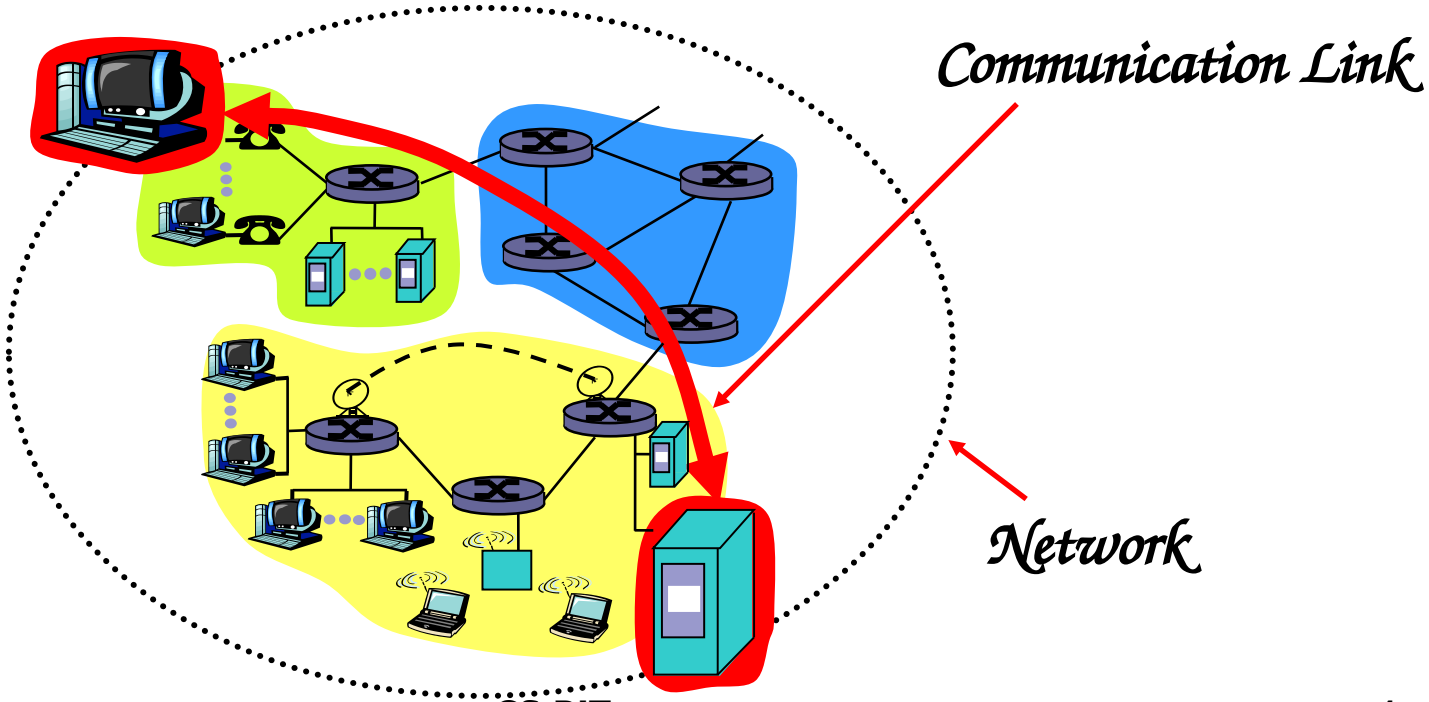
1.8 Architecture, OSI and TCP/IP Models

1.9 Internet History

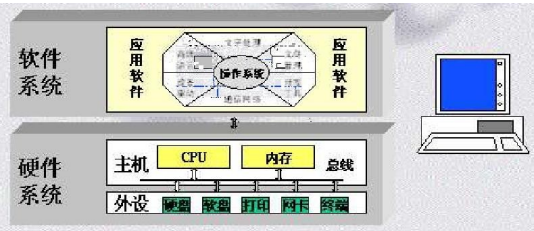
What is a computer network?

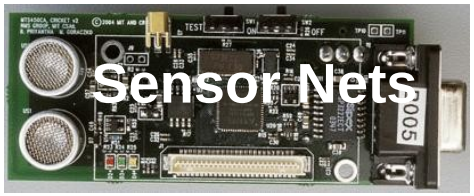
- **Tanenbaum's definition**

  Interconnected collection of autonomous computers connected by a communication technology



Autonomous computers



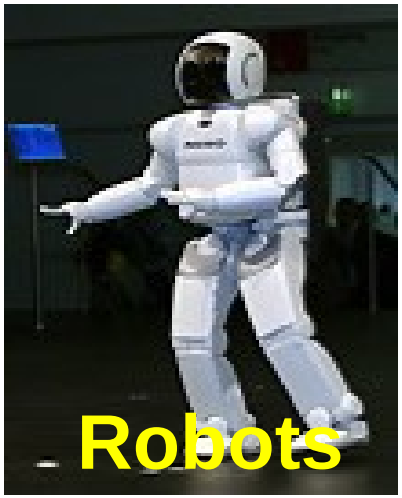
















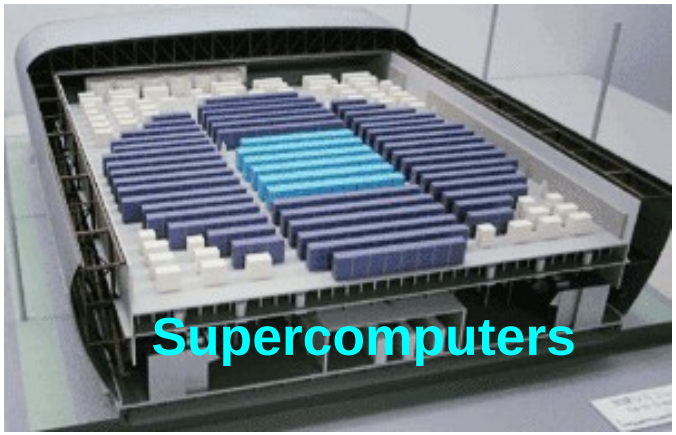




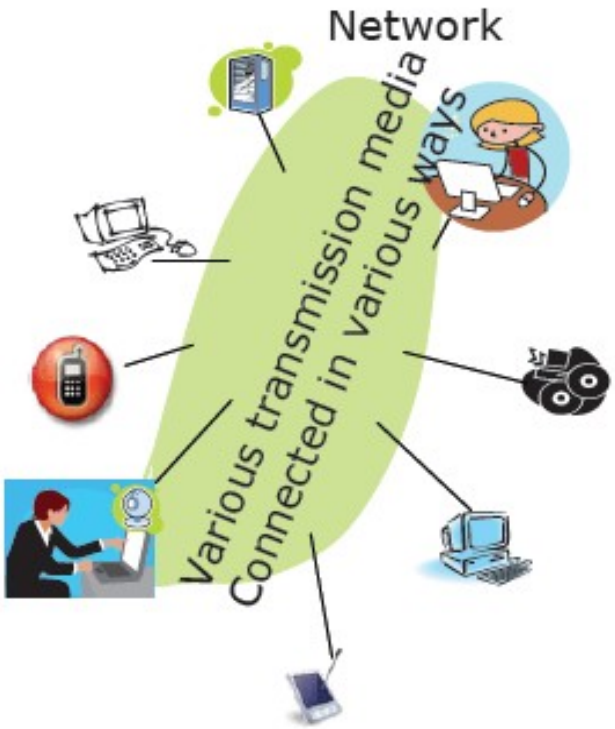


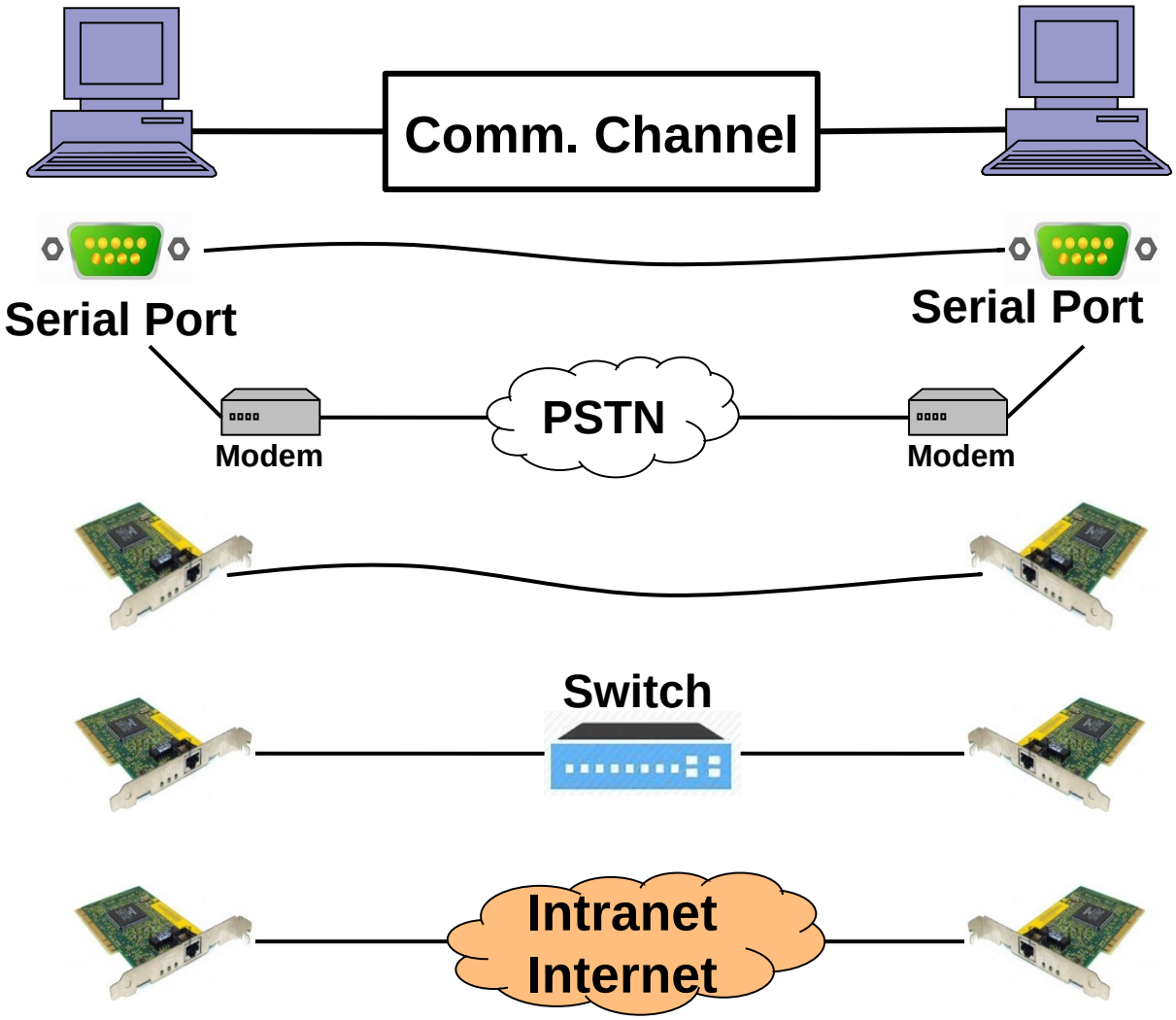




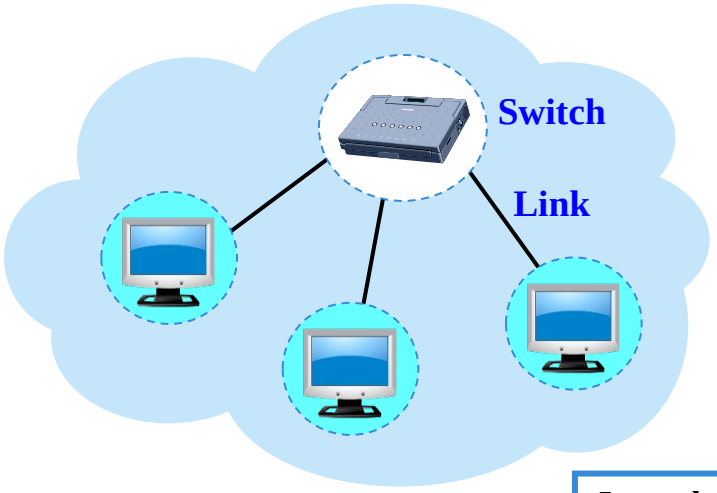


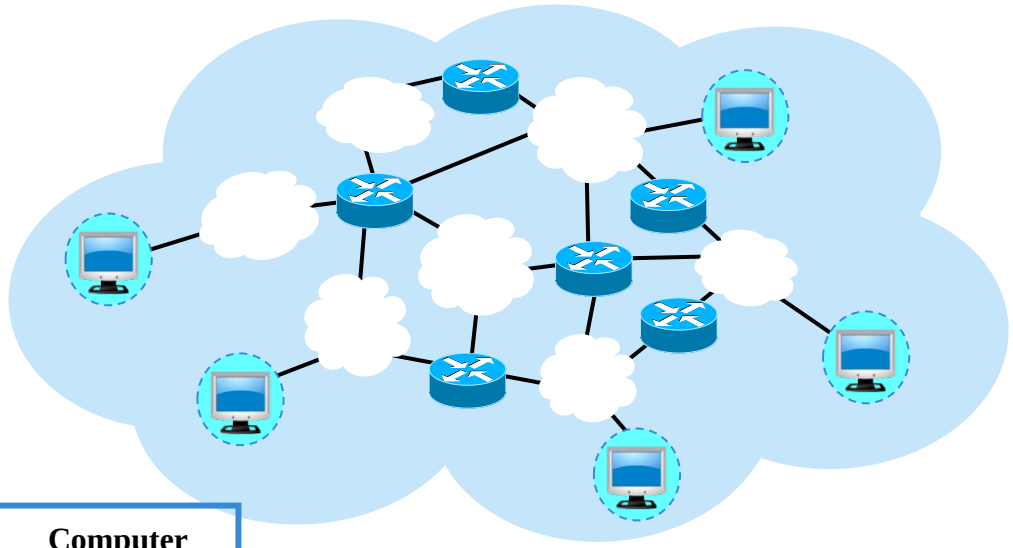
Connected by comm. technology

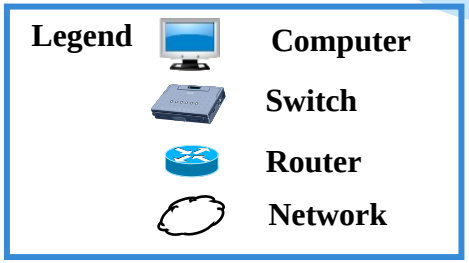




Interconnected collection







A simple computer network

An interconnected network

Services/Functions Provided by a computer network

Communication (通信)

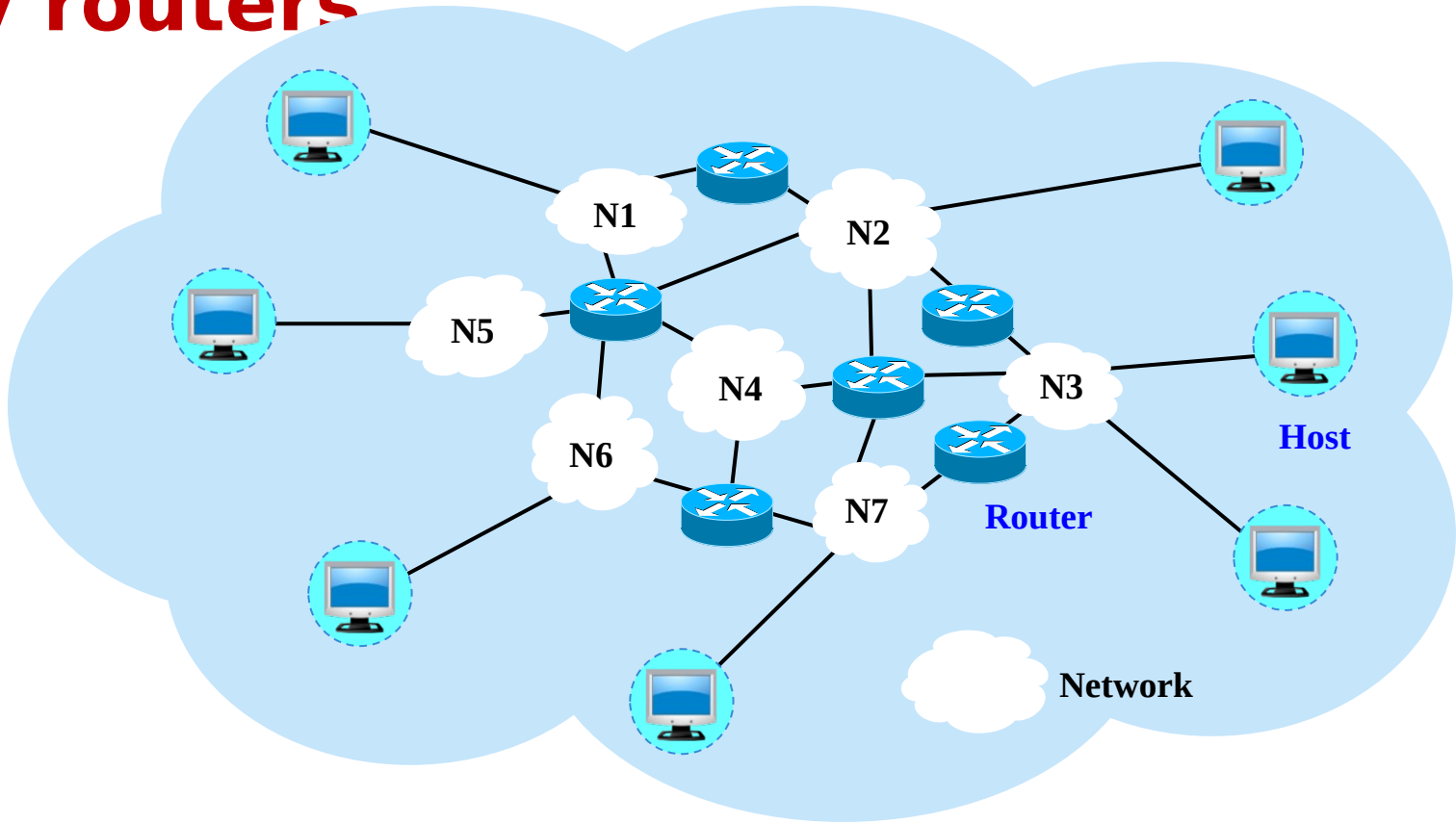
- **Allow users to exchange various information easily , quickly and economically.**
- **as if these user terminals are all directly connected to each other.**

Sharing (资源共享)

- **Sharing of information, software, and hardware, etc.**
- **Because of the existence of the network, these resources are as convenient to use as if they are right next to the user.**

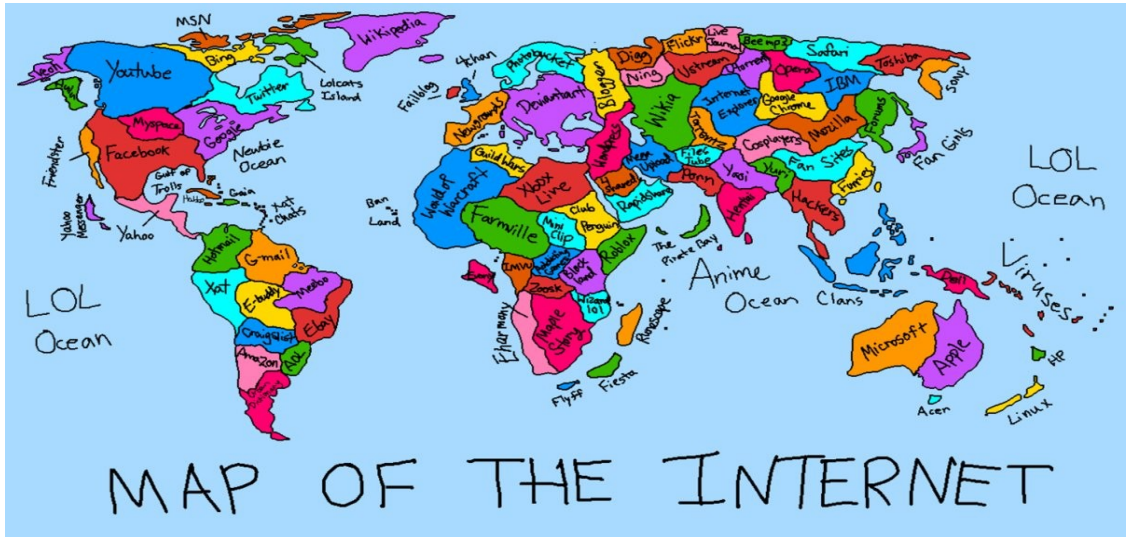
What is the Internet?

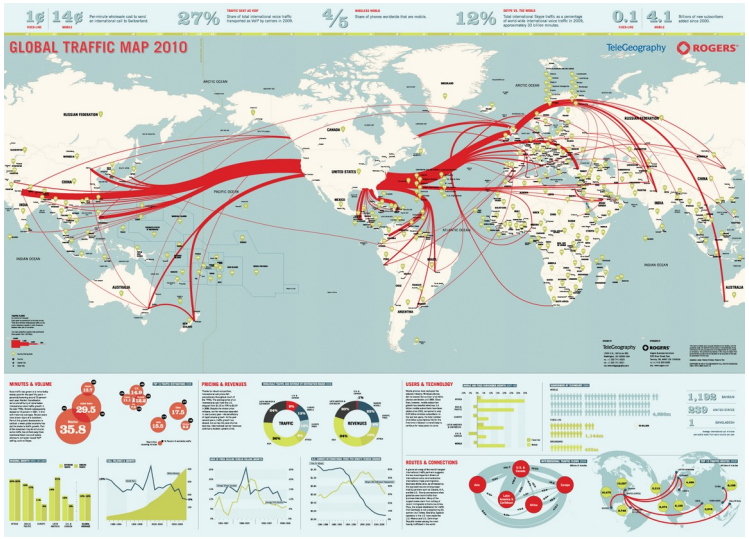
- "a network of networks"
- "collection of networks interconnected by routers"



What is the Internet?

- **Global scale, general purpose, heterogeneous-technologies, public, computer network.**

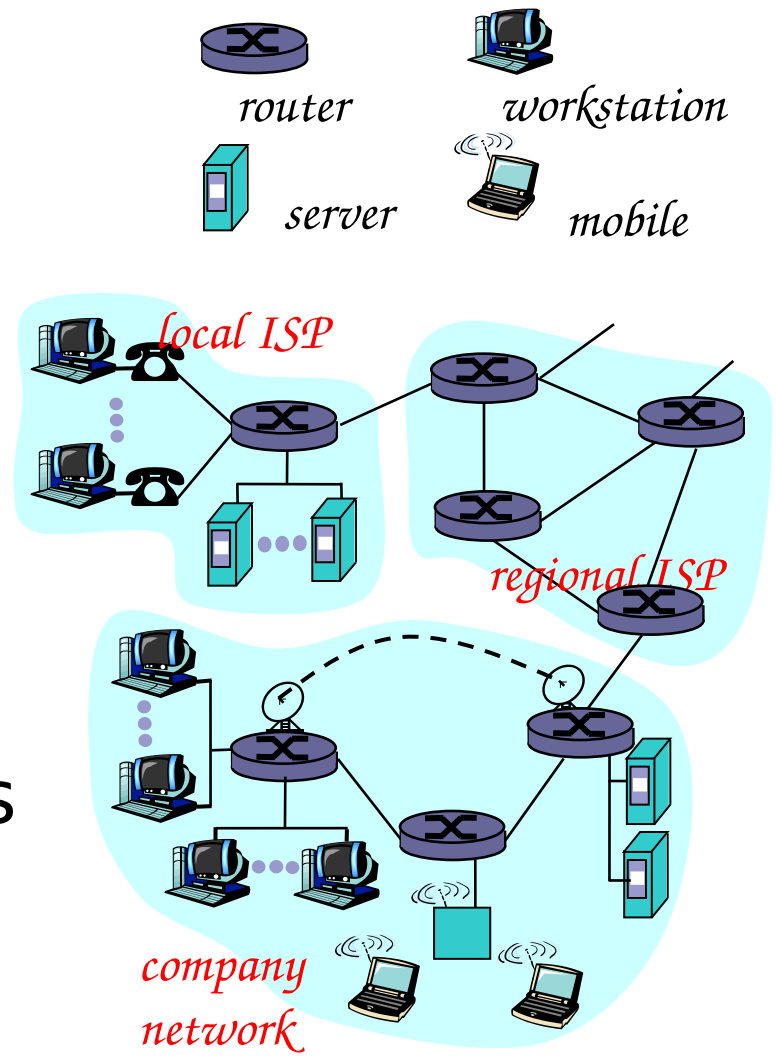




What's the Internet: "nuts and bolts" view

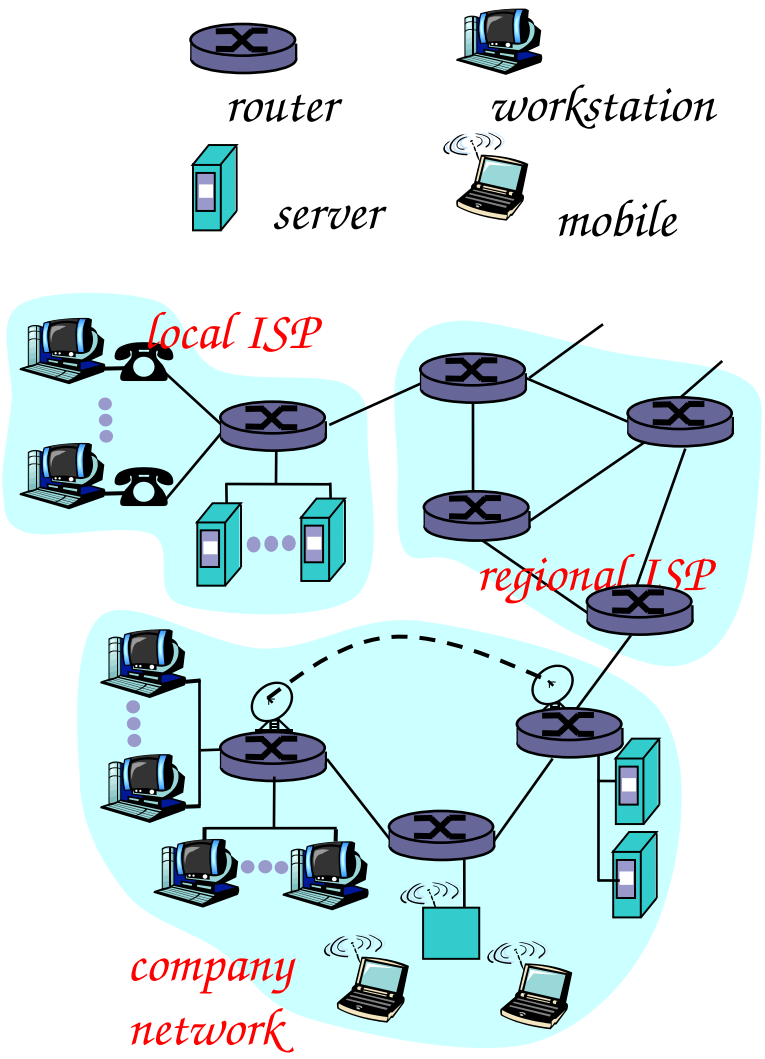
- **Millions of connected computing devices: *hosts, end-systems***
  - PCs, Servers
  - Tablets
  - Phones, Smart Phones
  - Sensors
  - Toasters

  running *network apps*



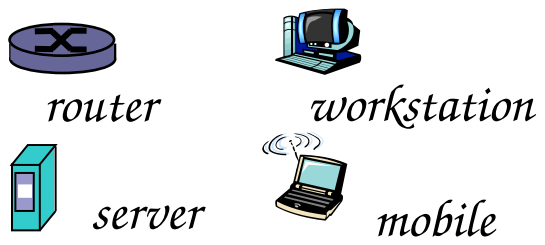
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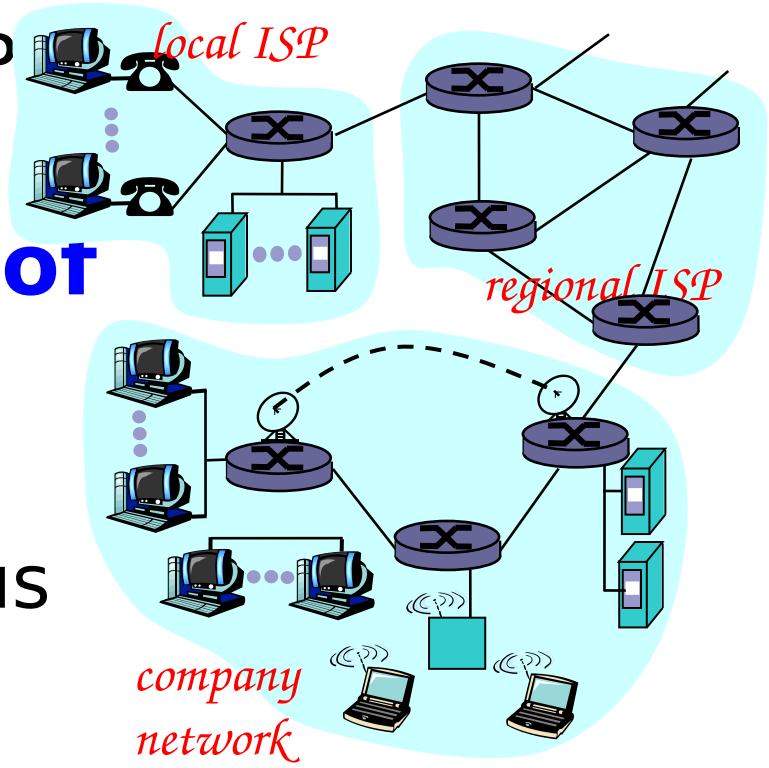
- ***Communication links***
  - fiber, copper,
  - radio, satellite
- ***Routers:***
  - **forward packets (chunks) of data thru network**



What's the Internet: "nuts and bolts" view

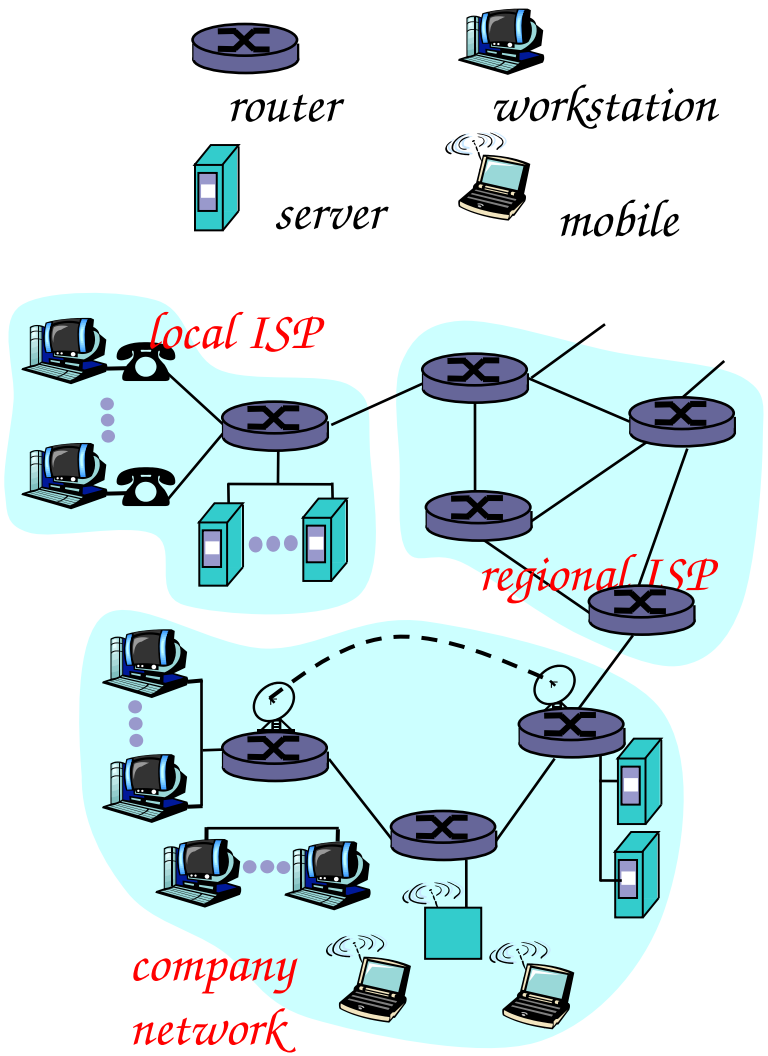
- **Protocols: control sending, receiving of messages**
  - e.g., TCP, IP, HTTP, FTP PPP
- ***Internet:* "network of networks"**
  - loosely **hierarchical**
  - **public Internet** versus **private intranet**





What's the Internet: a service view

- **communication *infrastructure*** enables distributed applications:
  - WWW, email, games, e-commerce, databases, voting, telephony, multimedia, IM, …
  - more?
- **communication services provided:**
  - **Datagram**



What's a protocol?

human protocols:

- **machines rather than humans**
- **all communication activity in Internet governed by protocols**

network protocols:

- **"What's the time?"**
- **"I have a question"**
- **introductions**

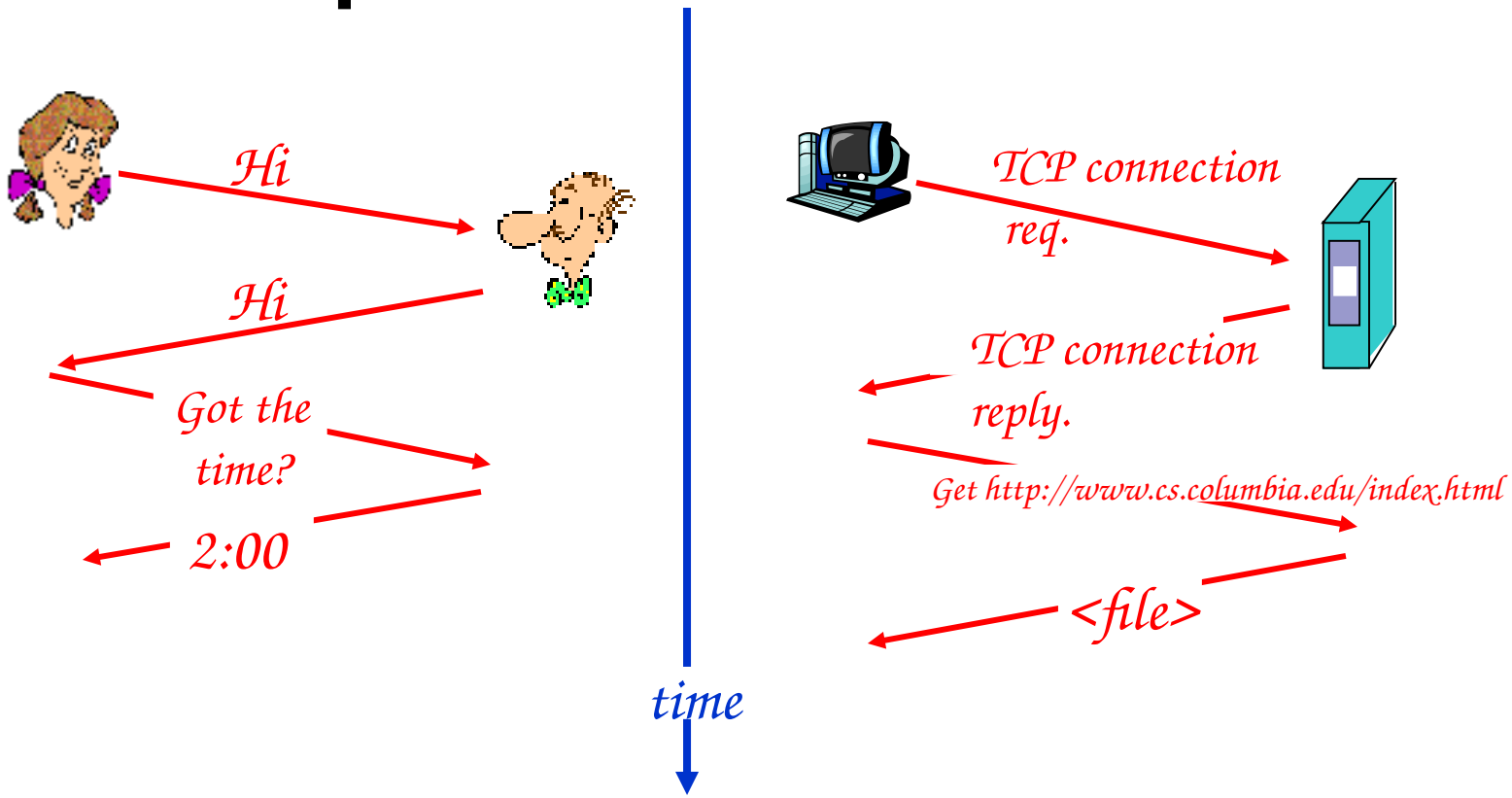
… specific msgs sent

… specific actions taken when msgs received, or other events

Protocols define format & order of messages sent and received among network entities, and actions taken on message transmission and

What's a protocol?

a human protocol and a computer network protocol:



Q: Other human protocol?

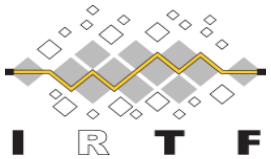
Who is Who in the Internet ?



- Internet Society (ISOC)



- Internet Engineering Task Force (IETF)

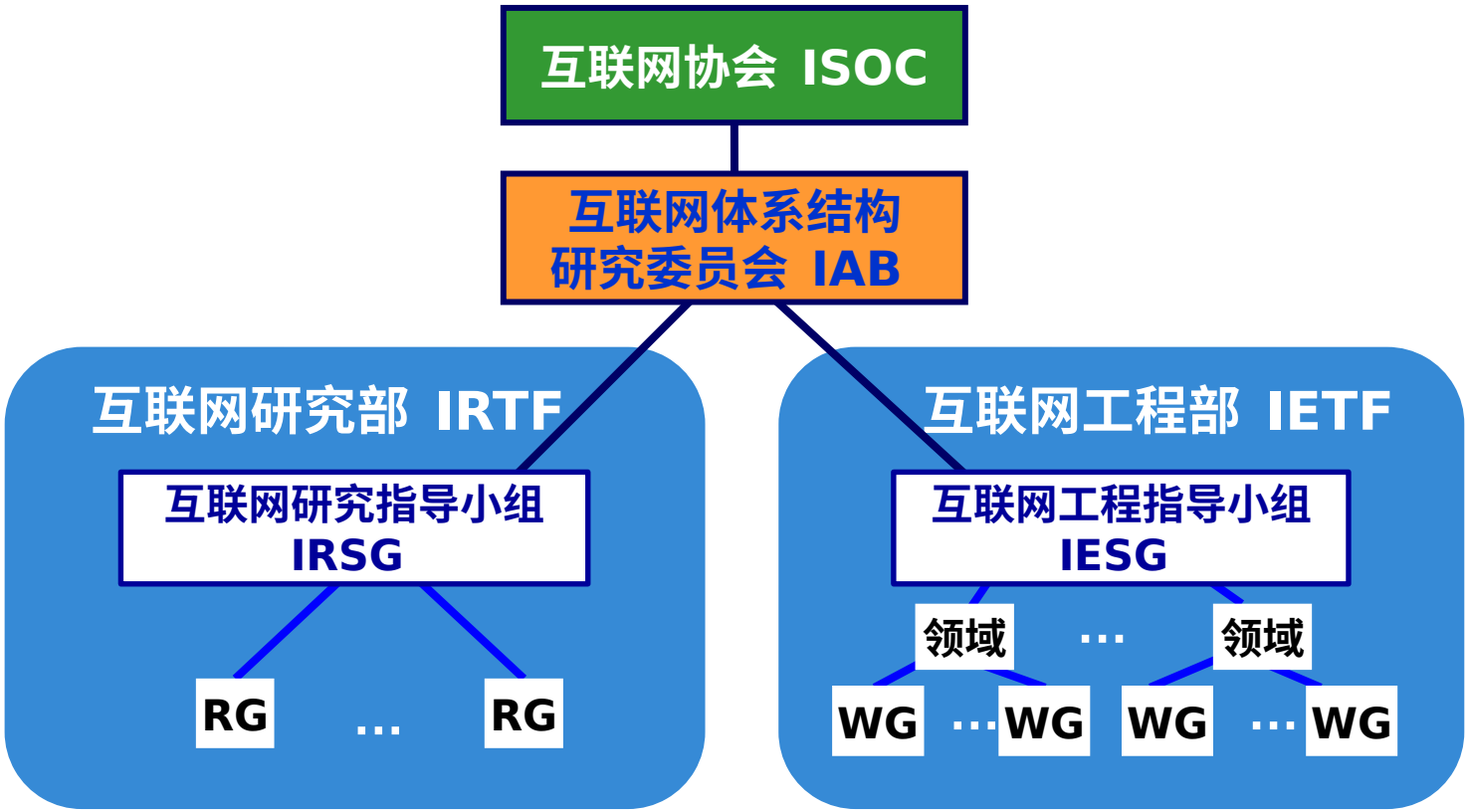


- Internet Research Task Force (IRTF)



- Internet Architecture Board (IAB)

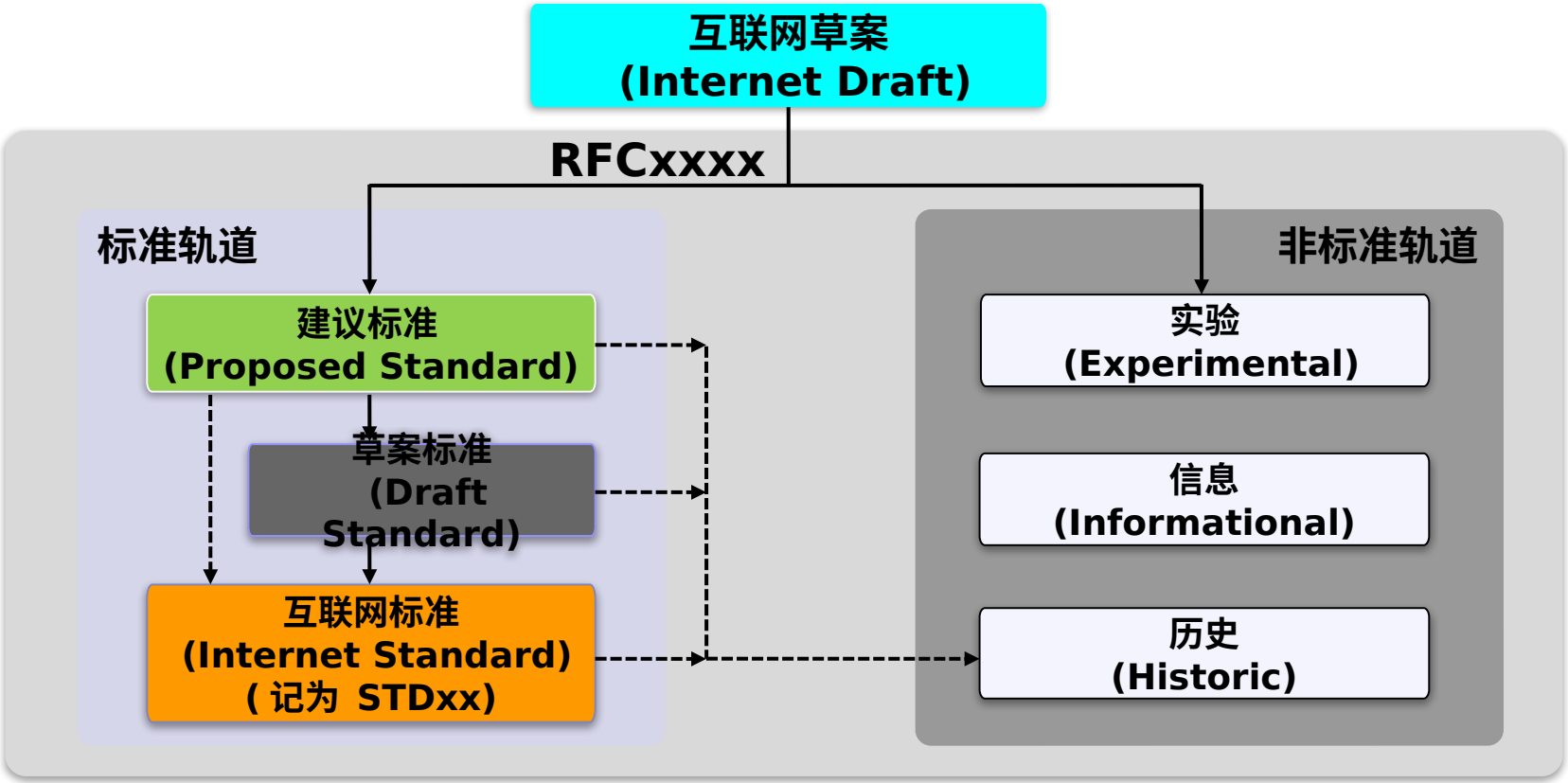
Who is Who in the Internet ?



组织架构

Internet Standardization Process

- **All standards of the Internet are published as RFC (Request for Comments). But not all RFCs are Internet Standards !**



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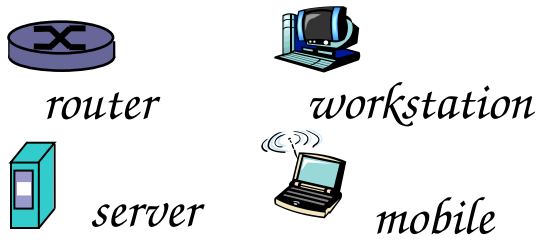
1.7 Protocol layers, service models

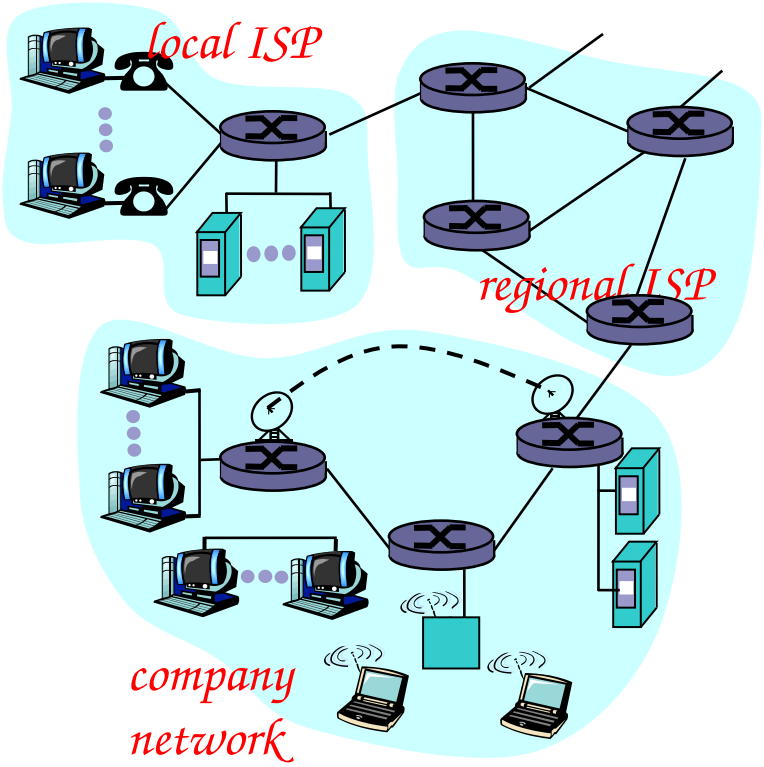
1.8 Architecture, OSI and TCP/IP Models

1.9 Internet History

A closer look at network structure

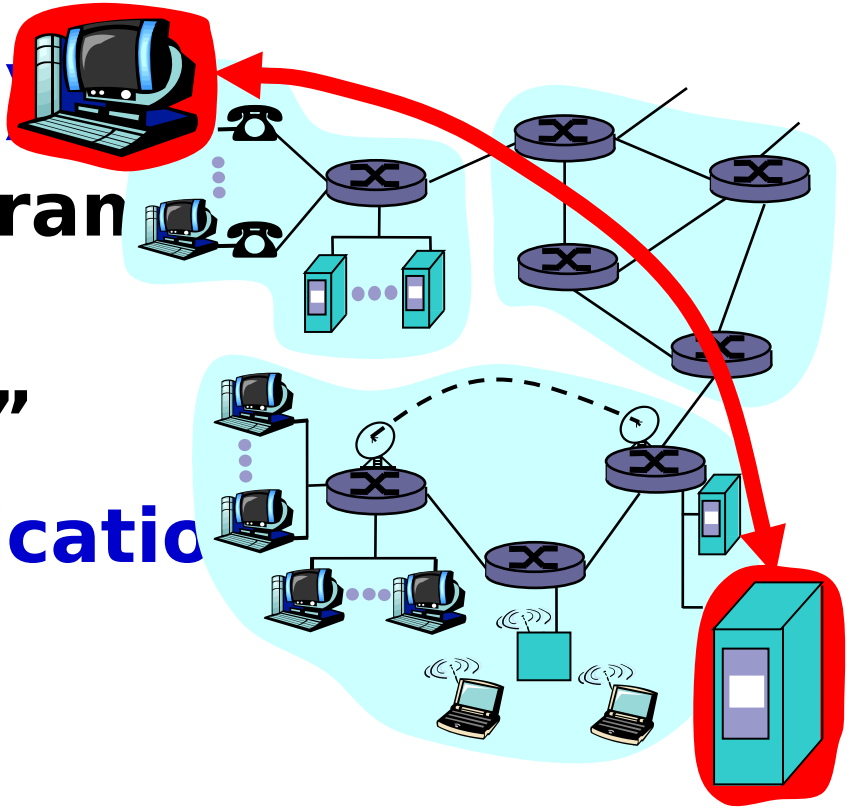
- **Network edge:**
  - applications and hosts
  - resources
- **Network core:**
  - routers
  - network of networks
  - communications
- **physical media, Access networks:**
  - communication links
  - access





The network edge

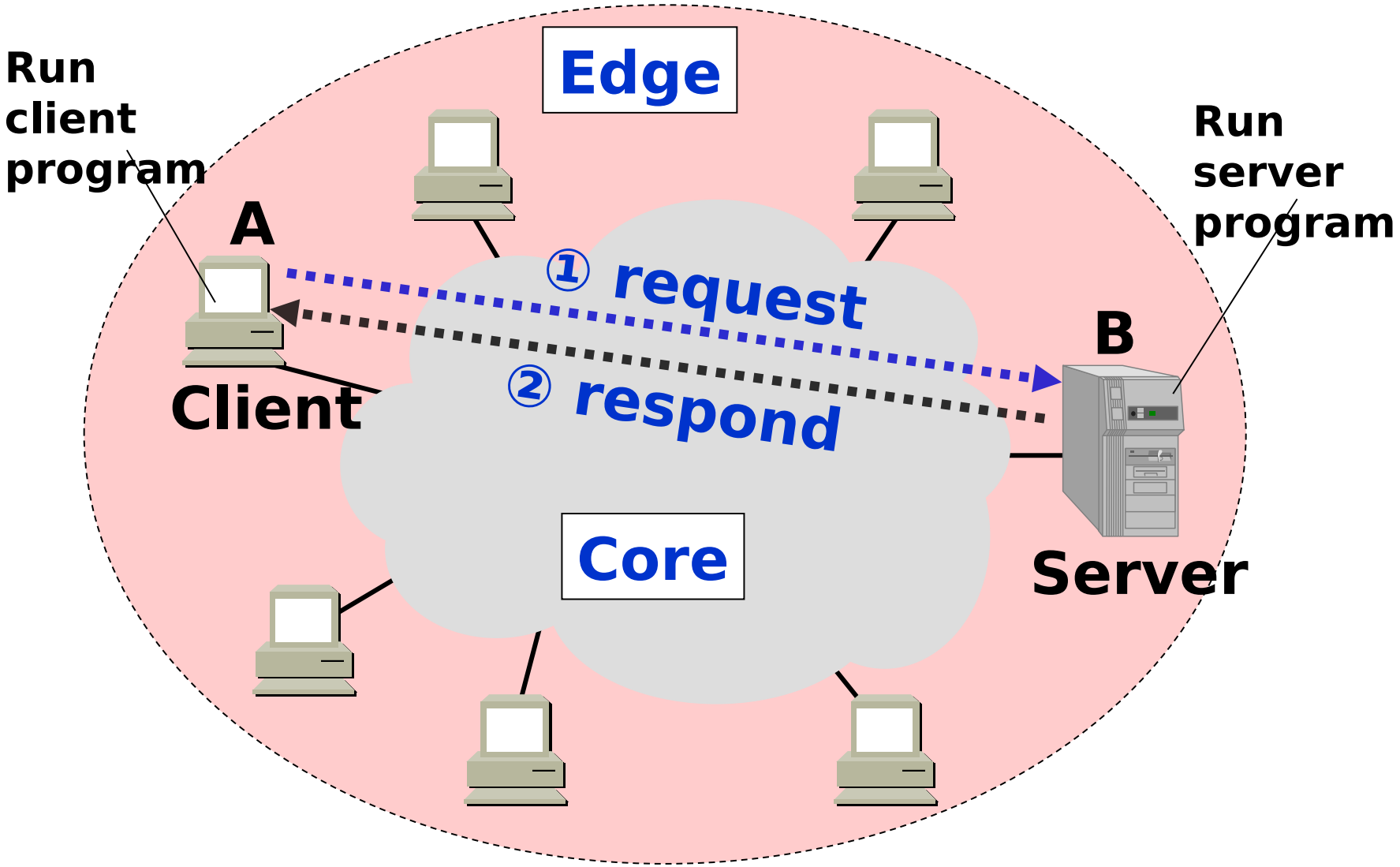
- **End Systems (hosts)**
  - run application program
  - e.g., WWW, email
  - at “edge of network”
- **End-to-End Communicatio**
  - client/server model
  - peer-to-peer model



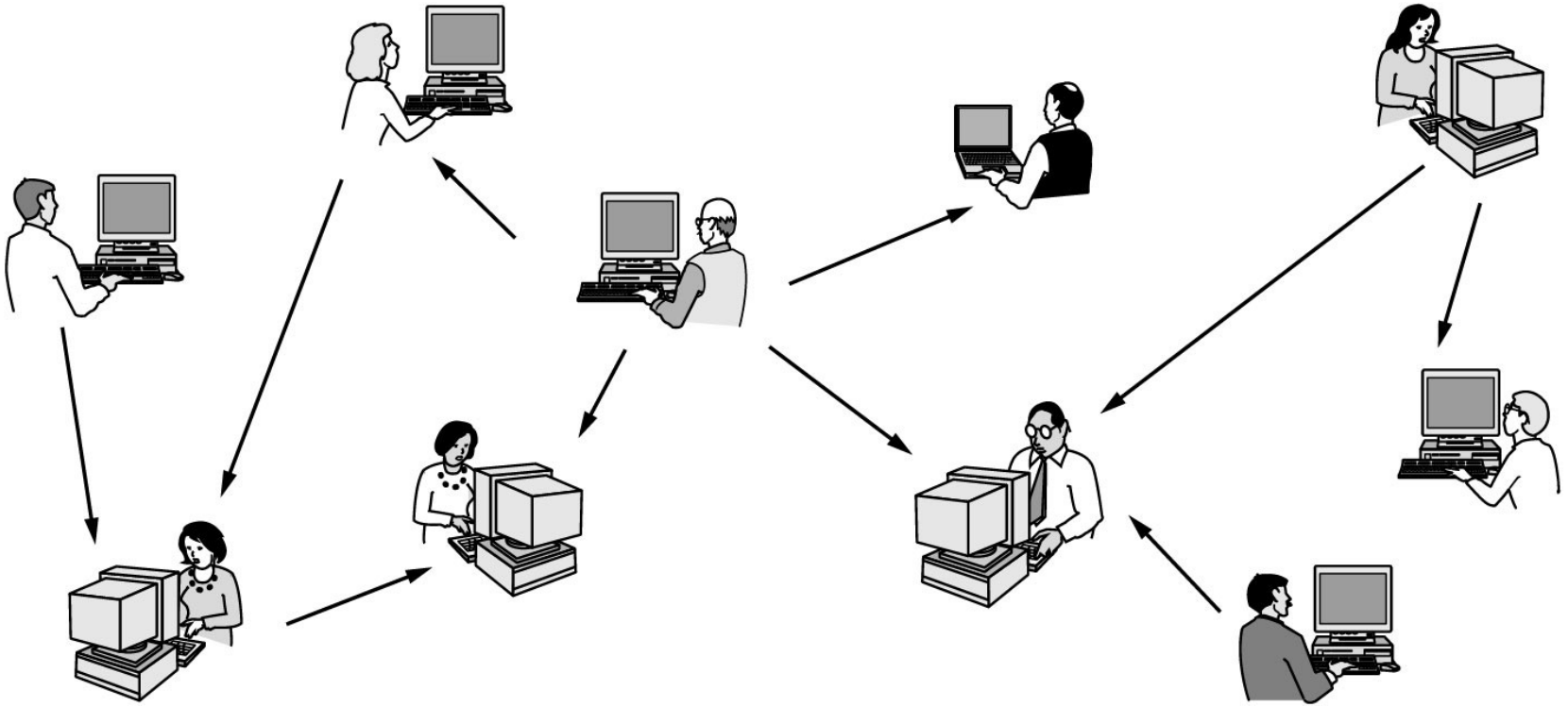
Host A communicate with host B actually means: "a program running on host A communicates with another program running on host B"

End-to-End Communication = process-to-process Communication

Client / Server Model

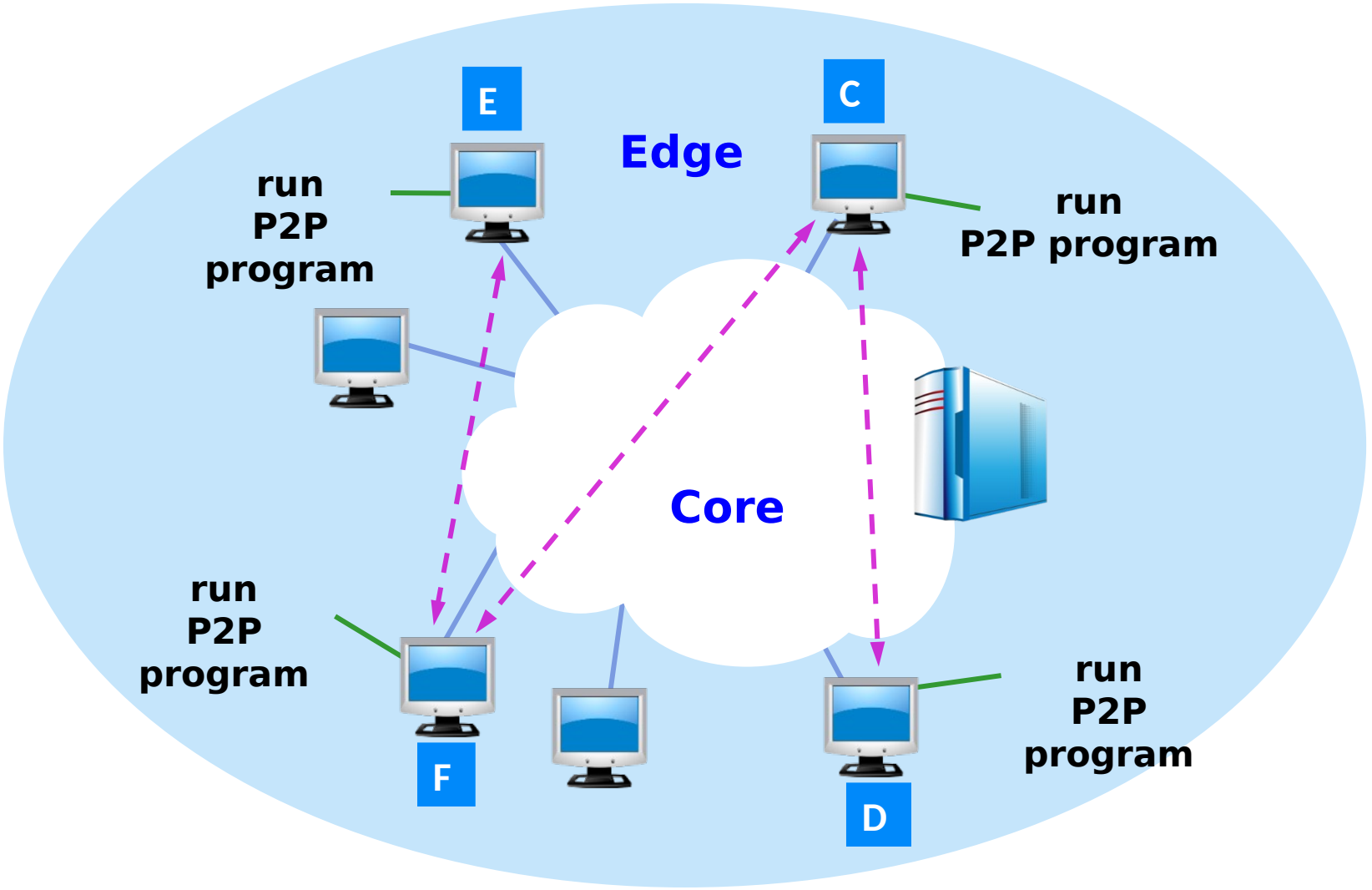


Peer-to-Peer model



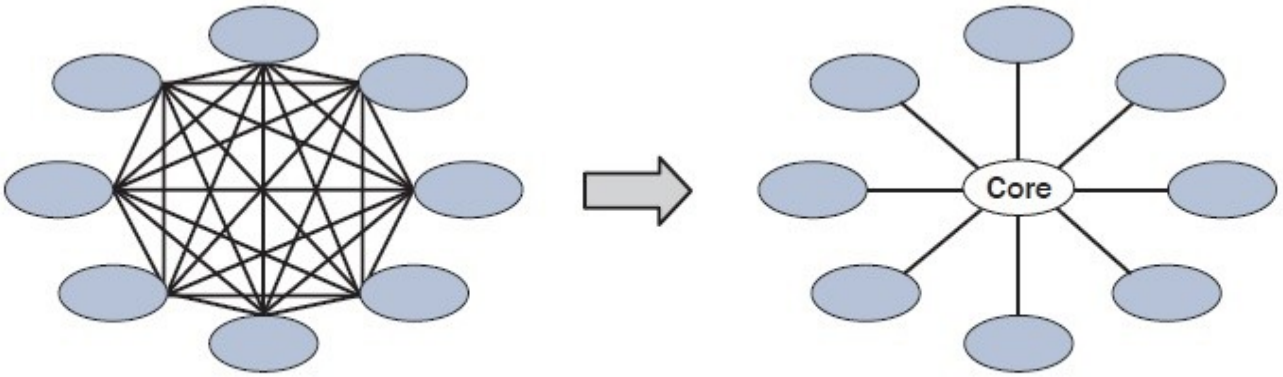
In a peer-to-peer system there are no fixed clients and servers.

Peer-to-Peer model



The Network Core

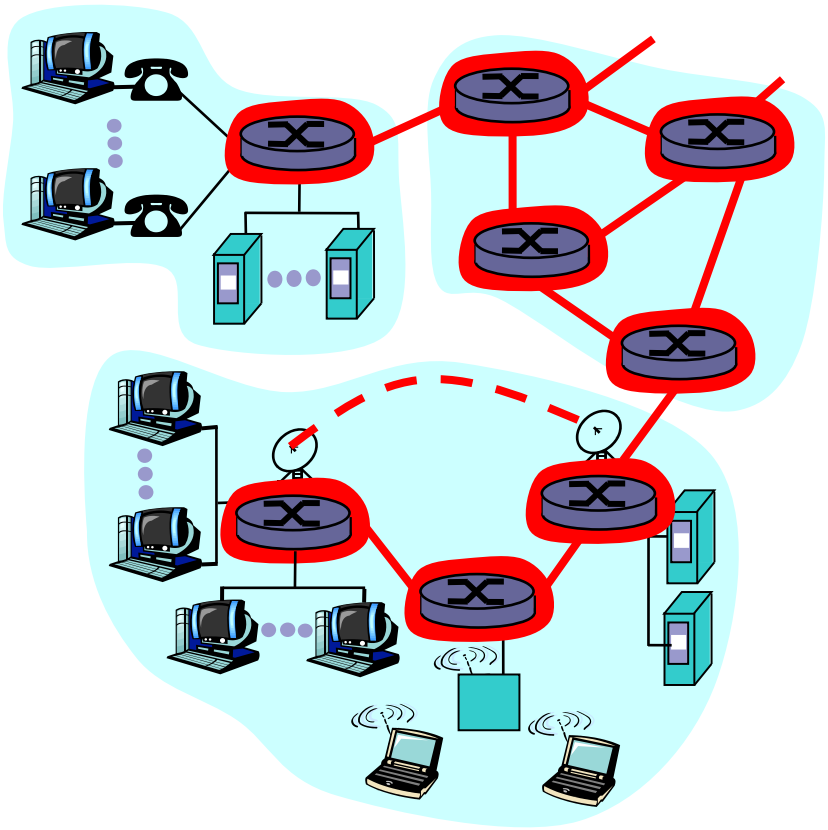
- **Network core provides the capability to scale the size of the network in a structured fashion that minimizes overall complexity when the size of the network grows**



Connectivity without core vs. with core

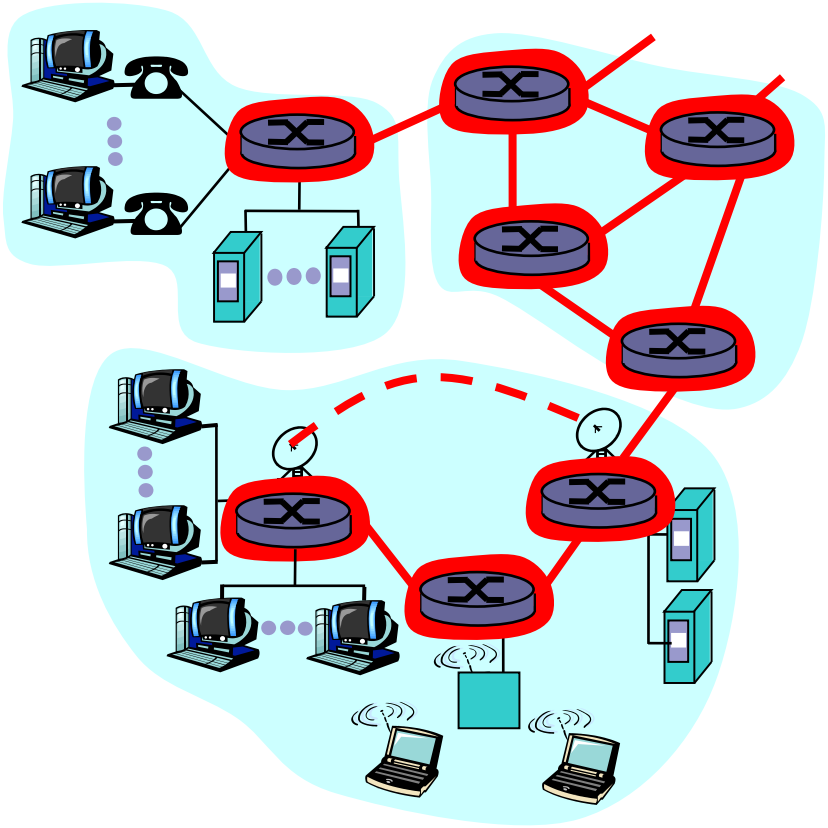
The Network Core

- Generally: mesh of interconnected routers
- Support end system communication.
- Q: How is data transferred through core?



The Network Core

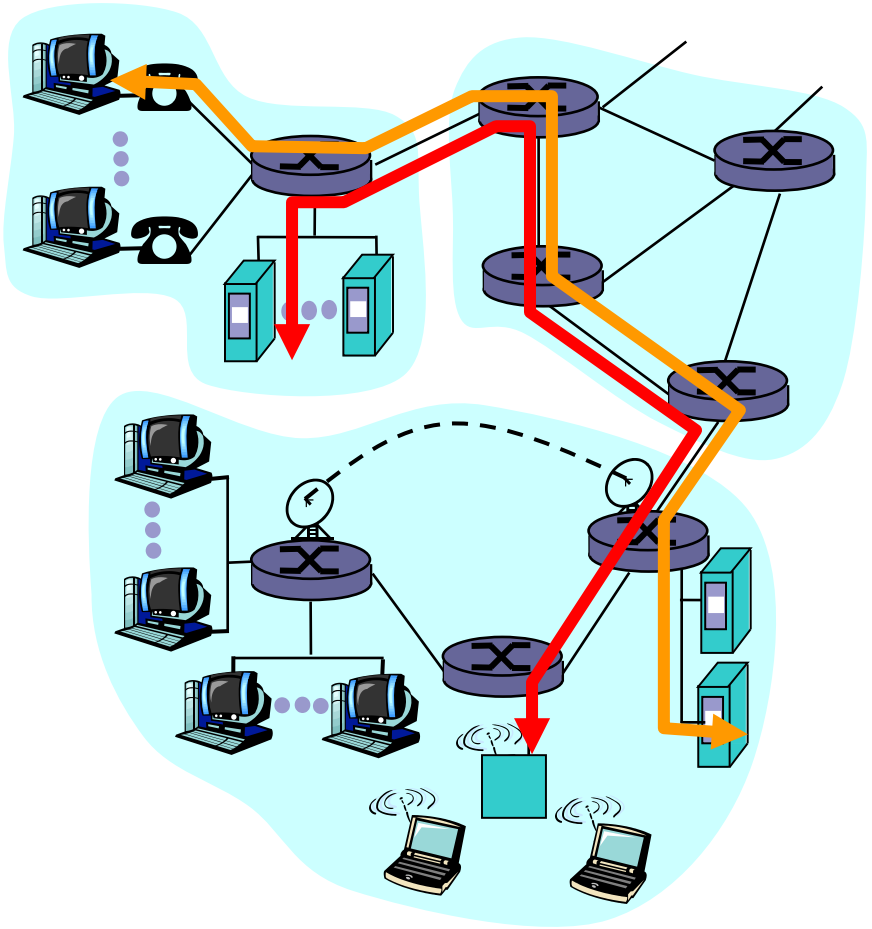
- **Solution**
  - **Circuit switching:** dedicated circuit per call: telephone net
  - **Message switching:** application units
  - **Packet-switching:** data sent thru net in discrete “chunks”



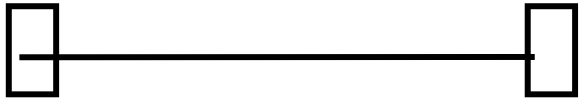
Network Core: Circuit Switching

End-end resources reserved for "call"

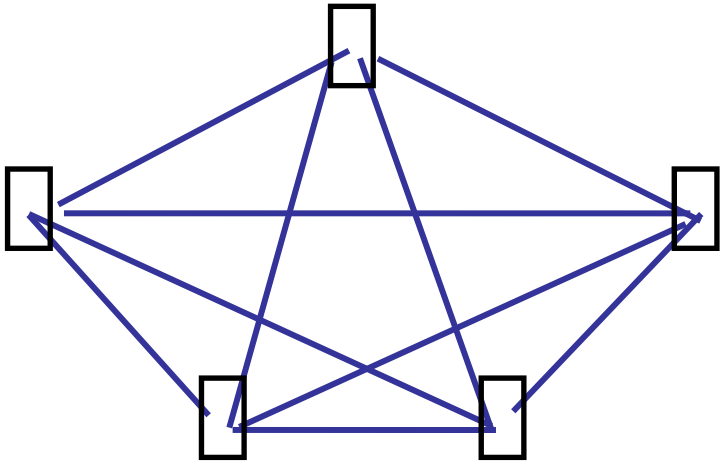
- **Resources: link bandwidth, switch capacity, etc.**
- **dedicated resources: no sharing**
- **circuit-like (guaranteed) performance**



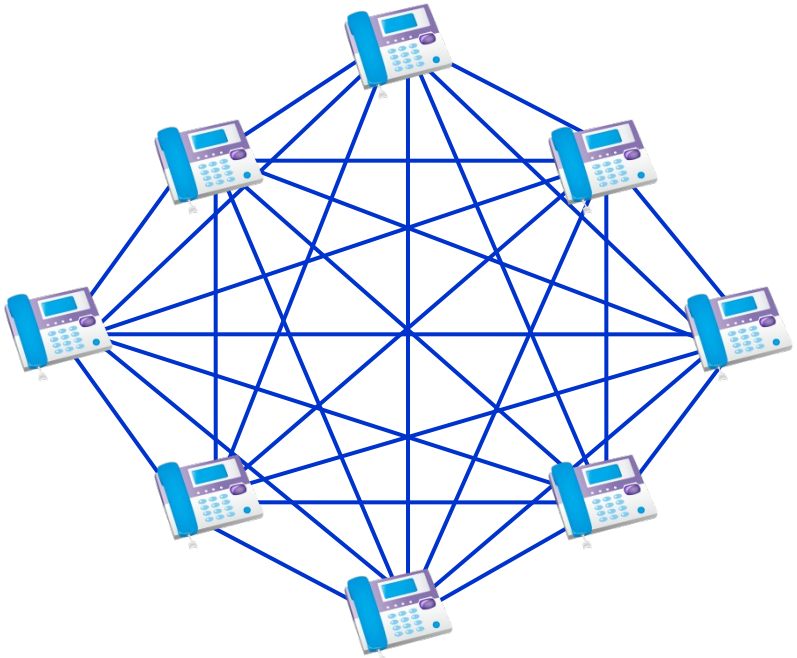
Telephone network uses circuit switching



Communication between 2 telephones: just one pair of wires



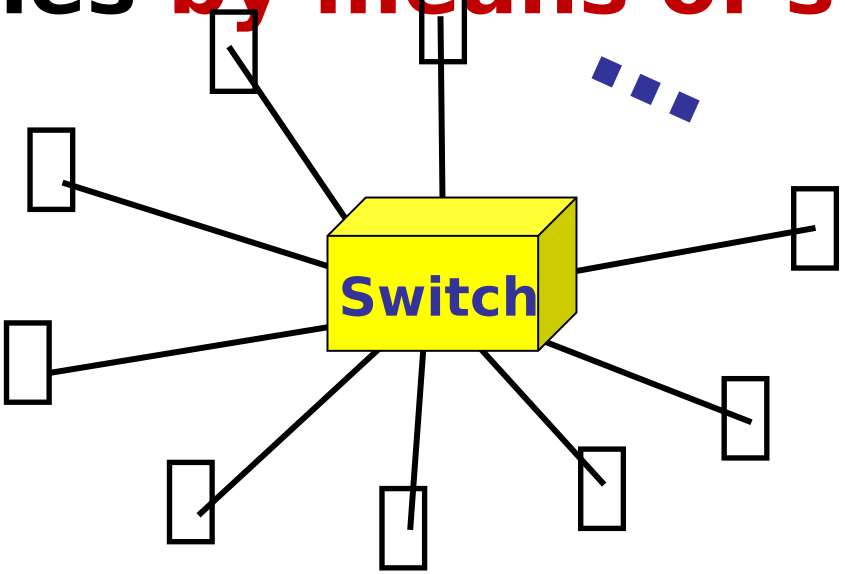
Communication between any 2 of 5 telephones: 10 pairs of wires required



Direct connection between any 2 of N telephones requires N(N – 1)/2 pairs of wires!!!

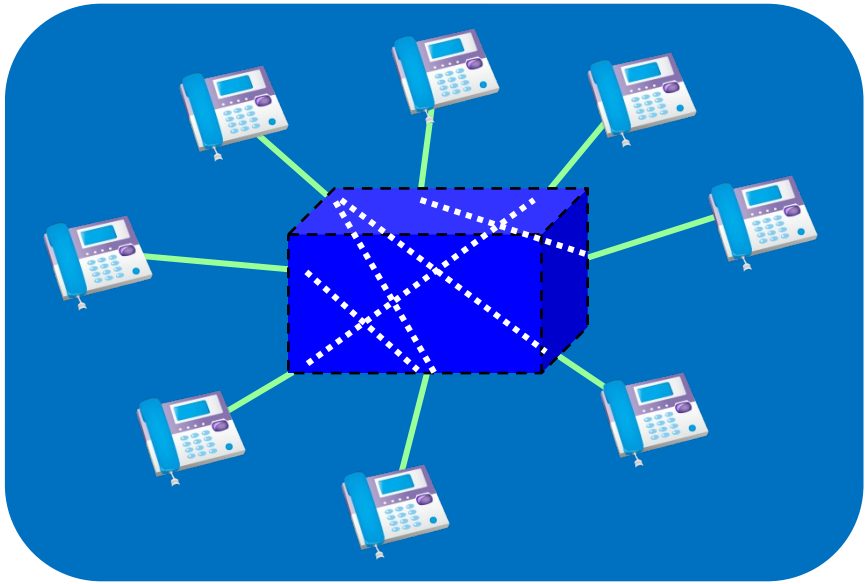
Telephone network uses circuit switching

- When the number of telephones increases, it is a more economical and efficient method to implement communication between telephones by means of switches.

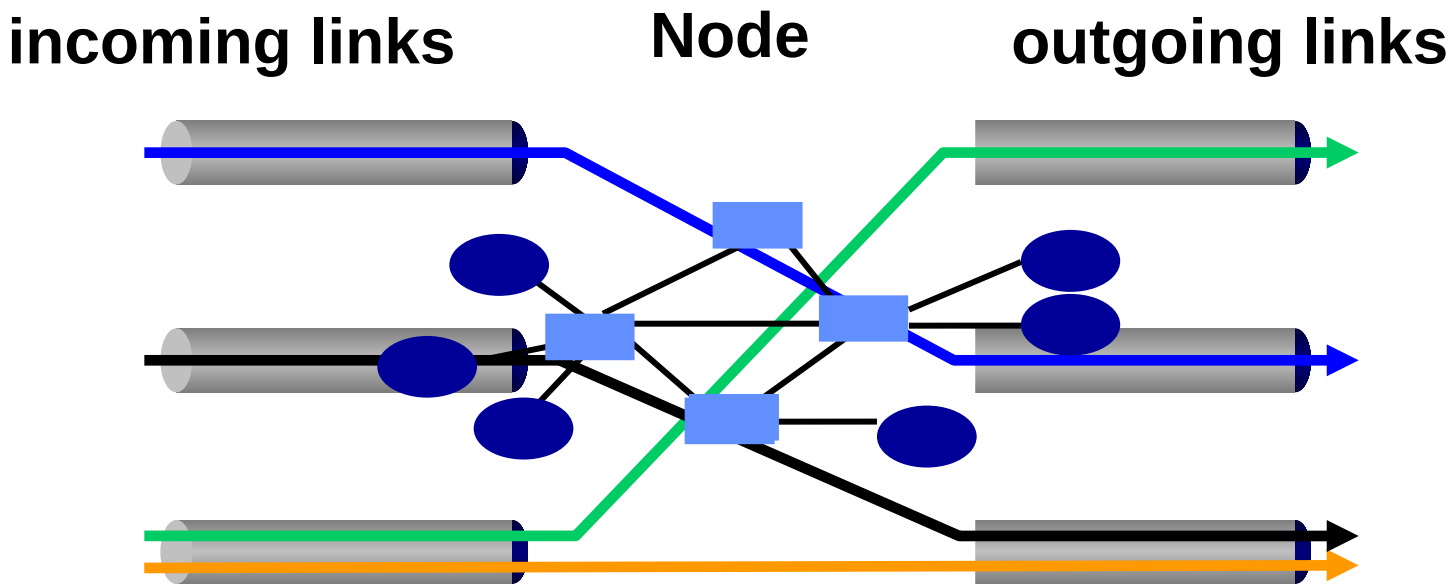


What does “switching" mean?

- **The transfer of one telephone line to another so that they are connected.**
- **From the perspective of the allocation of communication resources, "switching" is to dynamically allocate the resources of the transmission line in some way.**

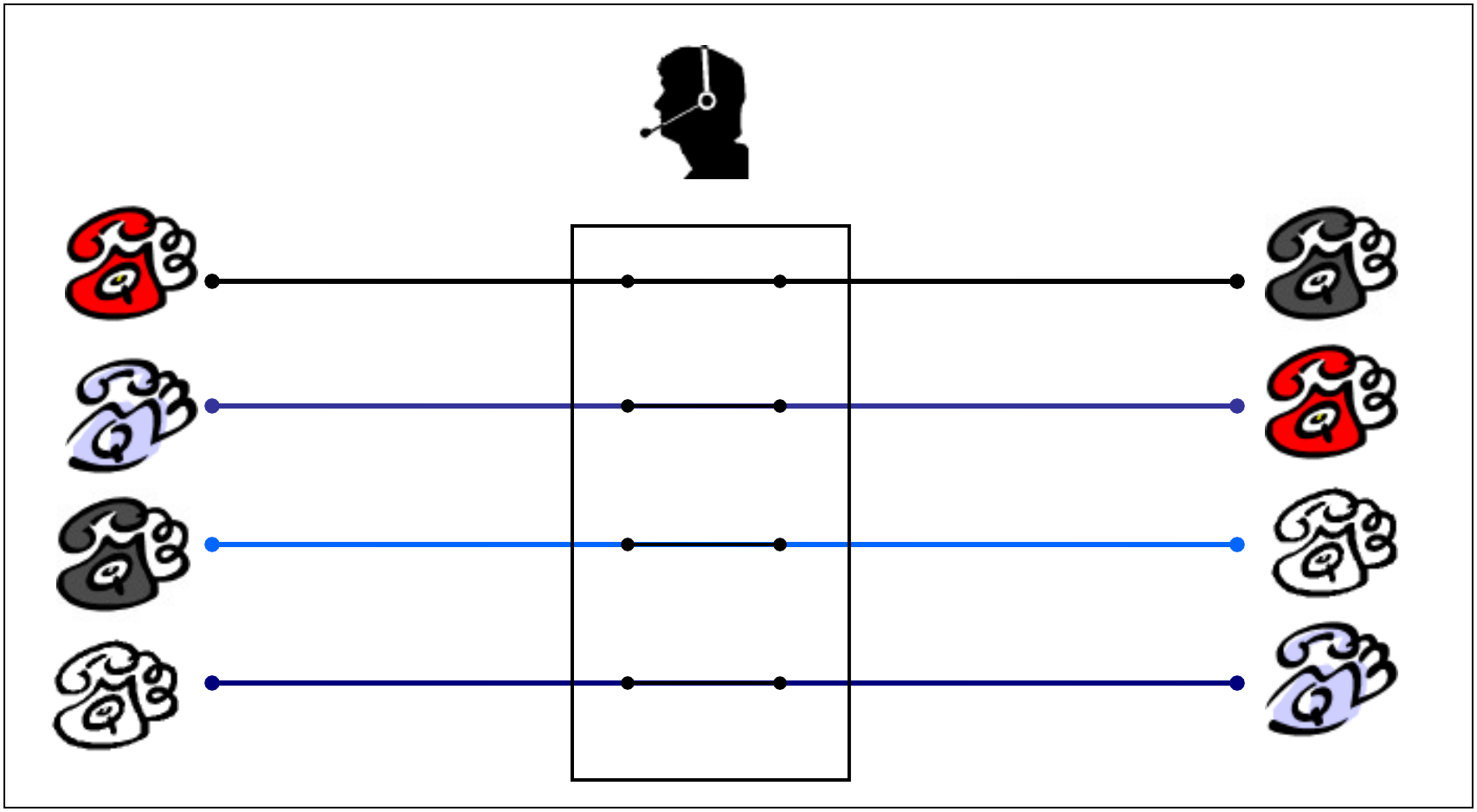


The switch in "circuit switching"



How does the node connect the incoming link to the outgoing link?

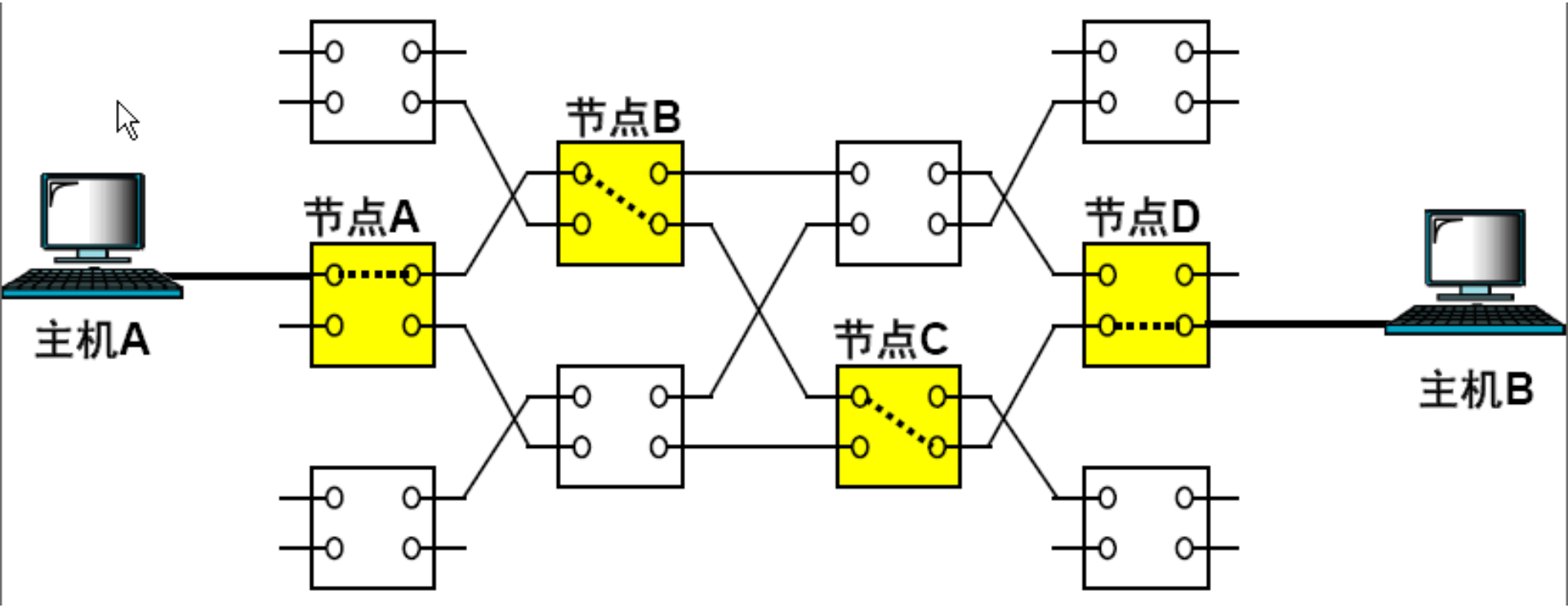
Circuit Switching With Human Operator



Computer network uses circuit switching

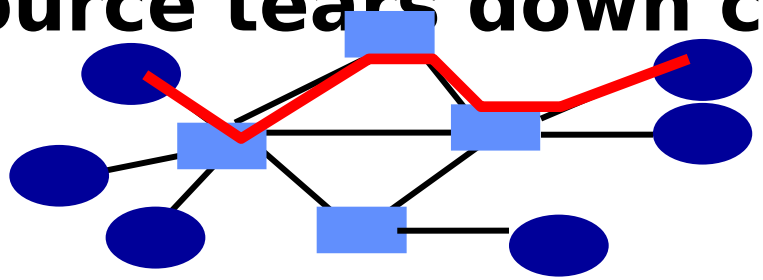
3 phases:

Circuit Establishment, Data Transfer, Circuit Release

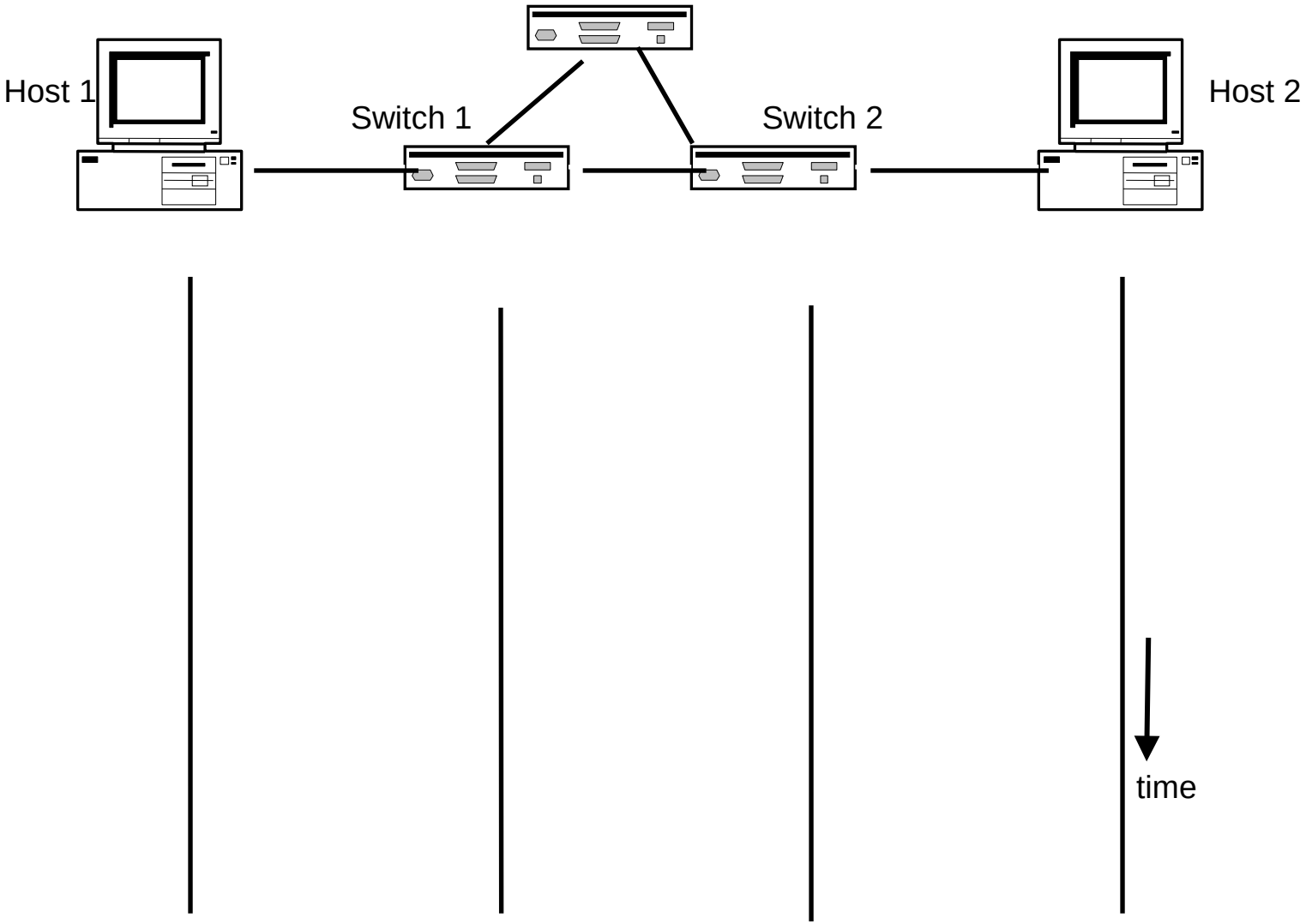


Computer network uses circuit switching

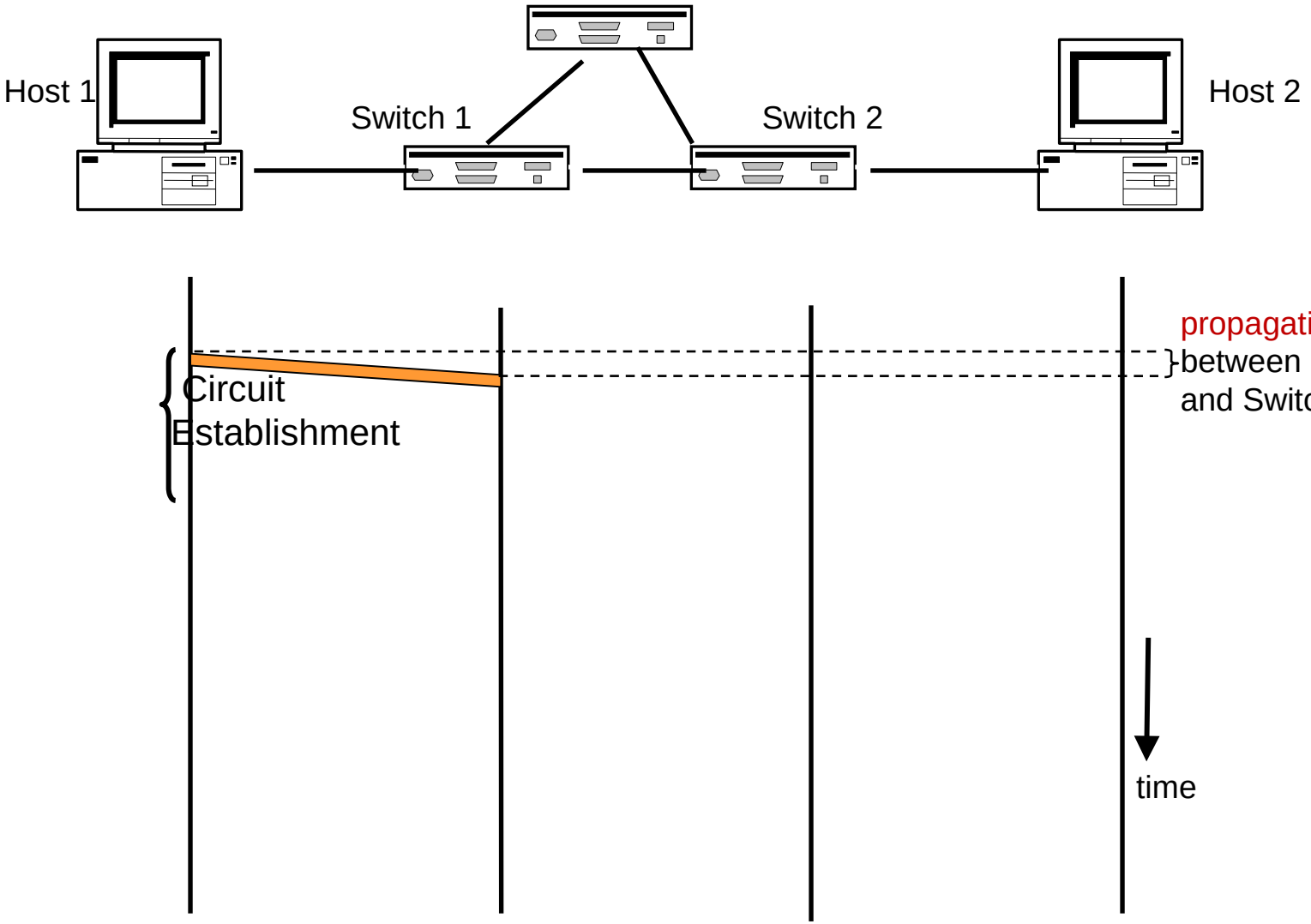
- **Establish: source creates circuit to destination**
  - Nodes along the path store connection info
  - And reserve resources for the connection
  - If circuit not available: “Busy signal”
- **Transfer: source sends data over the circuit**
  - No destination address, since nodes know path
  - Continual stream of data
- **Release: source tears down circuit when done**



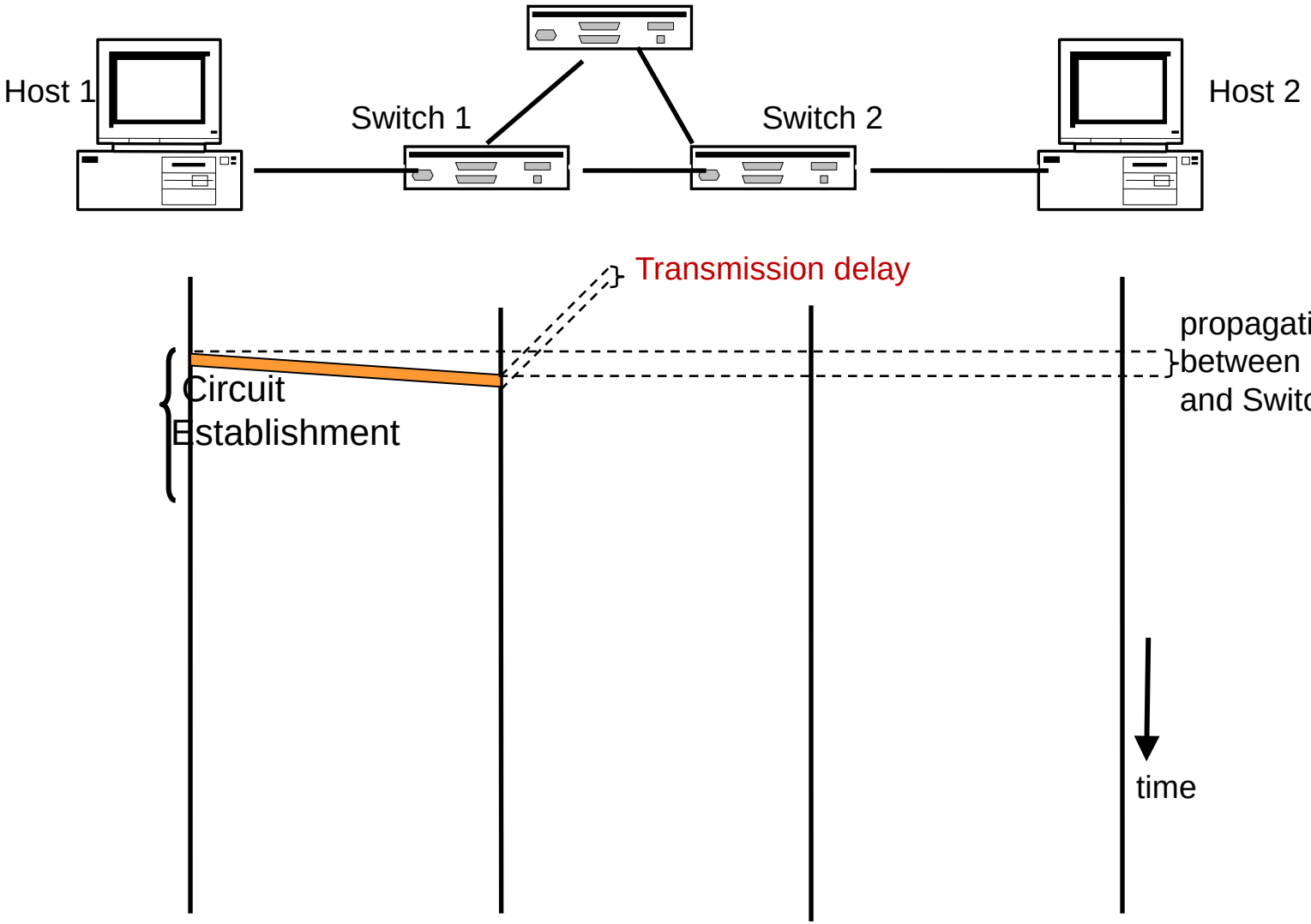
Timing in Circuit Switching



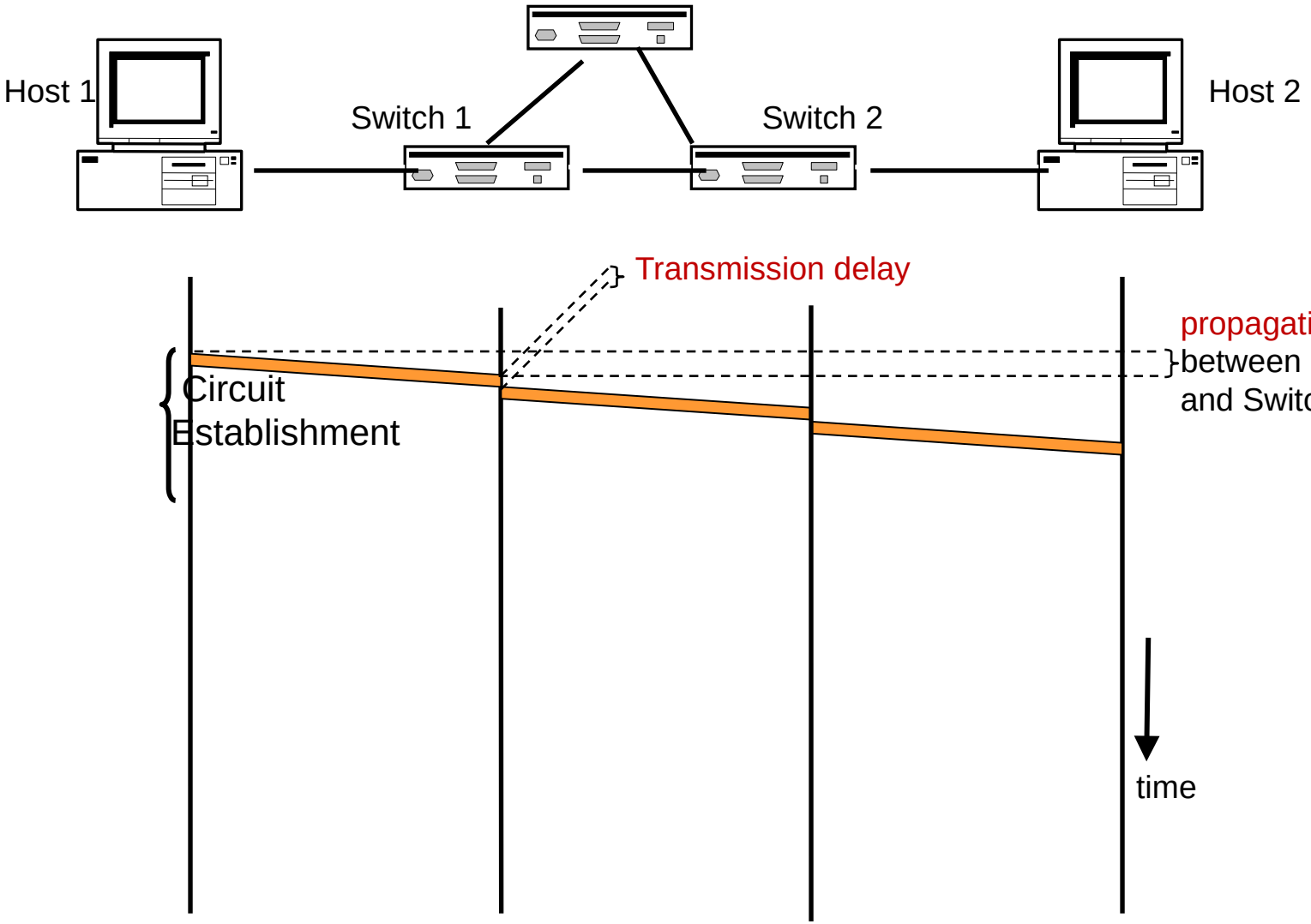
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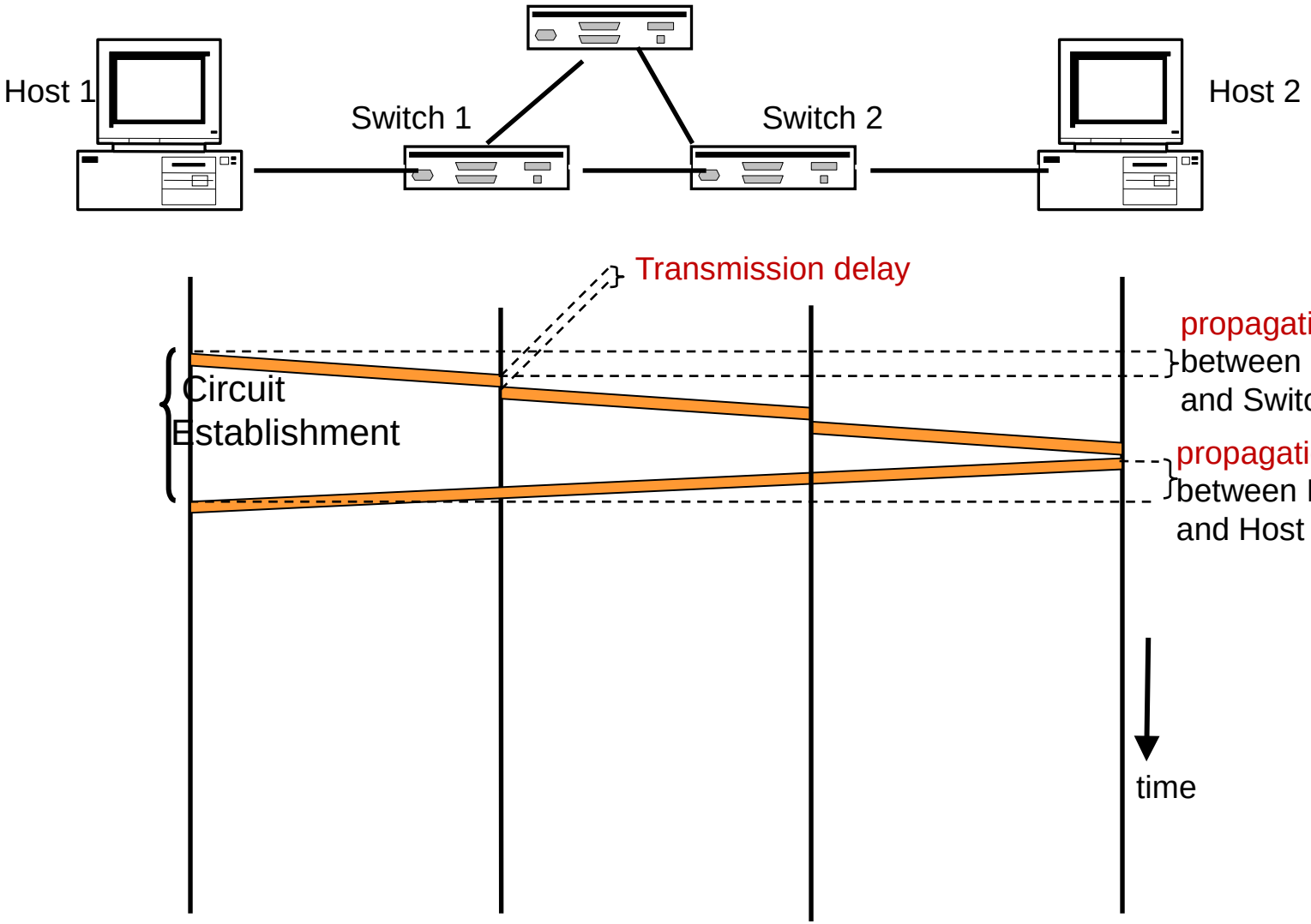
Timing in Circuit Switching



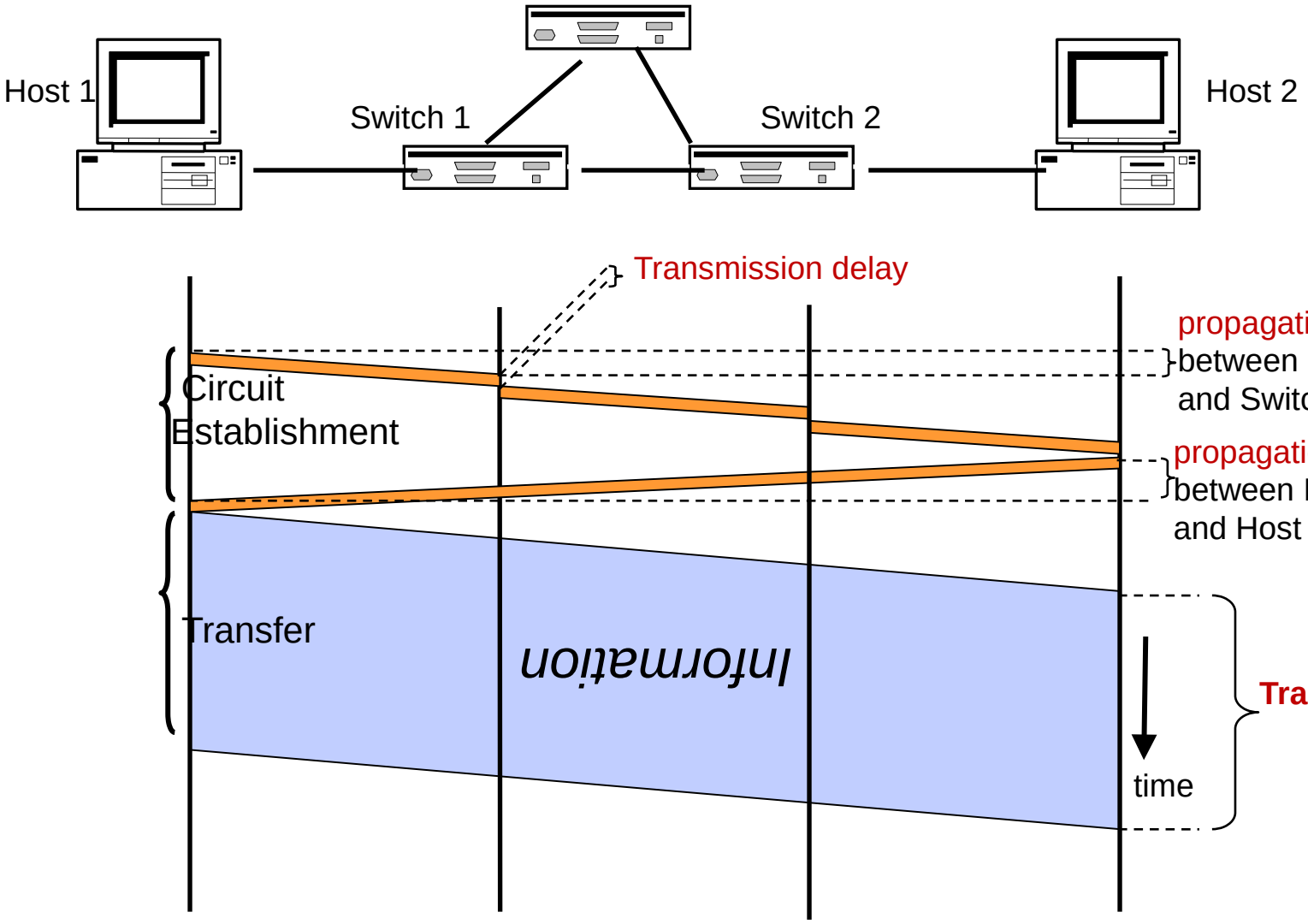
Timing in Circuit Switching



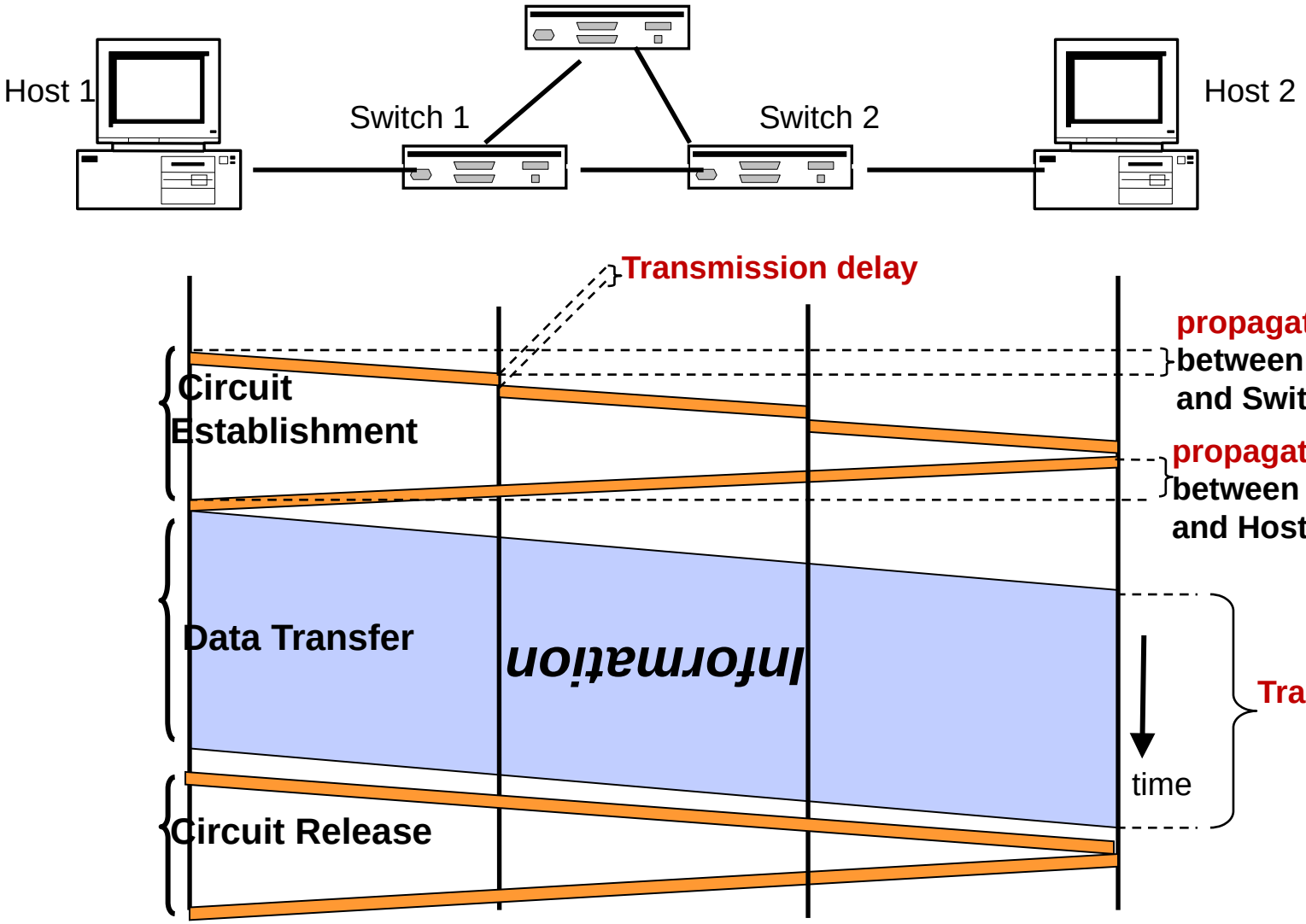
Timing in Circuit Switching



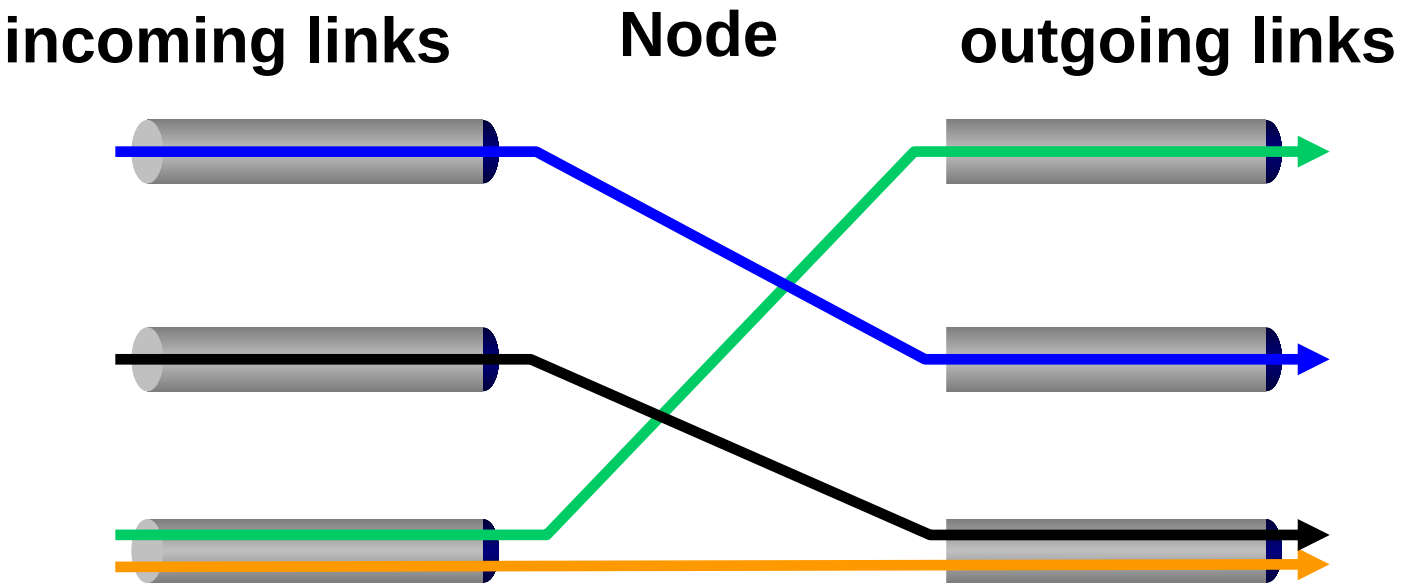
Timing in Circuit Switching



Timing in Circuit Switching



Circuit Switching: Sharing a link

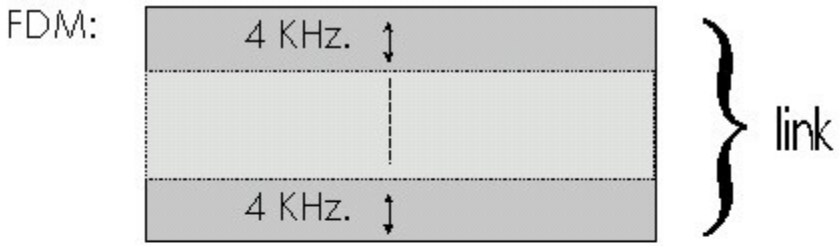


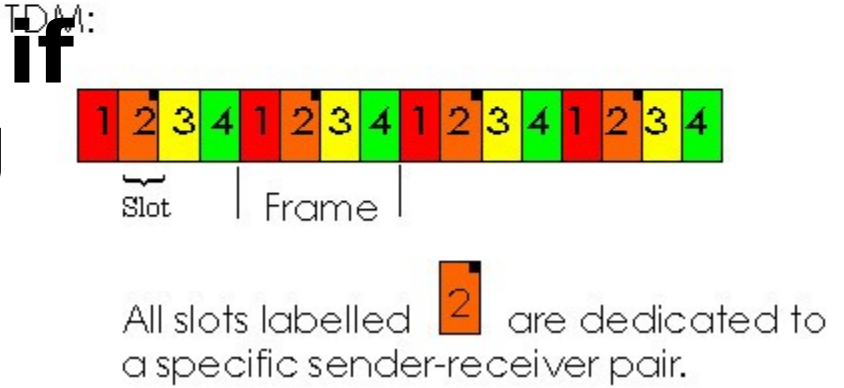
How do the black and orange circuits share the outgoing link?

Circuit Switching: Sharing a link

network resources (e.g., bandwidth) divided into "pieces"

- pieces allocated to calls
- resource piece *idle* if not used by owning call *(no sharing)*
- dividing link bandwidth into "pieces"
  - frequency division (FDM)





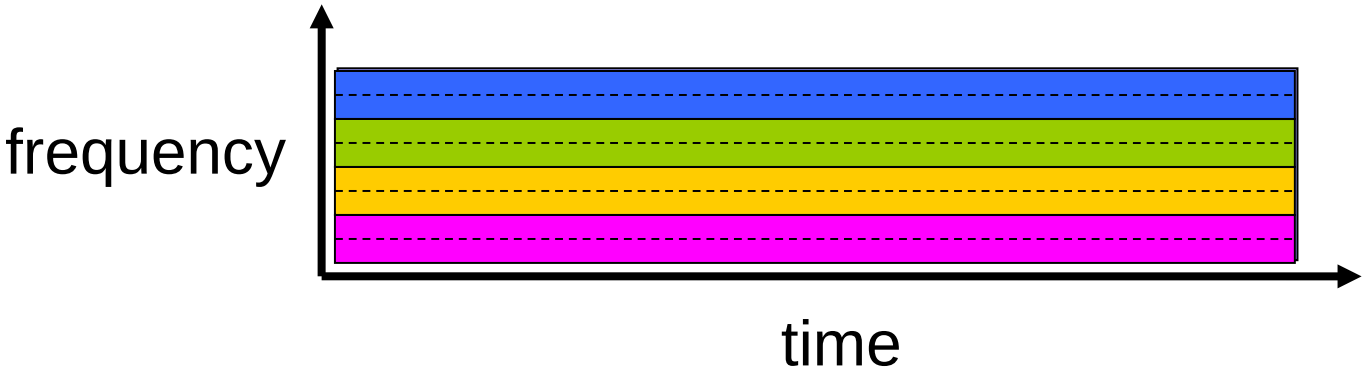
Circuit Switching: Sharing a link

Example:

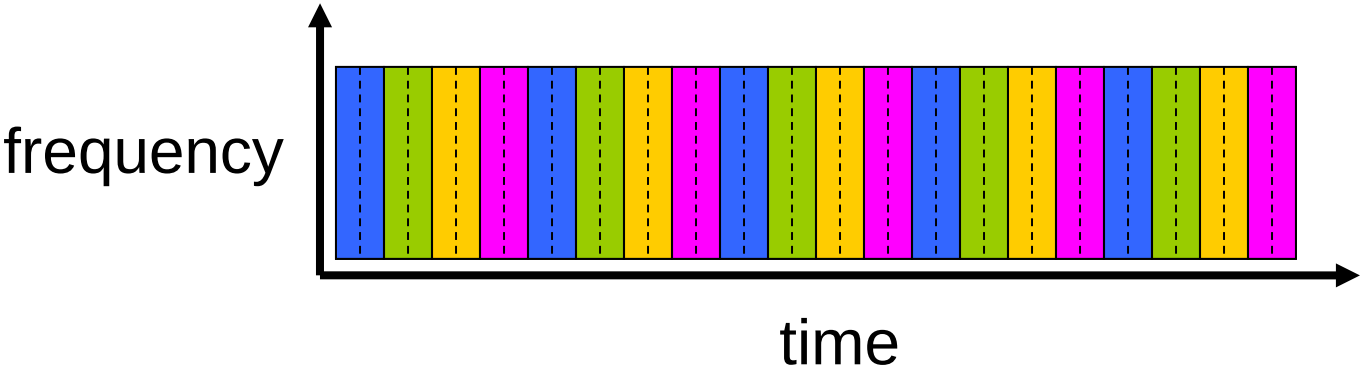
4 users



FDM



TDM



Network Core: Packet Switching

each end-end data stream divided into packets



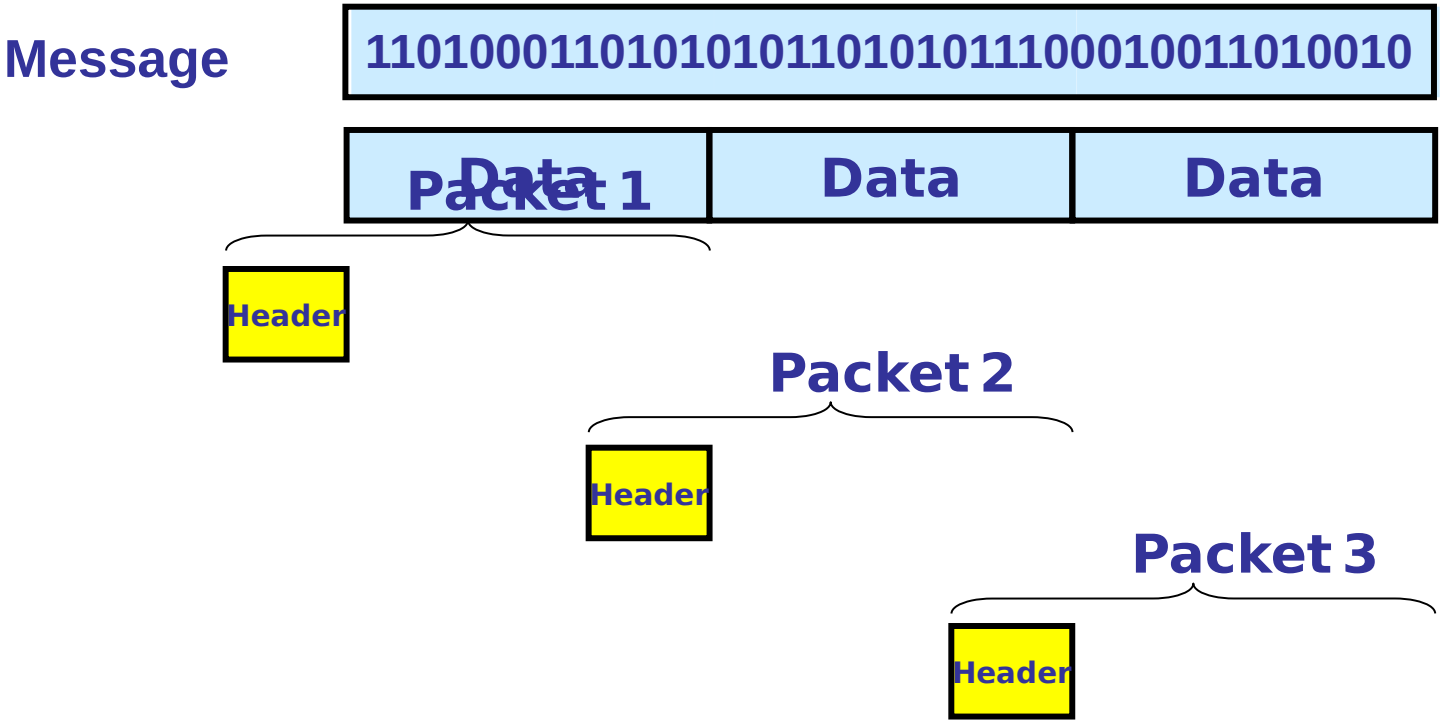
Header and Trailer carry control information (e.g., destination address, checksum)

- Each packet traverses the network from node to node along some path (**Routing**) based on header info.
- Usually, once a node receives the **entire** packet, it stores it (hopefully briefly) and then forwards it to the next node (**Store-and-Forward Networks**)

Build Packets

The sender first divides the message into shorter, fixed-length data segments

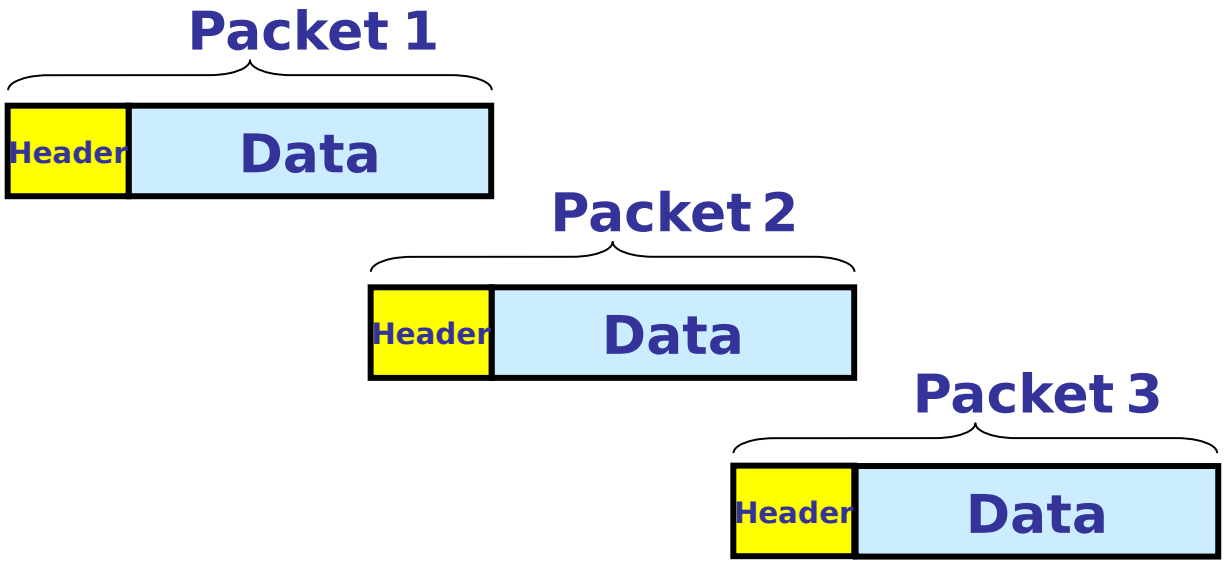
Then a header is added in front of each data segment to form a packet.



Send Packets

Send each packet to the receiver (assuming the receiver is on the left).

Packet-switched networks use "packets" as data transmission units.

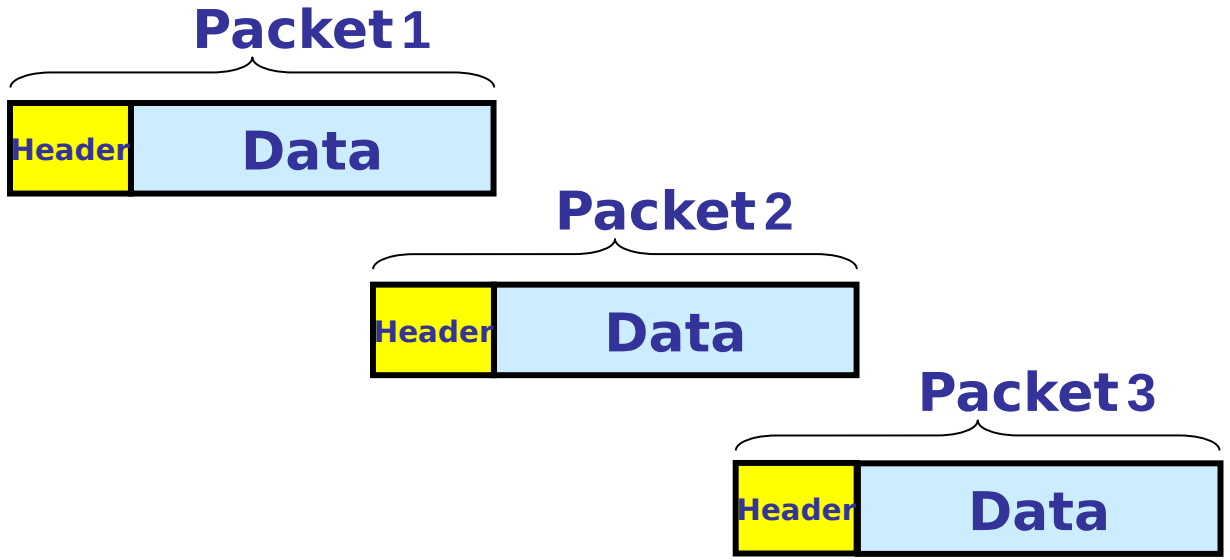


The header is very important

- **Headers are used to control the flow of packets through the network or over the communication link.**
- **The header contains control information, such as:**
  - **Addresses: source and destination,**
  - **Sequence number,**
  - **Protocol version, total length, etc.**
- **The switch in the packet-switched network forwards the packet to the next switch based on the address information in the header of the received packet.**

Restore to Message

After receiving the packet, the receiver strips off the header of the packet, and obtains the data, then restores data to a message.



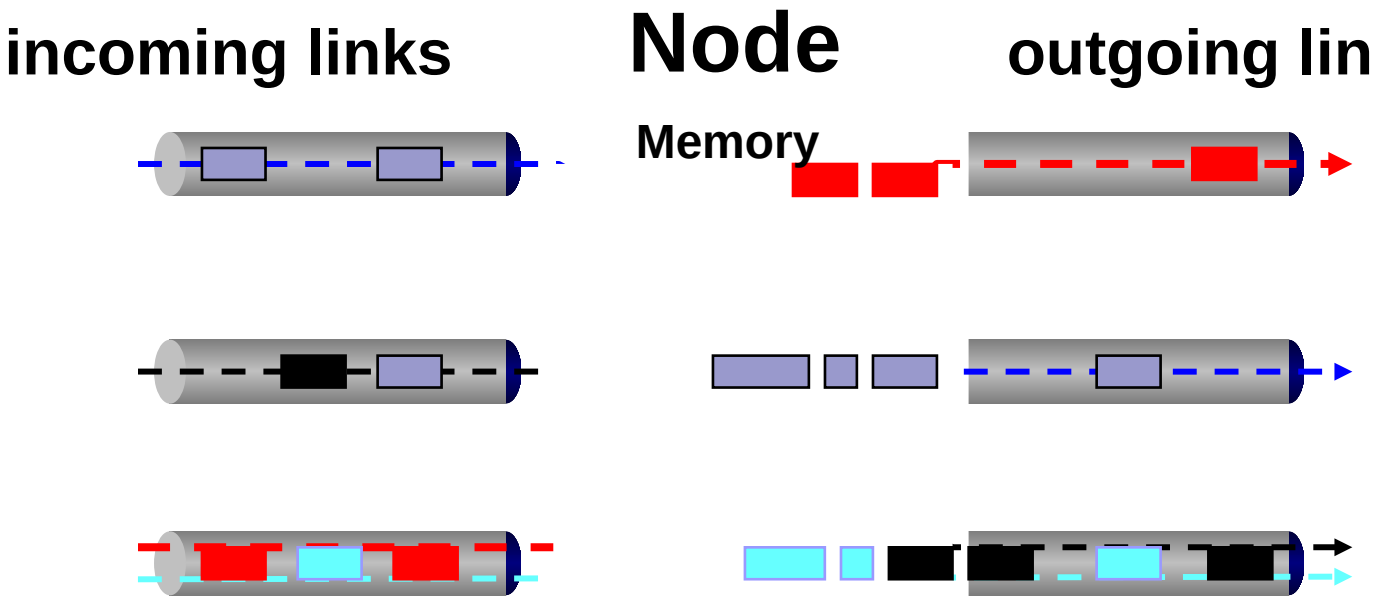
Data received

Message

110100011010101011010101110001001101001O

Network Core: Packet Switching

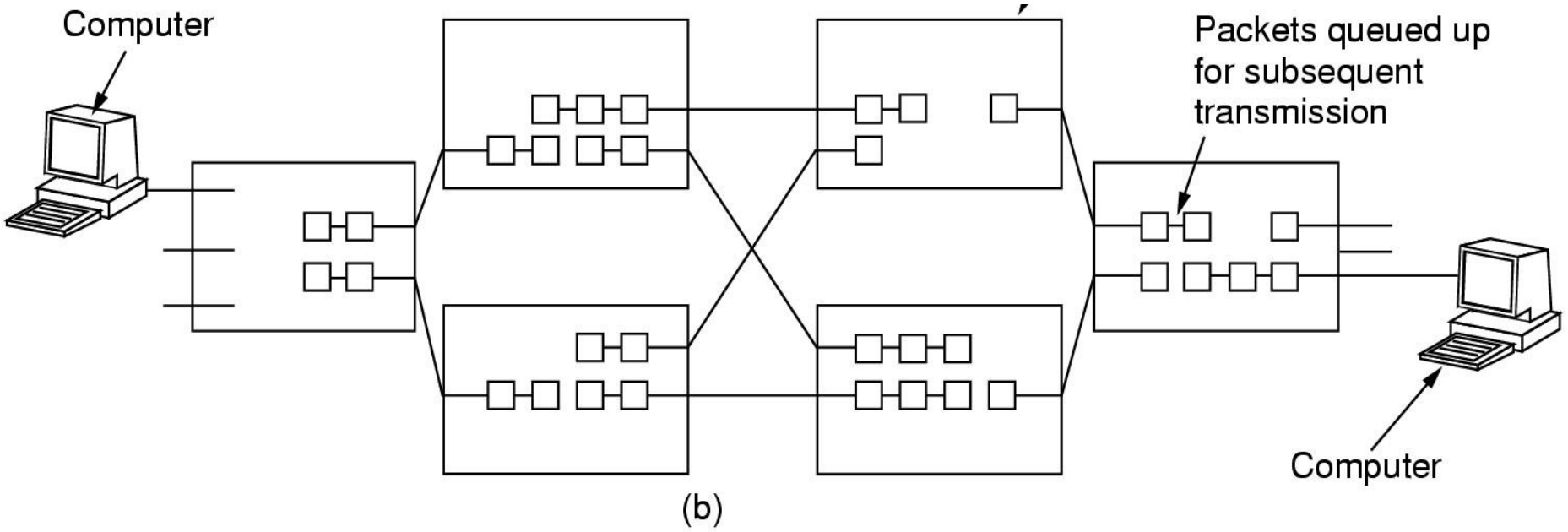
- **Node in a packet switching network**



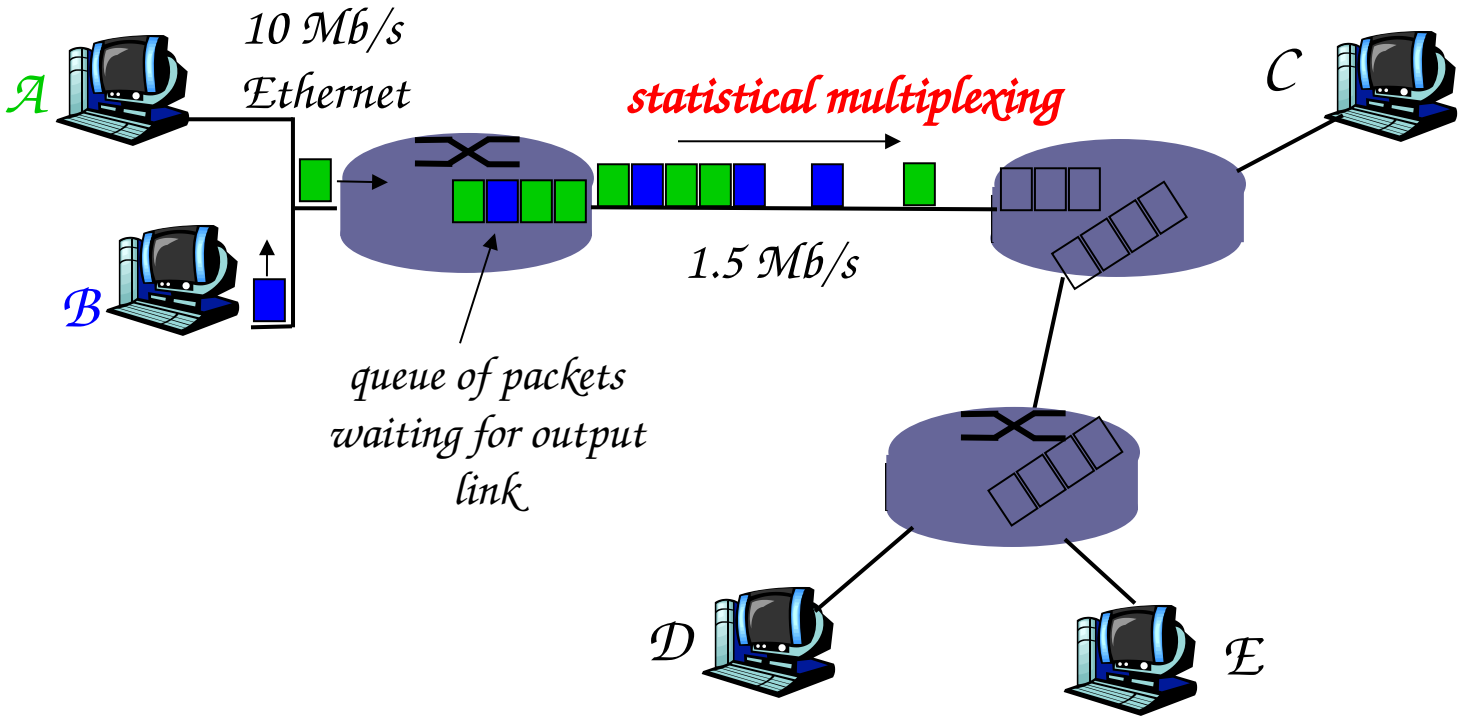
- user A, B packets *share* network resources
- each packet uses full link bandwidth
- resources used *as needed*

Network Core: Packet Switching

- Node in a packet switching network



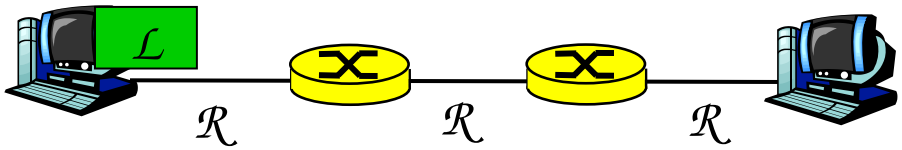
Packet Switching: Statistical Multiplexing



Sequence of A & B packets does not have fixed pattern ***statistical multiplexing***.

In TDM each host gets same slot in revolving TDM frame.

Packet-switching: store-and-forward

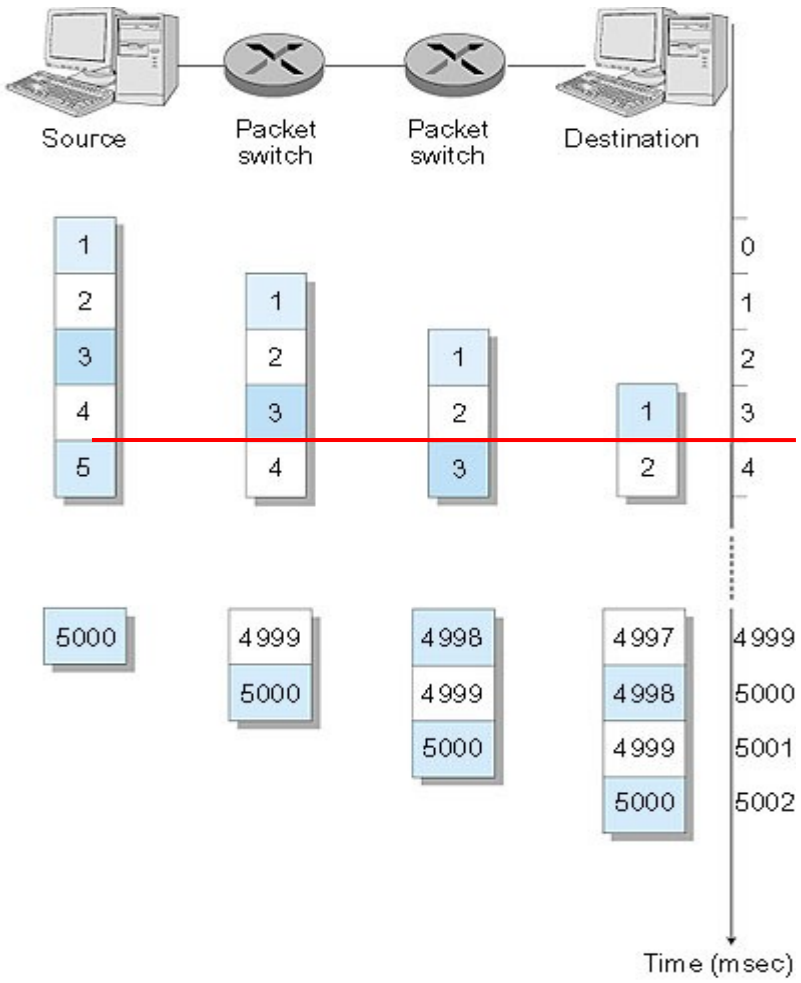


- Takes *L/R* seconds to transmit (push out) packet of *L* bits on to link or *R* b/s
- Entire packet must arrive at router before it can be transmitted on next link: ***store and forward***
- delay = *3*L/R*

Example:

- *L* = 7.5 Mbits
- *R* = 1.5 Mb/s
- delay = 15 sec

Packet Switching: Message Segmenting



Now break up the message into 5000 packets

- **Each packet 1,500 bits**
- **1 msec to transmit packet on one link**
- ***pipelining:* each link works in parallel**
- **Delay reduced from 15 sec to 5.002 sec**

If you were building a network….

- **Which would you choose?**
  - **Circuit switched?**
  - **Packet-switched?**

- **Let's review the strengths and weaknesses**

Advantages of Circuit Switching

- **Guaranteed bandwidth**
  - Predictable communication performance
  - Not “best-effort” delivery with no real guarantees
- **Simple abstraction**
  - Reliable communication channel between hosts
  - No worries about lost or out-of-order packets
- **Simple forwarding**
  - Forwarding based on time slot or frequency
  - No need to inspect a packet header
- **Low per-packet overhead**
  - Forwarding based on time slot or frequency
  - No headers on each packet

Disadvantages of Circuit Switching

- **Wasted bandwidth**
  - Bursty traffic leads to idle connection during silent period
  - Unable to achieve gains from “statistical multiplexing”
- **Blocked connections**
  - Connection refused when resources are not sufficient
  - Unable to offer “okay” service to everybody
- **Network state**
  - Network nodes must store per-connection information
  - Unable to avoid per-connection storage and state
  - This makes failures more disruptive!
- **Connection set-up delay**
  - No communication until the connection is set up
  - Unable to avoid extra latency for small data transfers

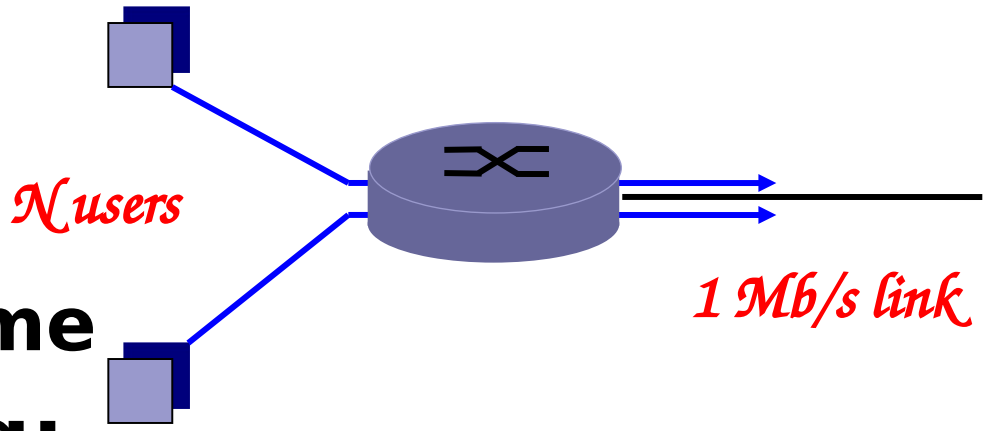
Packet-Switching vs. Circuit-Switching

- **Reliability advantage:** since routers don't know about individual conversations, when a router or link fails, it is
  easy to fail over to a different path
- **Efficiency advantage** of packet-switching over circuit switching: Exploitation of statistical multiplexing
- **Deployability advantage:** easier for different parties to link their networks together because they're not promising to reserve resources for one another
- **Disadvantage:** packet-switching must handle congestion
  - More complex routers (more buffering, sophisticated dropping)

Packet switching vs. circuit switching

Packet switching allows more users to use network!

- 1 Mb/s link
- each user:
  - 100 kb/s when "active"
  - active 10% of time
- circuit-switching:
  - 10 users
- packet switching:
  - with 35 users,



Packet switching vs. circuit switching

	circuit switching	packet switching
resource	**partitioned**	**not partitioned**
when using resource	**use a single partition bandwidth**	**use whole link bandwidth**
reservation/setup	**need reservation (setup delay)**	**no reservation**
resource contention	**busy signal (session loss)**	**congestion (long delay and packet losses)**
service guarantee	**yes**	**no**
charging	**time**	**packet**
header	**no header**	**per packet header**
fast path processing	**fast**	**per packet processing**

Packet switching vs. circuit switching

Is packet switching a "slam dunk winner?"

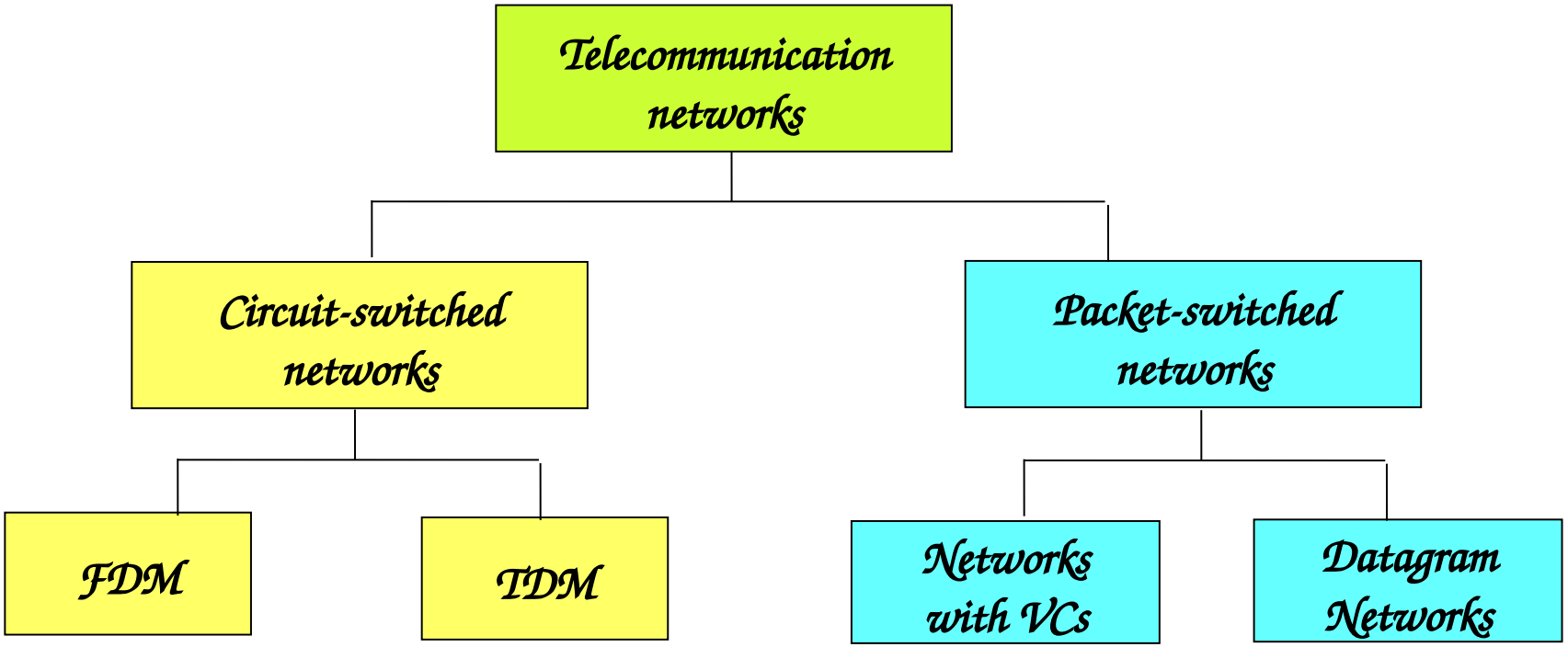
- **Great for bursty data**
  - **resource sharing - go as fast as you can and then get out of the way**
  - **simpler, no call setup**
- **Excessive congestion: packet delay and loss**
  - **protocols needed for reliable data transfer, congestion control**
- **Q: How to provide circuit-like behavior?**
  - **prioritization**
  - **bandwidth guarantees needed for audio/video apps still an unsolved problem**

Q: human analogies of reserved resources (circuit switching) versus on-demand allocation (packet-switching)?

Packet-switched networks: routing & forwarding

- ***Goal:*** move packets among routers from source to destination
  - we'll study several path selection algorithms.
- **datagram network:**
  - ***destination address*** determines next hop
  - **routes may change during session**
  - analogy: driving, asking directions
- **virtual circuit network:**
  - each packet carries tag (virtual circuit ID), *tag* determines next hop
  - **fixed path** determined at *call setup time*, remains fixed thru call
  - **routers maintain per-call state**

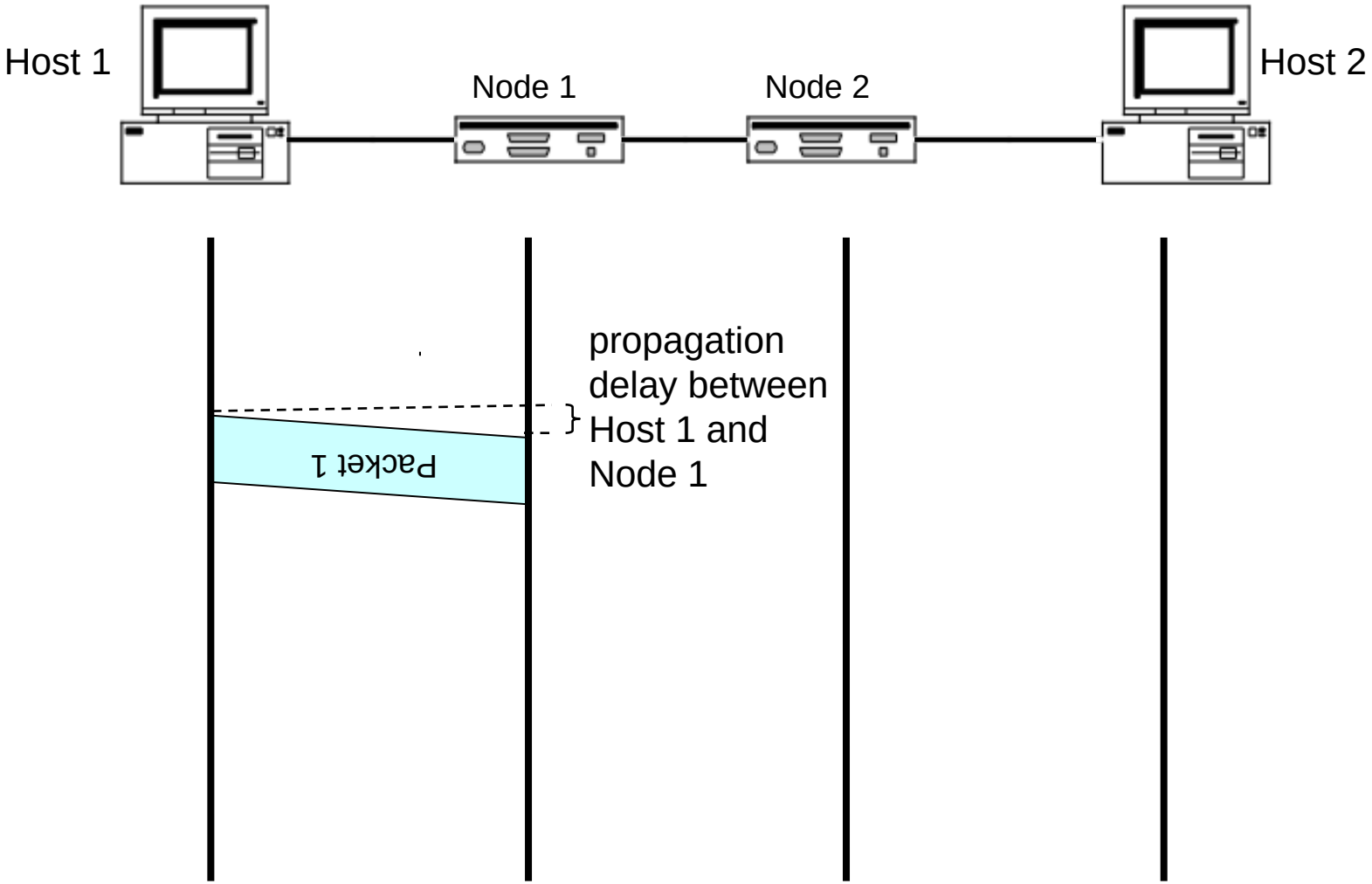
Network Taxonomy



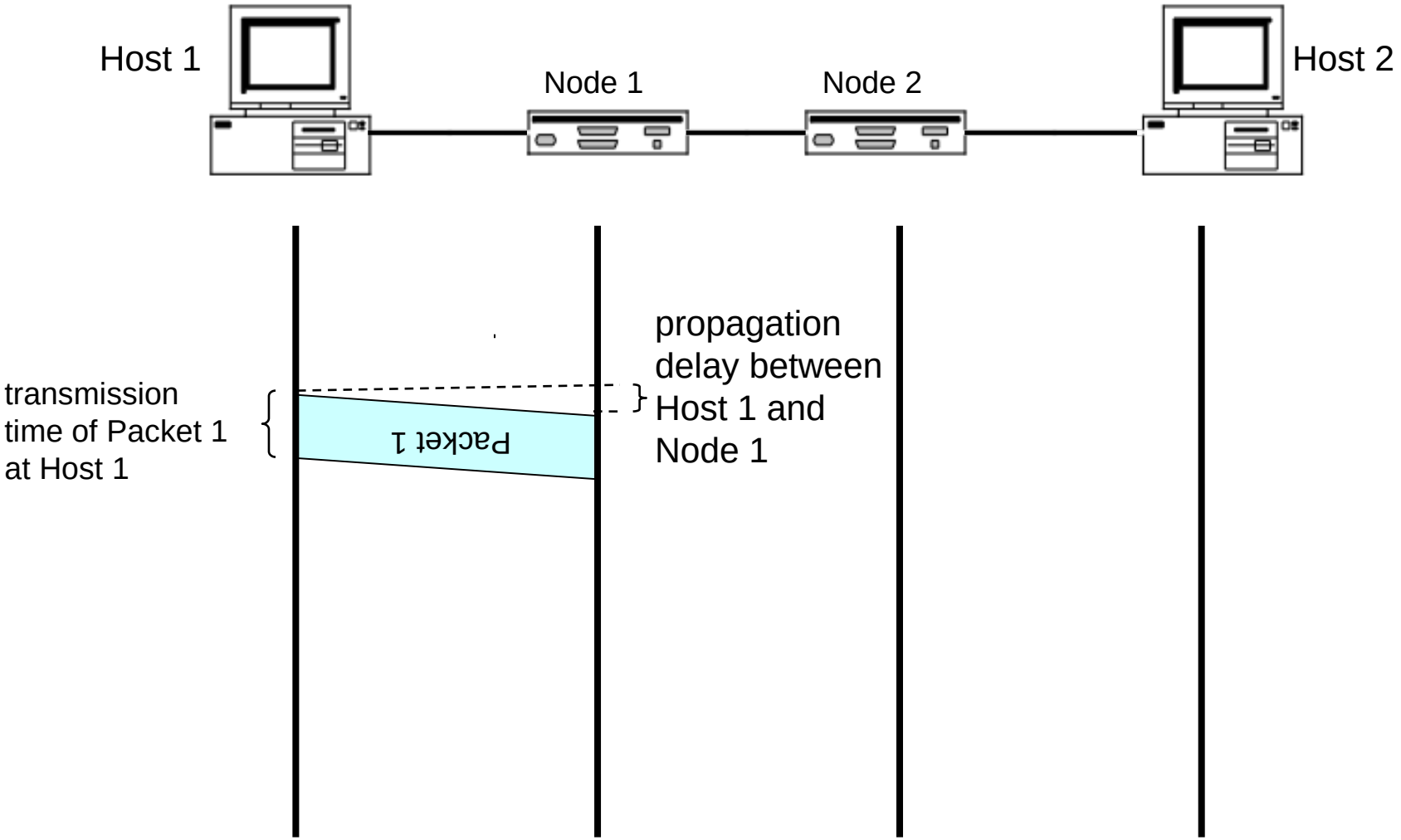
Datagram Packet Switching

- **Each packet is independently switched**
  - Each packet header contains full destination address
- **No resources are pre-allocated (reserved) in advance**
- **Leverages "statistical multiplexing"**
  - Gambling that packets from different conversations won't all arrive at the same time, so we don't need enough capacity for all of them at their peak transmission rate
  - *Assuming independence of traffic sources*, can compute probability that there is

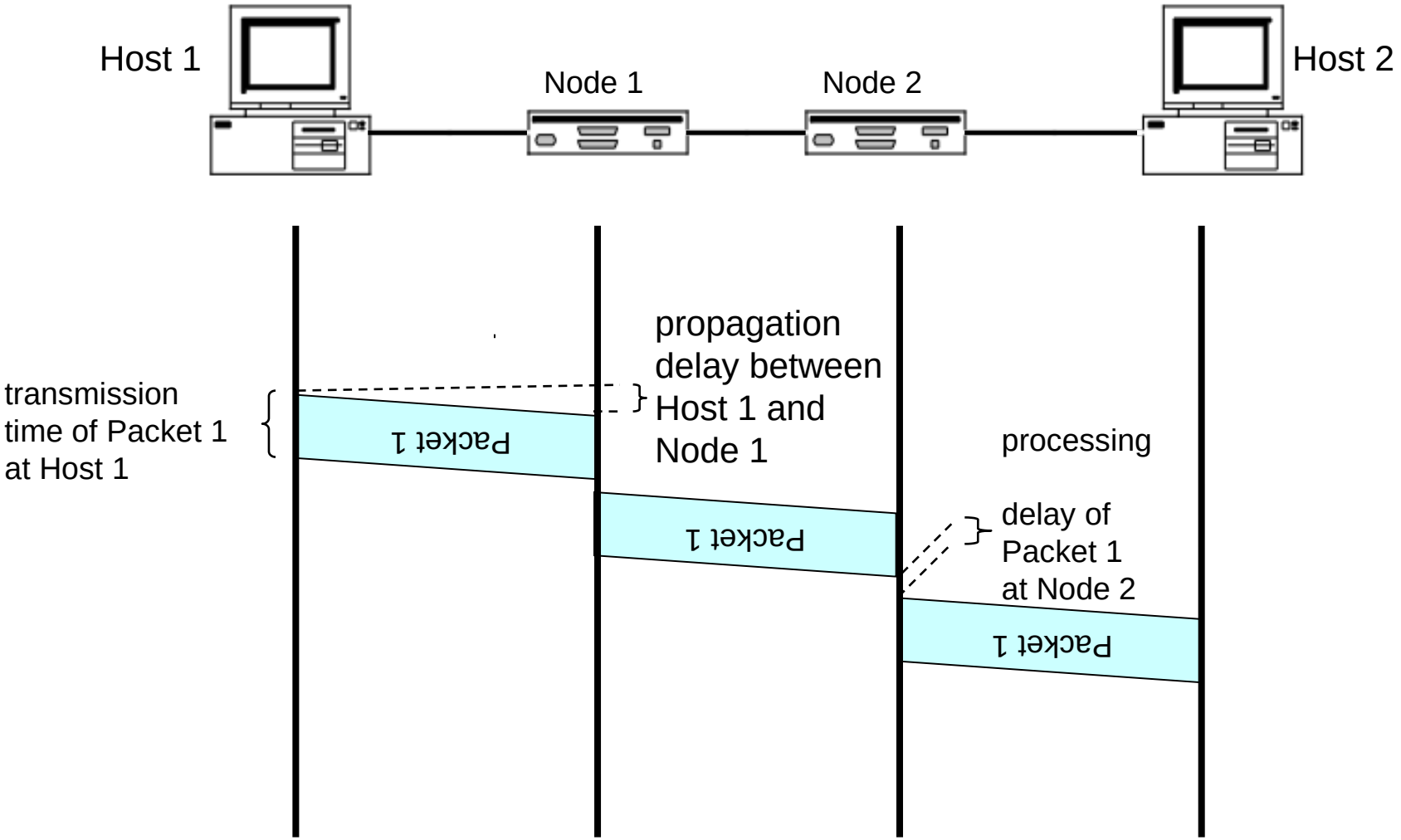
Timing of Datagram Packet Switching



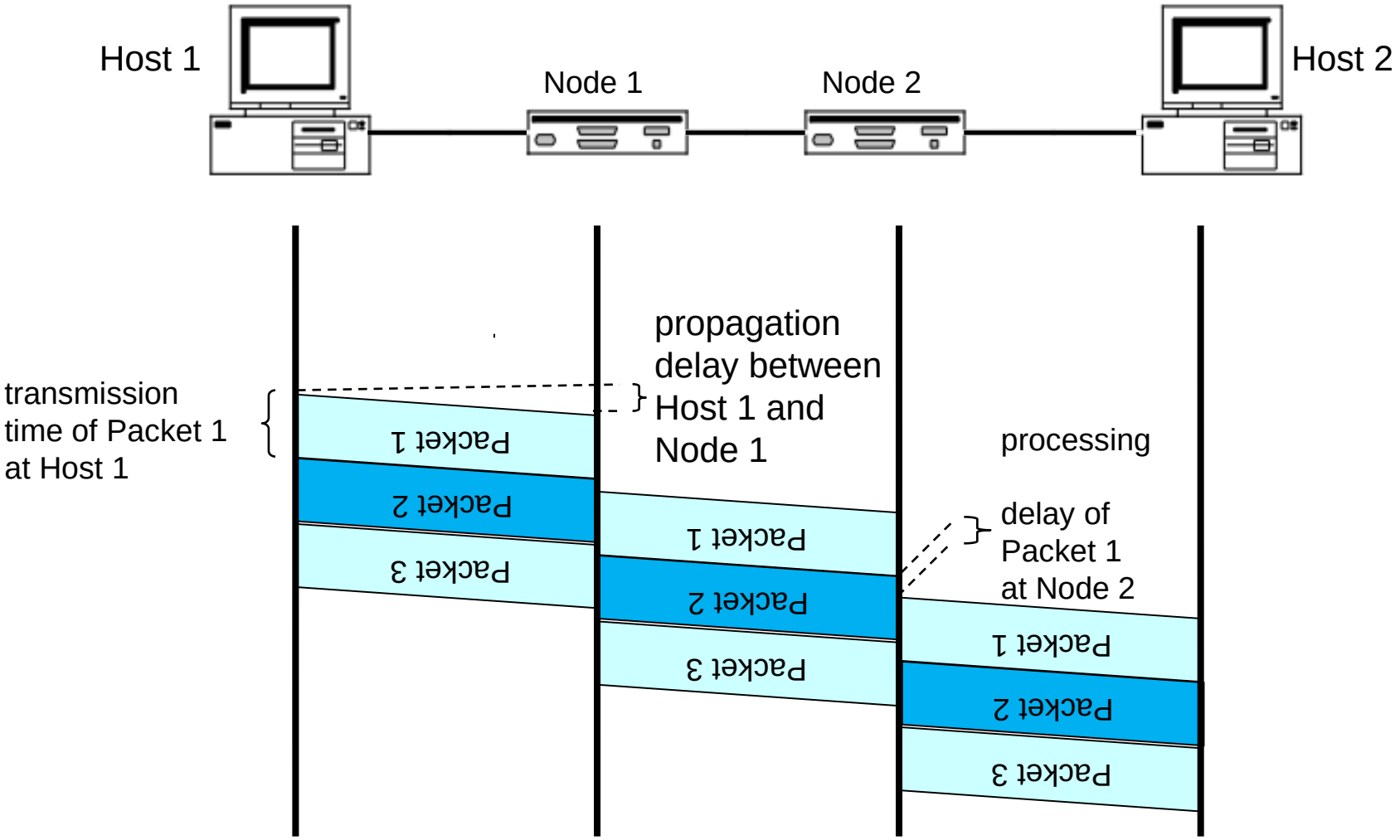
Timing of Datagram Packet Switching



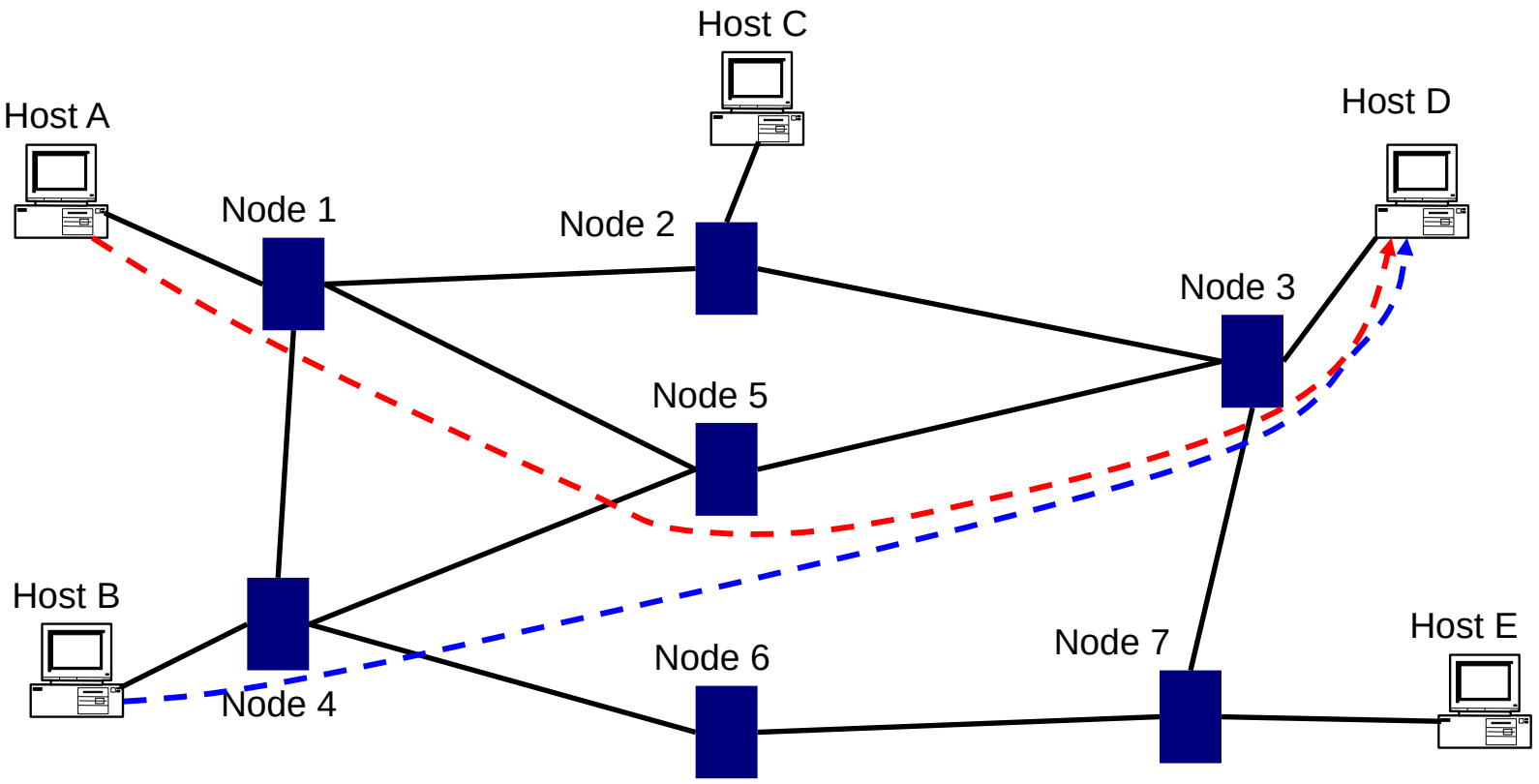
Timing of Datagram Packet Switching



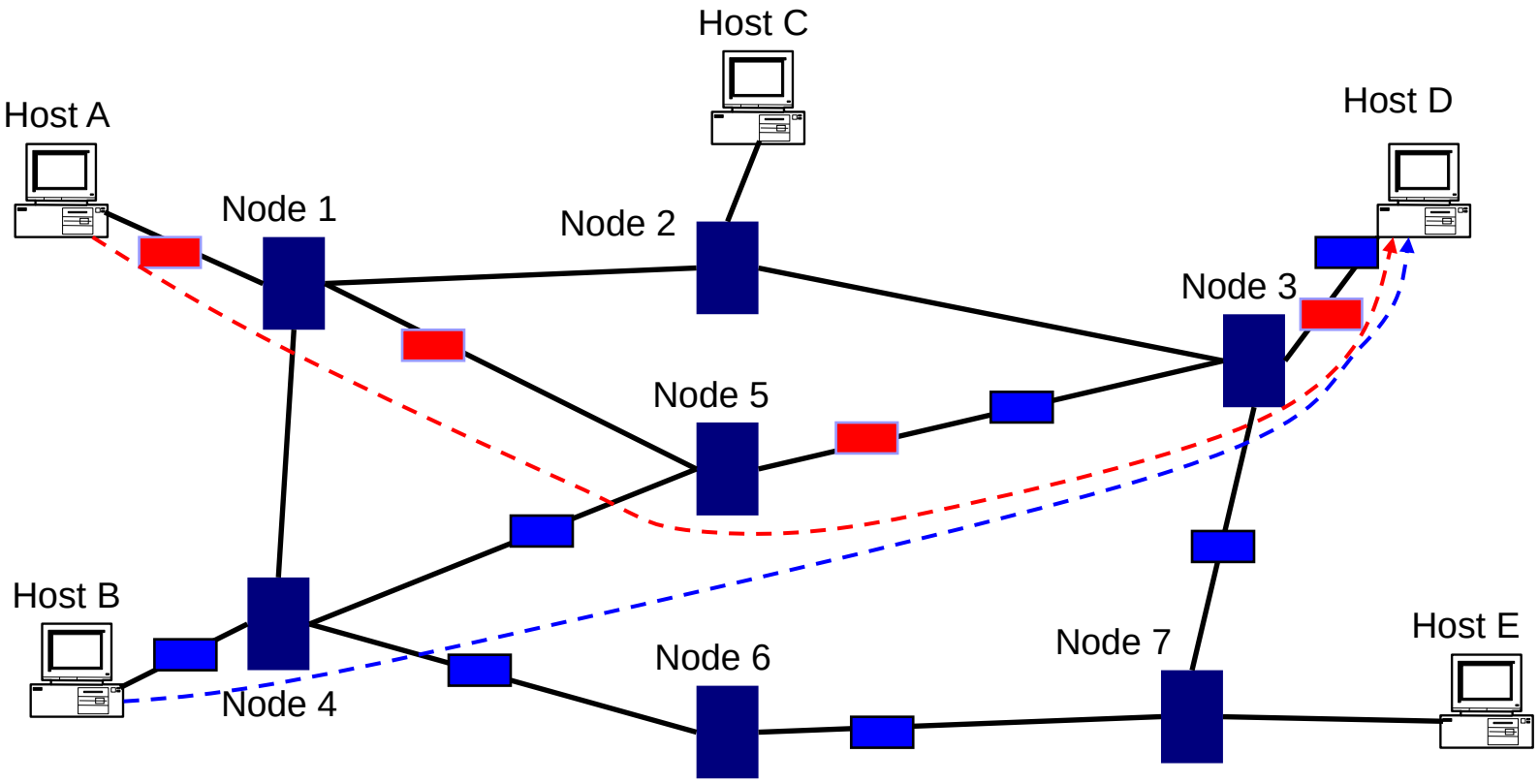
Timing of Datagram Packet Switching



Datagram Packet Switching



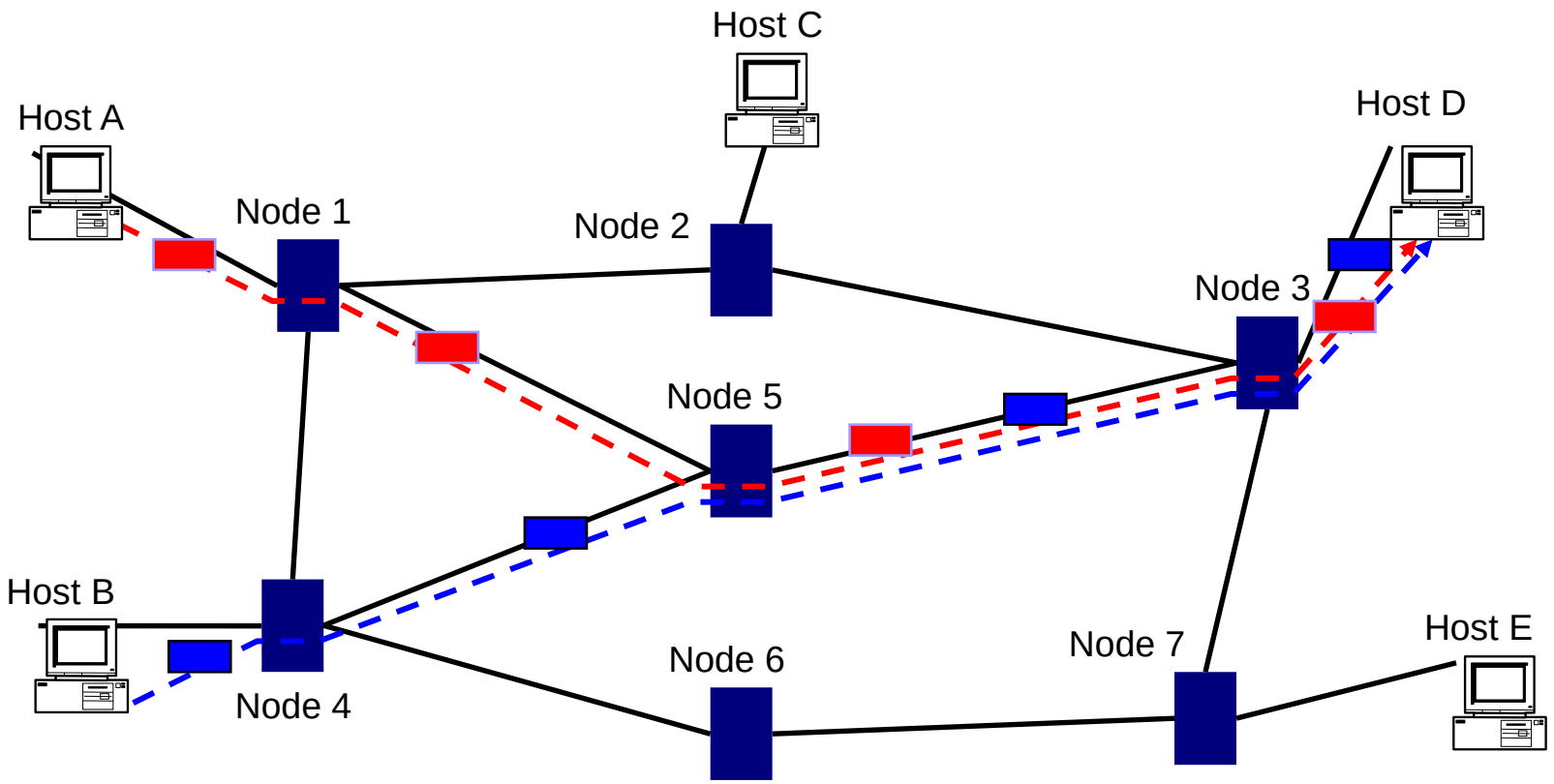
Datagram Packet Switching



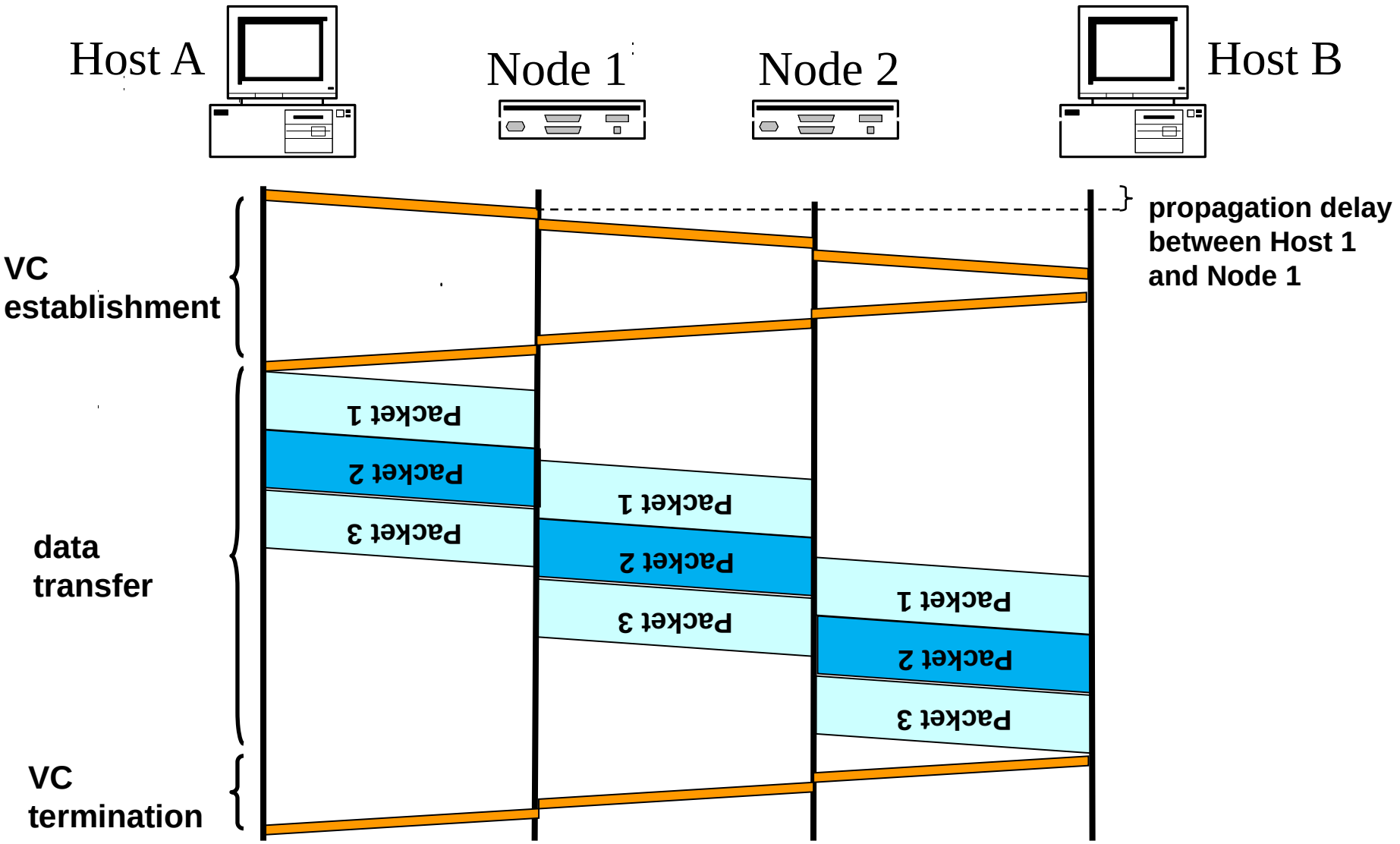
Virtual-Circuit Packet Switching

- **Fixed path** determined at *call setup time*, remains fixed thru call
- **Routers maintain per-call state**
- **3 phases:**
  1. VC establishment
  2. Data transfer
  3. VC disconnect

Virtual-Circuit Packet Switching



Timing Diagram of Virtual-Circuit Switching



Chapter 1: Roadmap

1.1 Computer network and the Internet

1.2 Network edge and core

1.3 Network access and physical media

1.4 Transmission Technology

1.5 Type and Topology

1.6 Network Performance

1.7 Protocol layers, service models

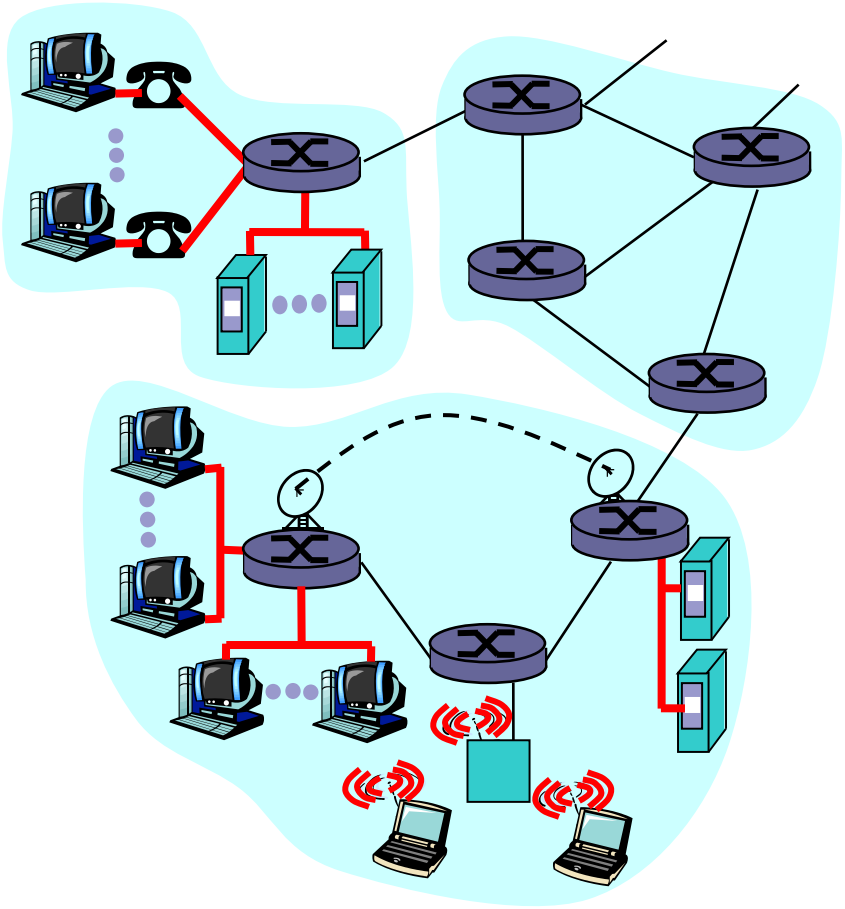
1.8 Architecture, OSI and TCP/IP Models

1.9 Internet History

Access networks and physical media

Q: How to connect end systems to edge router?

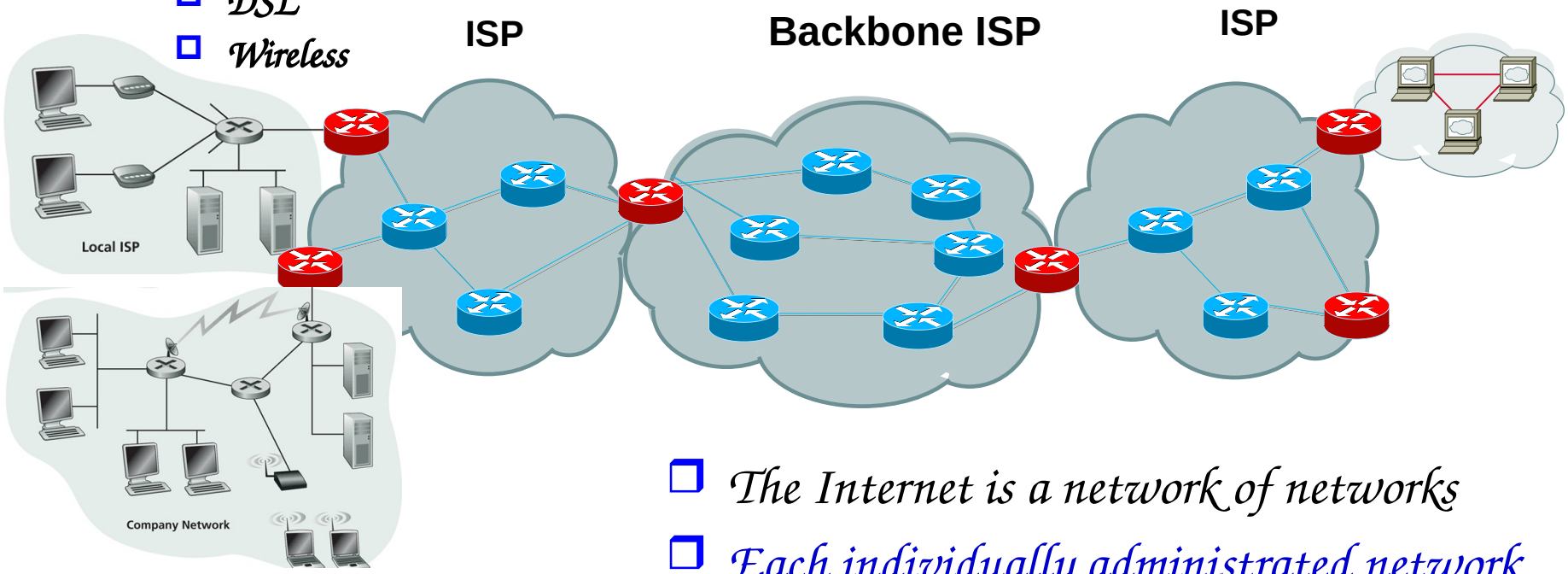
- residential access nets
- institutional access networks (school, company)
- mobile access networks



Internet Physical Infrastructure

Residential access

- *Cable*
- *Fiber*
- *DSL*
- *Wireless*



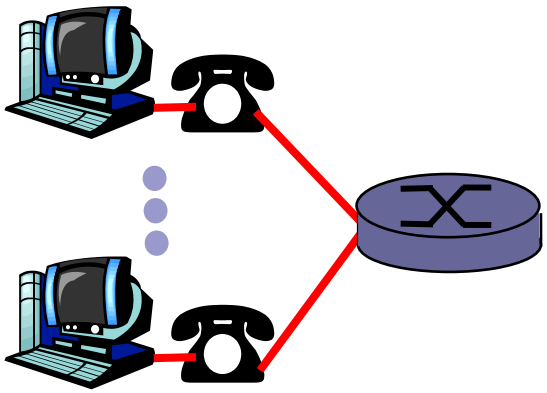
Campus access, e.g.,

- *Ethernet*
- *Wireless*

- *The Internet is a network of networks*
- *Each individually administrated network is called an **Autonomous System (AS)***

Residential access: Point-to-Point access

- *Dialup via modem*
  - *up to 56Kbps direct access to router (often less)*
  - *Can't surf and phone at same time: can't be "always on"*



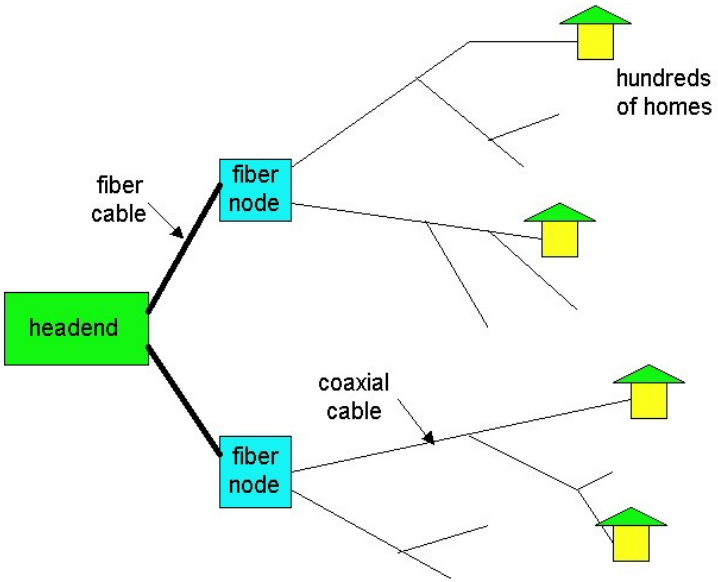
- *ADSL: asymmetric digital subscriber line*
  - *up to 1 Mb/s upstream (today typically < 256 kb/s)*
  - *up to 8 Mb/s downstream (today typically < 1 Mb/s)*
  - *FDM: 50 kHz - 1 MHz for downstream*

    4 kHz - 50 kHz for upstream

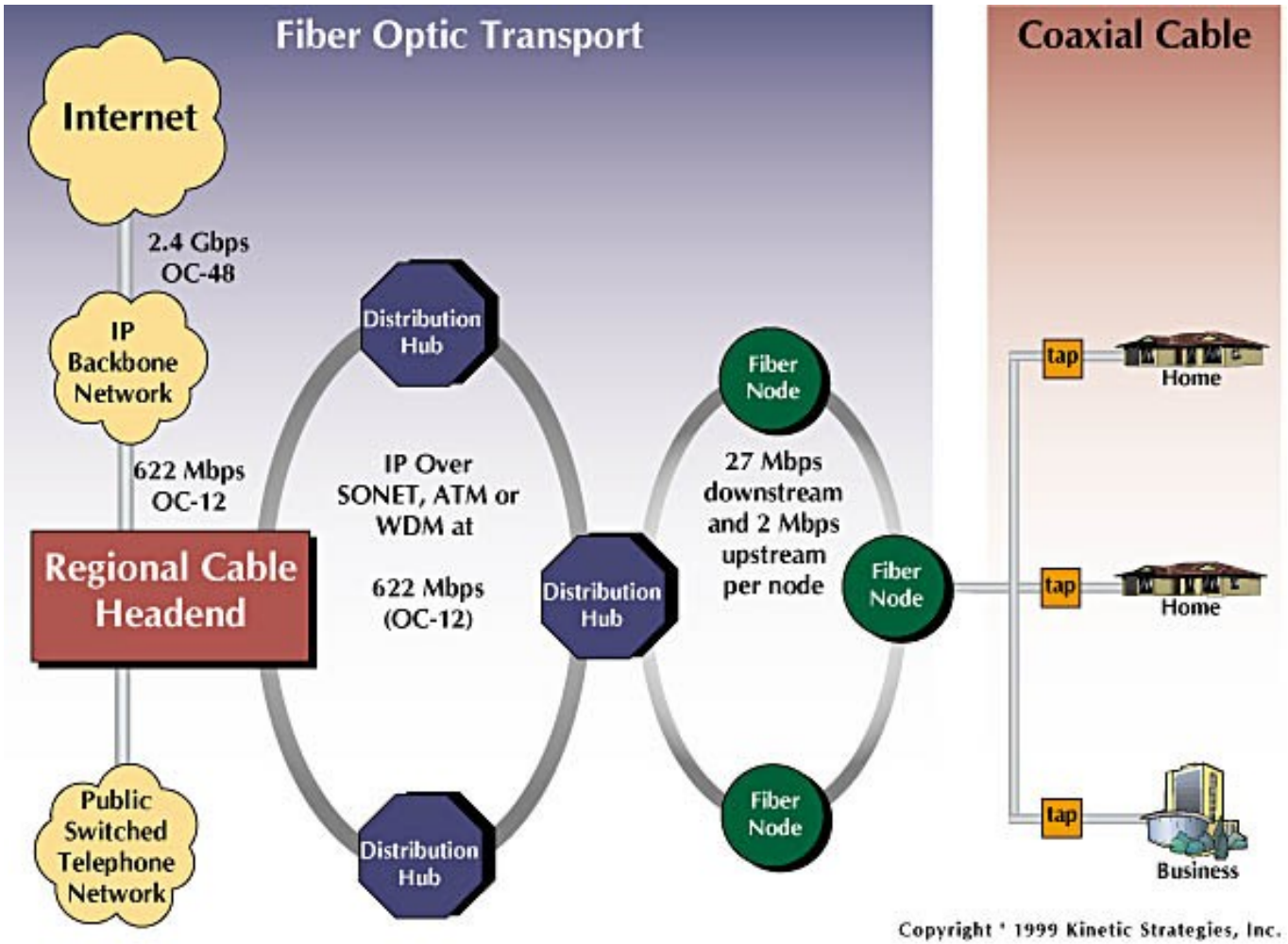
    0 kHz - 4 kHz for ordinary telephone

Residential access: Cable modems

- **HFC: hybrid fiber coax**
  - **asymmetric:** up to 30Mbps downstream, 2 Mbps upstream
- **network of cable and fiber attaches homes to ISP router**
  - **shared** access to router among home
  - **issues:** congestion, dimensioning
- **Deployment:** available via cable companies



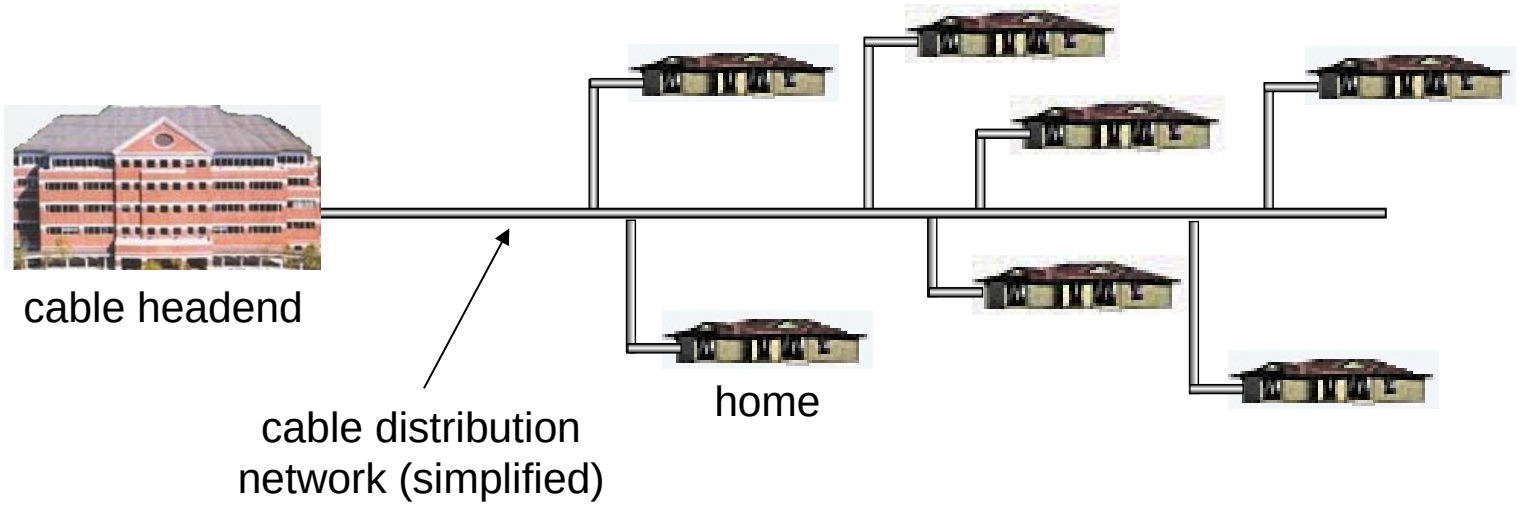
Residential access: Cable modems



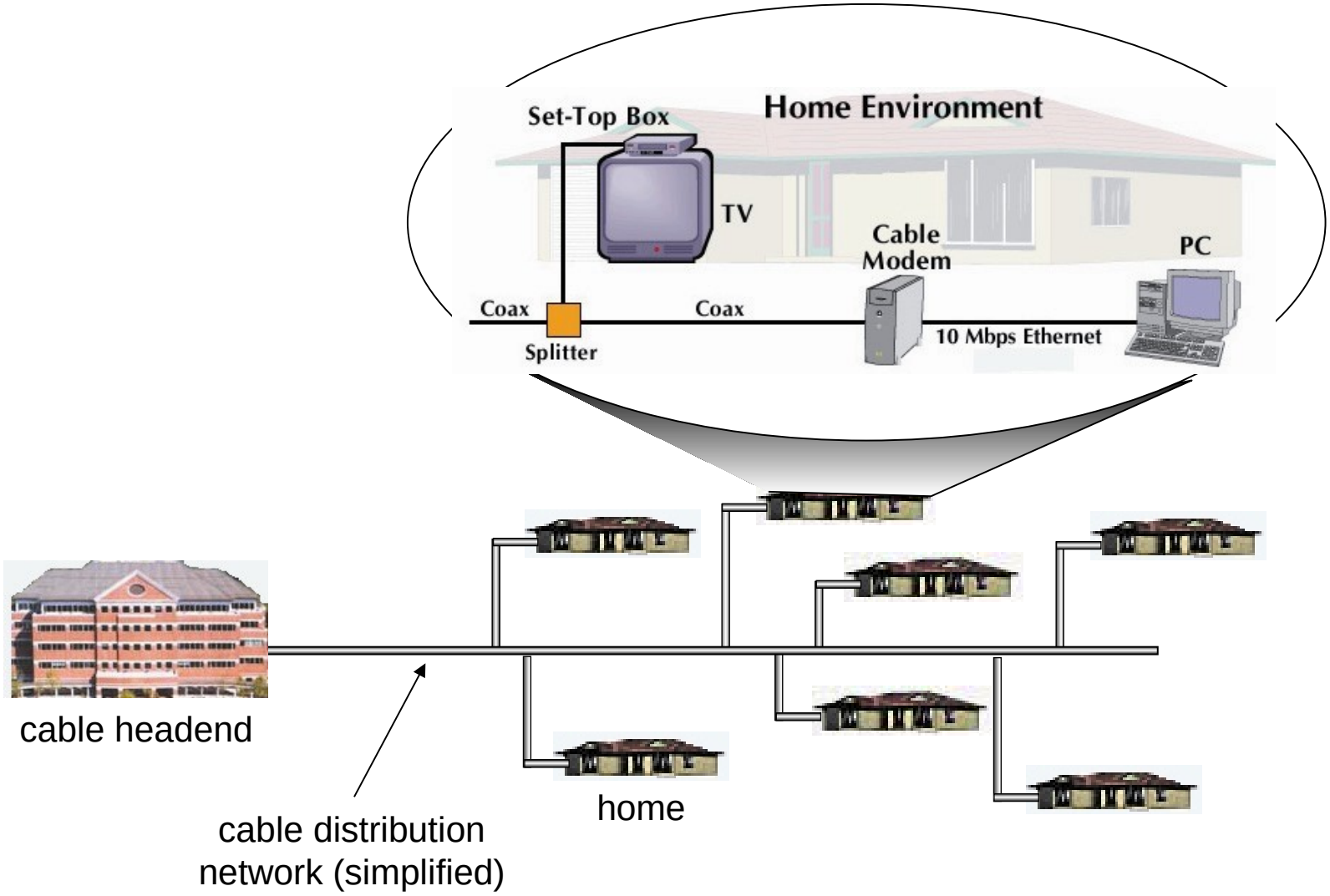
- Fiber node: 500 - 1K homes
- Distribution hub: 20K - 40 K homes
- Regional headend: 200 K - 400 K homes

Cable Network Architecture: Overview

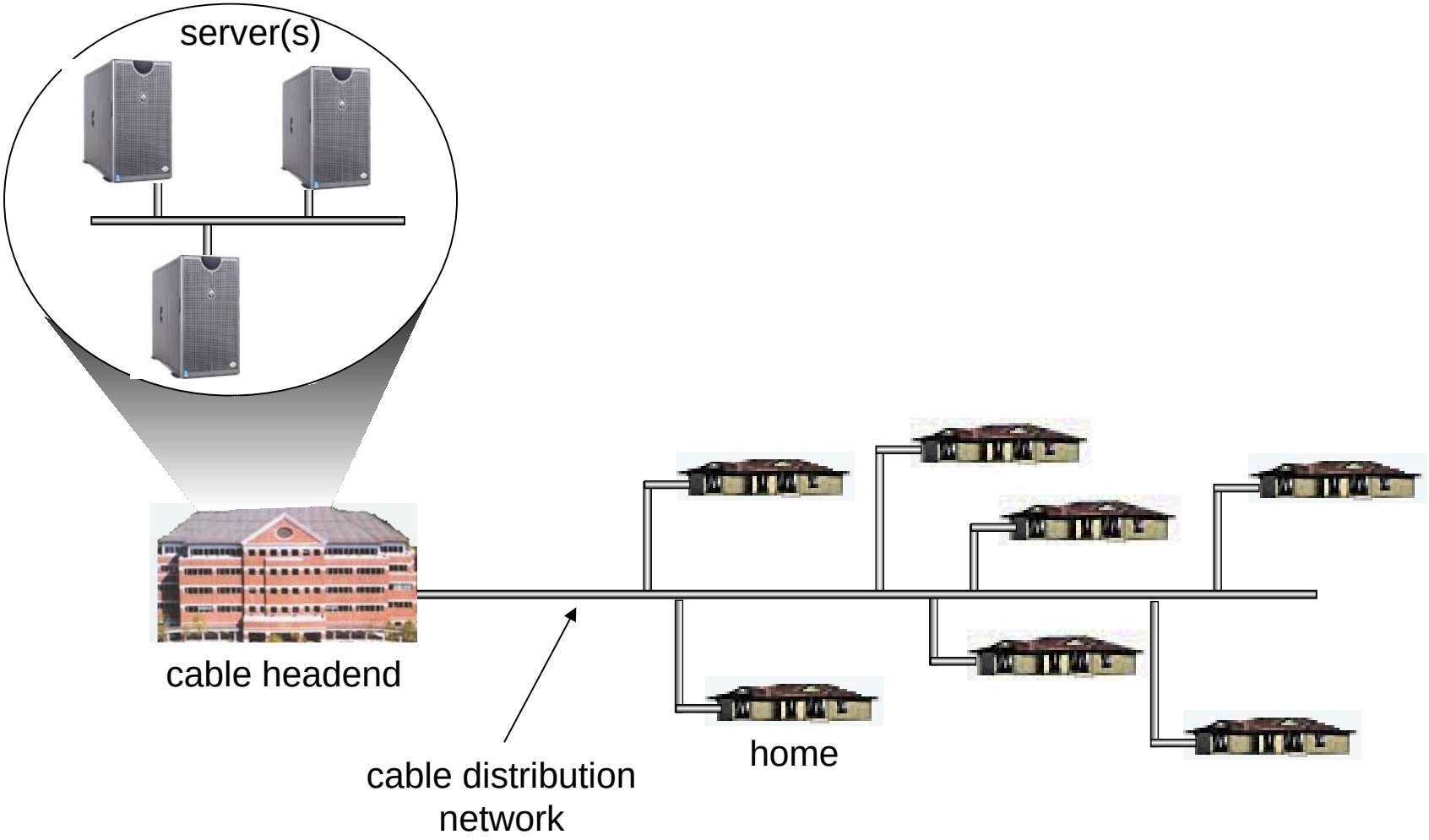
Typically 500 to 5,000 homes



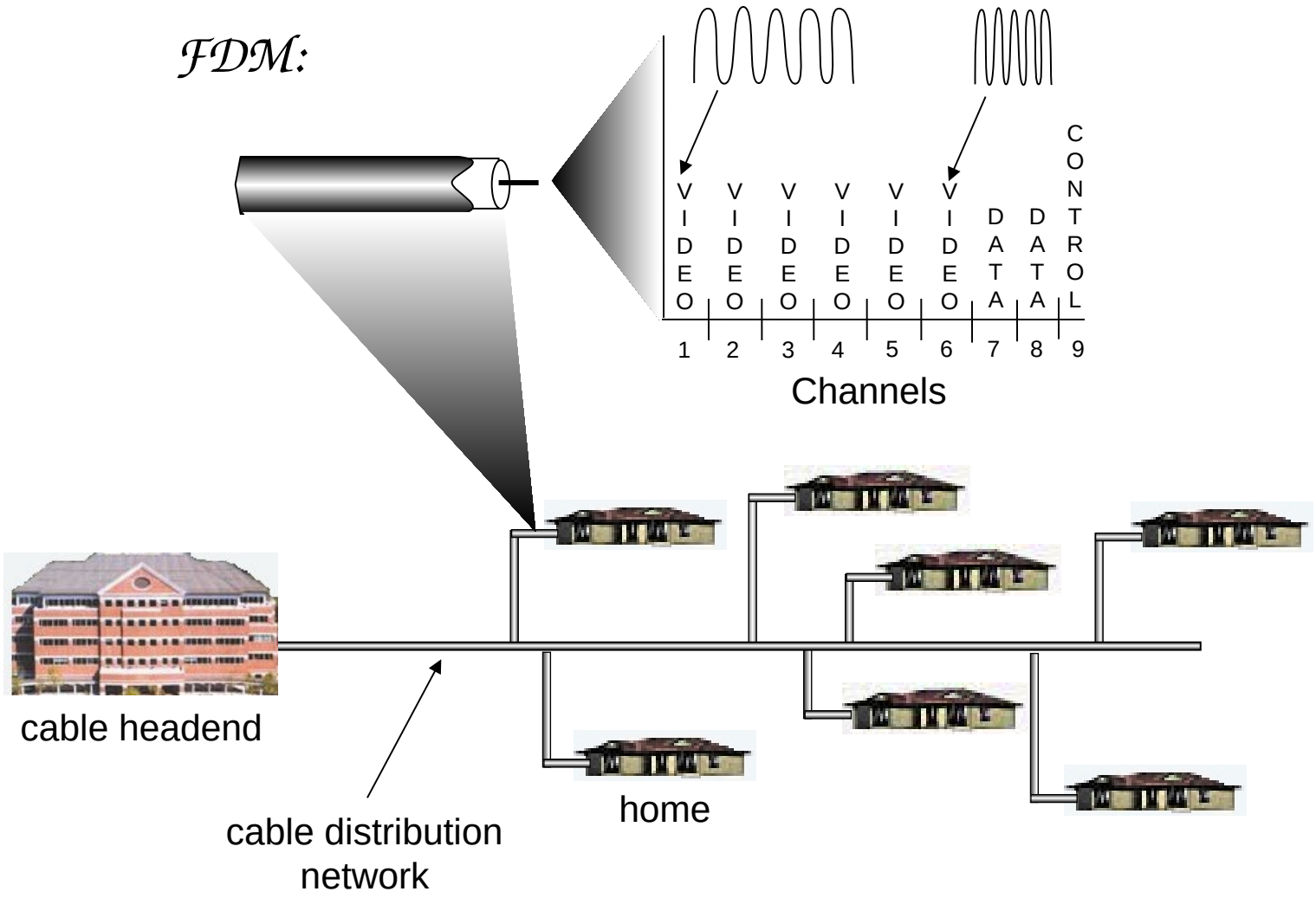
Cable Network Architecture: Overview



Cable Network Architecture: Overview



Cable Network Architecture: Overview



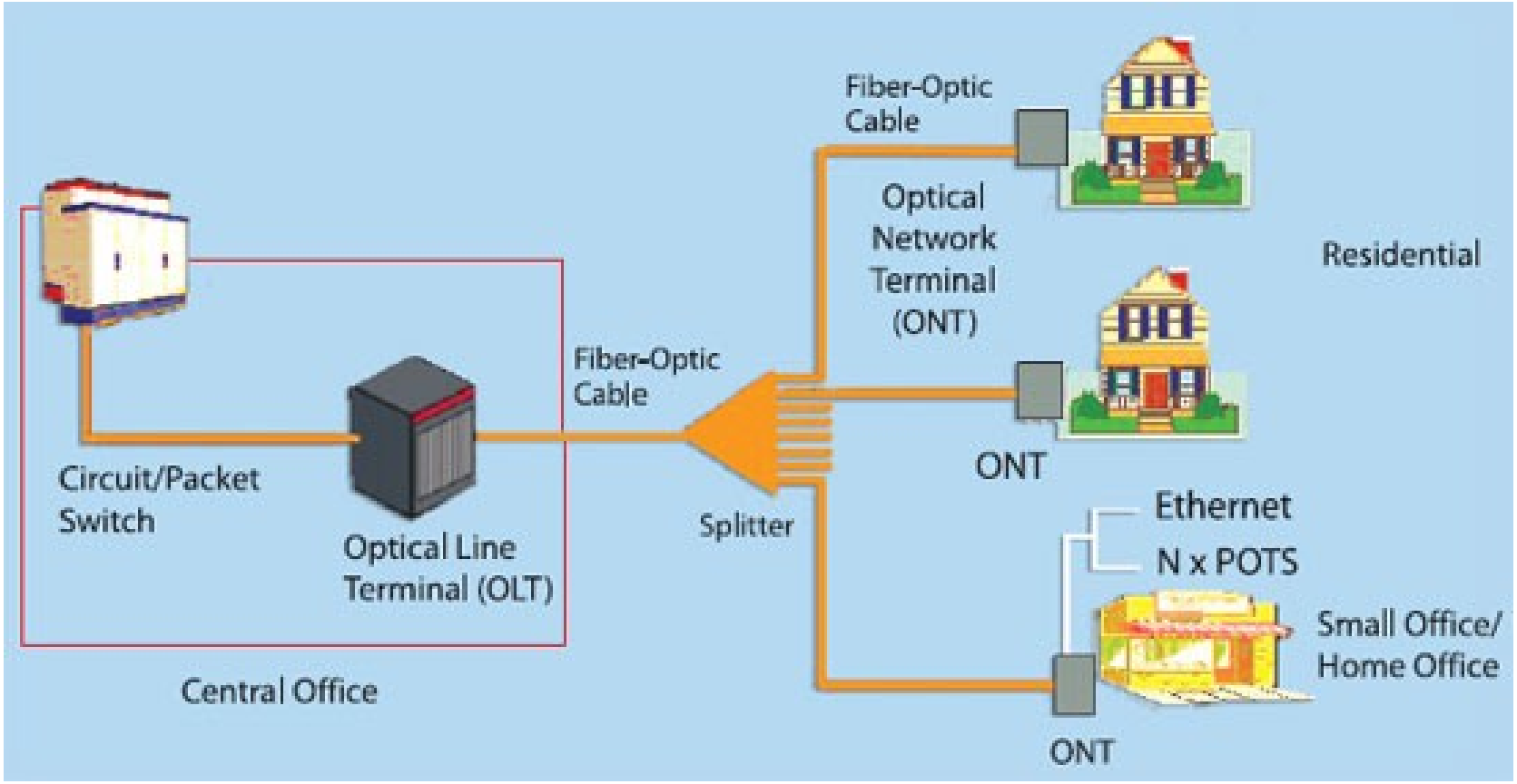
Residential Access: Fiber to the Premises (FTTP)

- Deployed by Verizon (all locations) and AT&T (new build areas)
- One of the largest comm. construction



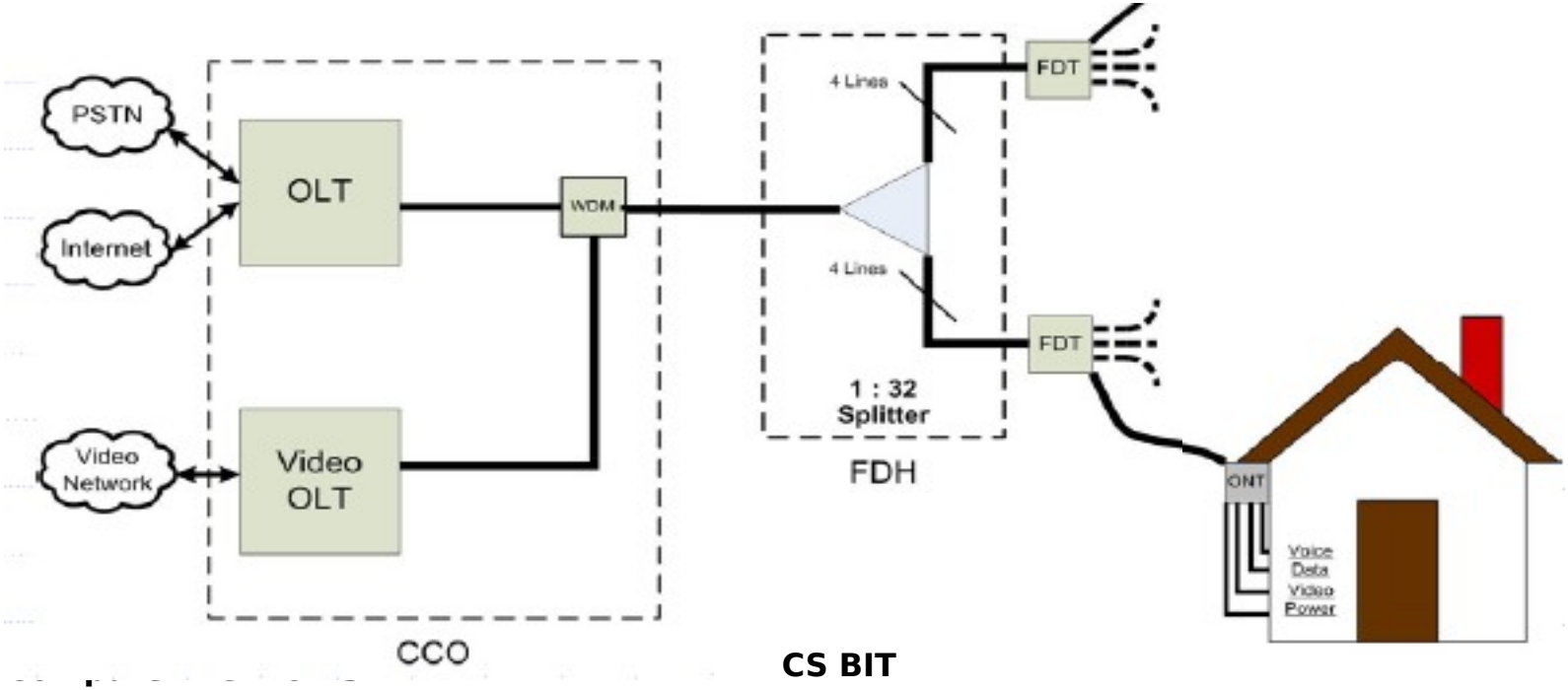


FTTP Architecture



FTTP Architecture

- **Optical Network Terminal box outside dwelling or business**
- **Fiber Distribution Terminal (FDT) in poles or pedestals**
- **Fiber Distribution Hub (FDH) at street cabinet**
- **Optical Line Terminal (OLT) at central office**



FTTP Architecture: Central to Fiber Distribution Hub (FDH)

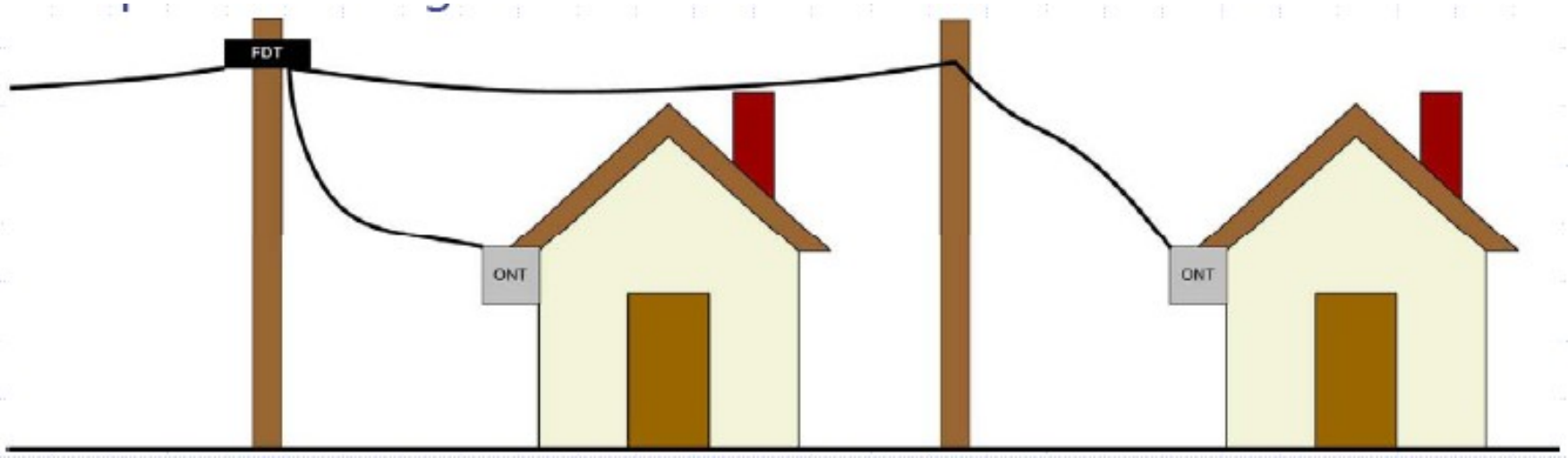


- *Backbone fiber ring on primary arterial streets (brown)*
- *Local distribution fiber plant (red) meets backbone at cabinet*



FDH

FTTP Architecture: Fiber Distribution Terminal (FDT)

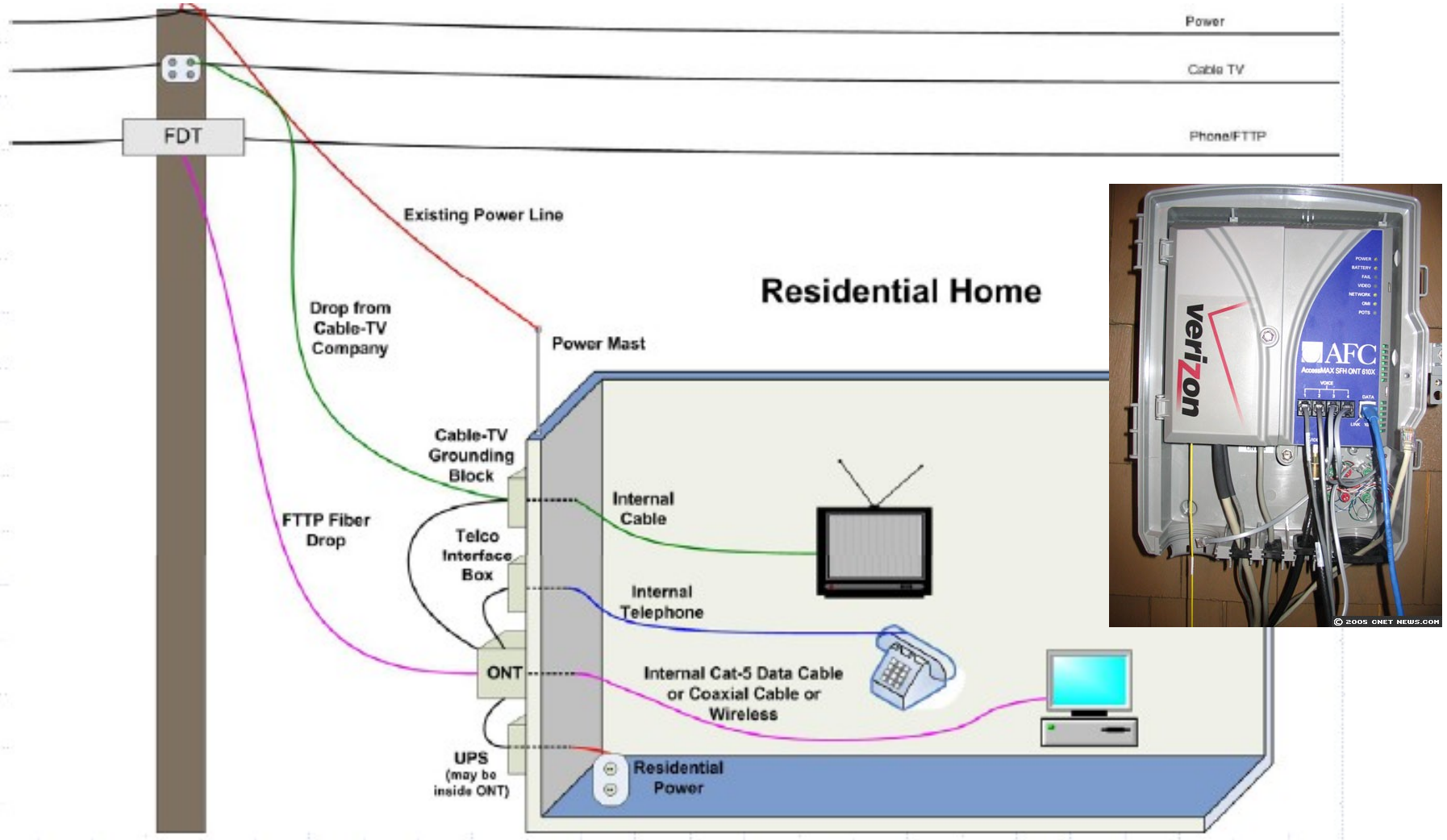








FTTP Architecture: To Home

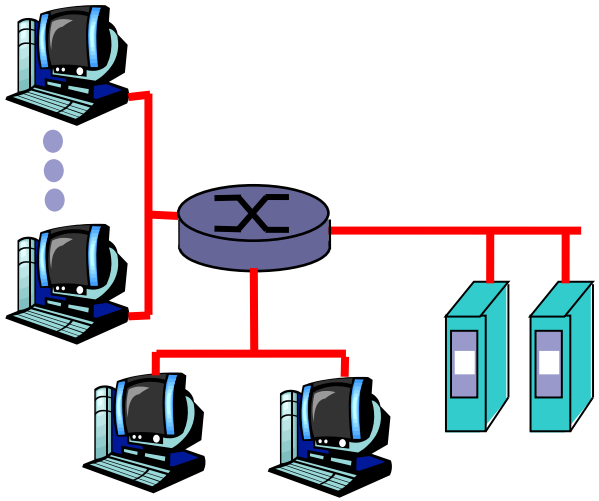


Residential Access: Fiber to the Premises (FTTP)

- **Highest theoretical capacity per user: 1 G bps**
  - currently 5 to 30 Mbps downstream, 2 to 5 Mbps upstream
- **Services**
  - analog and digital video
  - video viewable with and without set-top converter
  - voice
  - interfaces with existing phone, data, video cabling

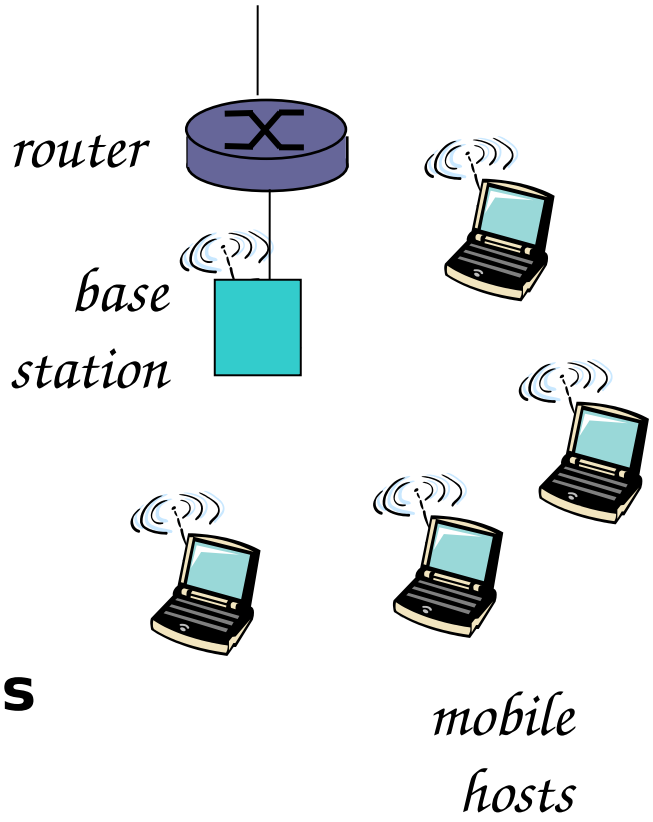
Institutional access: Local Area Networks

- company/university local area network (LAN) connects end system to edge router
- Ethernet:
  - shared or dedicated cable connects end system and router
  - 10 Mb/s, 100 Mb/s for PCs and enterprise networks
  - Gigabit Ethernet (GigE) for servers
  - 10 Gb/s for servers and backbones
- deployment: institutions, home LANs



Wireless access networks

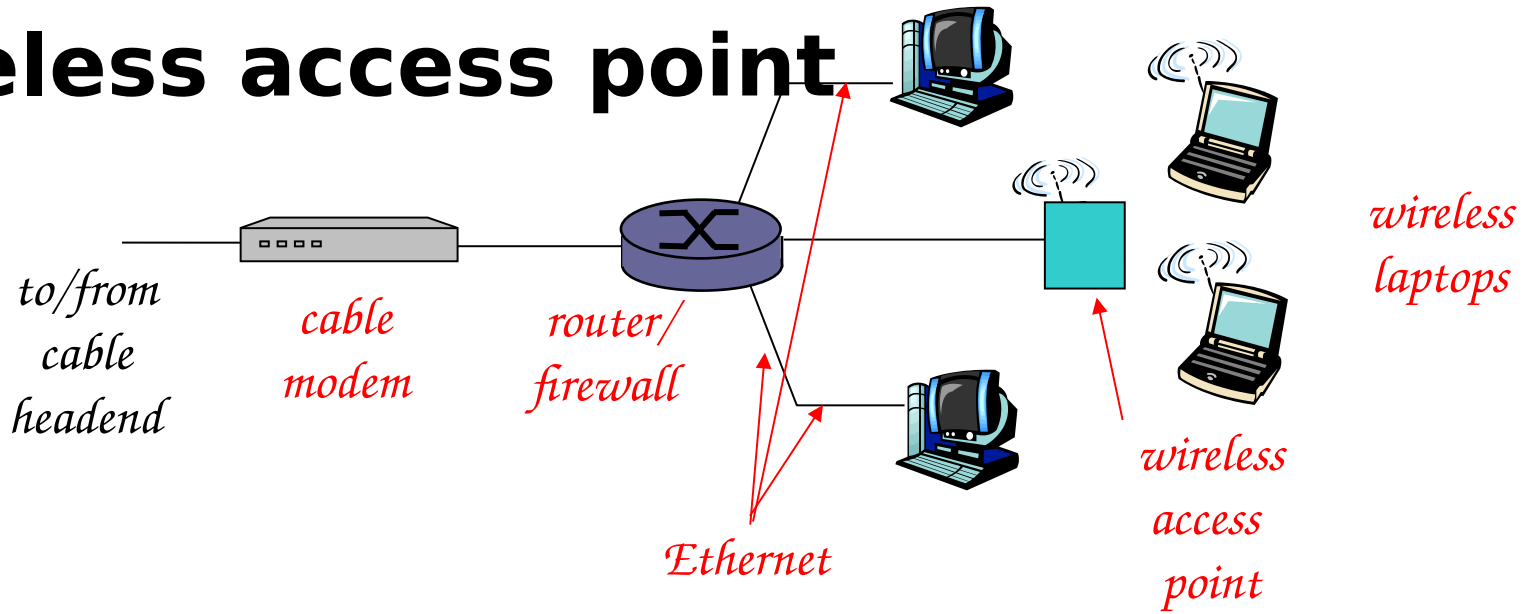
- **shared wireless access network connects end system to router**
- **wireless LANs:**
  - **radio spectrum replaces wire**
  - **e.g., 802.11 a/b/g/n**
    - **nominally, 1 to 100 Mb/s**
- **wider-area wireless access**
  - **through cell phone network**
  - **CDPD: old, slow wireless access to ISP router via cellular network (fill speech gaps) -- obsolete**
  - **EVDO, 1xRTT: emerging**



Home networks

Typical home network components:

- ADSL or cable modem
- router/firewall/NAT
- Ethernet
- wireless access point

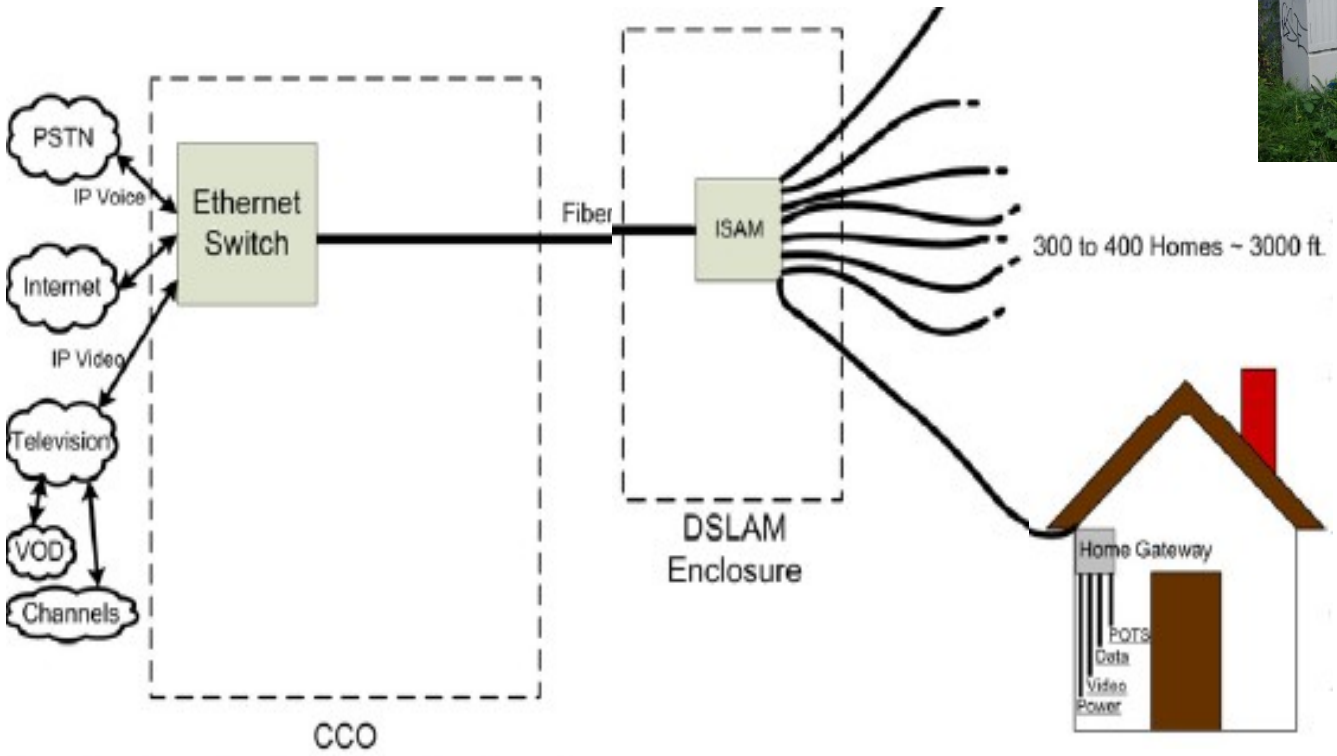


Access: DSL

- Compared with FTTP, copper from cabinet (DSLAM) to home



DSLAM



DSL Services to Customers

- Up to 25 Mbps per customer
- Currently a few Mbps downstream, and a few Mbps upstream
- Voice
  - converted to standard telephone signal at Home Gateway

Physical Media

- bit: propagates across link
- physical link: transmitted data bit
- guided media:
  - signals propagate in solid media: copper, fiber
- unguided media:
  - signals propagate freely, e.g., radio

Twisted Pair (TP)

two insulated copper wires

Category 3: traditional phone wires, 10 Mb/s Ethernet

Category 5 TP: 100 Mb/s Ethernet

Category 6 TP:

1 Gb/s Ethernet



Categories of Unshielded Twisted Pair

The EIA/TIA (Electronic Industry Association/Telecommunication Industry Association) has established standards of UTP.

Type	Use
Category 1	Voice Only (Telephone Wire)
Category 2	Data to 4 Mbps (LocalTalk)
Category 3	Data to 10 Mbps (Ethernet)
Category 4	Data to 20 Mbps (16 Mbps Token Ring)
Category 5	Data to 100 Mbps (Fast Ethernet)
Category 5e	Data to 100 Mbps (Fast Ethernet)
Category 6	Data to 250 MHz (Gigabit Ethernet?)

Ca a



(a) (b)

Twisted Pair Cable

- **RJ45 is a standard type of connector for network cables.**
- **RJ45 connectors are most commonly seen with Ethernet cables and networks.**

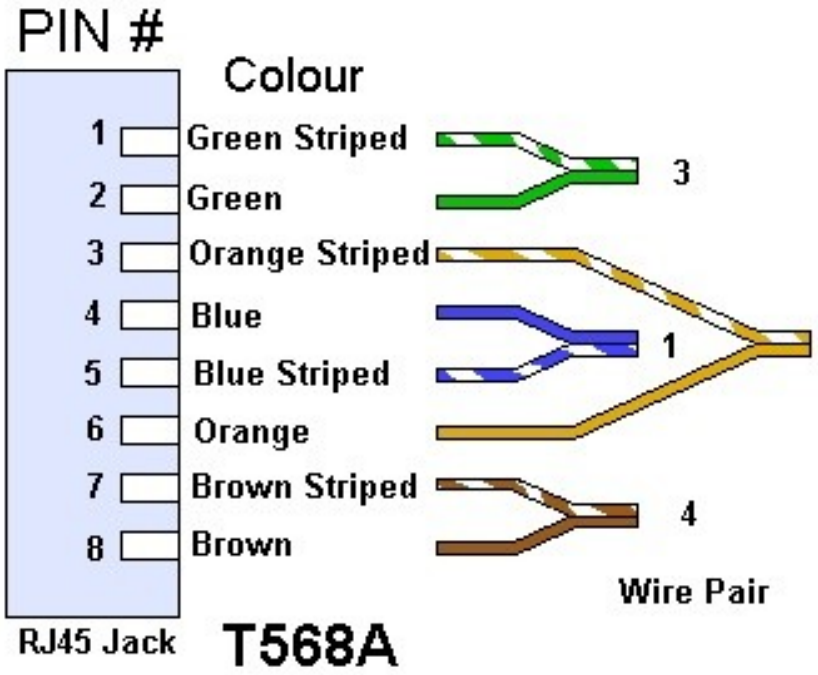


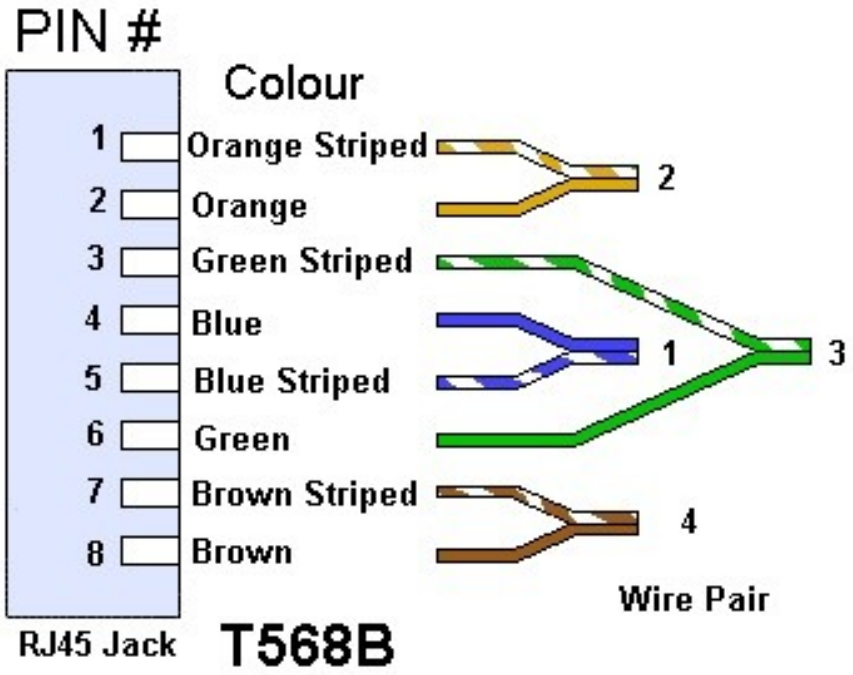




How Twisted-Pair Cables and Connectors Are Wired

- The most common standards are **T568A** and **T568B** with the second being of the most use.

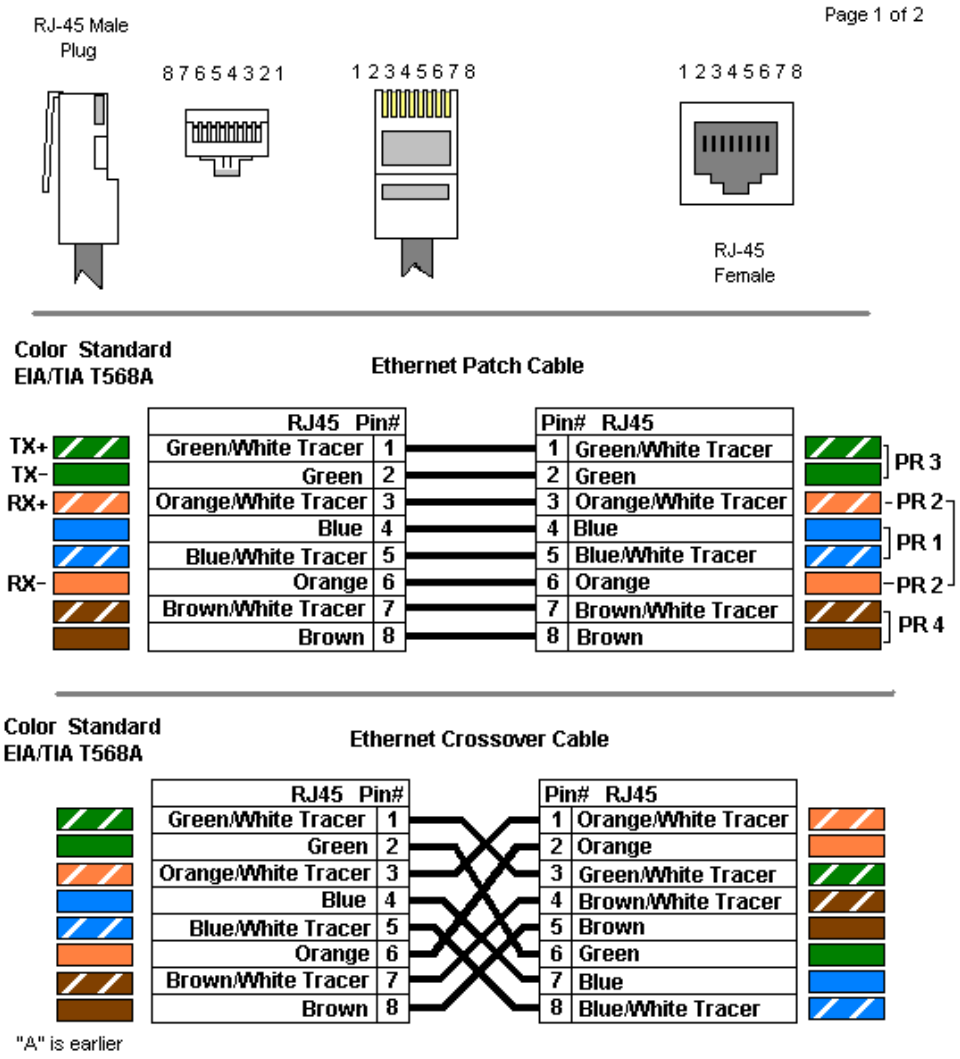




How Twisted-Pair Cables and Connectors Are Wired

Pin	100BaseT Purpose	T568A Wiring	T568B Wiring
1	Transmit+	Pair 3: White/green	Pair 2: White/orange
2	Transmit-	Pair 3: Green	Pair 2: Orange
3	Receive+	Pair 2: White/orange	Pair 3: White/green
4	(Used only on Gigabit Ethernet)	Pair 1: Blue	Pair 1: Blue
5	(Used only on Gigabit Ethernet)	Pair 1: White/blue	Pair 1: White/blue
6	Receive-	Pair 2: Orange	Pair 3: Green
7	(Used only on Gigabit Ethernet)	Pair 4: White/brown	Pair 4: White/brown
8	(Used only on Gigabit Ethernet)	Pair 4: Brown	Pair 4: Brown

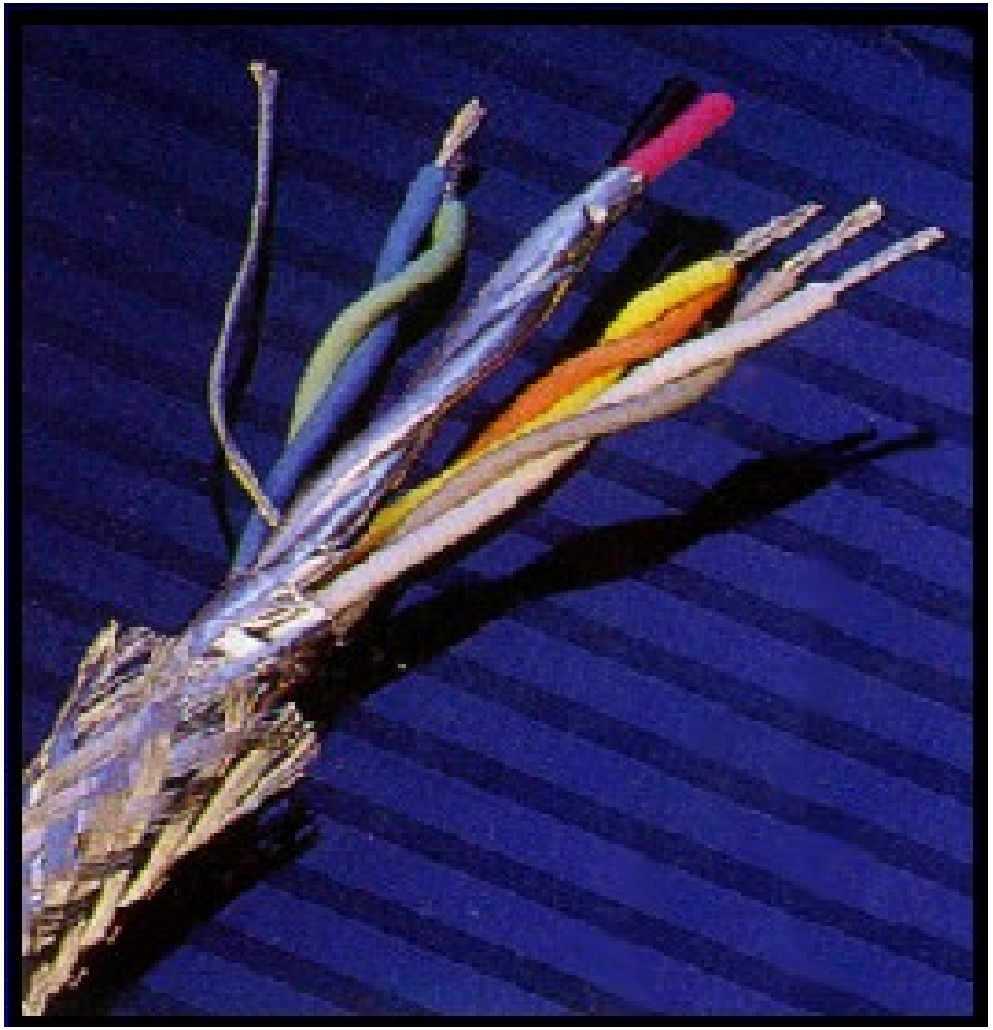
How Twisted-Pair Cables and Connectors Are Wired



Straight-through cable:
used to connect a computer to a switch or other network device.
Also called a **patch cable.**

Crossover cable:
used to connect two like devices such as a hub to a hub or a PC to a PC.
Transmit and receive lines are reversed

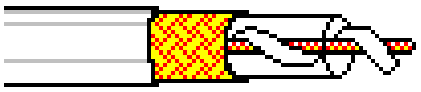
STP



Physical Media: coax, fiber

Coaxial cable:

- wire (signal carrier) within a wire (shield)
  - baseband: single channel on cable
  - broadband: multiple channel on cable
- bidirectional
- formerly common use in 10 Mb/s Ethernet



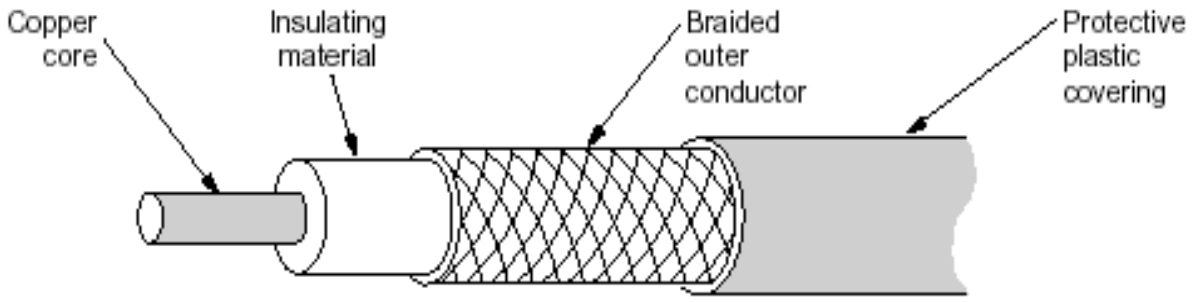
Fiber optic cable:

- glass fiber carrying light pulses
- typically dozens bundled into one cable
- high-speed operation:
  - 100 Mb/s Ethernet
  - high-speed point-to-point transmission (e.g., 40 Gb/s)
- low error rate: repeaters spaced far apart; immune to electromagnetic noise



Coaxial Cable

- **Exactly like the one you use for your TV Set:**

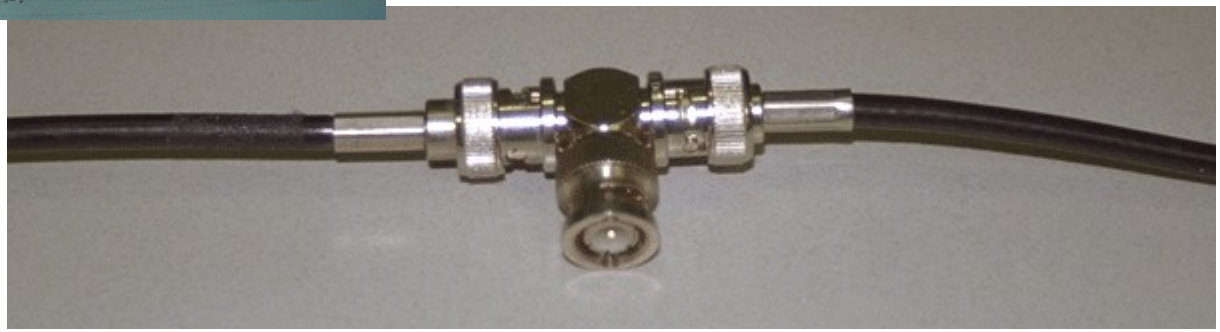


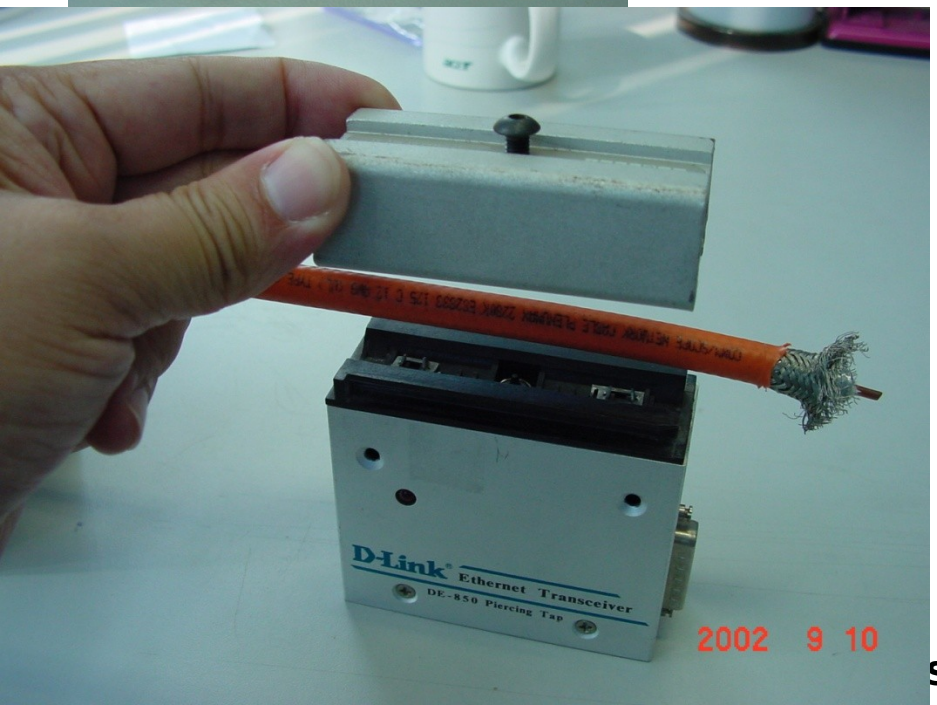
- **Baseband Coax**
  - **50-ohm cable for digital transmission**
  - **10Base-2, BNC, Thin-LAN, 185m/per segment**
  - **10Base-5, AUI, Thick-LAN, 500m/ per segment**
  - **At most 5 segments, up to 945m/2500m.**
- **Broadband Coax**
  - **75-ohm cable for analog transmission, like cable TV.**


2002 9 10








D-Link
Ethernet Transceiver
DE-850 Piercing Tap
2002 9 10


2002 9 10

For Notice & Materials



https://lexue.bit.edu.cn/course/view.php?id=17931

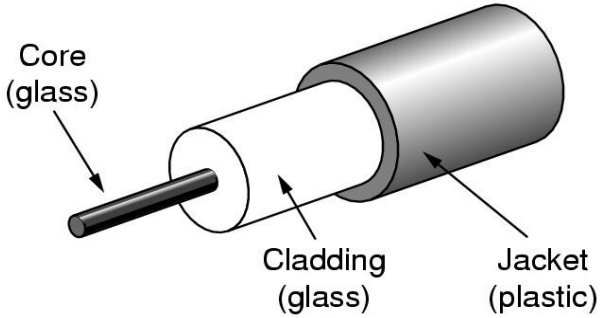
2025-2026 2nd Semester - Computer Networks (English)

Can also access from

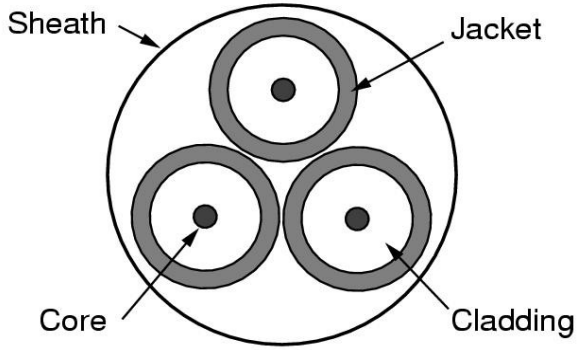
https://www.yanhekt.cn

Enroll Key: CN2026

Fiber Optics



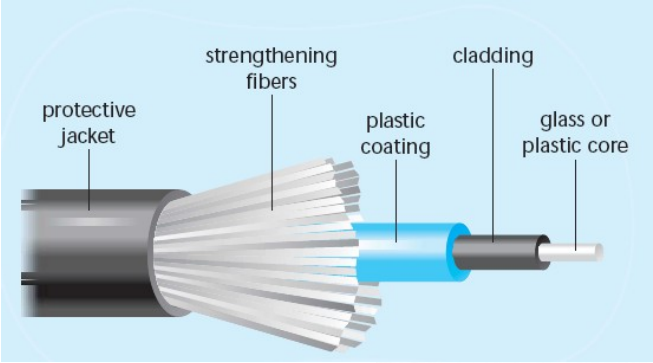
(a)



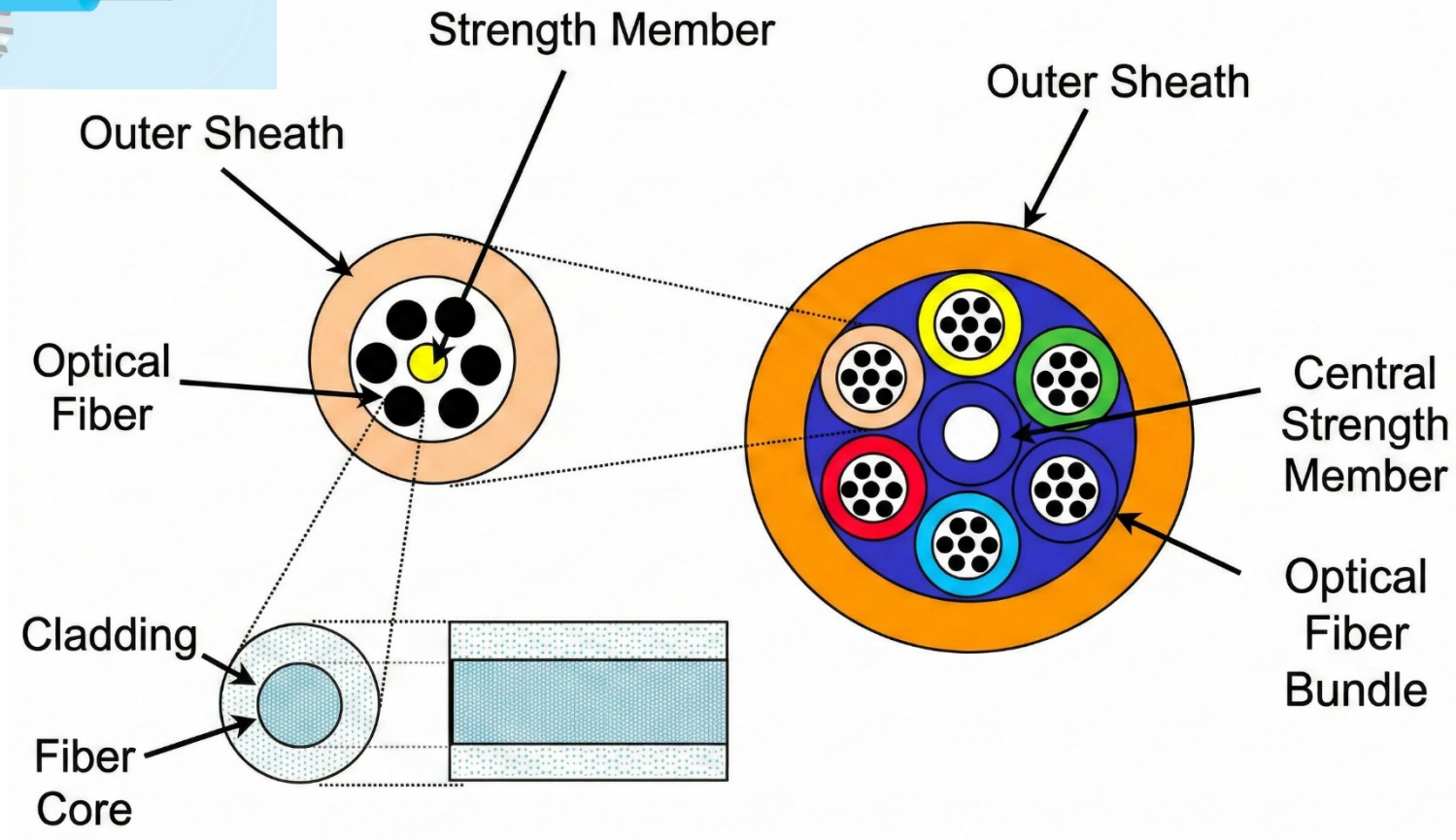
(b)

Item	LED	Semiconductor Laser
Data rate	Low	High
Mode	Multimode	Multimode or single mode
Distance	Short	Long
Lifetime	Long life	Short life
Temperature sensitivity	Minor	Substantial
Cost	Low cost	Expensive

Fiber Optics



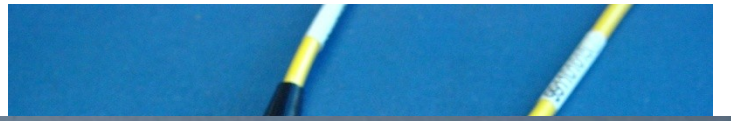
IGH-DENSITY MULTI-CORE OPTICAL CABLE CROSS-SECTION STRUCTURE

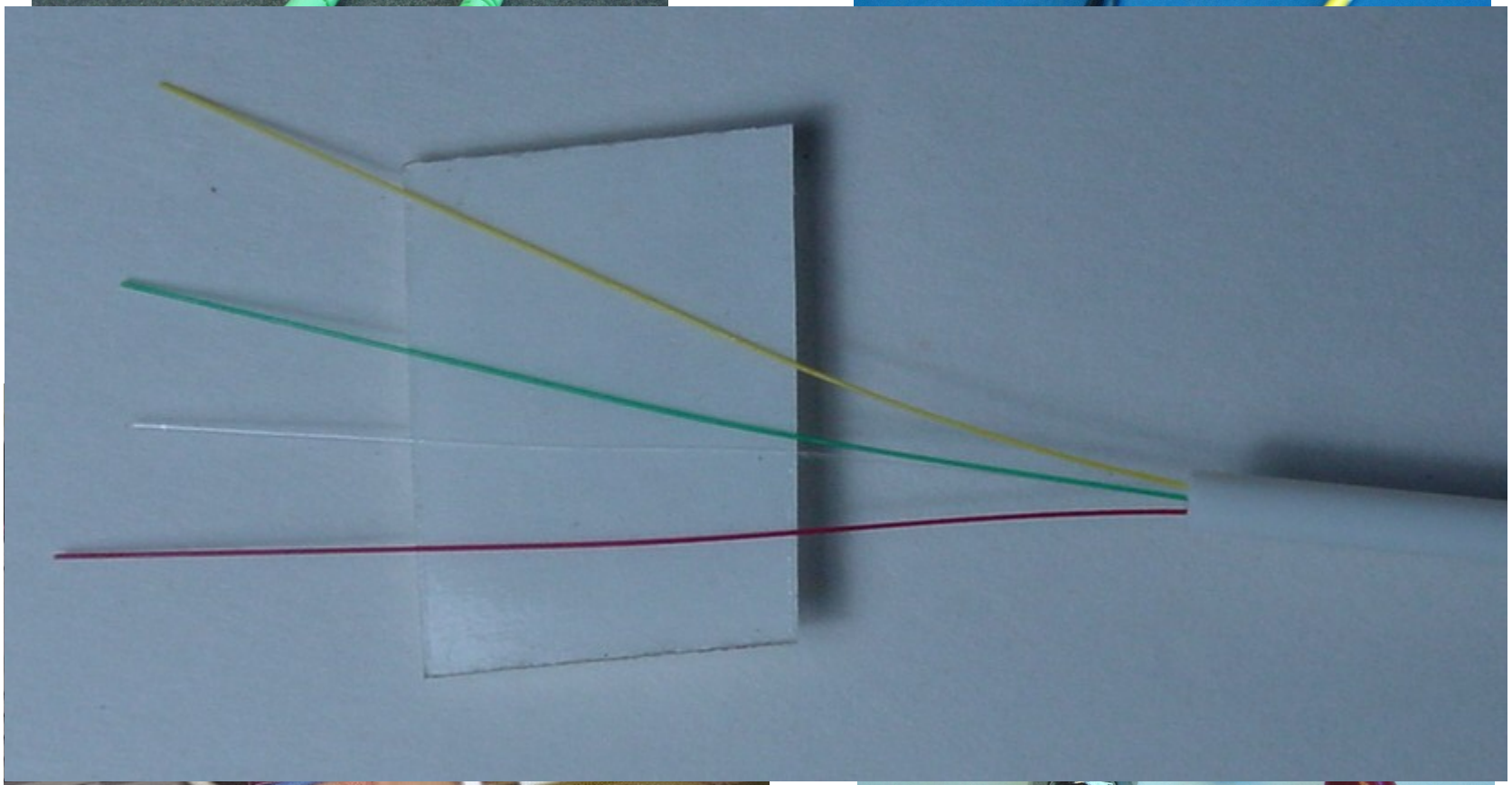


SC： 568A 标准，方形，插入锁定

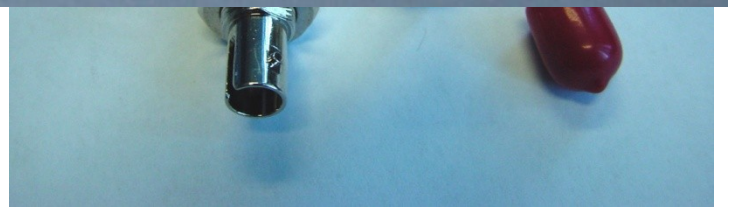


ST： 插入锁定

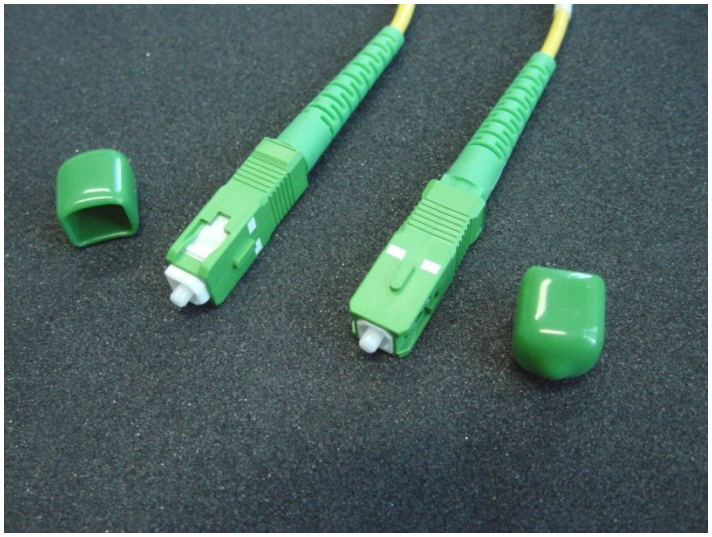




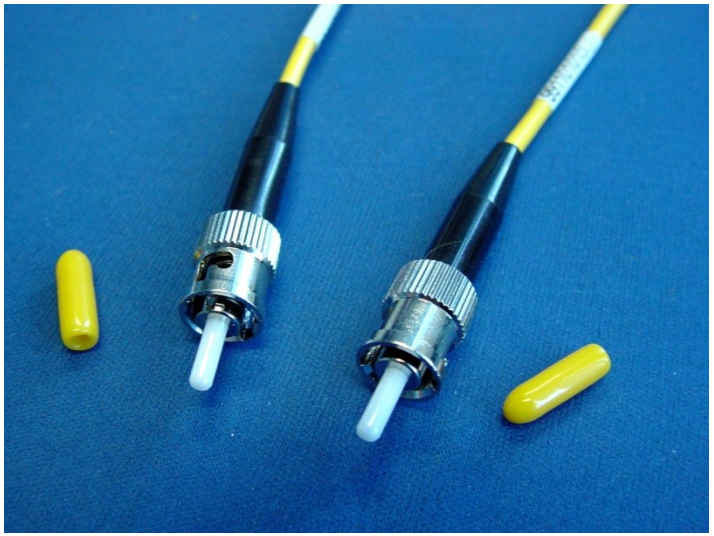


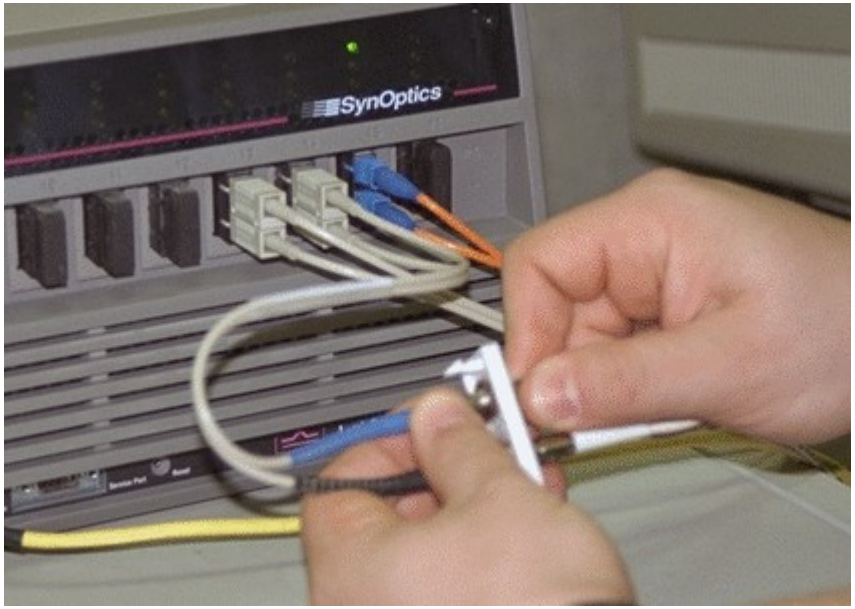


SC： 568A 标准，方形，插入锁定



ST： 插入锁定





光耦合器（ST）



- **Attenuation in decibels = $10\log_{10}$(transmitted power/received power)**

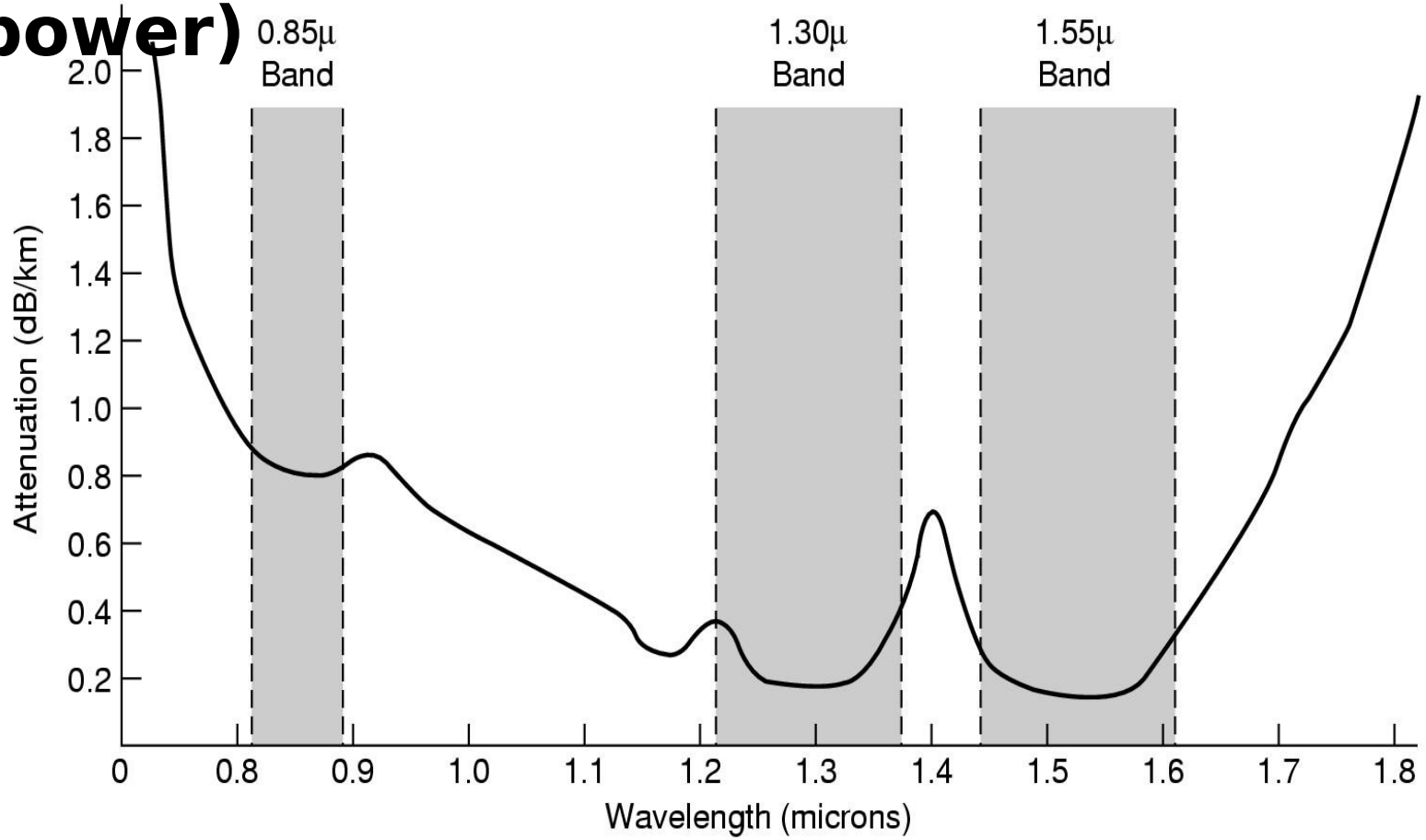


Fig. 2-6 Attenuation of light through fiber in the infrared region.

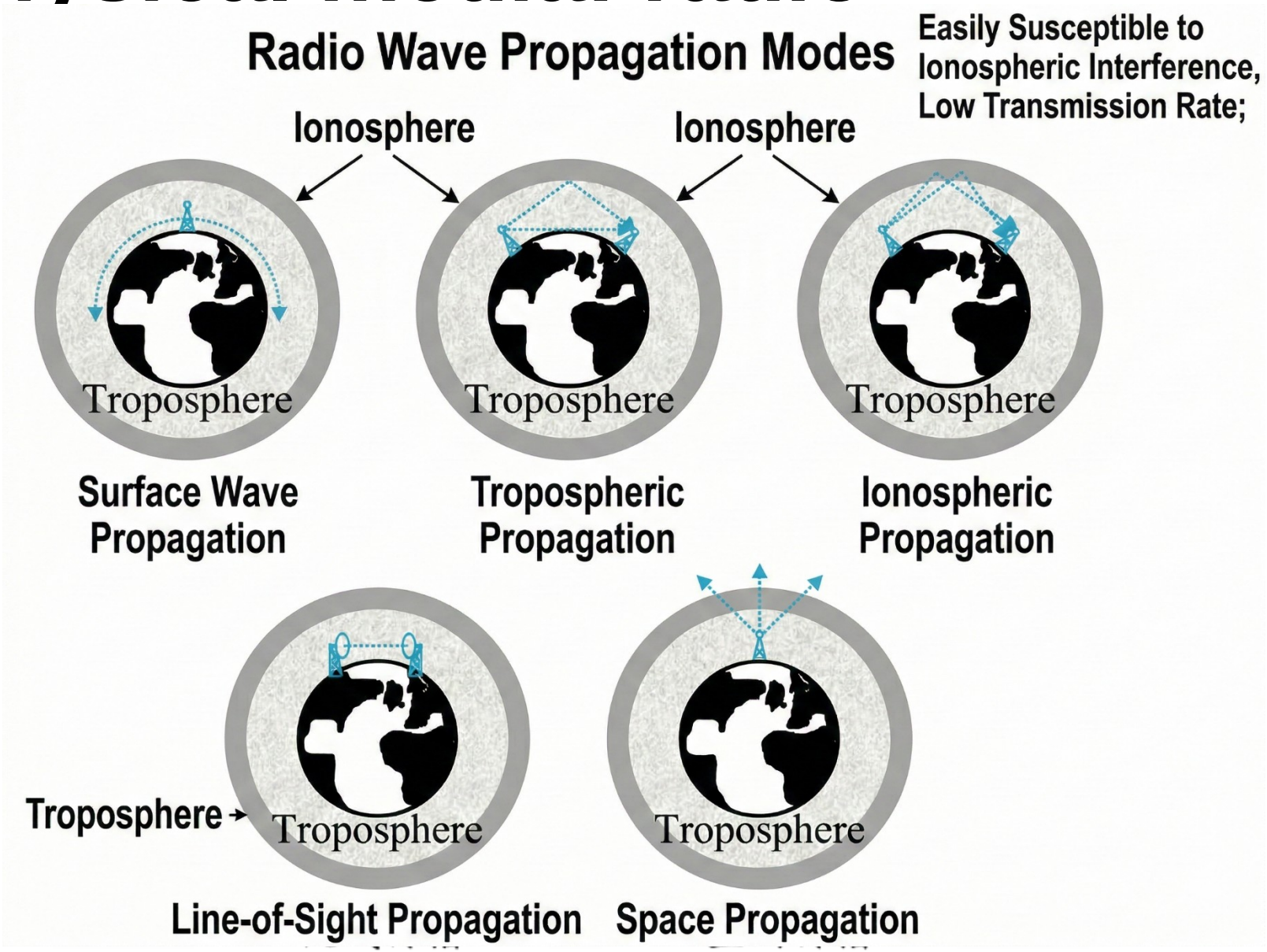
Physical media: radio

- signal carried in electromagnetic spectrum
- no physical "wire"
- bidirectional
- propagation environment effects:
  - reflection

Radio link types:

- terrestrial microwave
  - e.g. up to 45 Mb/s channels
- LAN (e.g., WiFi)
  - 2Mbps, 11Mb/s, 54Mbps
- wide-area (e.g., cellular)
  - e.g. 3G: hundreds of kb/s
- satellite
  - up to 50 Mb/s channel (or multiple smaller

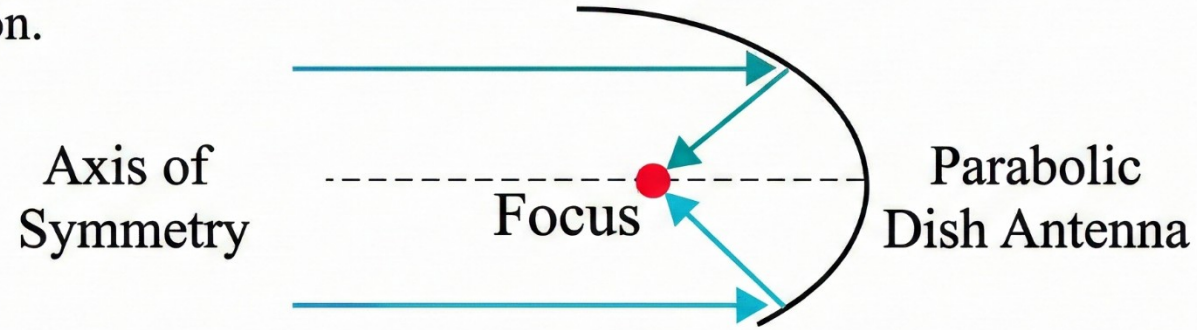
Physical media: radio



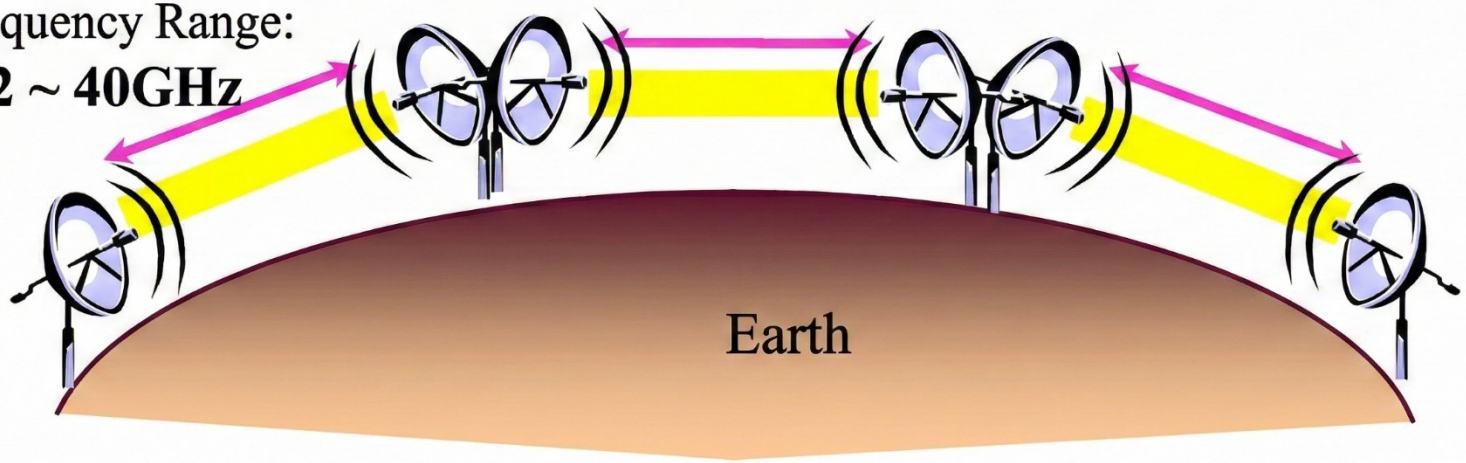
Physical media: radio

MICROWAVES

Line-of-sight propagation, requires relay stations (Tower height 100m, station spacing 100km);
Wide bandwidth, large capacity, resistant to interference, lower investment than cable communication.



Communication
Frequency Range:
2 ~ 40GHz

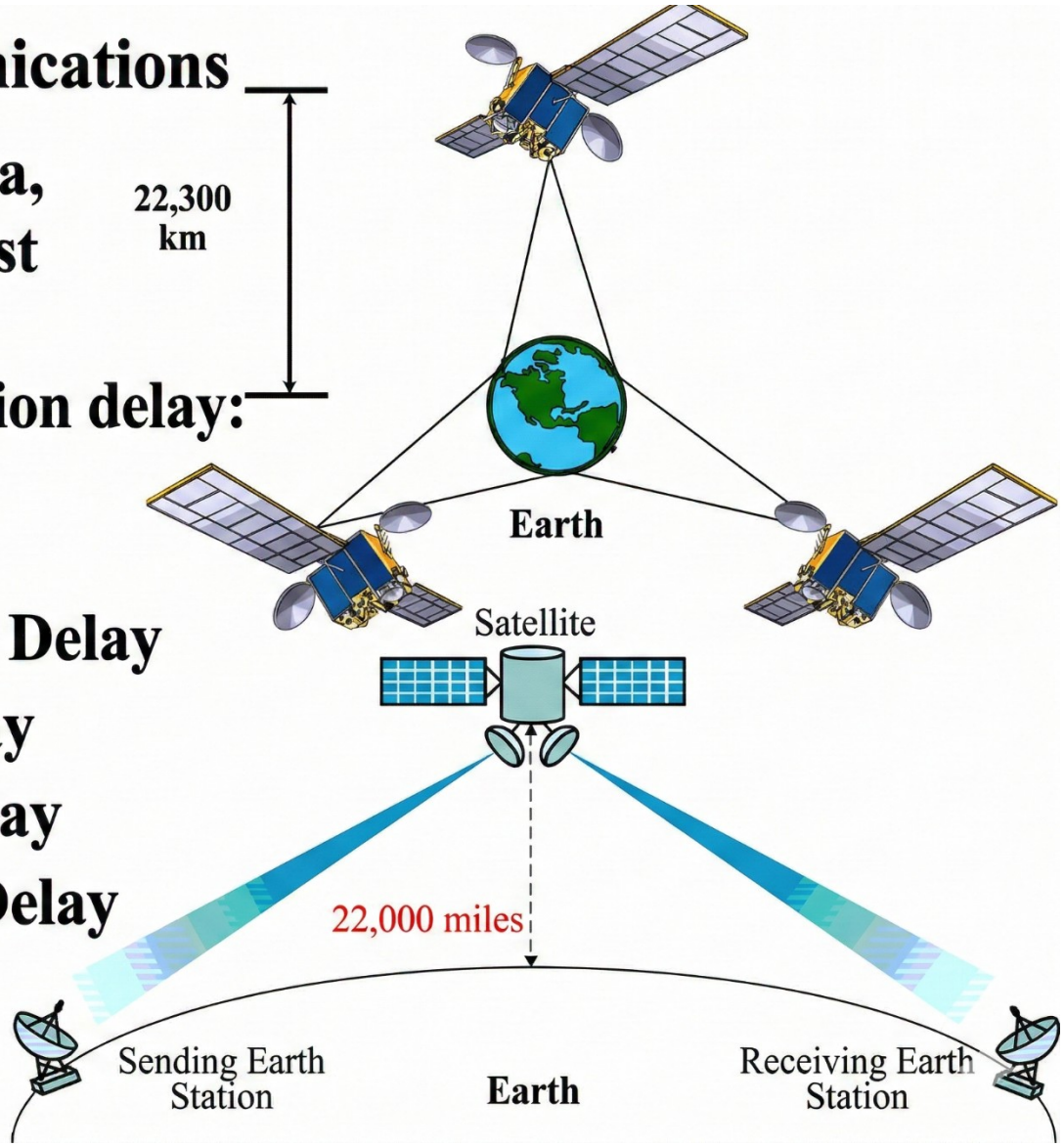


Physical media: radio

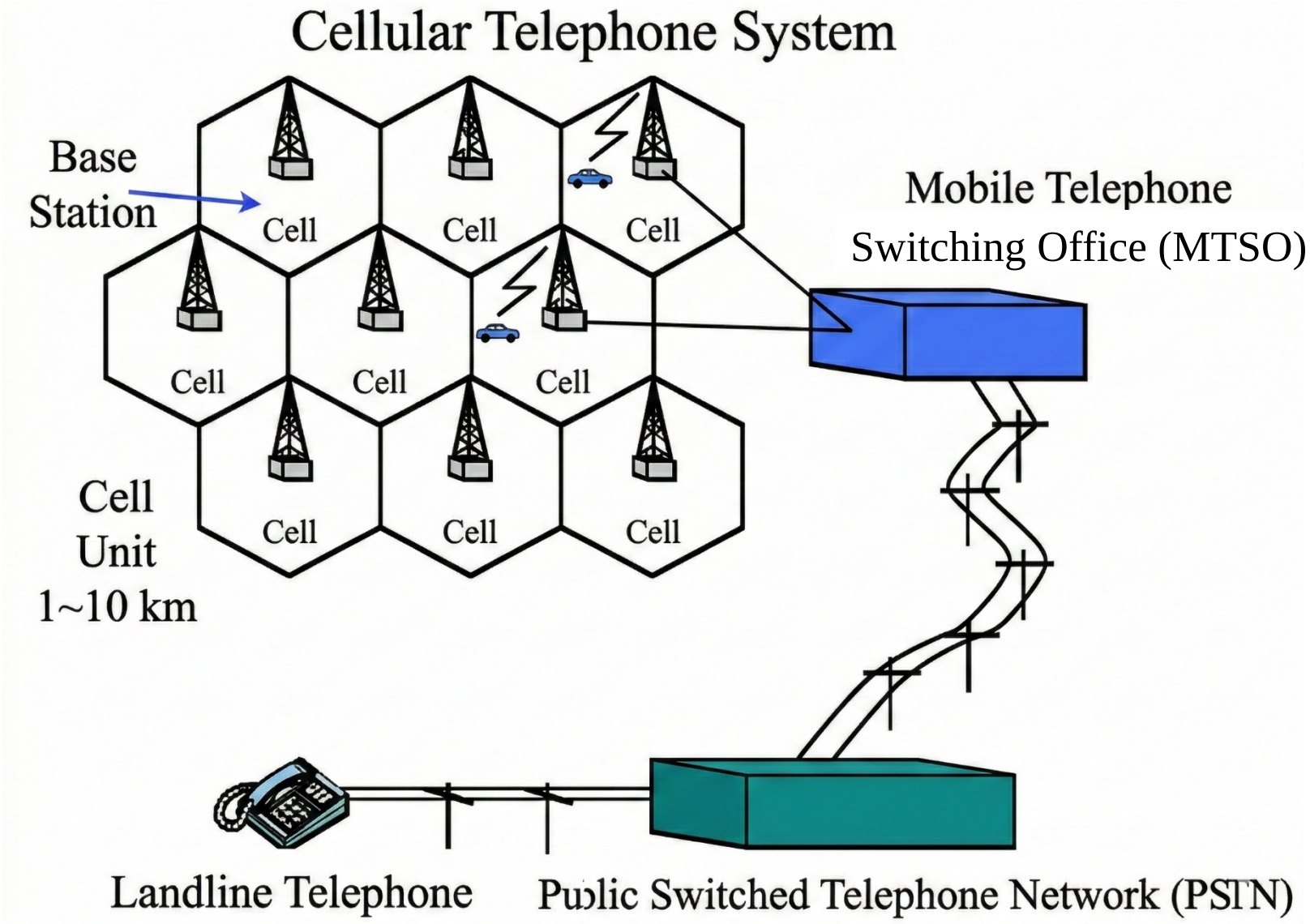
Satellite Communications

Wide coverage area, suitable for broadcast communication;
Significant propagation delay: 250ms

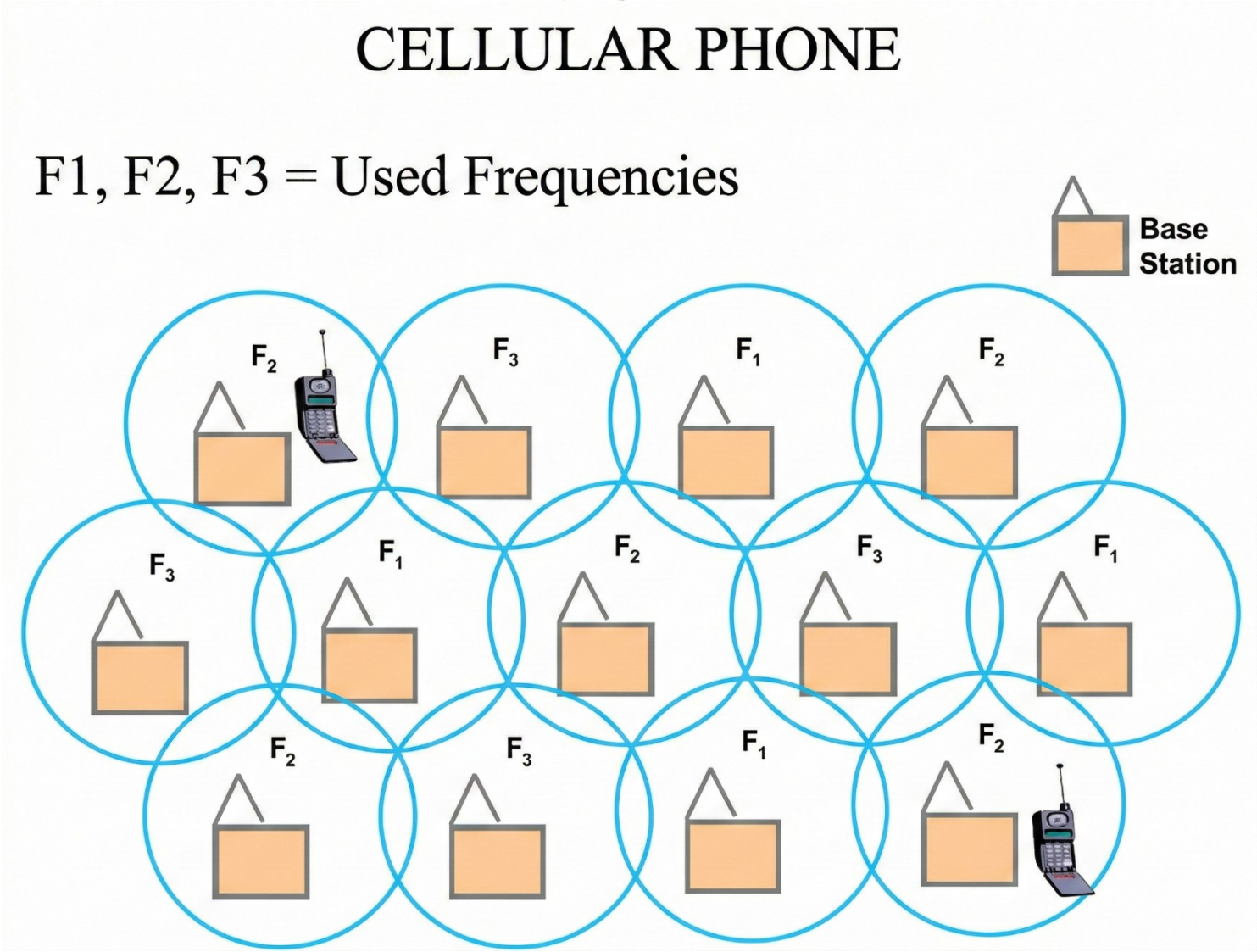
Total Data Transfer Delay
= Propagation Delay
+ Transmission Delay
+ Retransmission Delay



Physical media: radio



Physical media: radio



Other network media

- **Power lines**
- **Home phone lines – bus vs. star**
- **Infrared (IR)**
- **BlueTooth – *piconet***
- **Serial lines (RS232, RS485)**
- **Firewire (IEEE 1394)**

Comparison

Transmission Media	Rate	Transmission Distance	Performance (Antenna Interference Immunity)	Price	Application
Twisted Pair	10-1000Mb/s	Several dozen km	Fair	Low	Analog/Digital Transmission
50Ω Coaxial Cable	10Mb/s	Within 3km	Good	Slightly higher than Twisted Pair	Baseband Digital Signal
75Ω Coaxial Cable	300-450MHz	100km	Good	Relatively High	Analog TV Transmission, Data & Audio
Optical Fiber	Several dozen Gbps	30km up	Very Good	Relatively High	Long Distance Transmission
Short Wave	<50MHz	Global	Fairly Poor	Relatively Low	Long Range Low Speed Communication
Ground Microwave Relay	4-6GHz	Several hundred km	Good	Medium	Long Range Communication
Satellite	500MHz	18000km	Very Good	Irrelevant to Distance	Long Range Communication

Physical media: capacity

- **Capacity has theoretical limit**
  - **Shannon's Law:** capacity limit given by
  - **C = *B* $\log_2$ (1 + S/N)** with spectral bandwidth ***B***
  - **E.g., phone has *B* = 3000 Hz, S/N = 35 dB, *C* = 34**$10\log_{10} S/N$
  - **dB =** $10^{2.5} = 316$ **（Signal-to-Noise Ratio）**
  - **E.g., 25 dB =**
  - **Radio: typically 1 b/s per Hz of bandwidth**
- **Speed has physical limits:**
  - **c in free space, 0.66 c in fiber**

Chapter 1: Roadmap

1.1 Computer network and the Internet

1.2 Network edge and core

1.3 Network access and physical media

1.4 Transmission Technology

1.5 Type and Topology

1.6 Network Performance

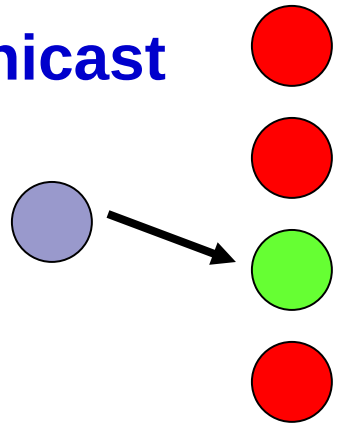
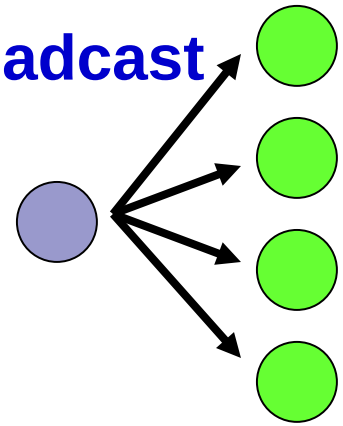
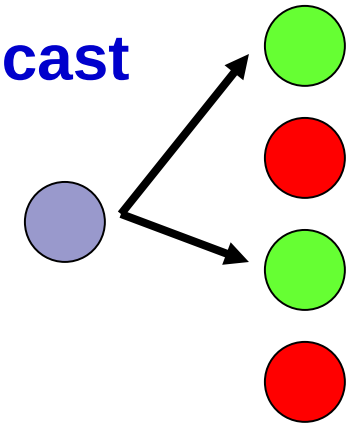
1.7 Protocol layers, service models

1.8 Architecture, OSI and TCP/IP Models

1.9 Internet History

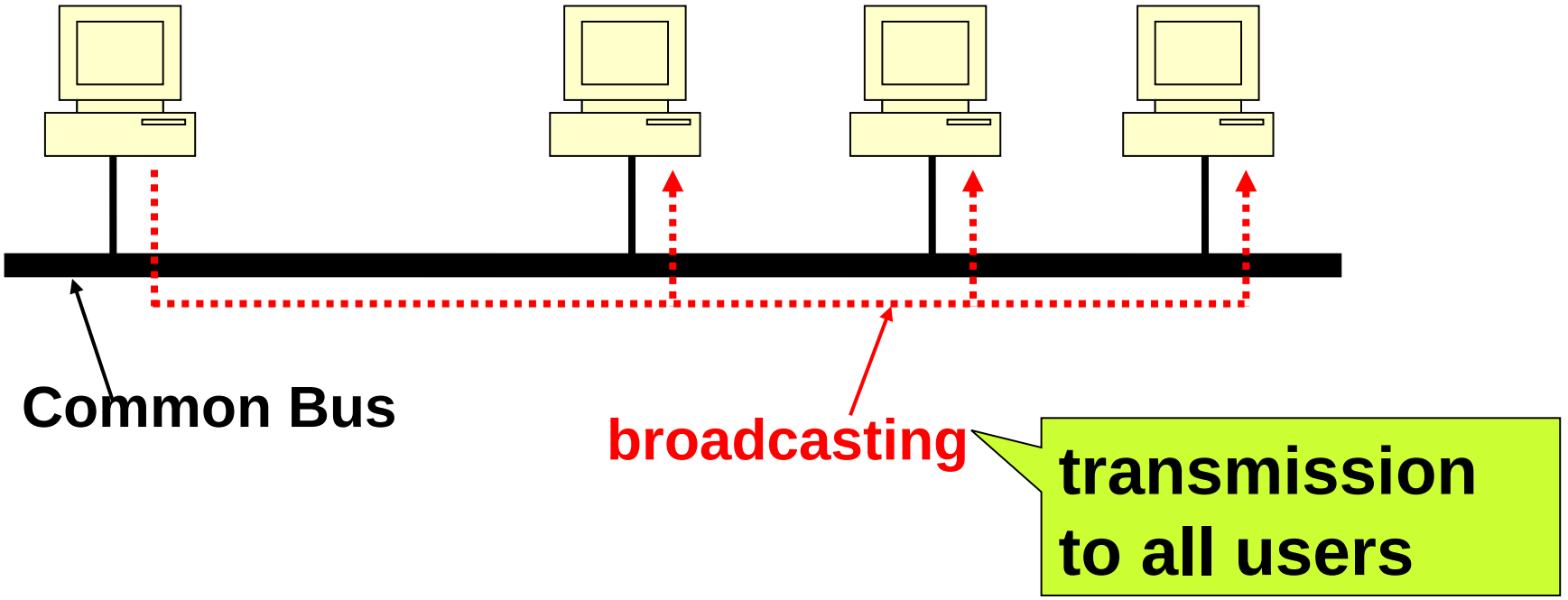
Transmission Technology

- A message can be unicast, multicast, or broadcast.

Unicast	Broadcast	Multicast
		
one source to one destination	One source to all nodes on (small, local) network	one source (potentially many sources) to many destinations

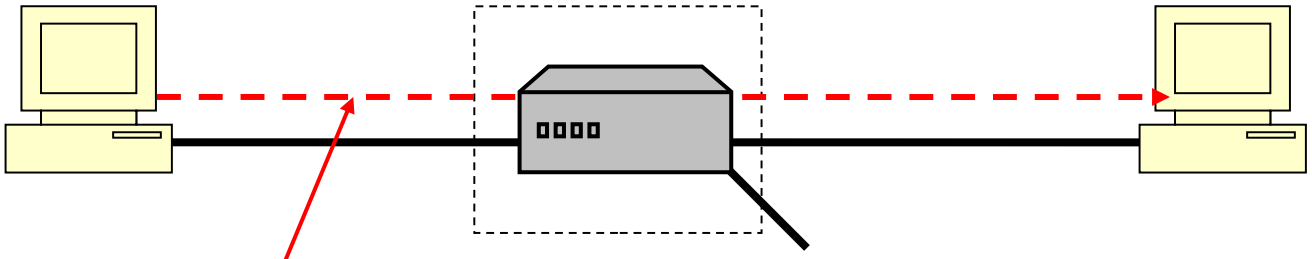
Broadcast

- A single communication channel is shared by all computers, that is, sending a message implies that all others receive it.



Unicast

- Computers are connected in pairs, that is, sending a packet goes strictly from the sender to the receiver (**unicasting**), possibly having to visit intermediate machines (*routing*).



unicasting

only one sender and one receiver

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1.6 Network Performance

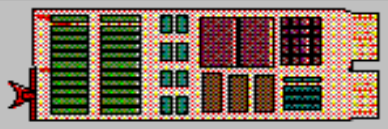
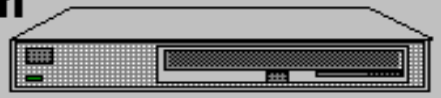
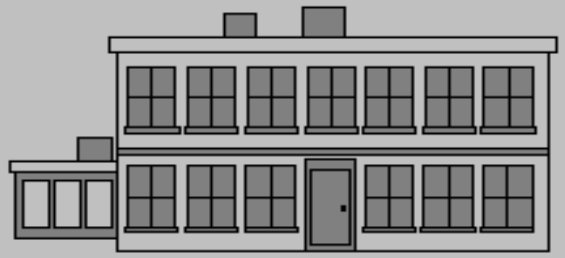
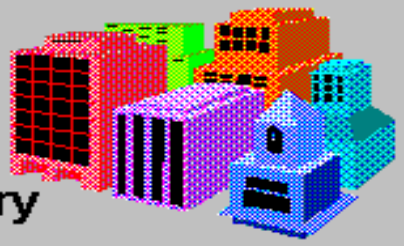
1.7 Protocol layers, service models

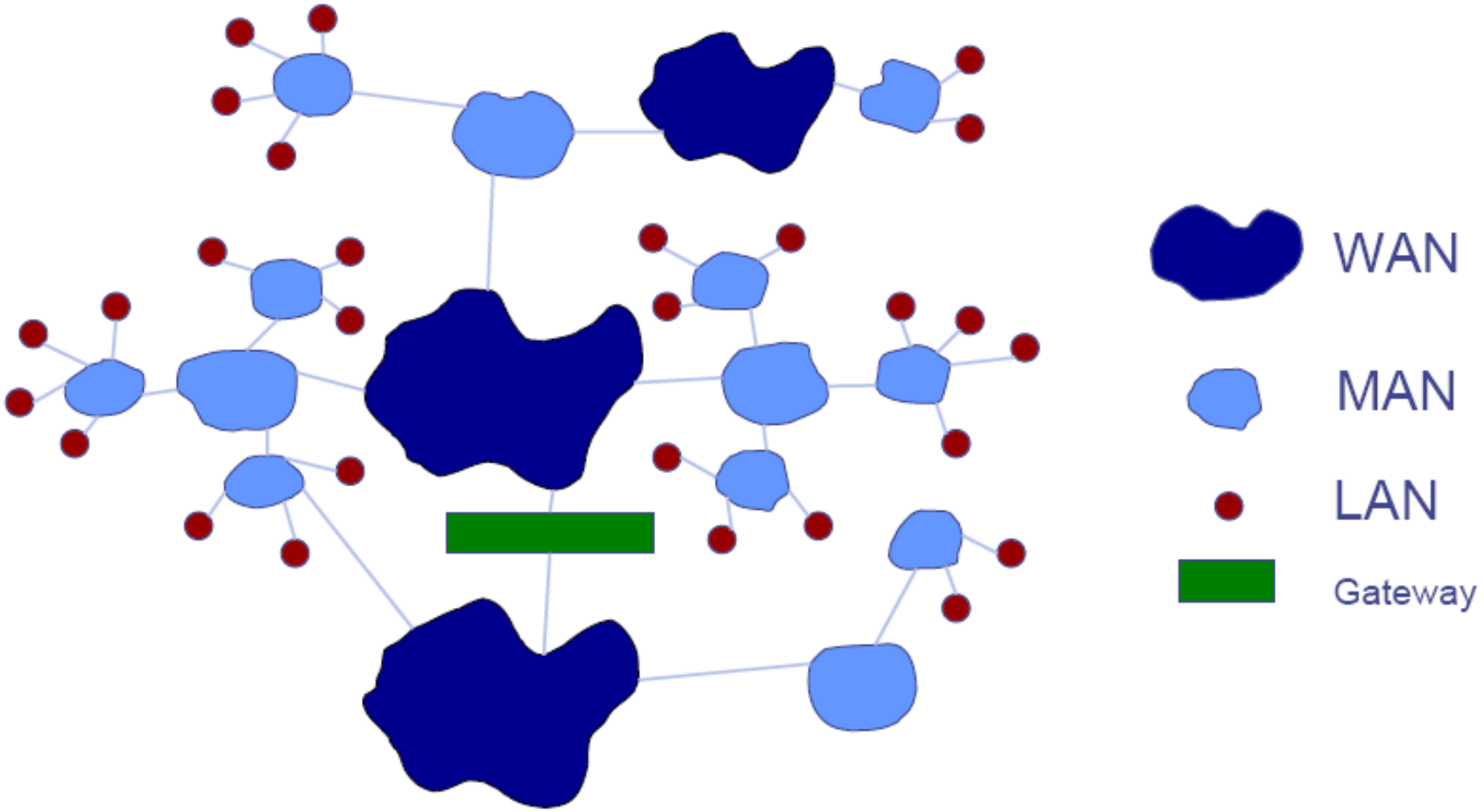
1.8 OSI and TCP/IP Architecture

1.9 Internet History

Types of Computer Networks

- **Geographical distance**
  - **Local Area Networks (LAN):**
    - Ethernet, Token ring, FDDI
  - **Metropolitan Area Networks (MAN):**
    - DQDB, SMDS
  - **Wide Area Networks (WAN):**
    - X.25, ATM, frame relay
- **Caveat:** LAN, MAN, WAN may mean different things

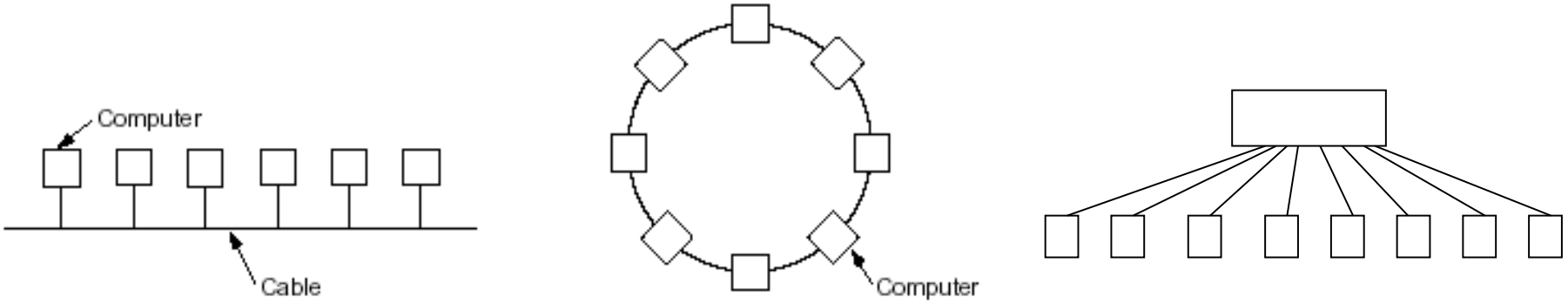
Distance	Processors located in same	Network Type
0.1 m	board 	Data Flow Machine
1 m	system 	Multiprocessor
10 m 100 m 1 km	 room, building, campus	L A N Local Area Network
10 km 100 km	city country 	W A N Wide Area Network
1000 km 10000 km	continent planet 	Interconection of WANs



Internetwork of WAN, MAN and LAN

Local Area Networks (LAN)

- **Apart from scale, LANs distinguish themselves from other networks by (generally) using broadcast technology, and having simple**



(a) Bus (b) Ring (c) Star

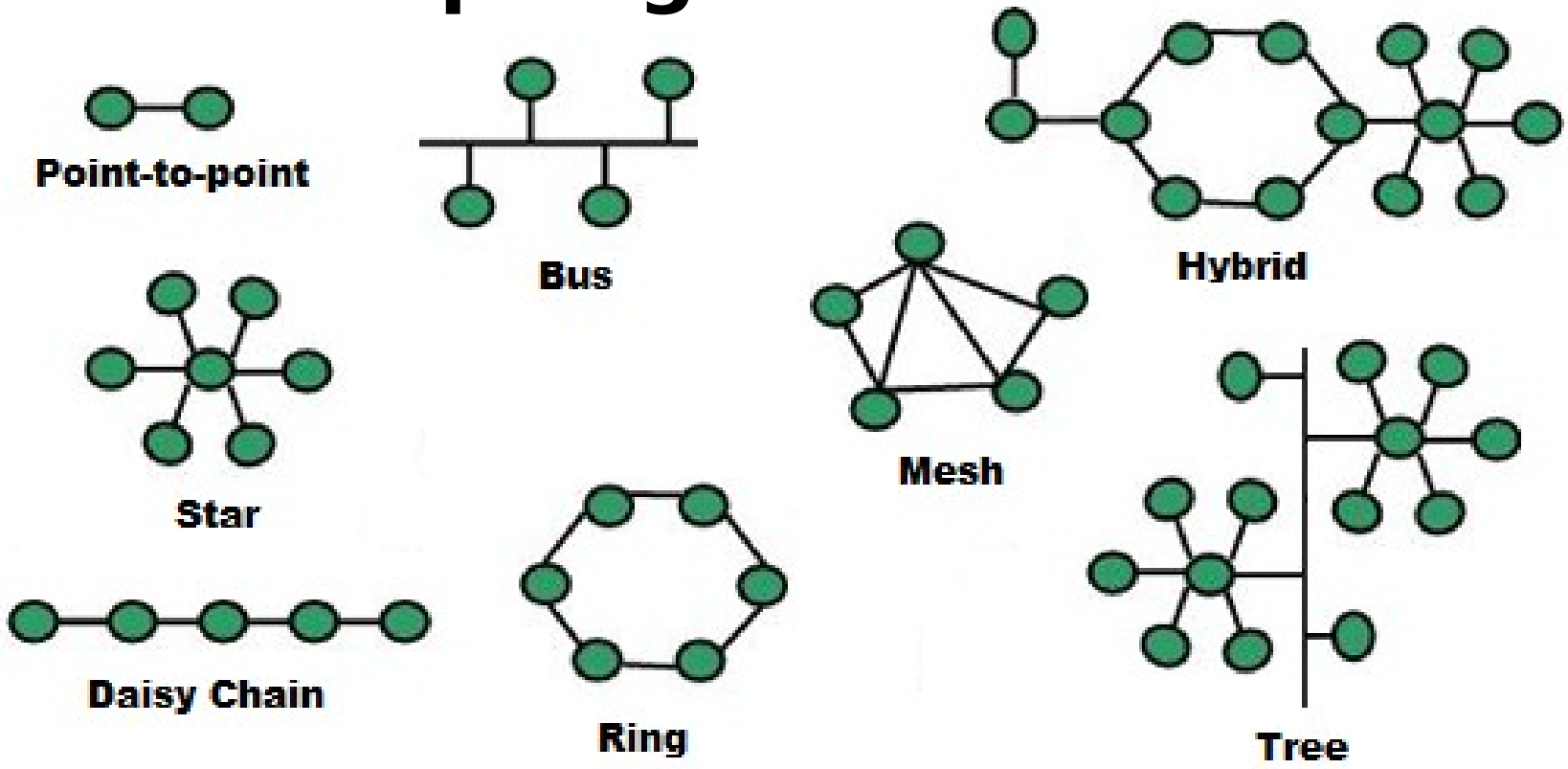
broadcast networks. (a) Bus. (b) Ring. (c) Star

Network Topology

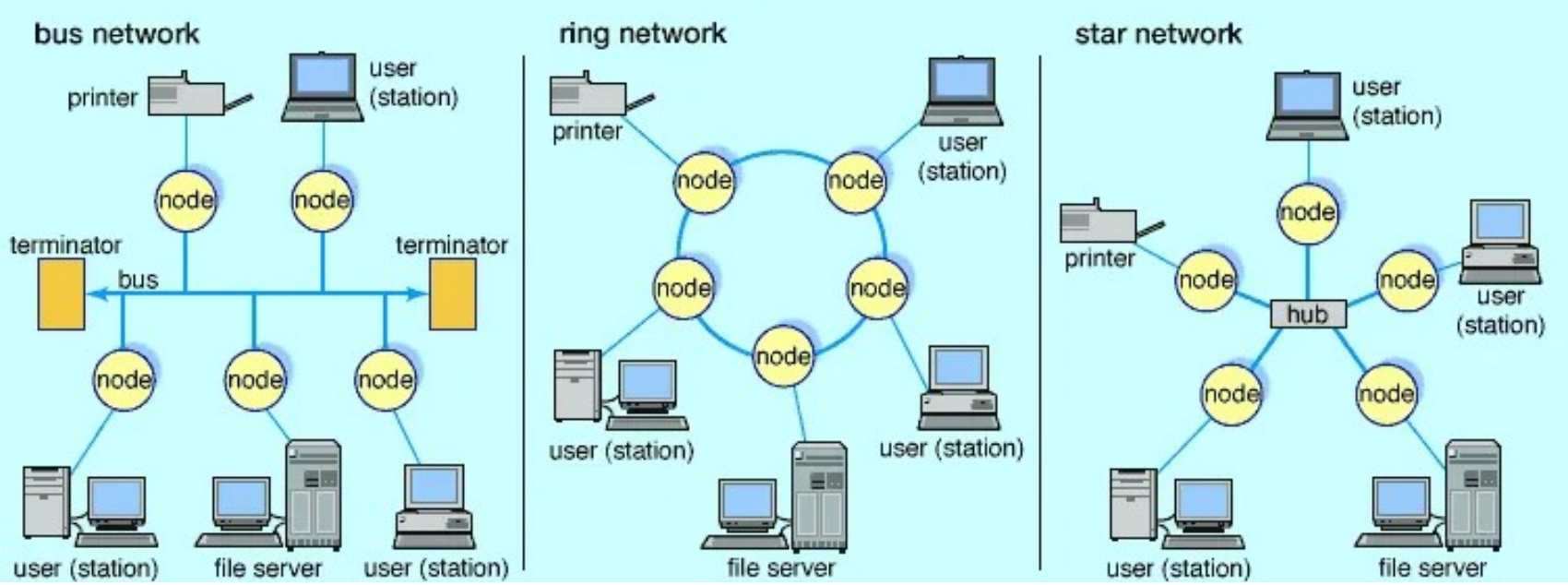
- **Computer network topology** is the way various components of a network (like nodes, links, peripherals, etc.) are arranged.
- **Network topologies** define the layout, virtual shape or structure of network, not only physically but also logically.

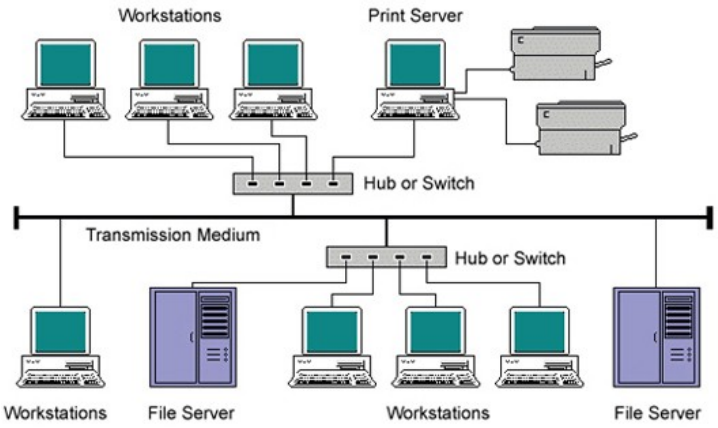
Network Topology

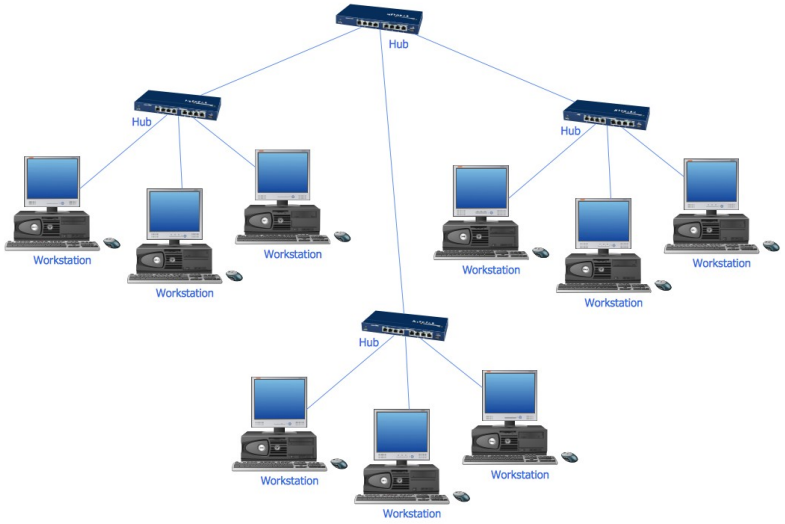
- 8 basic topologies:



Local Area Networks (LAN)

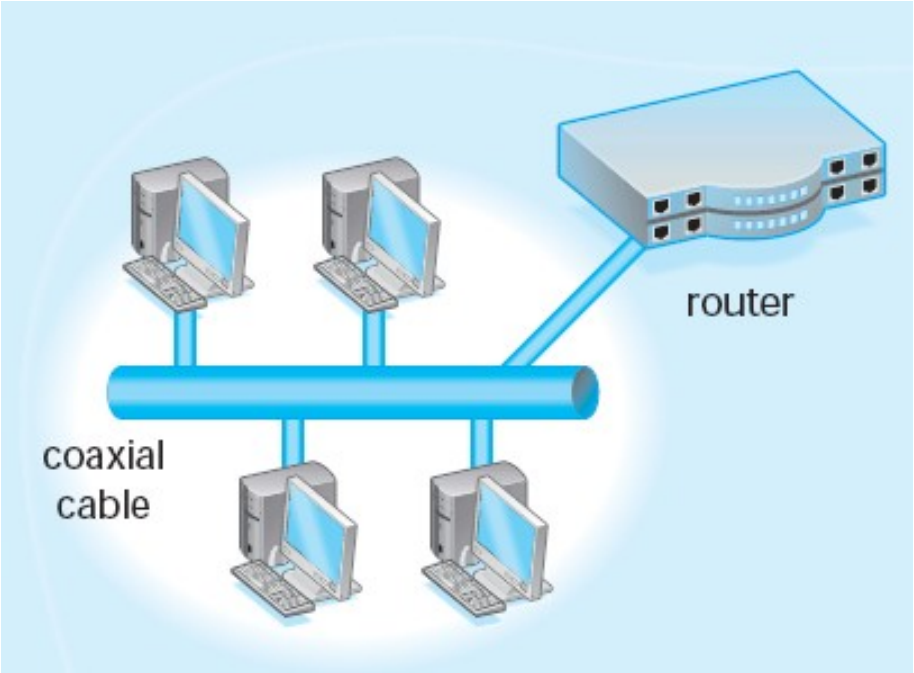






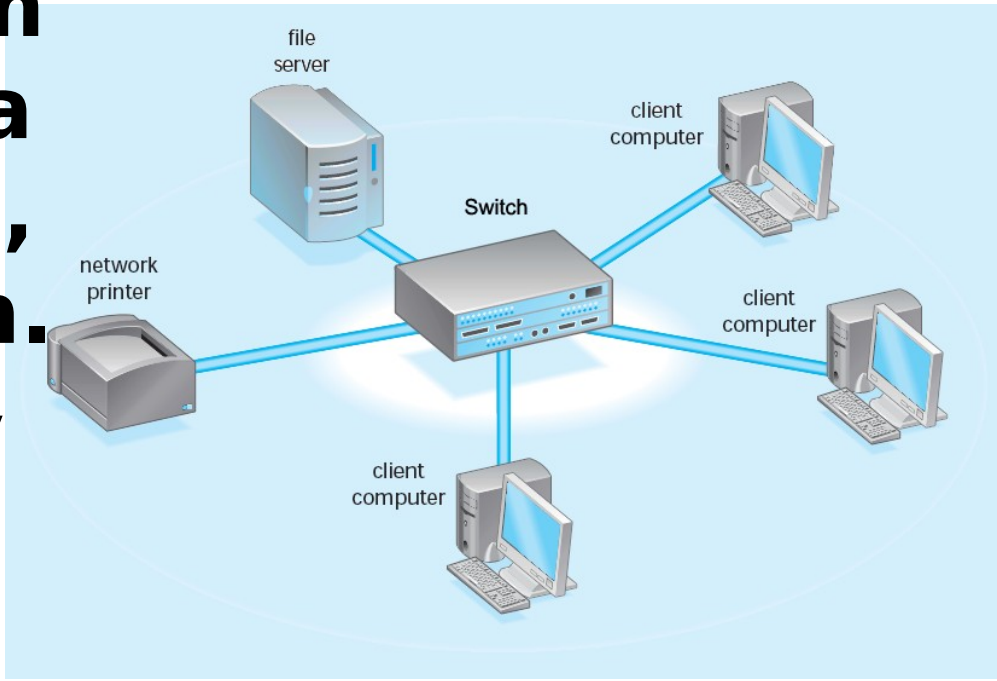
Bus Topology

- All network devices connected to a common cable in logical linear fashion.
- Transmissions are sent along the length of the bus segment.
- Adding hosts to the network requires breaking the network.
- Failure of one host can cause failure of network.

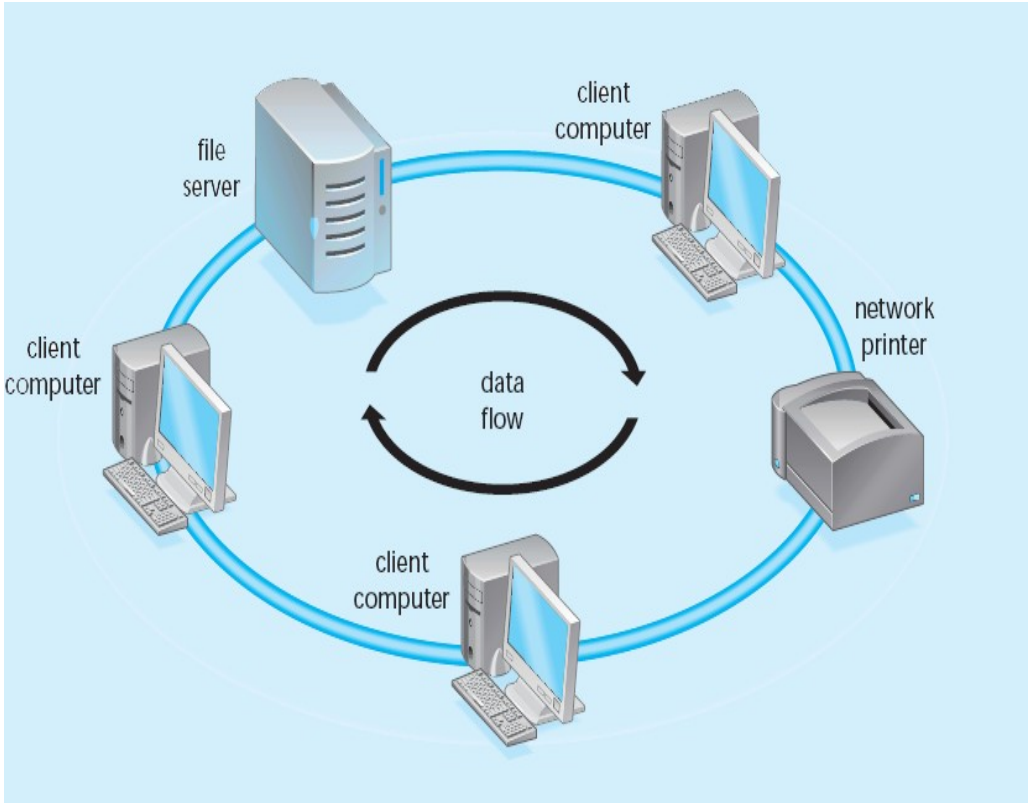


Star Topology

- Connection from each device to a central location, usually a switch.
- Most commonly used physical topology.
- Failure of one cable does not bring down network.

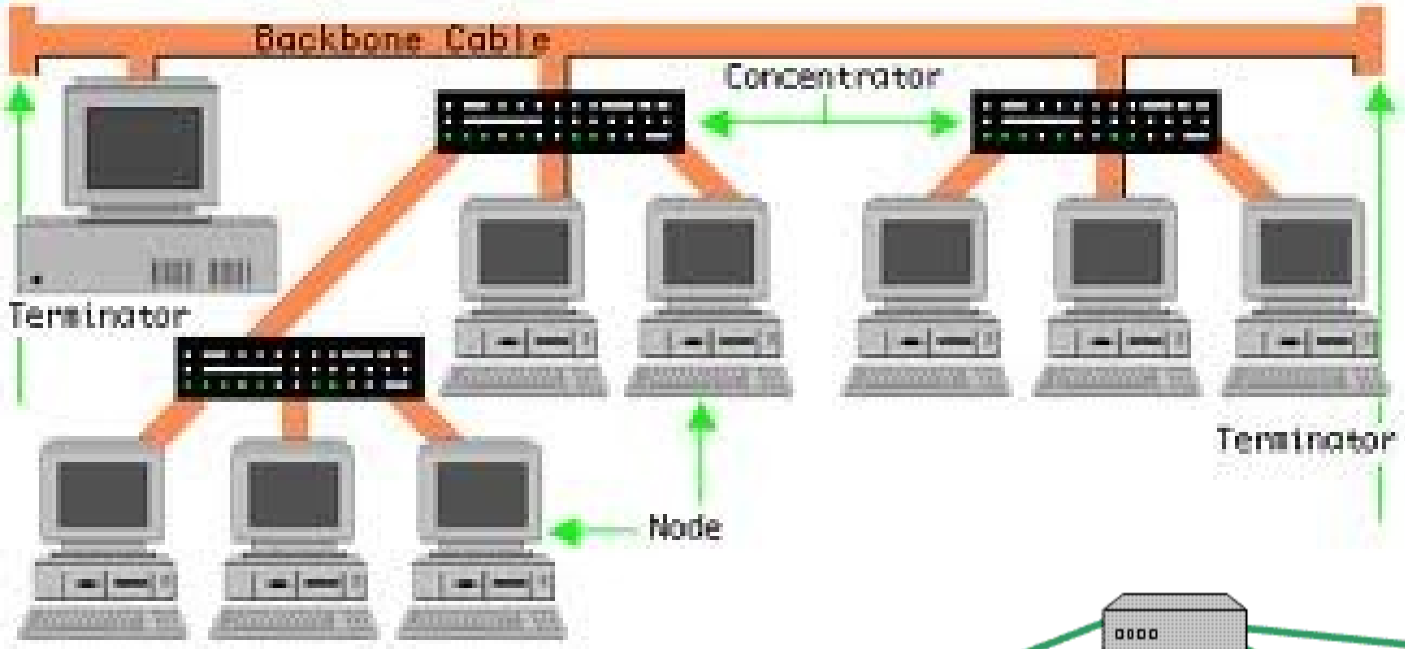


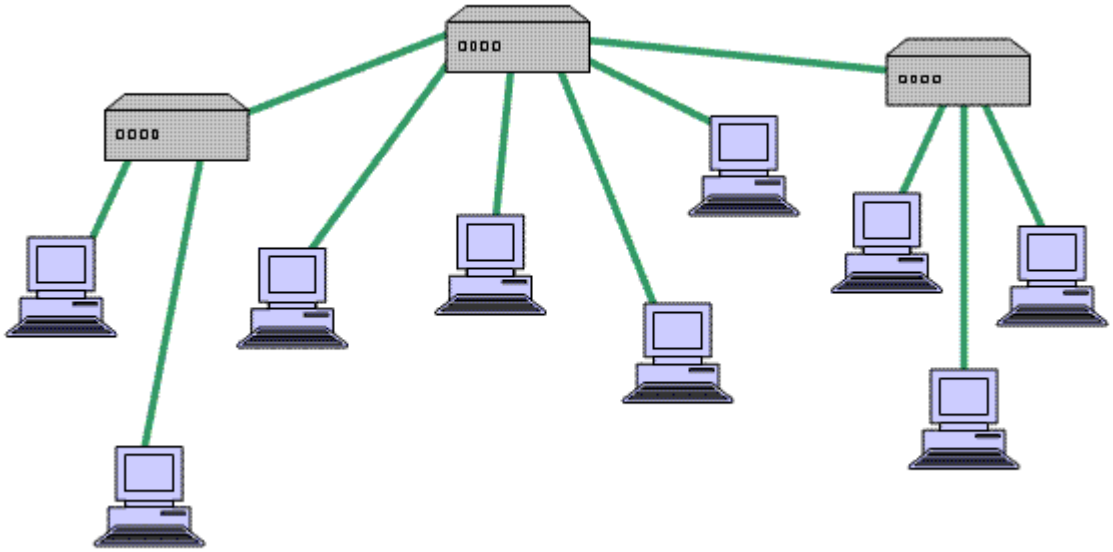
Ring Topology



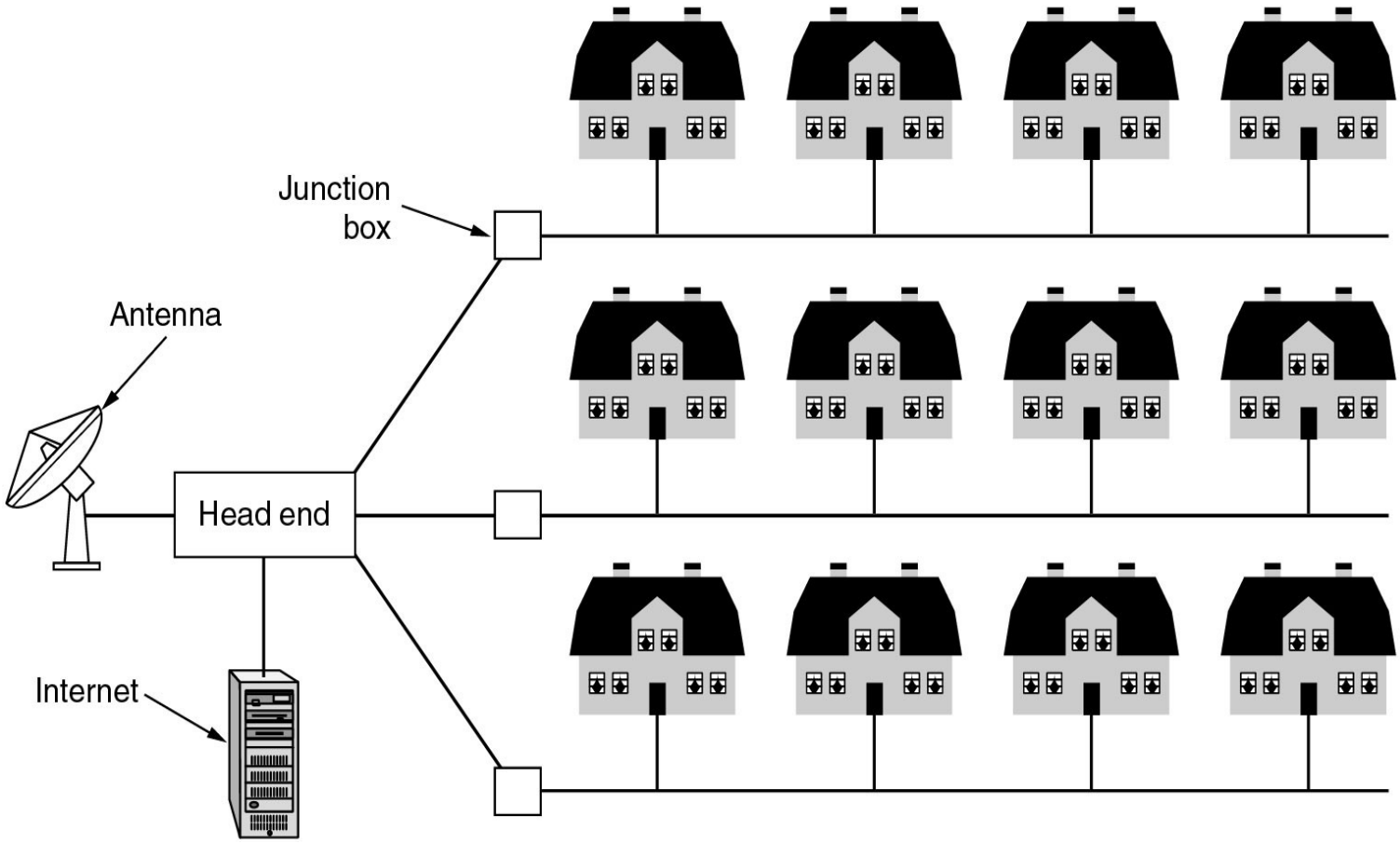
- Network is connected in an endless loop.
- No termination required.
- Uncommon topology today, more common in

Tree Topology





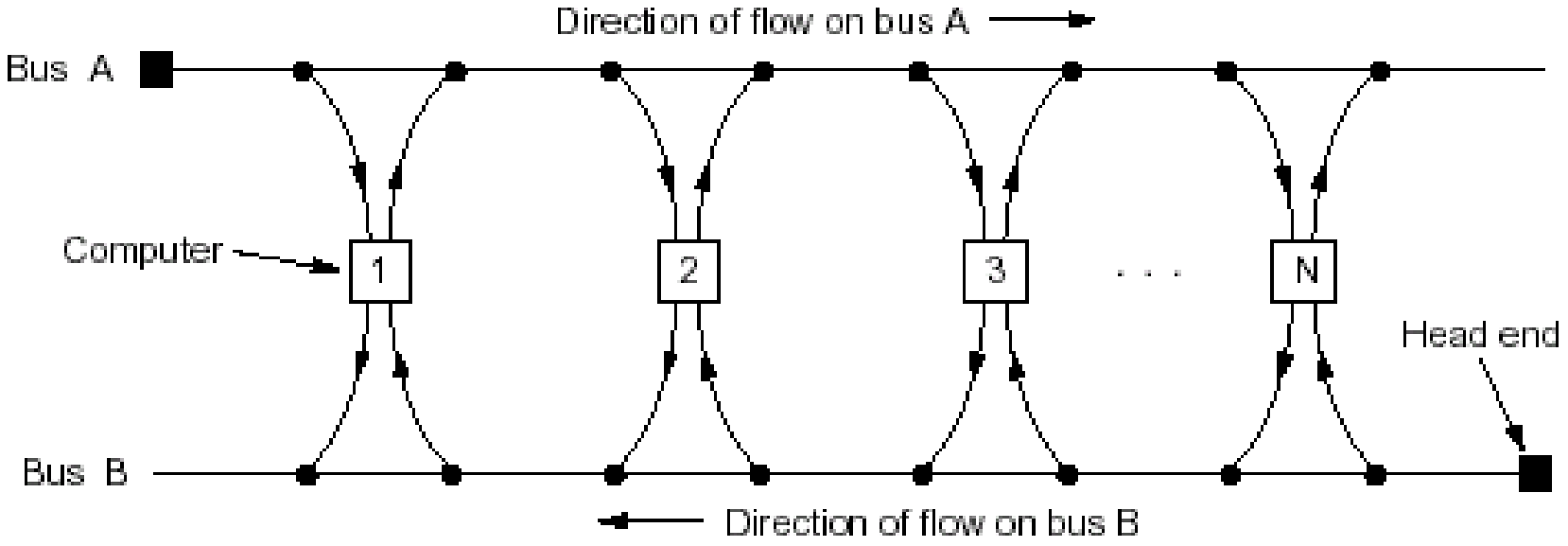
Metropolitan Area Networks (MAN)



A metropolitan area network based on cable TV.

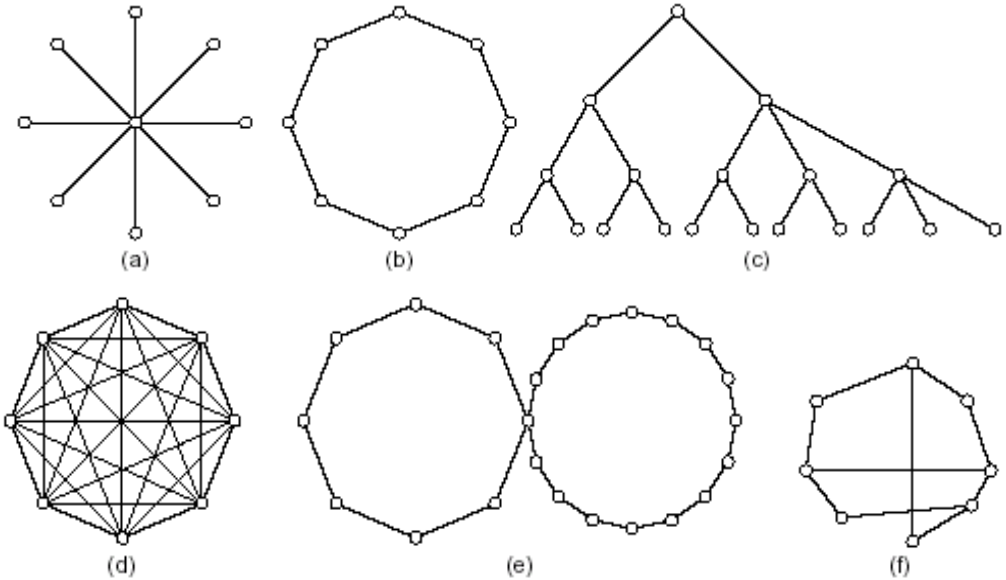
MAN: DQDB

- Basically, MANs are very similar to LANs in the sense that they also use broadcasting technology. **However,** to achieve scalability, a different mechanism is used: Distributed Queue,



WAN Topologies

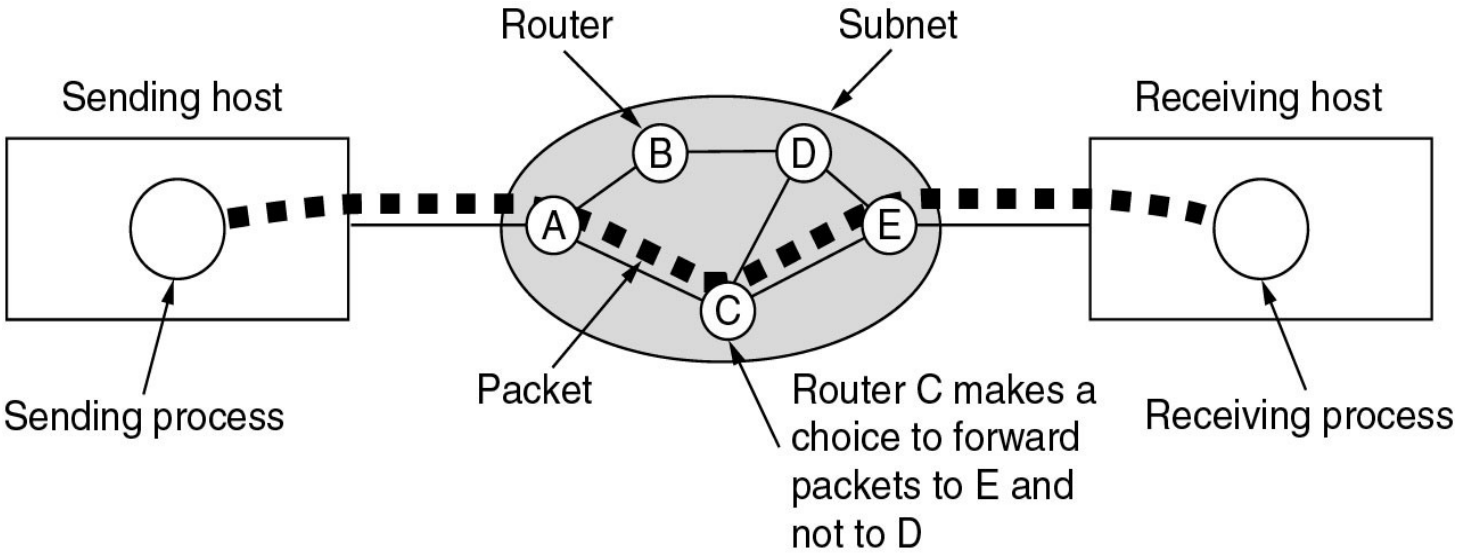
- In contrast to LANs and MANs, the organization of a WAN in terms which hosts are interco



- Most often you'll see **arbitrary topologies**; the others are used in application-specific ways (mostly star, ring, and tree)

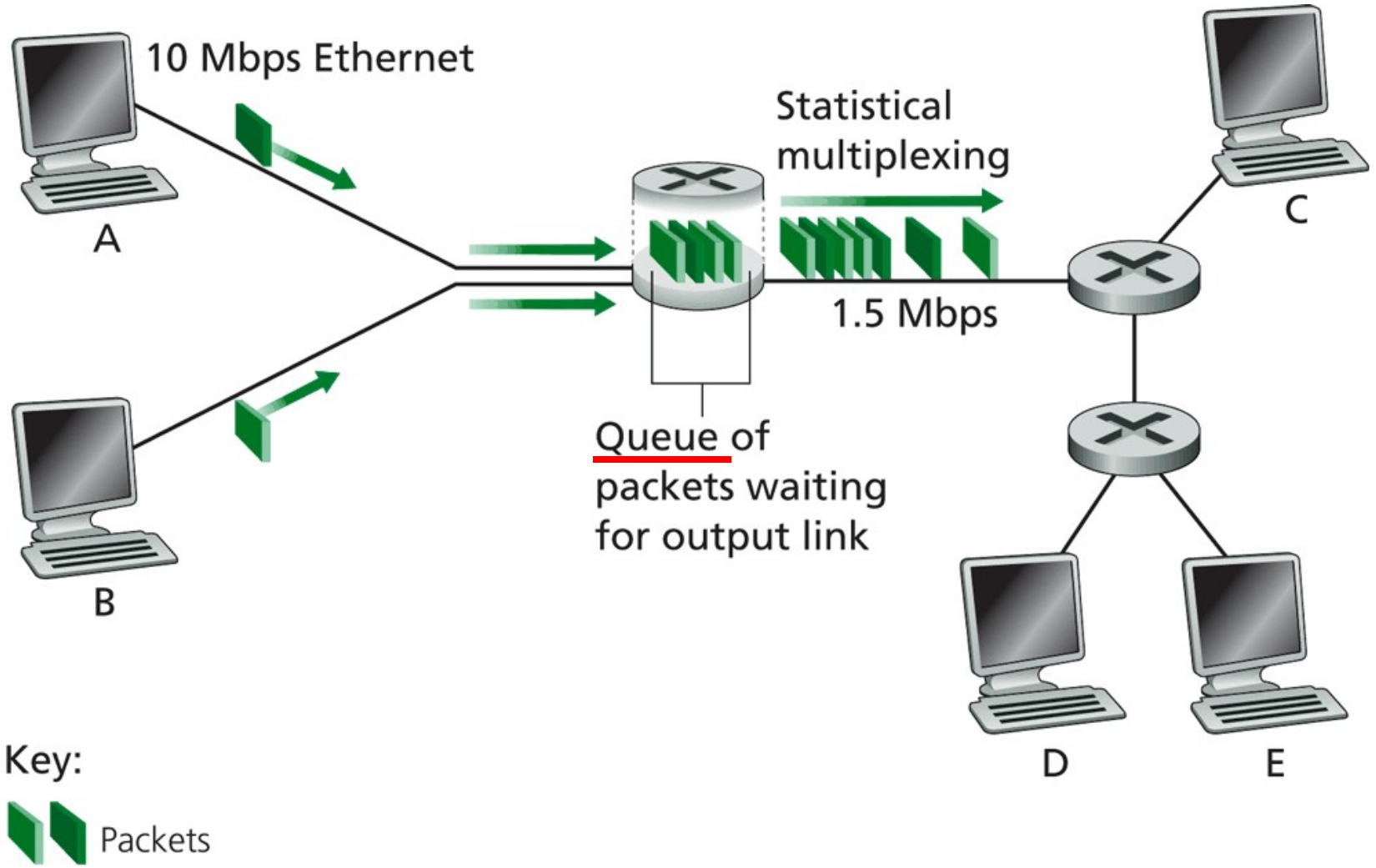
Wide Area Networks (WAN)

- In a WAN, hosts are connected to a subnet, which in turn consists of routers (switching elements) and trunks.
- Routers generally adhere to a **store-and-forward principle**



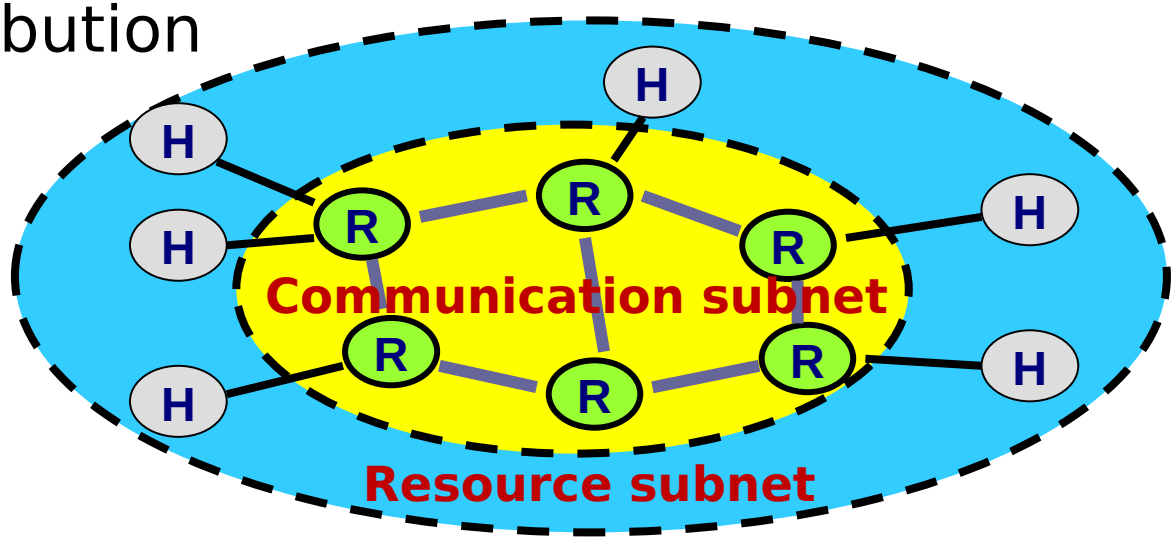
A stream of packets from sender to receiver.

Wide Area Networks (WAN)



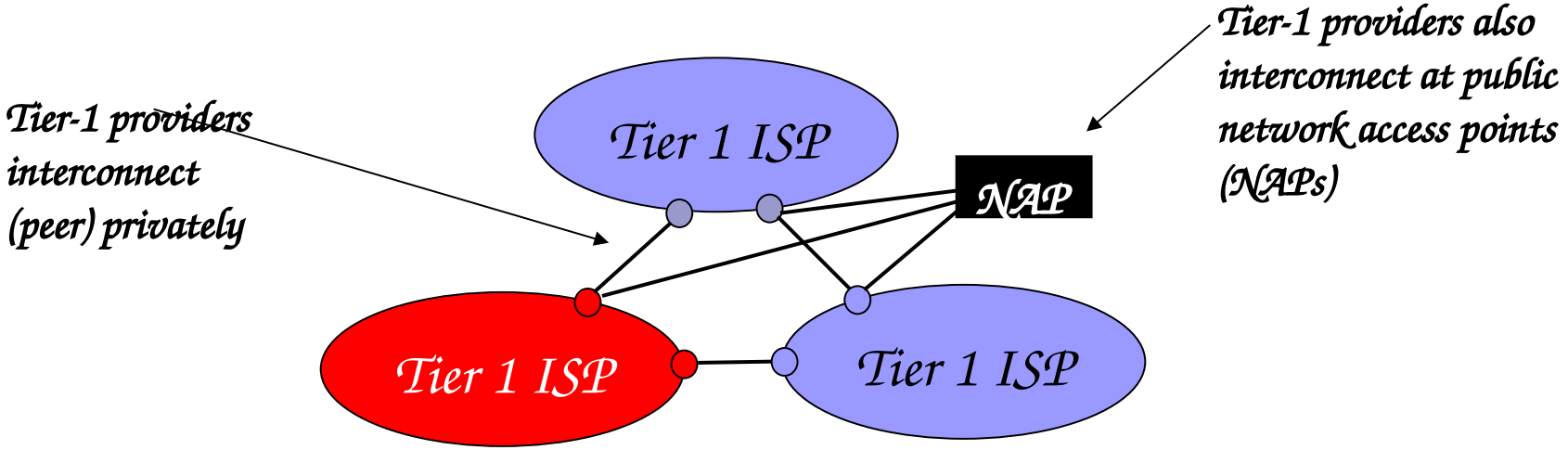
WAN consists of two subnets

- **Resource subnet**
  - Includes computers, terminals, programs, etc.
  - Responsible for information processing
- **Communication subnet**
  - Includes transportation lines, switching elements (routers and switches, etc.)
  - Responsible for information delivery and distribution



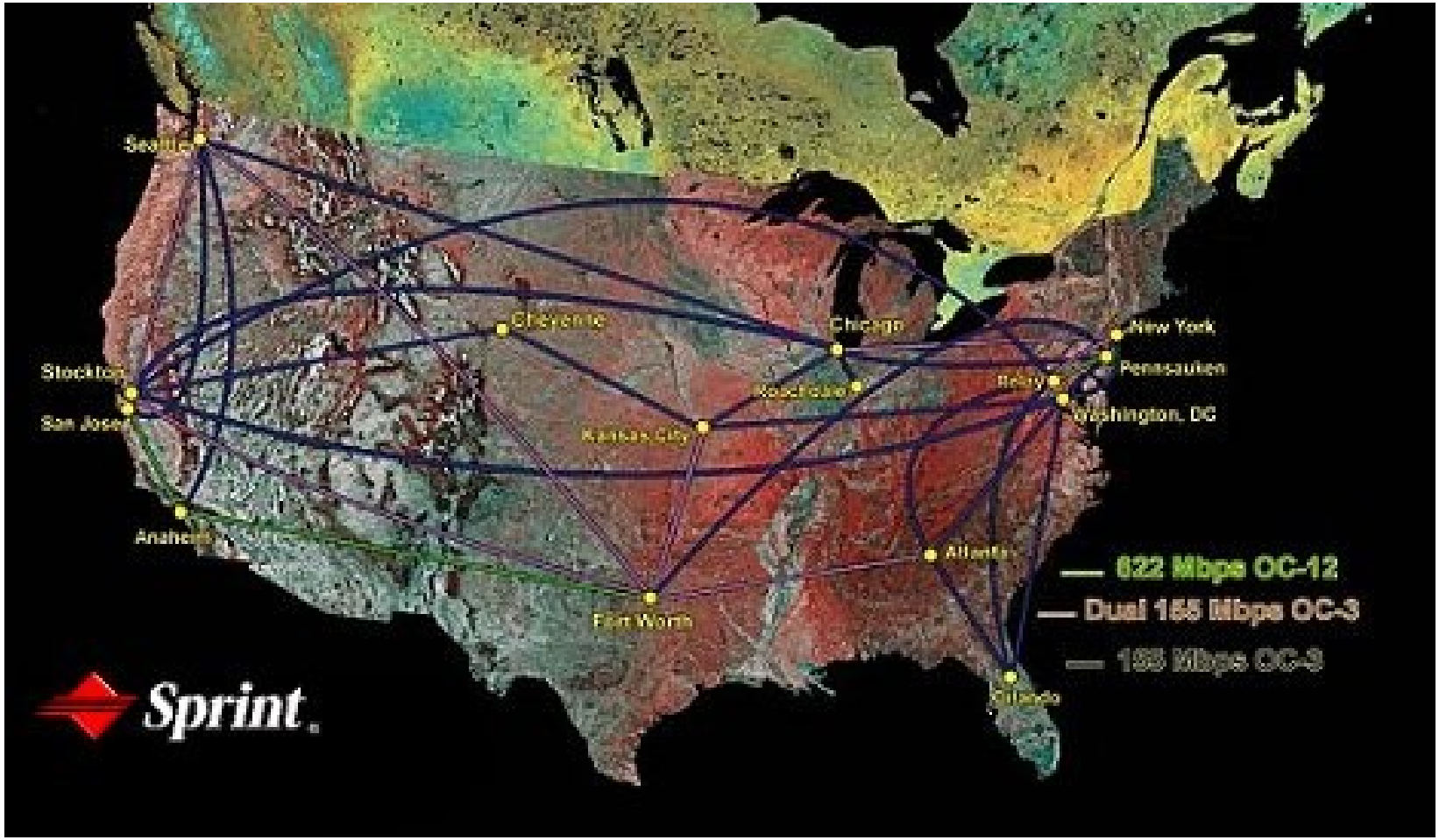
Internet topology: network of networks

- roughly hierarchical
- at center: "tier-1" ISPs (e.g., UUNet, BBN/Genuity, Sprint, AT&T), national/international coverage
  - treat each other as equals



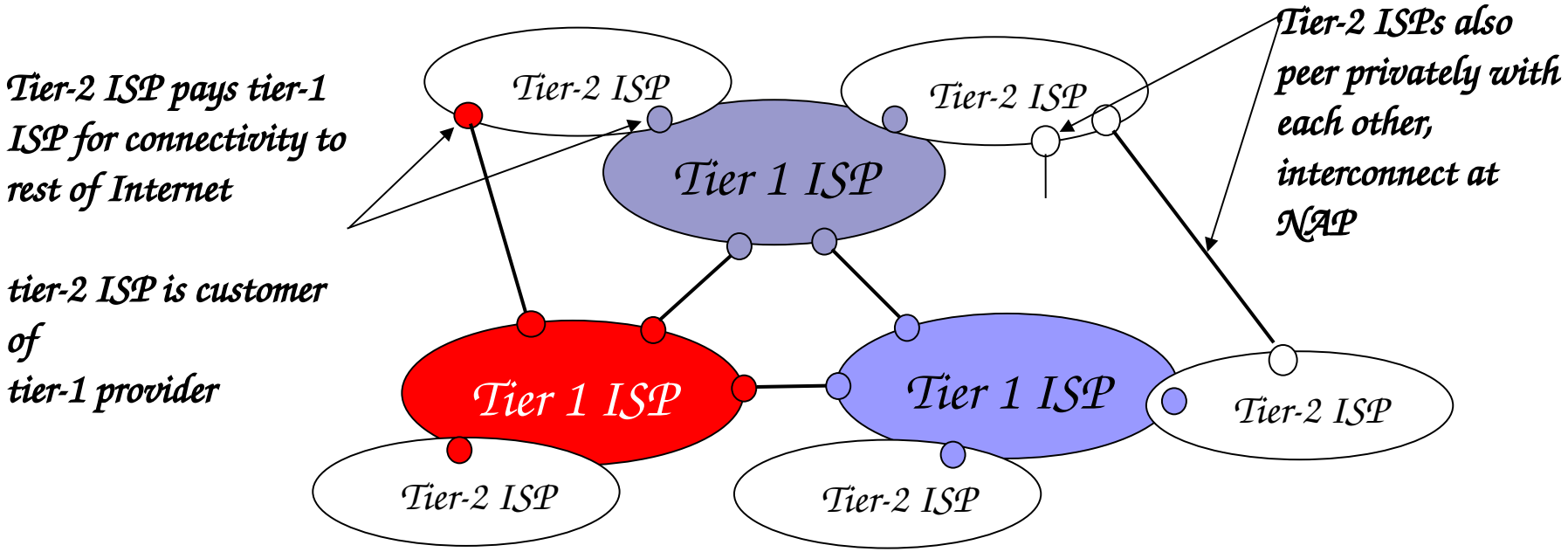
Tier-1 ISP: e.g., Sprint

Sprint US backbone network



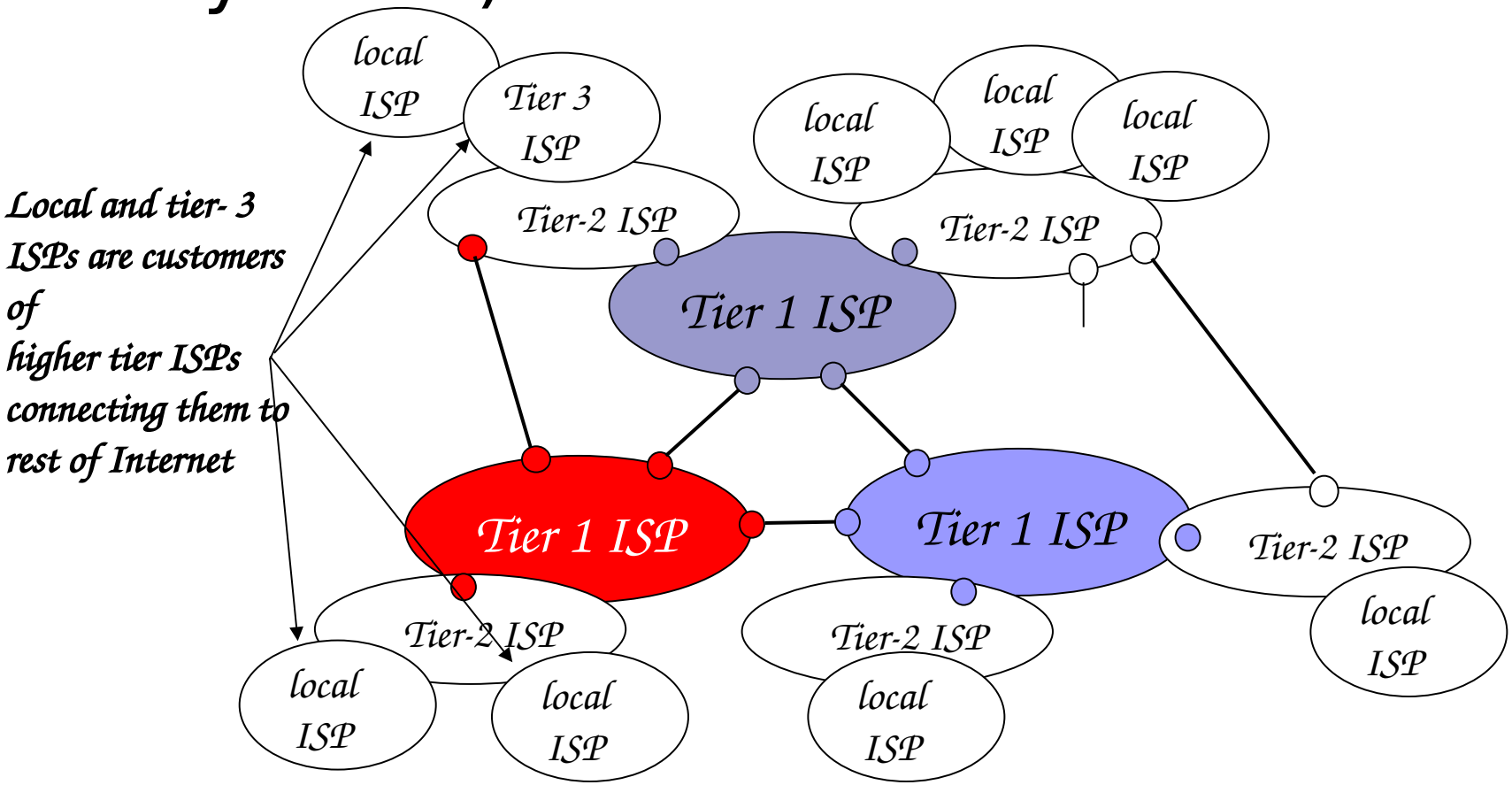
Tier-2 ISP: smaller ISP

- Connect to one or more tier-1 ISPs, possibly other tier-2 ISPs



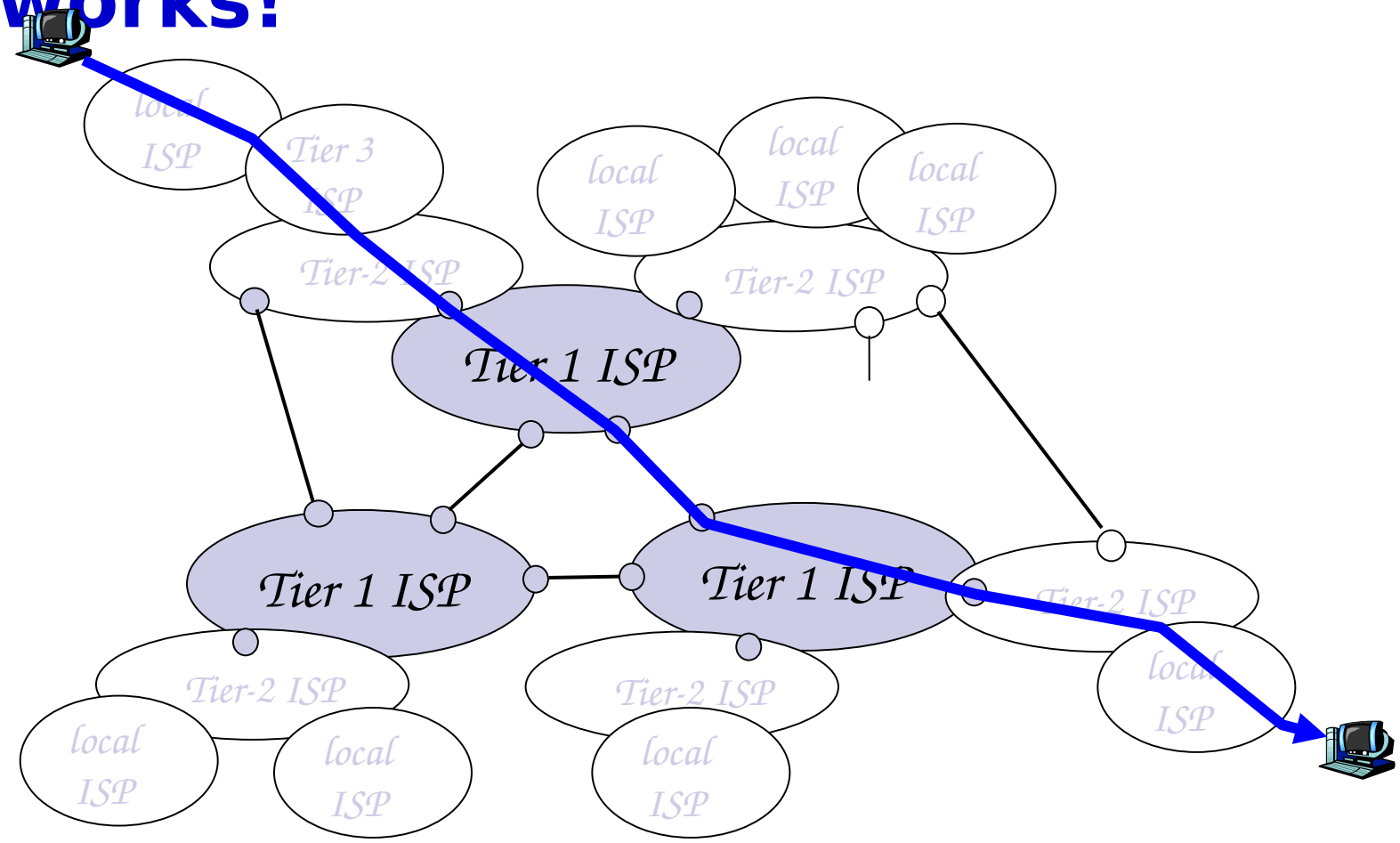
Tier-3 ISPs and local ISPs

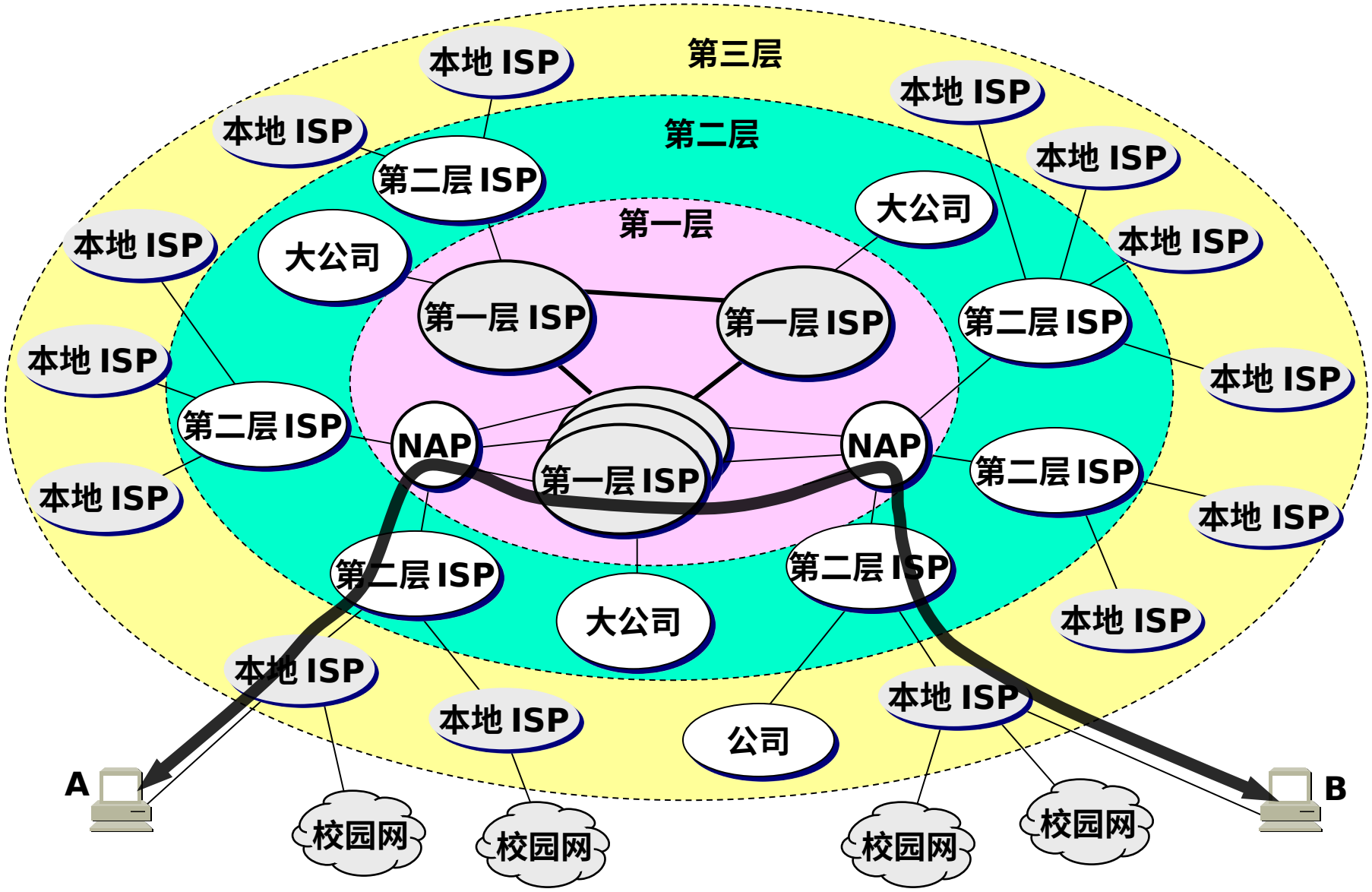
- last hop ("access") network (closest to end systems)



Internet topology: network of networks

- a packet passes through many networks!





主机 A → 本地 ISP → 第二层 ISP → NAP → 第一层 ISP → NAP → 第二层 ISP → 本地 ISP → 主机

Chapter 1: Roadmap

1.1 Computer network and the Internet

1.2 Network edge and core

1.3 Network access and physical media

1.4 Transmission Technology

1.5 Type and Topology

1.6 Network Performance

1.7 Protocol layers, service models

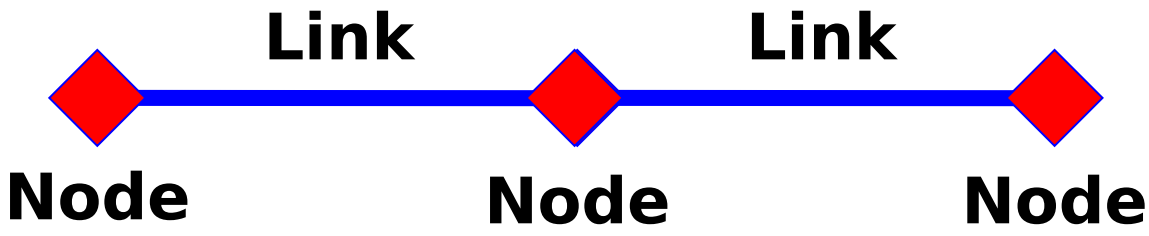
1.8 Architecture, OSI and TCP/IP Models

1.9 Internet History

Nodes and Links

- **Link: transmission technology**
  - **Twisted pair, optical, radio, whatever**

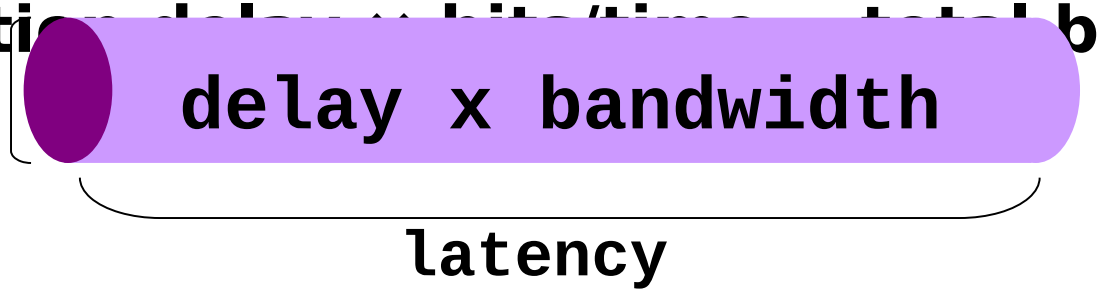
- **Node: computational devices on end of links**
  - **Host: general-purpose computer**
  - **Network node: switch or router**



Properties of Links

- **Latency (delay)**
  - Time for data sent along the link
  - Corresponds to the “length” of the link
- **Bandwidth (capacity)**
  - Amount of data sent (or received) per unit time
  - Corresponds to the “width” of the link
- **Bandwidth-delay product: (BDP)**
  - Amount of data that can be “in flight” at any time
  - Propagation delay x bits/time = total bits in link

bandwidth



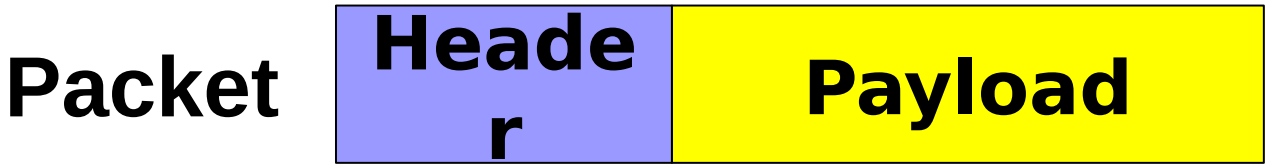
Examples of Bandwidth-Delay

- **Same city over slow link:**
  - Bandwidth: ~100mbps
  - Latency: ~ 0.1msec
  - BDP: ~ 10^4 bits / ~ 1.25 MBytes
- **Cross-country over fast link:**
  - Bandwidth: ~10Gbps
  - Latency: ~ 10msec
  - BDP: ~ 10^8 bits / ~ 12.5 GBytes
- **Another example where the Internet has to be prepared for a wide range of conditions!**

Utilization

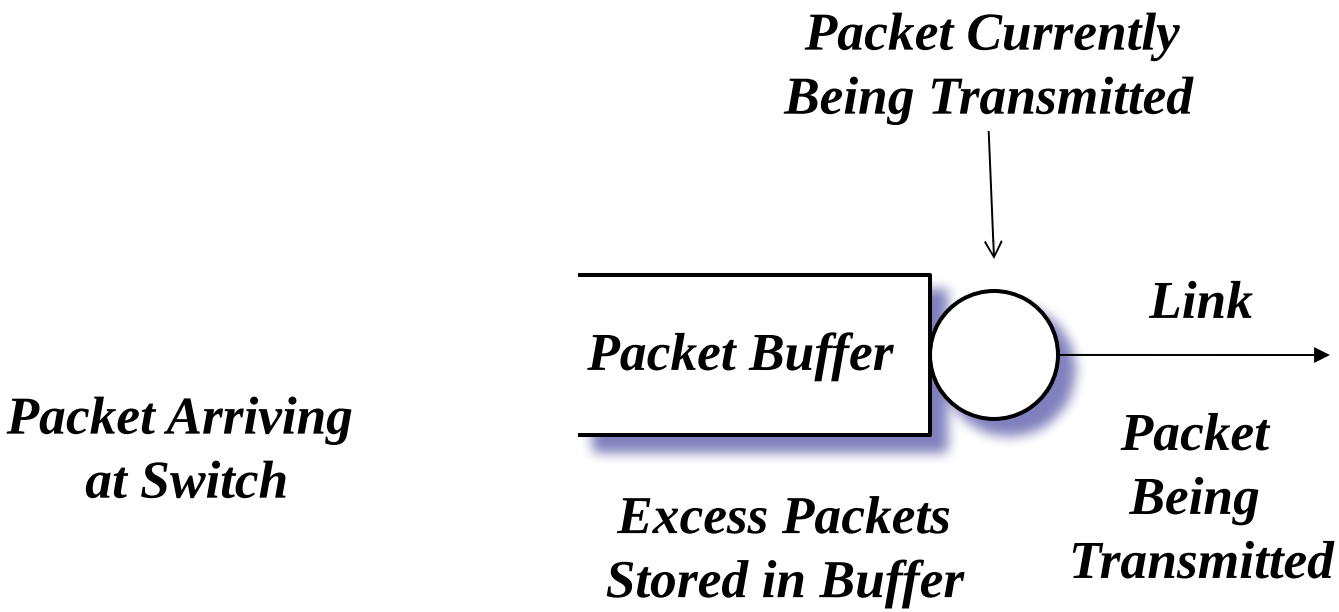
- **Fraction of time link is busy transmitting**
  - **Often denoted by ρ**
- **Ratio of arrival rate to bandwidth**
  - **Arrival: A bits/sec on average**
  - **Bandwidth: B bits/sec**
  - **Utilization ρ = A/B**

Packet Switching



- **Header**
  - **Instructions to the network for how to handle packet**
  - **Think of the header as an interface!**
- **Payload (Body)**
  - **Data being transferred**

The Lifecycle of Packets



The Delays of Their Lives



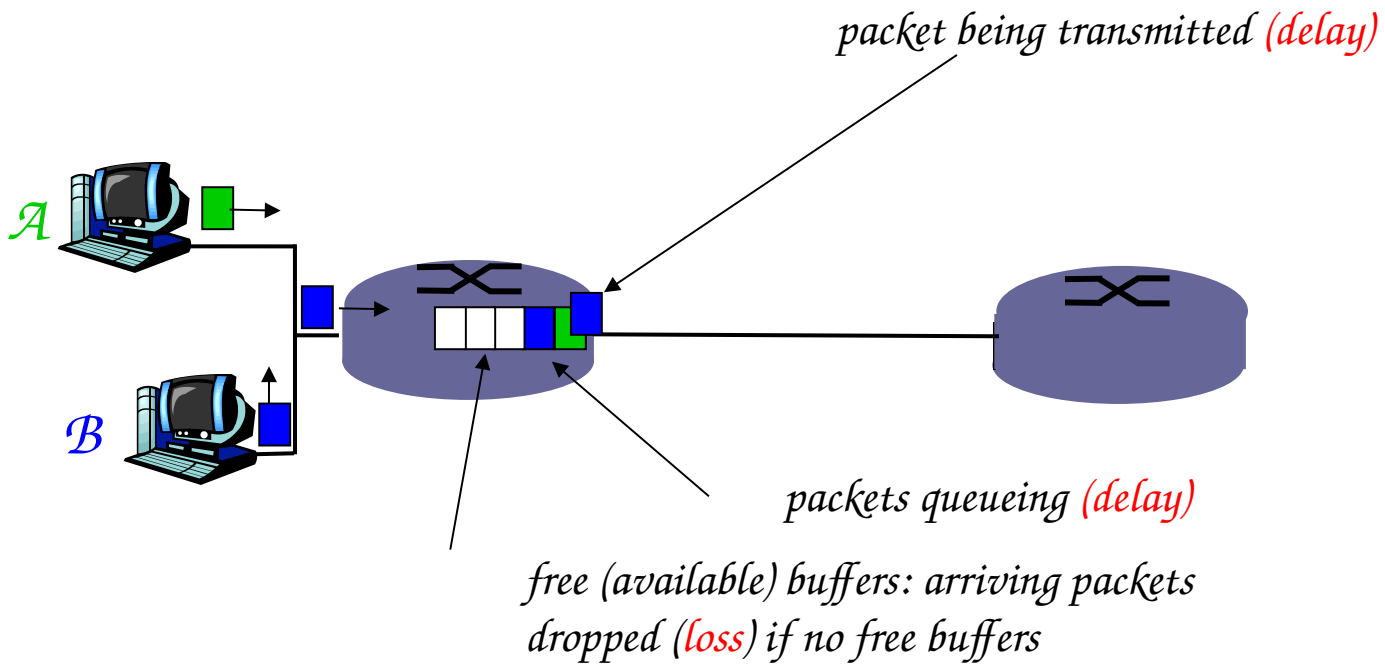
Propagation Delay is how long it takes to reach the next switch after transmission

Round-Trip Time (RTT) is the time it takes a packet to reach the destination and the response to return to the sender

How do loss and delay occur?

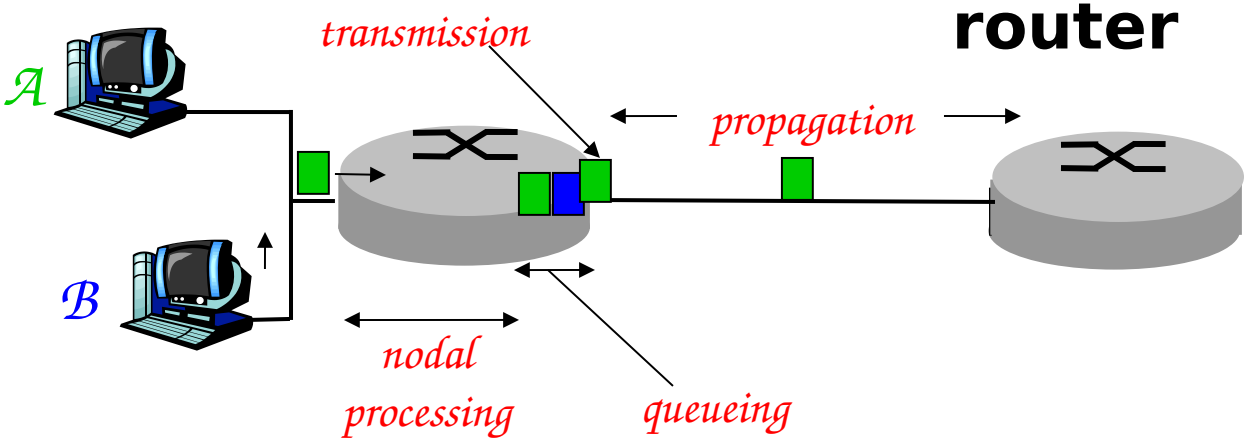
packets *queue* in router buffers

- **packet arrival rate to link exceeds output link capacity**
- **packets queue, wait for turn**



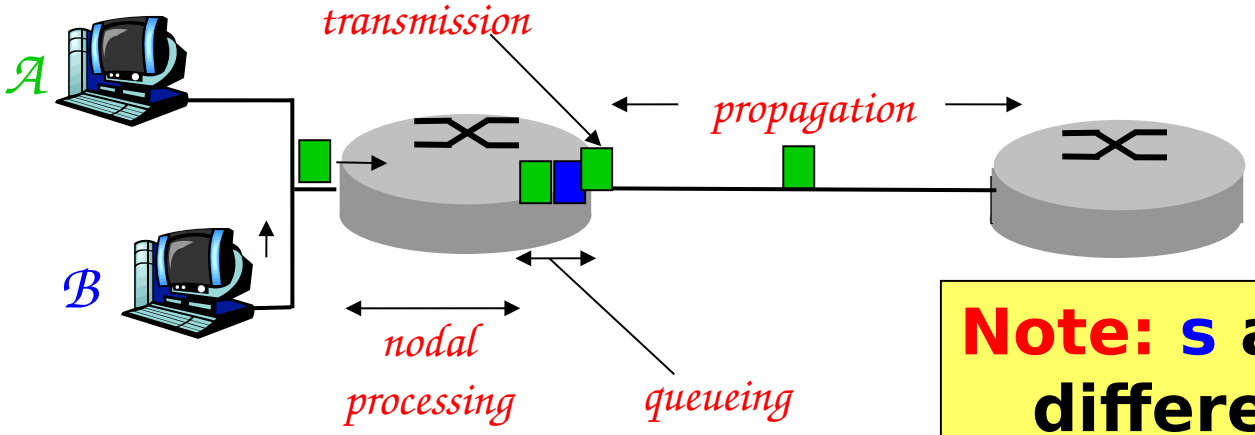
Four sources of packet delay

- **1. nodal processing:**
  - check bit errors
  - determine output link
- **2. queueing**
  - time waiting at output link for transmission
  - depends on congestion level of router



Delay in packet-switched networks

- 3. Propagation:
  - d = length of physical link
  - s = propagation speed in medium (~2×10^8 m/sec)
  - time to send bits into link = L/R
  - propagation delay = d/s



Note: s and R are *very* different quantities!

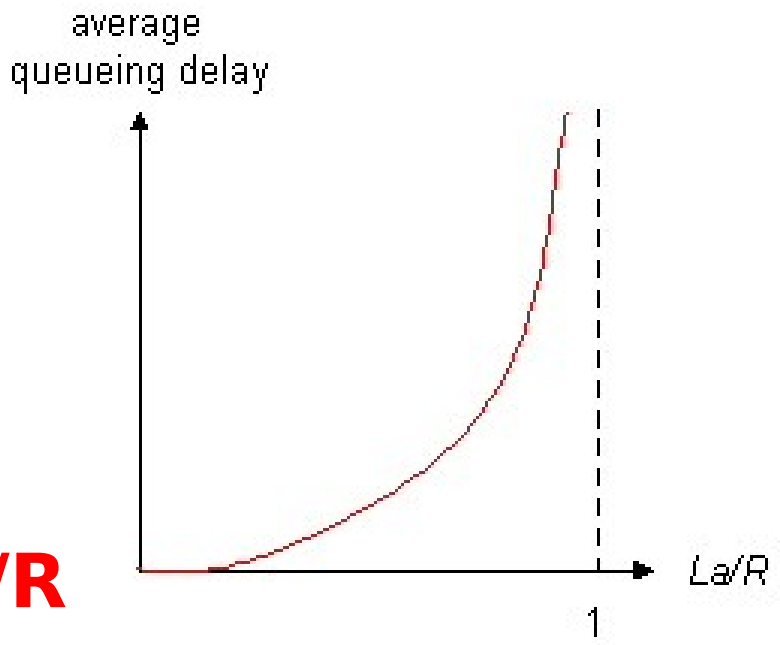
Nodal delay

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

- d_{proc} = processing delay
  - typically a few microseconds or less
- d_{queue} = queuing delay
  - depends on congestion
- d_{trans} = transmission delay
  - = L/R, significant for low-speed links
- d_{prop} = propagation delay
  - a few microsecs to hundreds of msecs

Queueing delay (revisited)

- R=link bandwidth (b/s)
- L=packet length (bits)
- a=average packet arrival rate

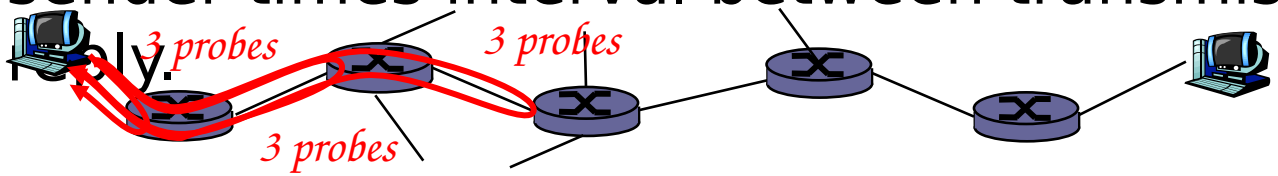


traffic intensity = La/R

- La/R ~ 0: average queueing delay small
- La/R 1: delays become large
- La/R > 1: more “work” arriving than can be serviced, average delay infinite!

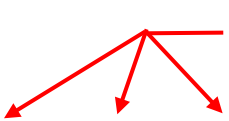
“Real” Internet delays and routes

- **What do “real” Internet delay & loss look like?**
- Traceroute **program:** **provides delay measurement from source to router along end-end Internet path towards destination. For all *i:***
  - sends three packets that will reach router *i* on path towards destination
  - router *i* will return packets to sender
  - sender times interval between transmission and reply.



“Real” Internet delays and routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr



Three delay measements from gaia.cs.umass.edu to cs-gw.cs.umass.edu

```
1  cs-gw (128.119.240.254)  1 ms  1 ms  2 ms
2  border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)  1 ms  1 ms  2 ms
3  cht-vbns.gw.umass.edu (128.119.3.130)  6 ms 5 ms 5 ms
4  jn1-at1-0-0-19.wor.vbns.net (204.147.132.129)  16 ms 11 ms 13 ms
5  jn1-so7-0-0-0.wae.vbns.net (204.147.136.136)  21 ms 18 ms 18 ms
6  abilene-vbns.abilene.ucaid.edu (198.32.11.9)  22 ms  18 ms  22 ms
7  nycm-wash.abilene.ucaid.edu (198.32.8.46)  22 ms  22 ms  22 ms
8  62.40.103.253 (62.40.103.253)  104 ms 109 ms 106 ms
9  de2-1.de1.de.geant.net (62.40.96.129)  109 ms 102 ms 104 ms
10  de.fr1.fr.geant.net (62.40.96.50)  113 ms 121 ms 114 ms
11  renater-gw.fr1.fr.geant.net (62.40.103.54)  112 ms  114 ms  112 ms
12  nio-n2.cssi.renater.fr (193.51.206.13)  111 ms  114 ms  116 ms
13  nice.cssi.renater.fr (195.220.98.102)  123 ms  125 ms  124 ms
14  r3t2-nice.cssi.renater.fr (195.220.98.110)  126 ms  126 ms  124 ms
15  eurecom-valbonne.r3t2.ft.net (193.48.50.54)  135 ms  128 ms  133 ms
16  194.214.211.25 (194.214.211.25)  126 ms  128 ms  126 ms
17  * * *
18  * * *
19  fantasia.eurecom.fr (193.55.113.142)  132 ms  128 ms  136 ms
```



trans-oceanic link



** means no reponse (probe lost, router not replying)*

Packet loss

- queue (aka buffer) preceding link in buffer has finite capacity
- when packet arrives to full queue, packet is dropped (aka lost)
- lost packet may be retransmitted by previous node, by source end system, or not retransmitted at all

Example



- **Node A needs to send a data block to Node B over a 1000 km fiber-optic link. Assume processing and queuing delays are negligible. Calculate the total delay for each scenario below and verify whether the statement "The higher the transmission rate, the smaller the total delay" is correct.**
  - **（1） Data block size: 100 MB, Link bandwidth: 1 Mbit/s**
  - **（2） Data block size: 100 MB, Link bandwidth: 100 Mbit/s**
  - **（3） Data block size: 1 B, Link bandwidth: 1 Mbit/s**

Example

- **Solution:**

Propagation delay = 1000 km / 2.0×10^5 km/s = 5 ms.

(1) Transmission delay = $100 \times 2^{20} \times 8 \div 10^6 = 838.9$ s,

Total delay = 838.9 + 0.005 ≈ **838.9 s**,

(2) Transmission delay = $100 \times 2^{20} \times 8 \div 10^8 = 8.389$ s

Total delay = 8.389 + 0.005 = **8.394 s**, Reduced to nearly 1/100 of (1).

(3) Transmission delay = $1 \times 8 \div 10^6 = 8 \times 10^{-6}$ s = 8 μs,

Total delay = 0.008 + 5 = **5.008 ms**,

(4) Transmission delay = $1 \times 8 \div 10^9 = 8 \times 10^{-9}$ s = 0.008 μs

Total delay = 0.000008 + 5 = **5.000008 ms**, No significant eduction compared to (3).

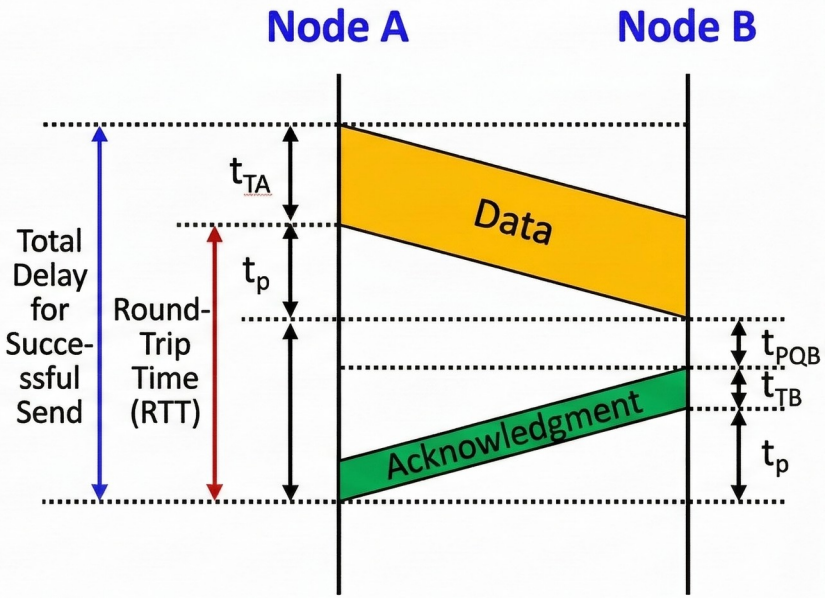
It cannot be broadly assumed that: "The higher the data transmission rate, the smaller the total transmission delay".

Round-Trip Time （RTT)

- Represents the total time from the sender finishing data transmission to the sender receiving the acknowledgment from the receiver.

Round-Trip Time (RTT) =
+ Propagation delay from Node A to B (t_p)
+ Processing and queuing delay at Node B (t_{PQB})
+ Transmission delay at Node B (t_{TB})
+ Propagation delay from Node B to A (t_p)

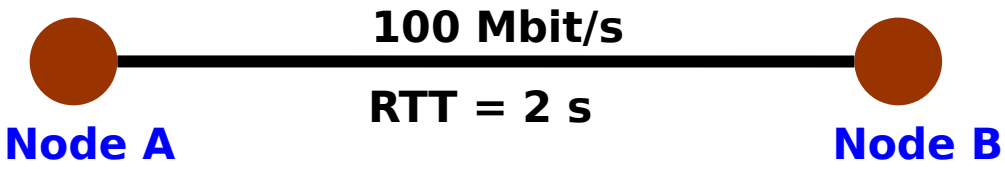
= 2 × Propagation delay (t_p)
+ Processing and queuing delay at Node B (t_{PQB})
+ Transmission delay at Node B (t_{TB})



t_{PQB}

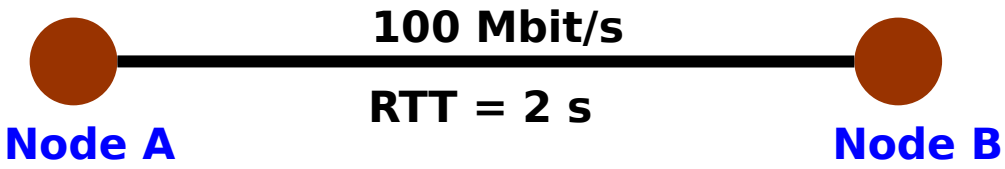
+ 结点 B 发送时延 t_{TB}

Example



- **Node A needs to send 100 MB of data to Node B at a transmission rate of 100 Mbit/s. Once Node B correctly receives all the data, it immediately sends an acknowledgment back to Node A. Assume that Node A can only continue sending data to Node B after it receives this acknowledgment, and that the acknowledgment message is very short. Calculate the effective data rate for Node A**

Example



- **Solution :**

$$\text{Transmission Delay} = \frac{\text{Data Length}}{\text{Trans. Rate}} = \frac{100 \times 2^{20} \times 8}{100 \times 10^{6}} \approx 8.39 \text{ s}$$

$$\text{Effective Data Rate} = \frac{\text{Data Length}}{\text{Trans. Time + RTT}} = \frac{100 \times 2^{20} \times 8}{8.39 + 2} \approx 80.7 \text{ Mbit/s}$$

Chapter 1: Roadmap

1.1 Computer network and the Internet

1.2 Network edge and core

1.3 Network access and physical media

1.4 Transmission Technology

1.5 Type and Topology

1.6 Network Performance

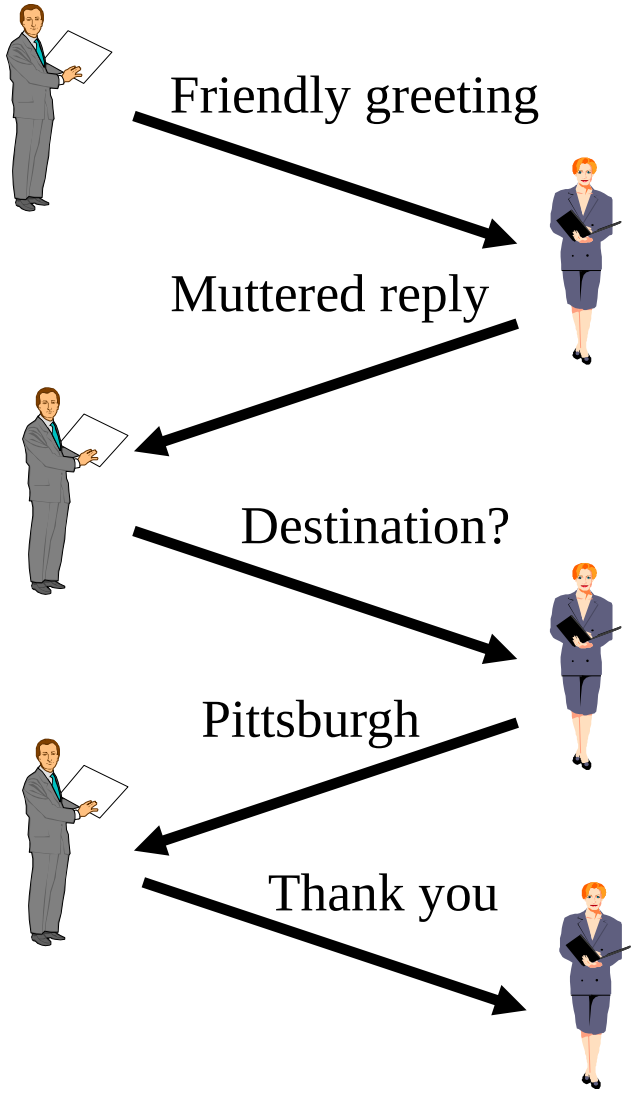
1.7 Protocol layers, service models

1.8 Architecture, OSI and TCP/IP Models

1.9 Internet History

Protocols

- **Protocol:** an agreement on how to communicate
  - To exchange data
  - To coordinate sharing of resources
- Protocols specifies ***syntax*** and ***semantics.***
- **Syntax:** how protocol is structured
  - Format, order messages are sent and received
- **Semantics:** what these bits mean
  - How to respond to various messages, events, etc.



Protocol “Layers”

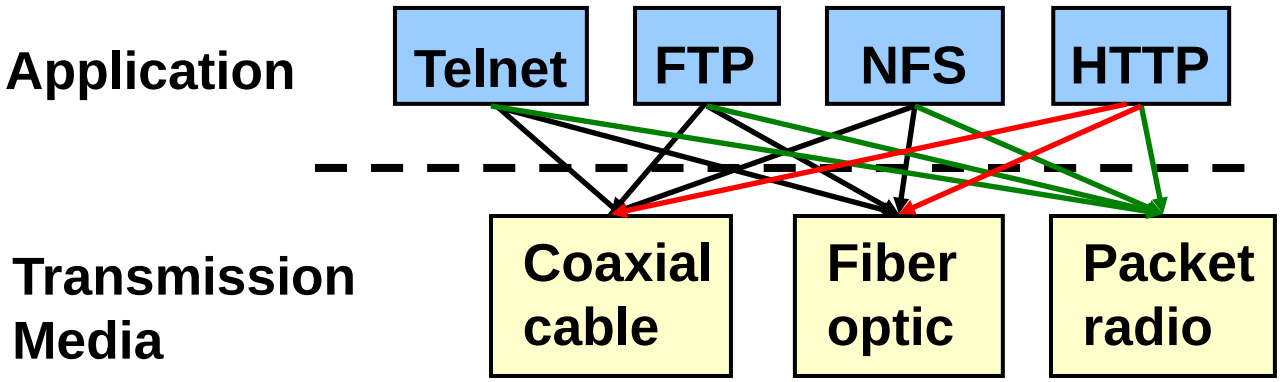
Networks are complex!

- many “pieces”:
  - hosts
  - routers
  - links of various media
  - applications

What is Layering?

- **A technique to organize a network system into a succession of logically distinct entities, such that the service provided by one entity is solely based on the service provided by the previous (lower level) entity**

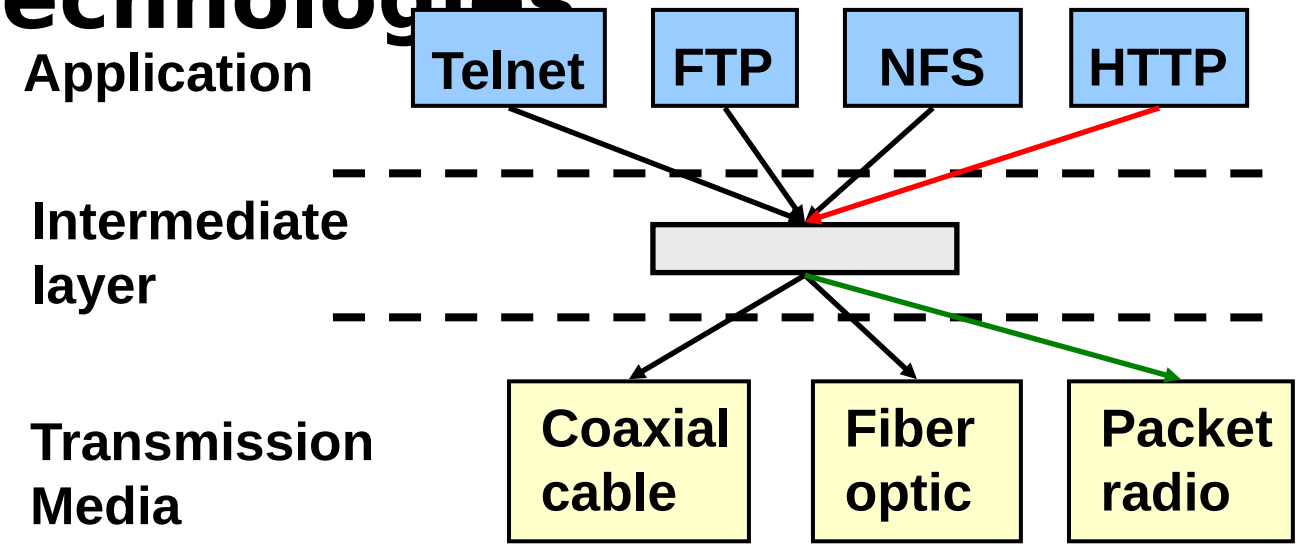
Why Layering?



- **No layering: each new application has to be re-implemented for every network technology!**

Why Layering?

- **Solution:** introduce an intermediate layer that provides a **unique** abstraction for various network technologies



Avoids NxM Implementation Problem

Why layering?

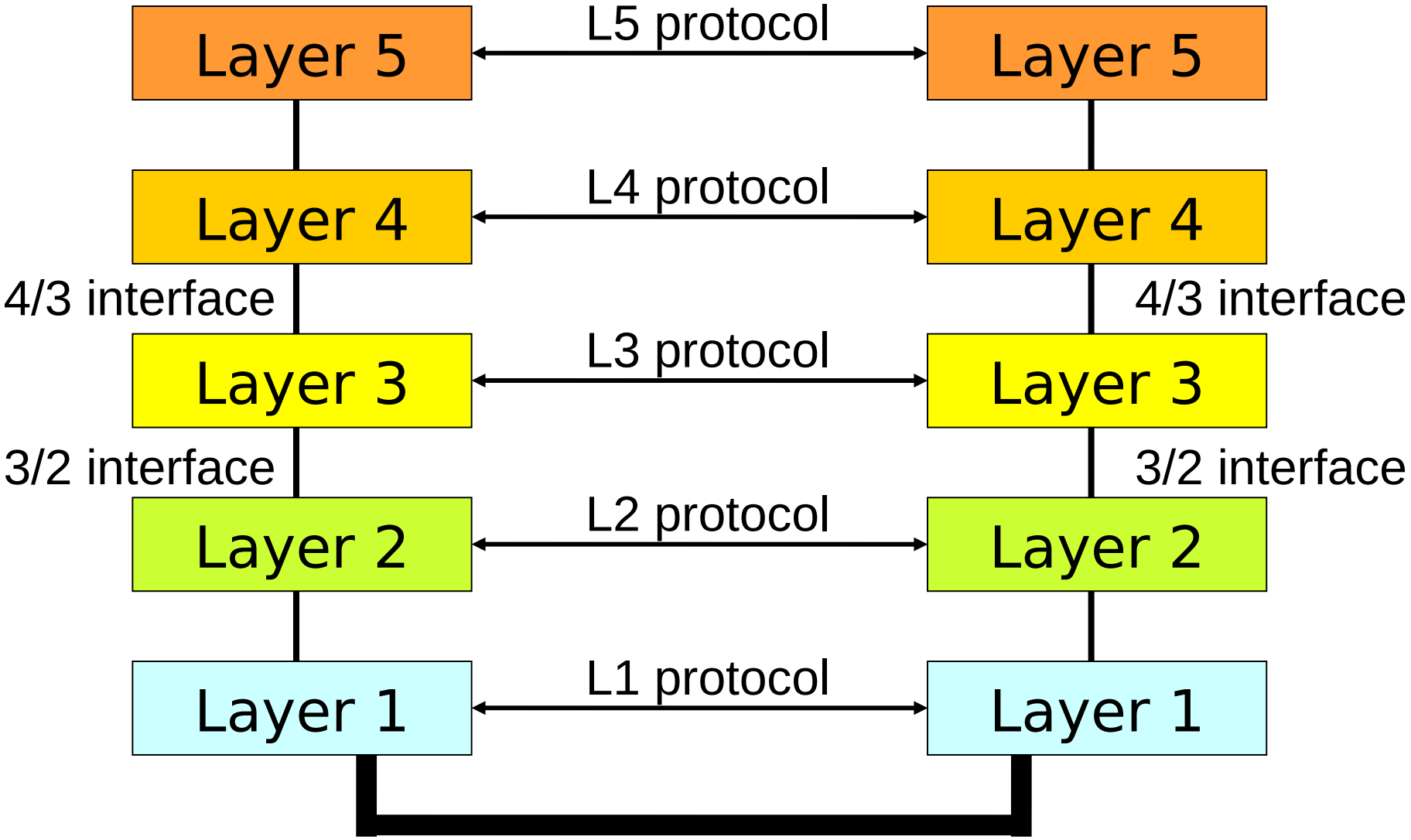
Dealing with complex systems:

- explicit structure allows identification, relationship of complex system's pieces
  - layered **reference model** for discussion
- modularization eases maintenance, updating of system
  - change of implementation of layer's service transparent to rest of system
  - e.g., change in gate procedure doesn't affect rest of system
  - also found in operating systems and languages
- **Q: layering considered harmful?**

Why layering?

- **Advantages**
  - ☑ **Modularity – protocols easier to manage and maintain**
  - ☑ **Abstract functionality –lower layers can be changed without affecting the upper layers**
  - ☑ **Reuse – upper layers can reuse the functionality provided by lower layers**
- **Disadvantages**
  - ☑ **Information hiding – inefficient implementations**

Layered Protocol Architecture



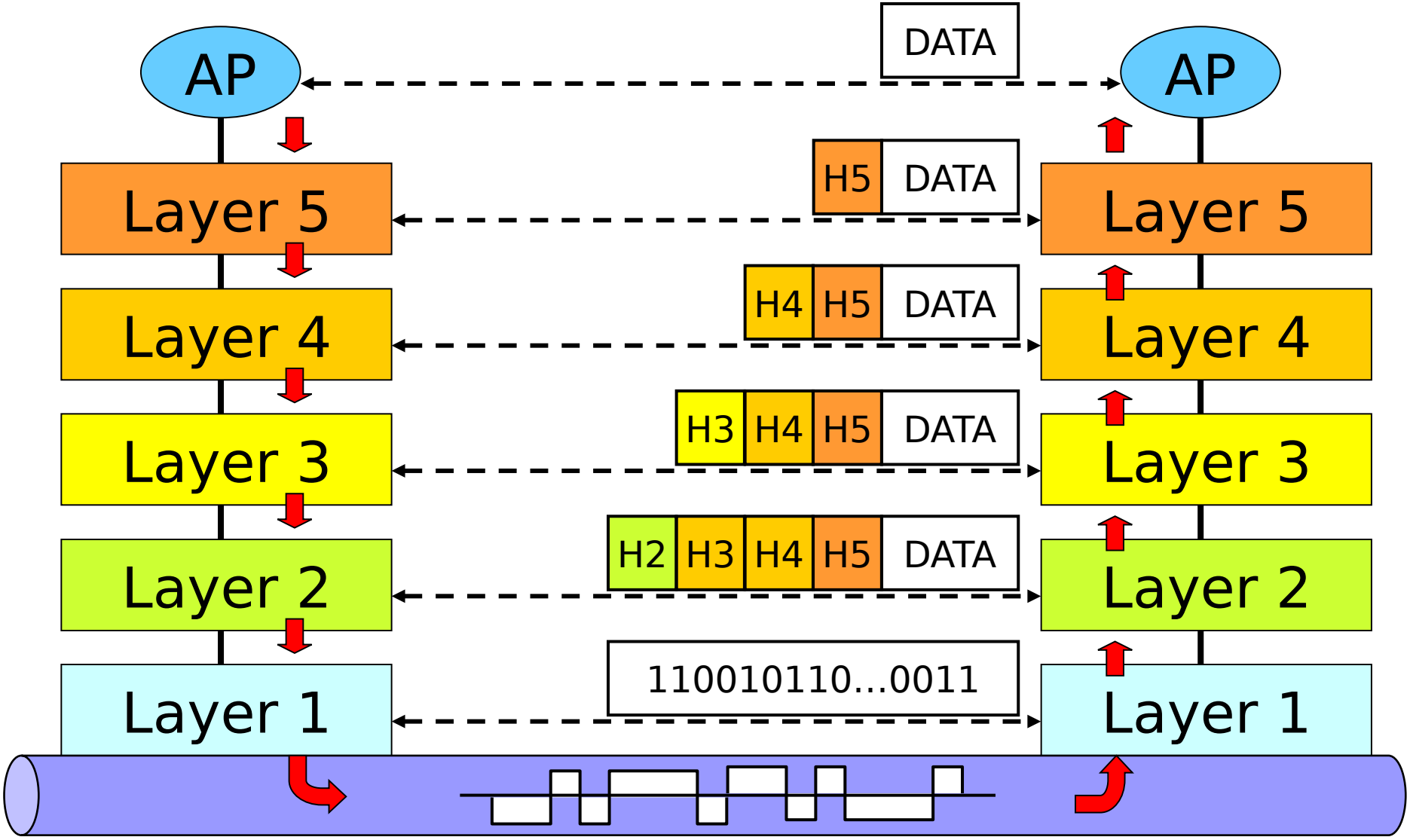
Layered Protocol Architecture

- To reduce design complexity, most networks are organized as a series of layers, each layer offers certain services to its upper layer.
- Layer *n* on one machine carries on a conversation with layer *n* on another machine.
- The rules and conventions used in this conversation are collectively known as the layer *n* protocol.

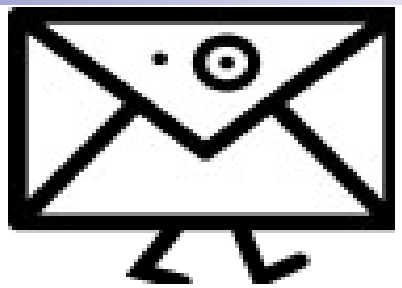
Characterizing the Layers

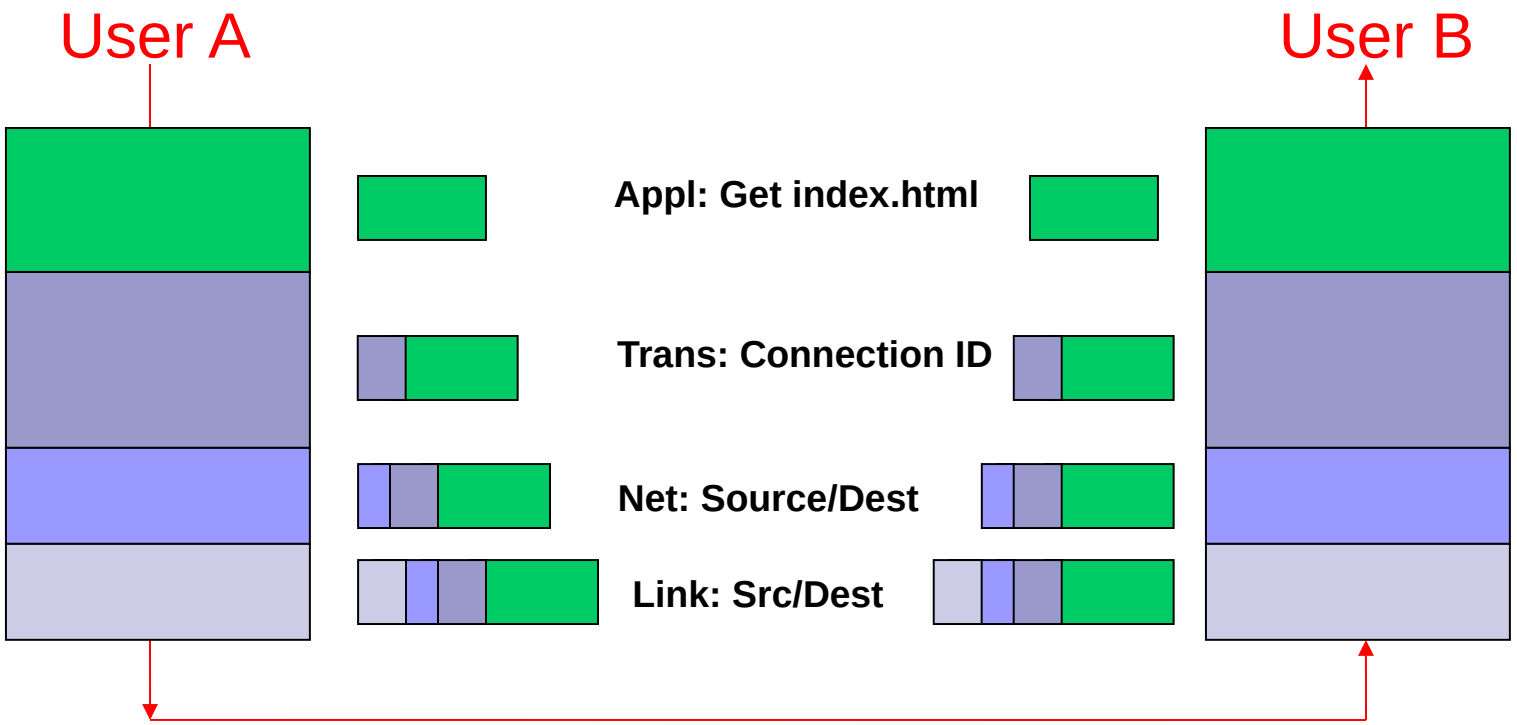
- **Service: what a layer *does***
- **Service interface: how to access the service**
  - Interface for layer above
- **Protocol: how peers communicate**
  - Protocol interface: set of rules and formats that govern the communication between network elements
  - Determines how the peers achieve the service
  - Does not govern implementation on a single machine, but how the layer is implemented between machines
- **Peer: the entities comprising the corresponding layers on different**

Layer Encapsulation



Layer Encapsulation

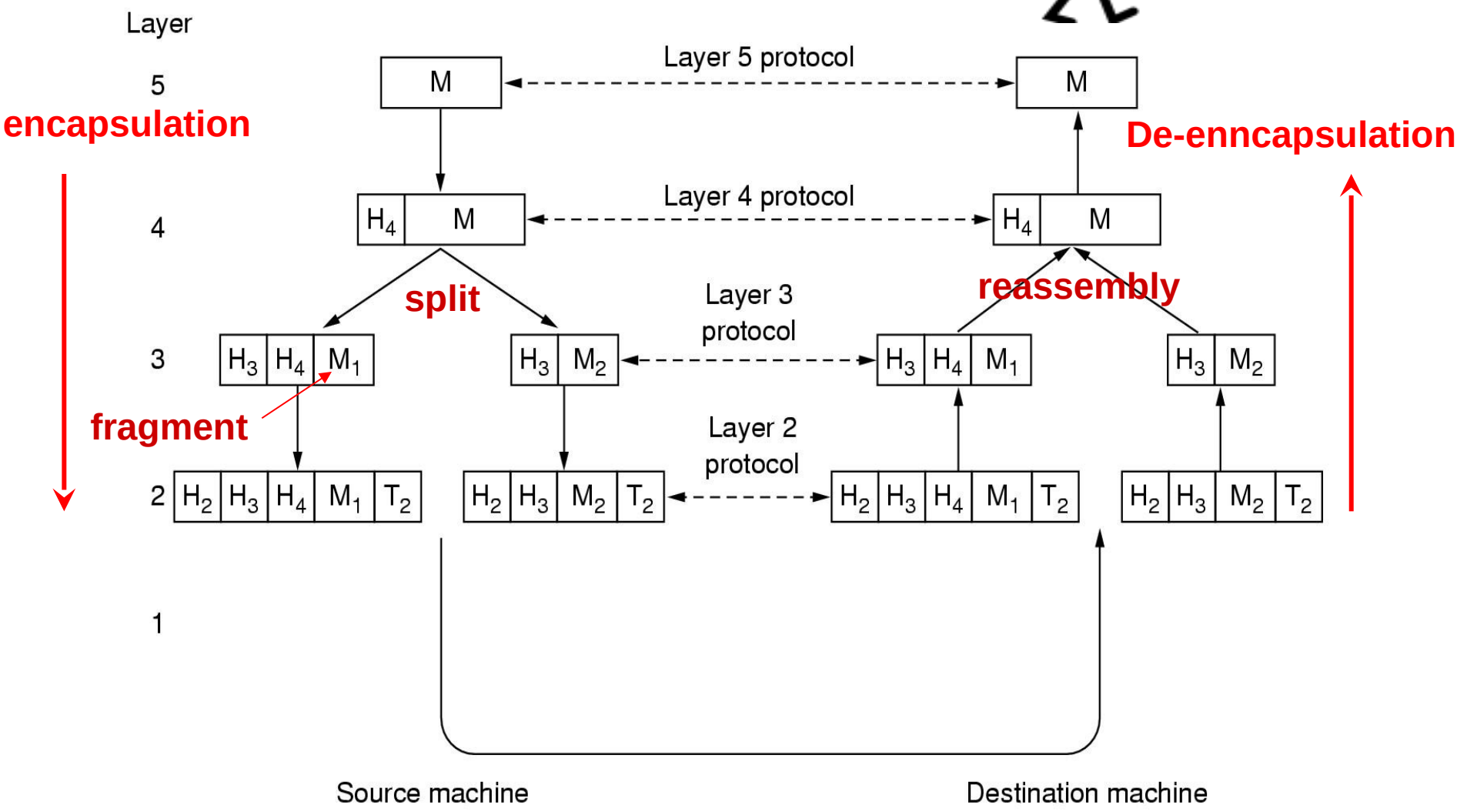




Common case: 20 bytes TCP header + 20 bytes IP header + 14 bytes Ethernet header = *54 bytes overhead*

Layer Encapsulation



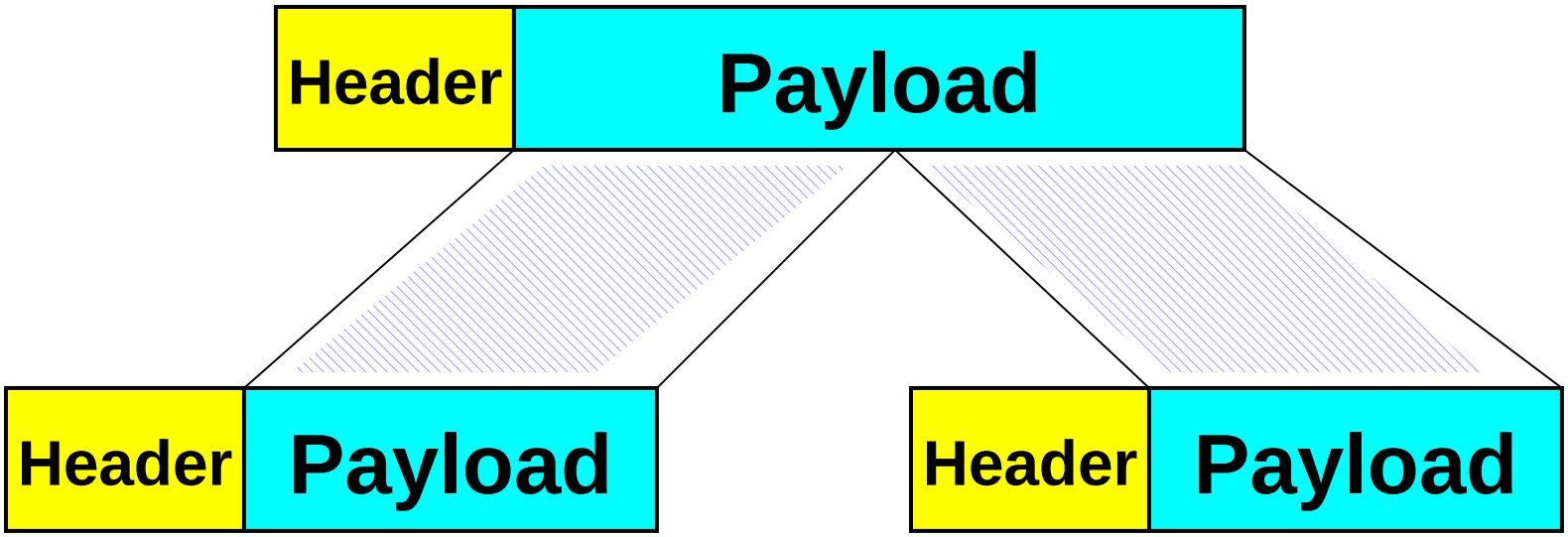


Information flow supporting virtual communication in layer 5

Fragmentation / Reassembly

- **Observation:** In a protocol stack, *Layer k* puts its entire packet as **data** into a *Layer k-1* packet, that is, the *k-1* packet **payload**; the latter may add a header and/or a trailer.
- **Note:** It may even occur that *Layer k* data has to be **split** across several *Layer k-1* packets, i.e., **fragmentation**. At the destination, **reassembly** is needed to recover *Layer k* data.

Fragmentation / Reassembly



Services

- **What is service?**
  - **A set of communication abilities and operations provided by lower layer to its higher layer.**
- **Types**
  - **Connection-oriented Service**
  - **Connectionless Service**

Services

- **Connection-oriented Service**
  - **Like the telephone model: you first establish a connection, then do a lot communication, and finally release the connection. Three phases.**
  - **Circuit switching**
  - **Virtual Circuit packet switching**
- **Connectionless Service**
  - **Like the postal model: your data is put into some kind of envelope called packet, on which the destination address has been written. The packet is independently routed in communication subnet, until gets to the destination, and that's it.**

Services

- **Each service can provide some quality:**
  - ***Is data delivered in the order it was sent?*** **With connections, this is generally the case.**
  - ***Is data transmission reliable?*** **Generally offered with connections, but not always with connectionless services. Reliability requires sending acknowledgements, so that performance may degrade.**

Six different types of service

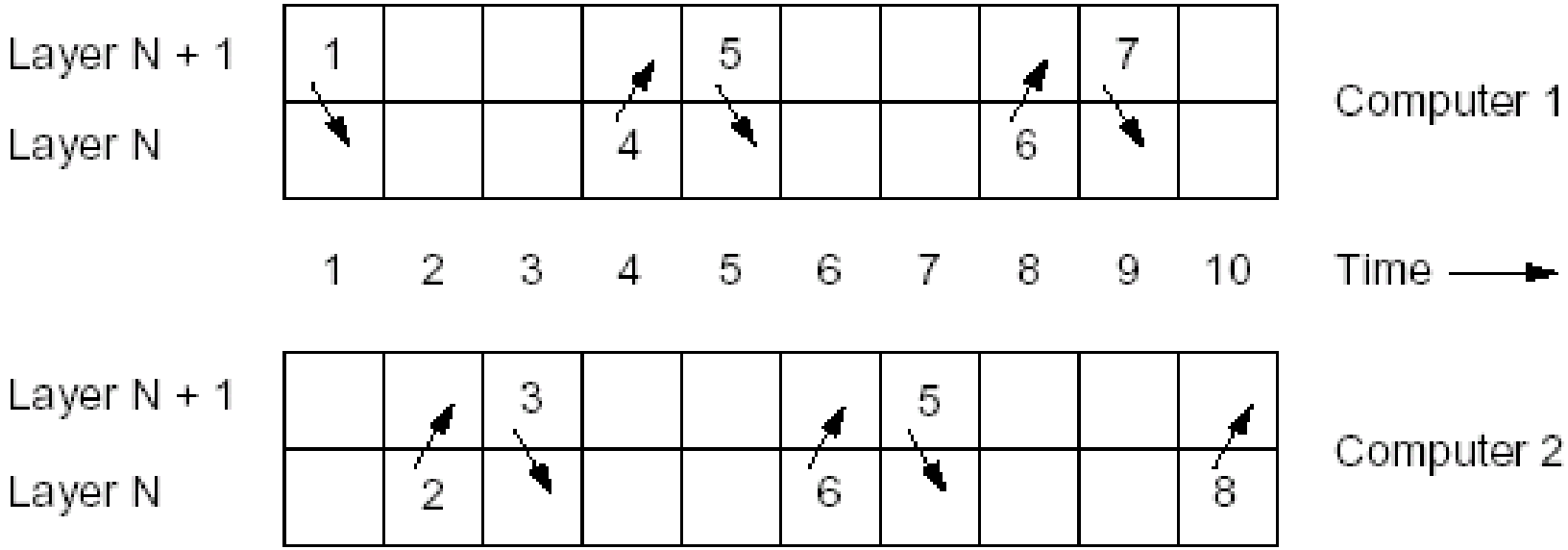
	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Remote login
	Unreliable connection	Digitized voice
Connectionless	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Registered mail
	Request-reply	Database query

Service Primitives

- Services are generally specified by a set of **primitives** (operations), which tell the service to perform some action or report on an action taken by a peer entity. Primitives are normally system

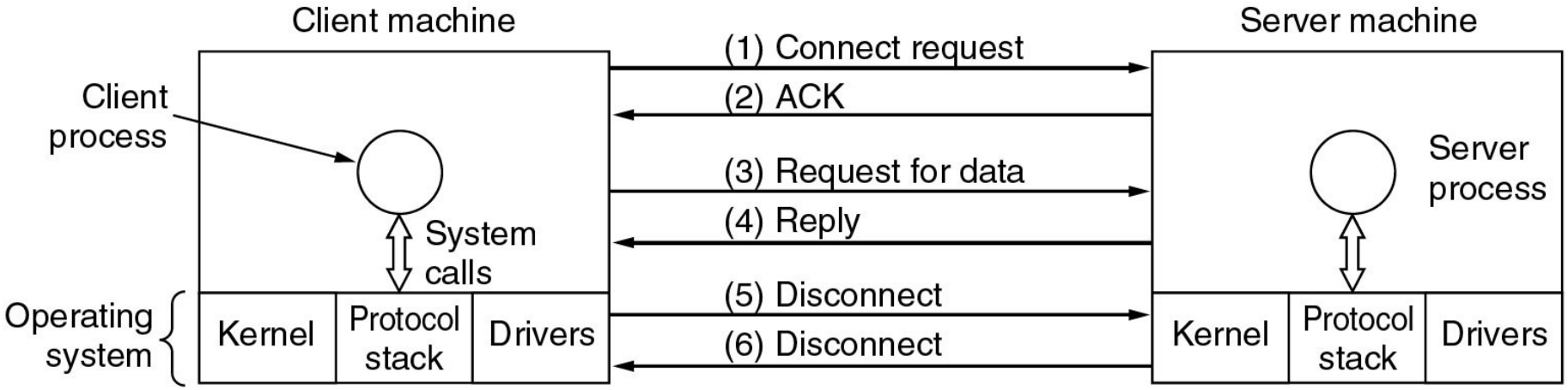
Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

1. **CONNECT.request: request for establishing a connection (*dial a phone number*).**
2. **CONNECT.indication: signal the callee (*phone rings*).**
3. **CONNECT.response: reaction by callee to indication (*pick up the phone*).**
4. **CONNECT.confirm: tell caller whether call was accepted (*caller hears ringing stop*).**
5. **DATA.request: request data to be sent (*say something*)**
6. **DATA.indication: signal arrival of data (*callee hears you*)**
7. **DISCONNECT.request: request release of connection (*caller hangs up*)**
8. **DISCONNECT.indication: signal release of connection (*callee*



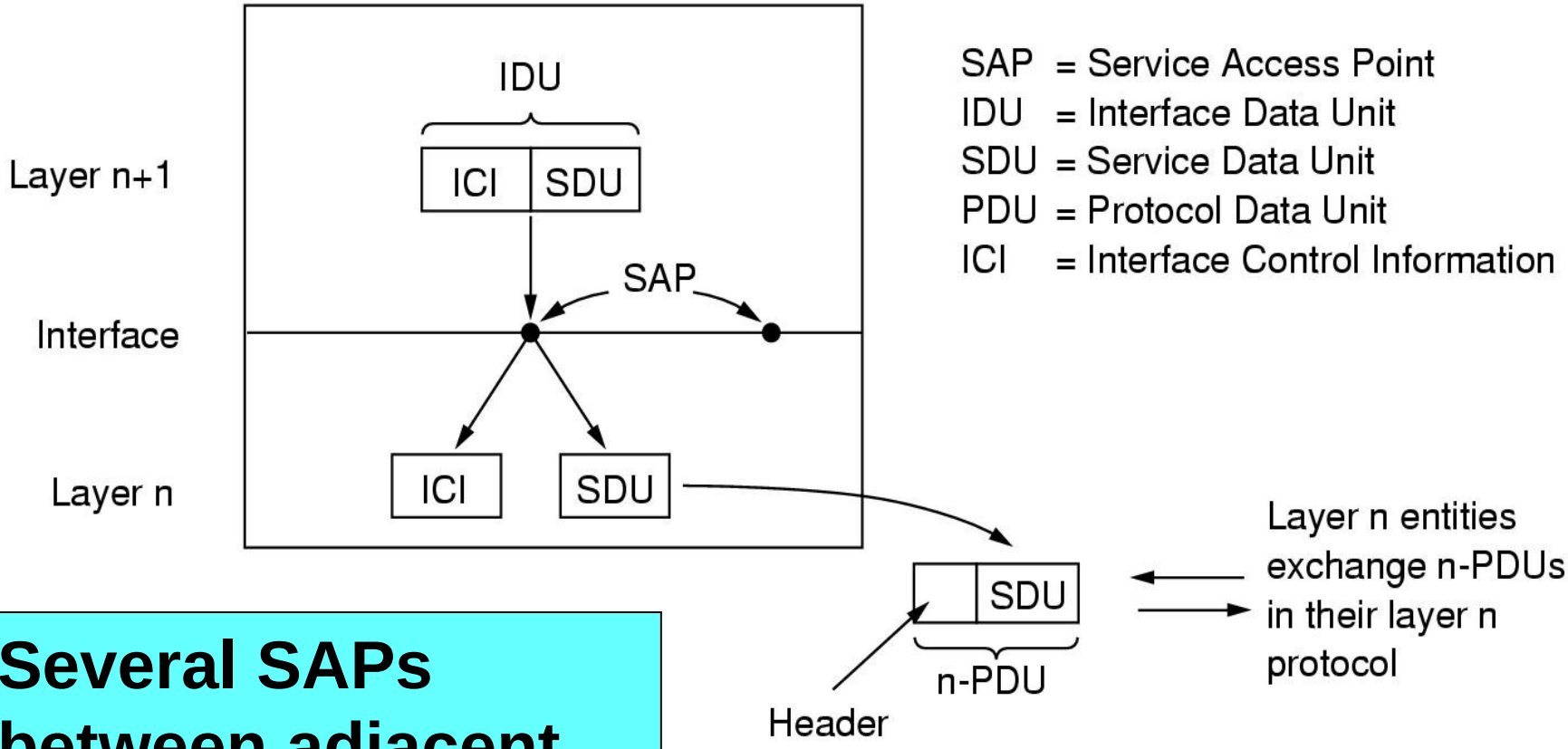
Service Primitives

- **Packets sent in a simple client-server interaction on a connection-oriented network.**



Interfaces and Services

Relation between layers at an



Several SAPs between adjacent layers.

Interfaces and Services

- A **Service Access Point (SAP)** is identified by an address, and forms the interface to a set of services.
- The **Service Data Unit** contains the data you want to send.
- The **Interface Control Information** contains info needed to send the SDU, e.g. number of bytes.
- The **Protocol Data Unit** is the data that is sent across the network, containing your SDU as well as protocol-specific data.

Interfaces and Services

(N)PCI	(N) User Data (optional)

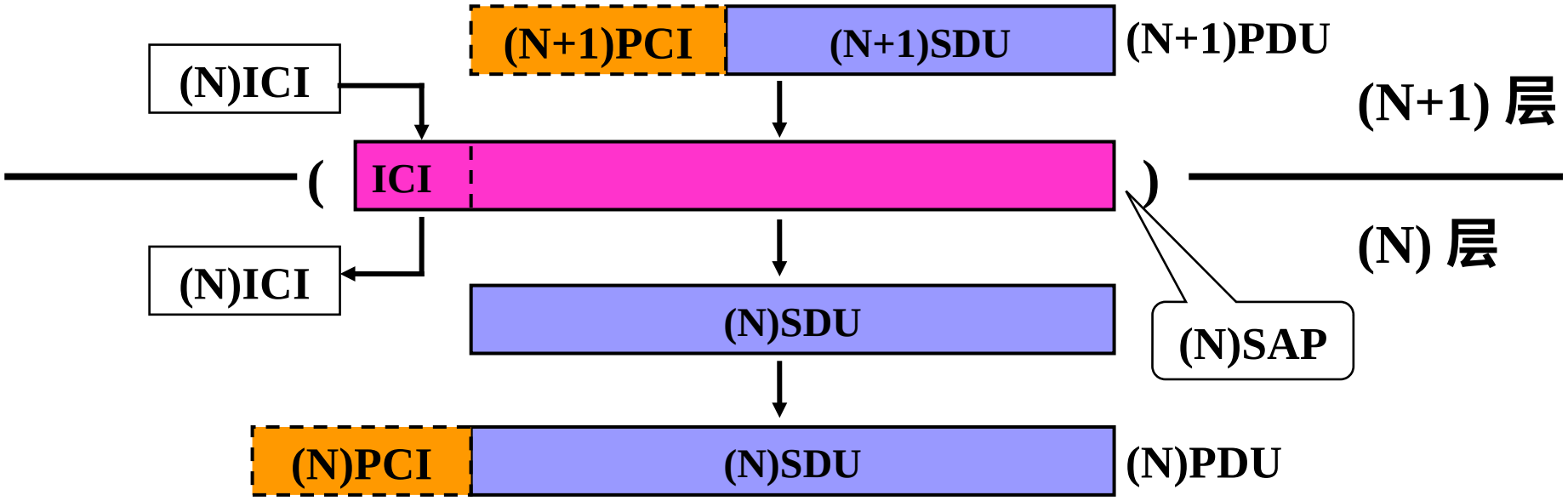
(N)PCI: Protocol Control Information. Header

a. (N)PDU Format

(N)ICI	(N+1) PDU

(N)ICI: Interface Control Information. Only useful for PDUs over the interface.

b. (N)IDU Format

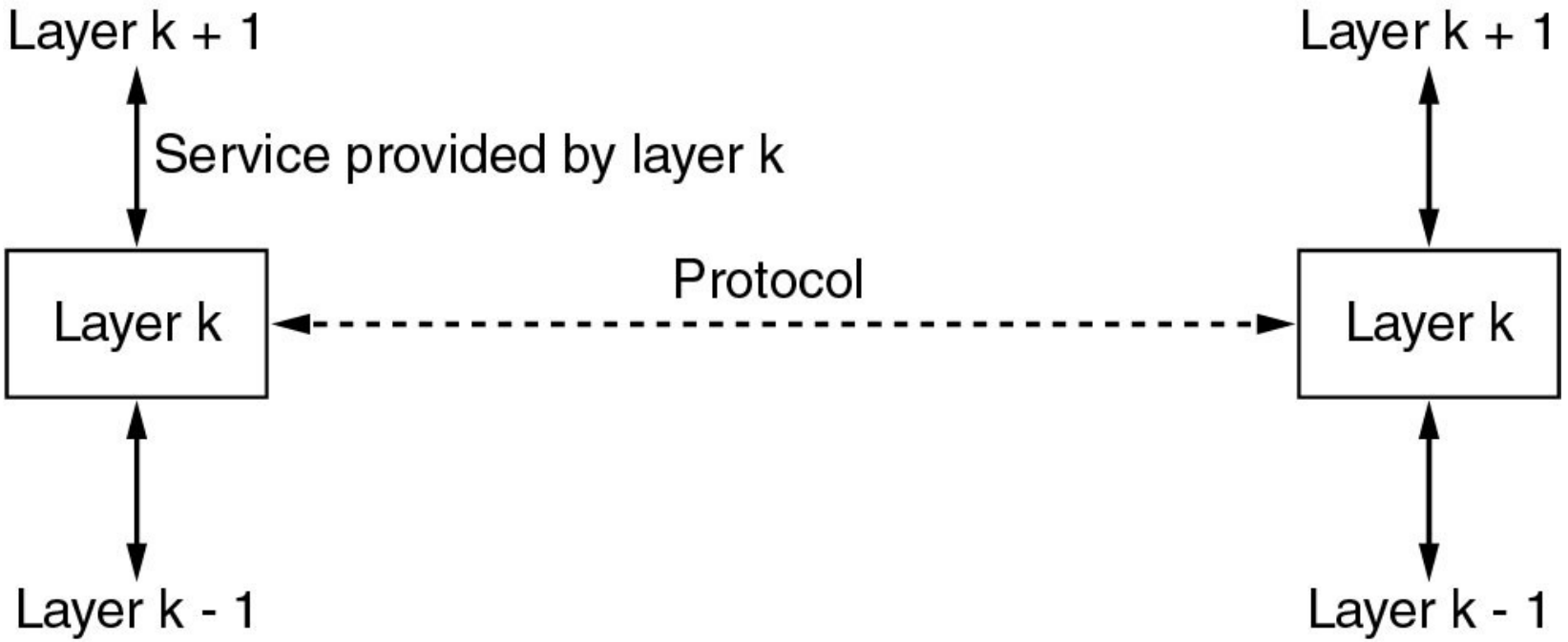


SAP and Data Units

The Relationship of Services to Protocols

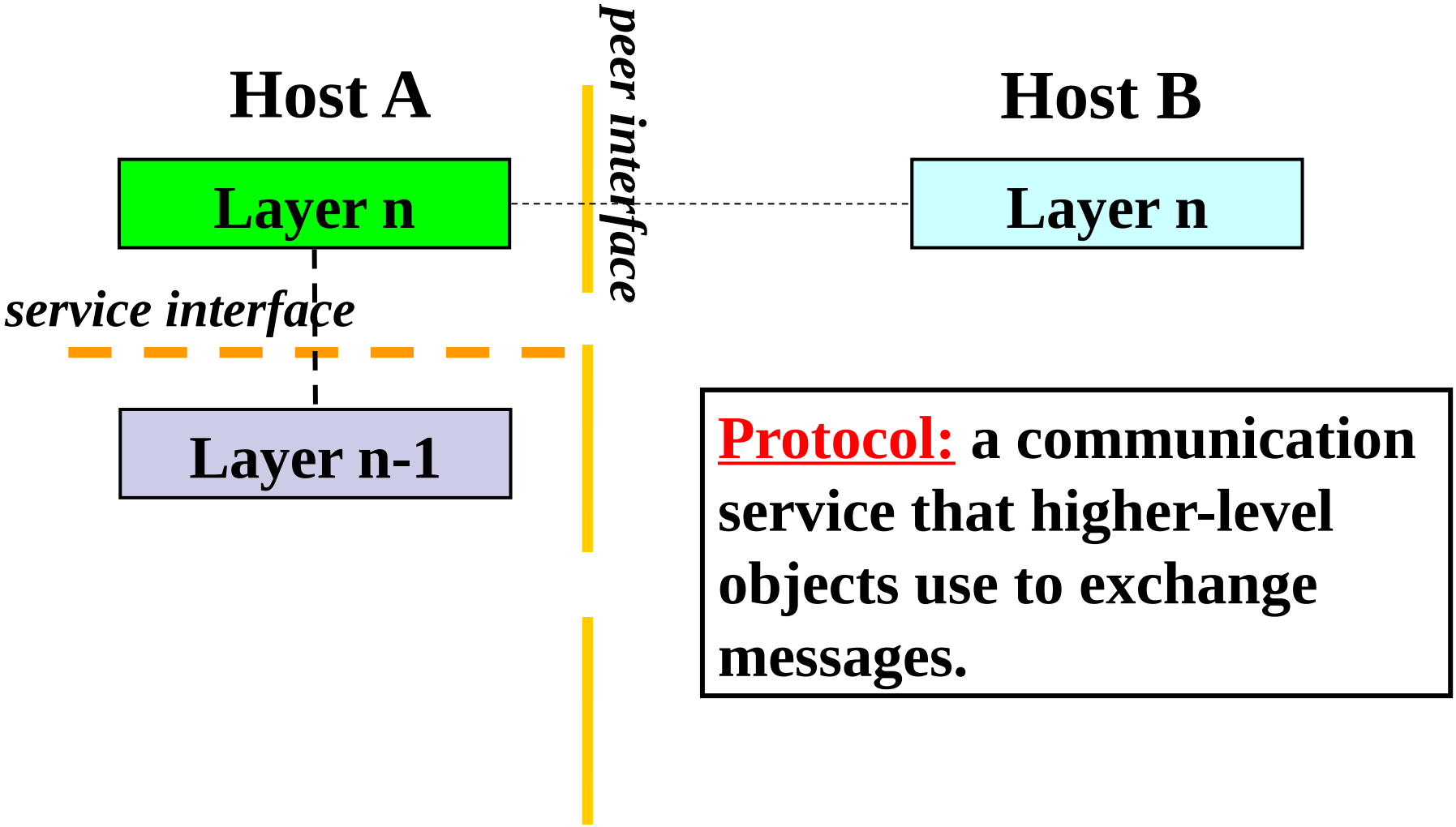
- **Service**
  - **is a set of primitives (operations) that a layer provides to the layer above it, defining what operations the layer is prepared to perform on behalf of its users, but nothing about how these operations are implemented.**
- **Protocol**
  - **is a set of rules governing the format and meaning of the packets, or messages that are exchanged by the peer entities within a layer. Entities use protocols to implement their service definitions.**

The Relationship of Services to Protocols



The relationship between a service and a protocol

Protocols, Layers, and Interfaces

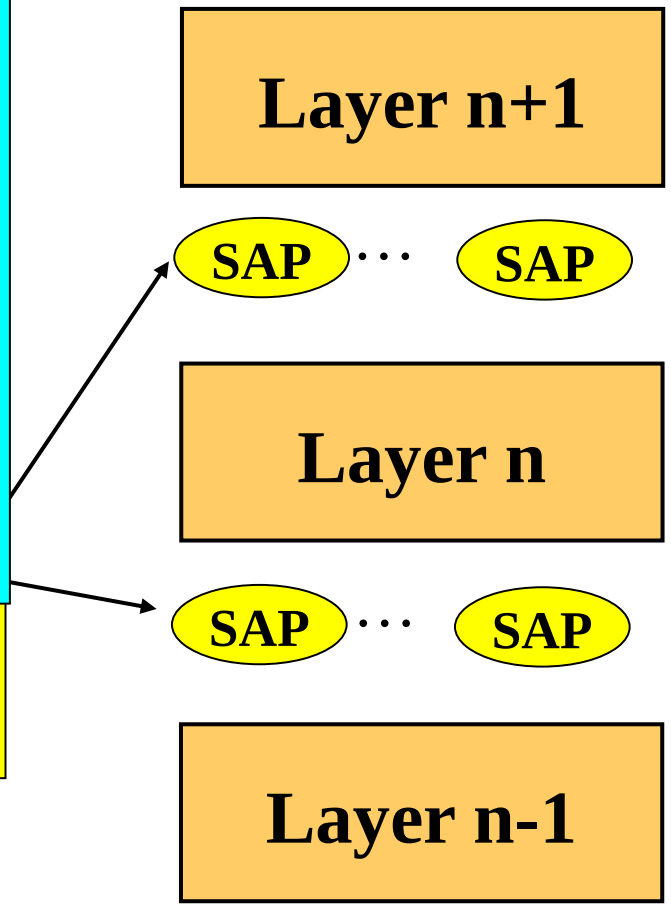


Protocol: a communication service that higher-level objects use to exchange messages.

Communication Between Layers within a Host

It's important to specify the services offered to higher layers in the hierarchy.
What they are + how to use them = *interface*.

Note: This is ISO terminology.



Chapter 1: Roadmap

1.1 Computer network and the Internet

1.2 Network edge and core

1.3 Network access and physical media

1.4 Transmission Technology

1.5 Type and Topology

1.6 Network Performance

1.7 Protocol layers, service models

1.8 Architecture, OSI and TCP/IP Models

1.9 Internet History

Some terminologies

- **Network Architecture**
  - **A set of layers and protocols, together with interfaces between layers, is called Network Architecture.**
- **protocol hierarchies**
  - **Fundamental to *all* software that makes a computer network run, is the notion of protocol hierarchies: structuring the services that a network must offer in terms of layers.**
- **Protocol stack**
  - **A list of protocols used by a certain system, one protocol per layer, is called a protocol stack.**

Back to Modularity

- **Modularity in programming:**
  - **Set of abstractions**

- **Modularity in networking:**
  - **Distributed nature of system requires deciding where abstractions are implemented**

Three Basic Architectural Decisions

- How to break system into modules?
- Where modules are implemented?
- Where state is stored?

Network Architectures

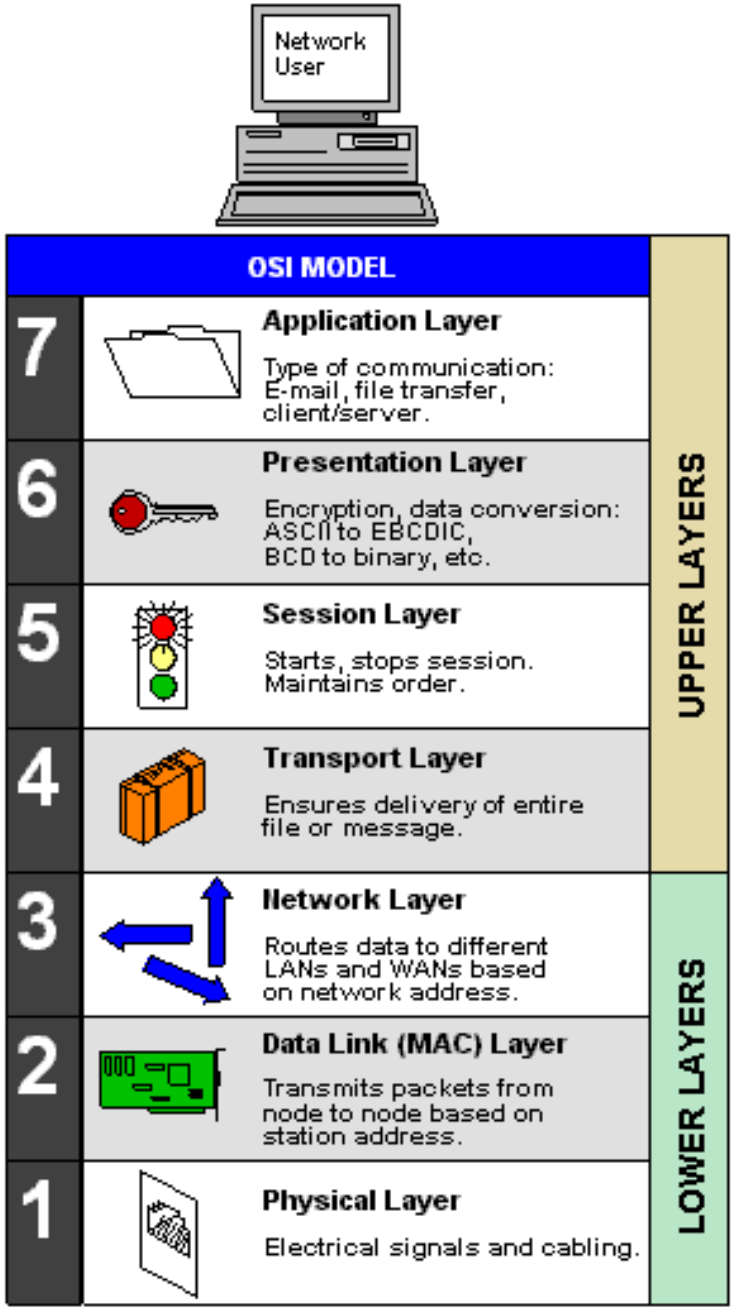
- **Network Architecture**
  - **A set of layers and protocols, together with interfaces between layers, is called Network Architecture**
  - **Describe how networks are organized**
- **Reference Model = Architecture**
- **2 typical Reference Models**
  - **ISO/OSI**
  - **TCP/IP**

ISO OSI Reference Model

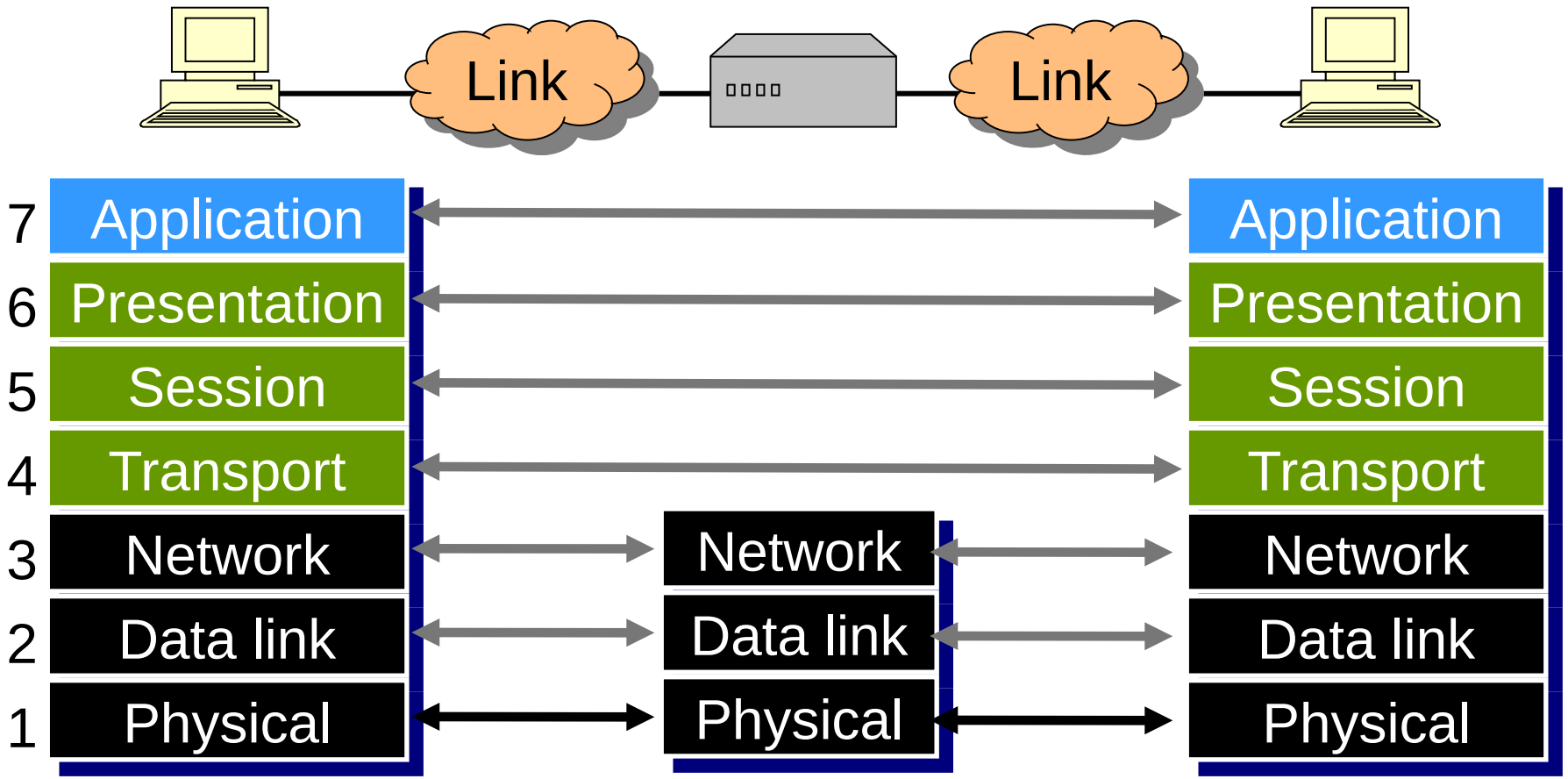
- ISO – International Standard Organization
- **OSI – Open System Interconnection**
- Started to 1978; first standard 1979
  - ARPANET started in 1969; TCP/IP protocols ready by 1974
- ***Goal*:** a general open standard
  - Allow vendors to enter the market by using their own implementation and protocols

ISO OSI Refere

- Model consists of a 7-layer stack:



ISO OSI Reference Model



OSI Functions

- **(1) Physical: transmission of a bit stream.**
- **(2) Data link: flow control, framing, error detection.**
- **(3) Network: switching and routing.**
- **(4) Transport: reliable end to end delivery.**
- **(5) Session: managing logical connections.**
- **(6) Presentation: data transformations.**
- **(7) Application: specific uses, e.g. mail, file transfer, telnet, network management**

Multiplexing takes place in multiple layers

Physical Layer (1)

- **Service:** move ***bits*** between two systems connected by a *single physical link*
- **Interface:** specifies how to send and receive *bits*
  - E.g., require quantities and timing
- **Protocols:** coding scheme used to represent *bits*, voltage levels, duration of a bit
- **Examples:** Optical, wireless, etc.

(Data) Link Layer (2)

- **Service:** Enable hosts to exchange messages
  - Using abstract (but local) addresses
  - Not just direct physical connections
- **Interface:** send messages (frames) to other end hosts; receive messages addressed to end host
- **Protocols:**
  - Routing
  - Media Access Control (MAC)
- **Examples:**
  - Ethernet, 802.11, Frame Relay, ATM

(Inter) Network Layer (3)

- **Service:** Deliver **packets** to specified destination
  - Using abstract global addresses
  - Inter-network = across multiple layer-2 networks
  - E.g., from Ethernet to 802.11 to Frame Relay to ATM
- **Interface:** send packets to specified internetwork destinations; receive packets destined for end host
- **Protocols:** construct routing tables
- **Examples: IP**
  - Why no other examples?

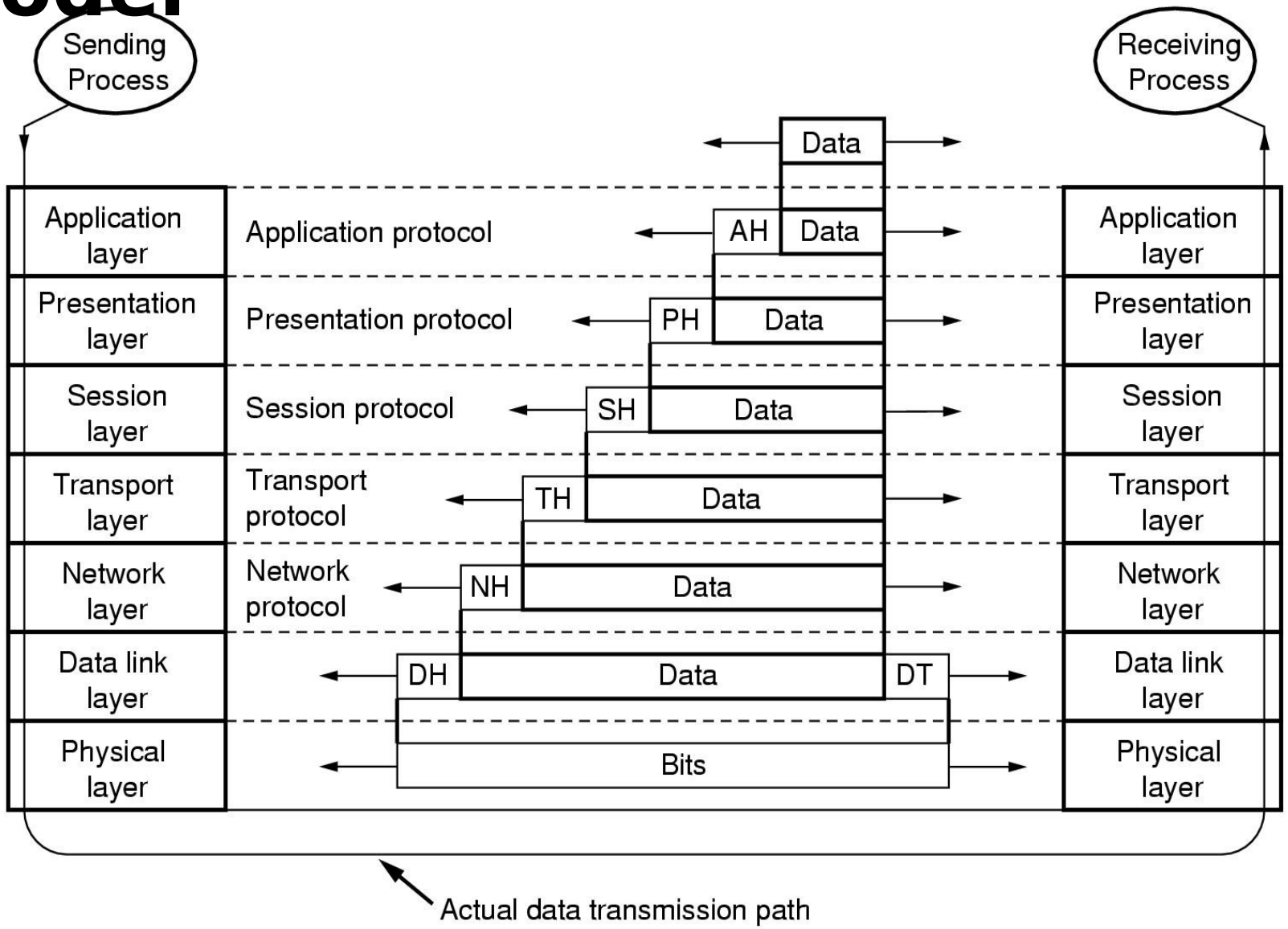
Transport Layer (4)

- **Service:** Communication between processes
  - Demultiplexing of communication between hosts
  - Perhaps: reliable transport, rate adaptation
- **Interface:**
  - send message to specific process at given destination
  - local process receives messages sent to it
- **Protocols:** reliability, flow control, packetization of large messages, framing
- **Examples:** TCP and UDP
  - SCTP, T/TCP, DCCP,

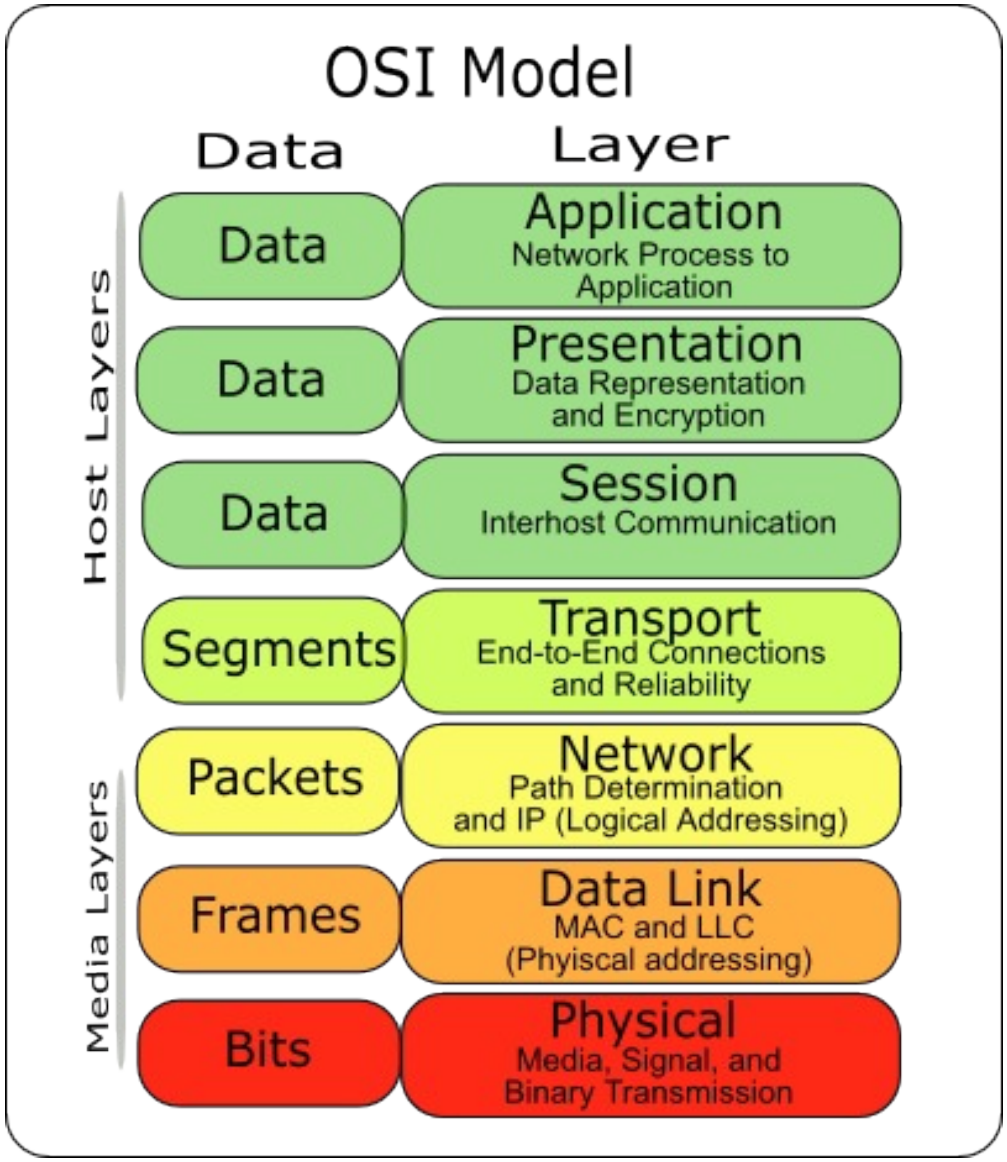
Application Layer (7)

- **Service:** any service provided to the end user
- **Interface:** depends on the application
- **Protocols:** depends on the application
- **Examples:** Skype, SMTP (email), HTTP (Web), Halo, BitTorrent …

Data Transmission in the OSI Model



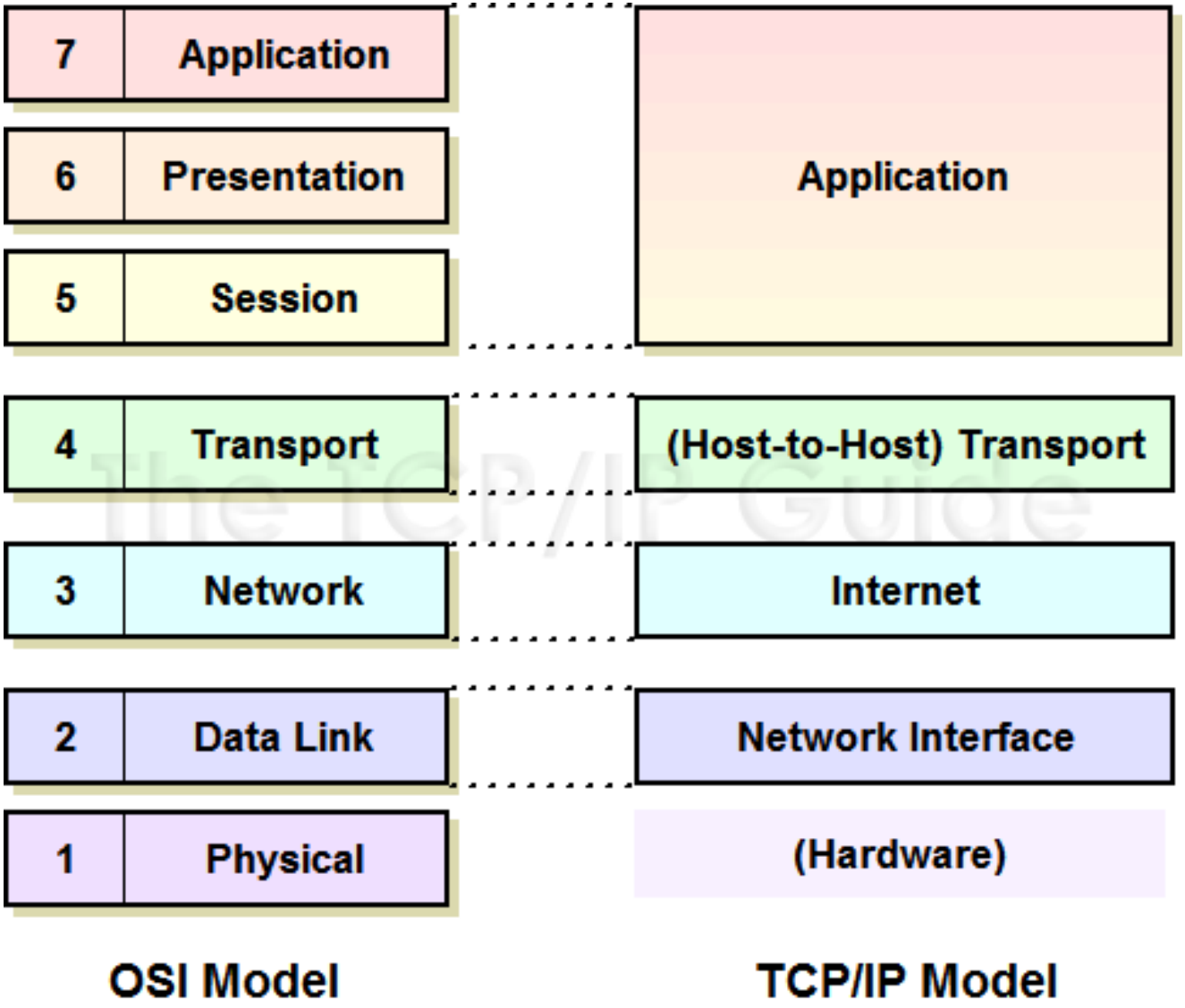
ISO OSI Reference Model

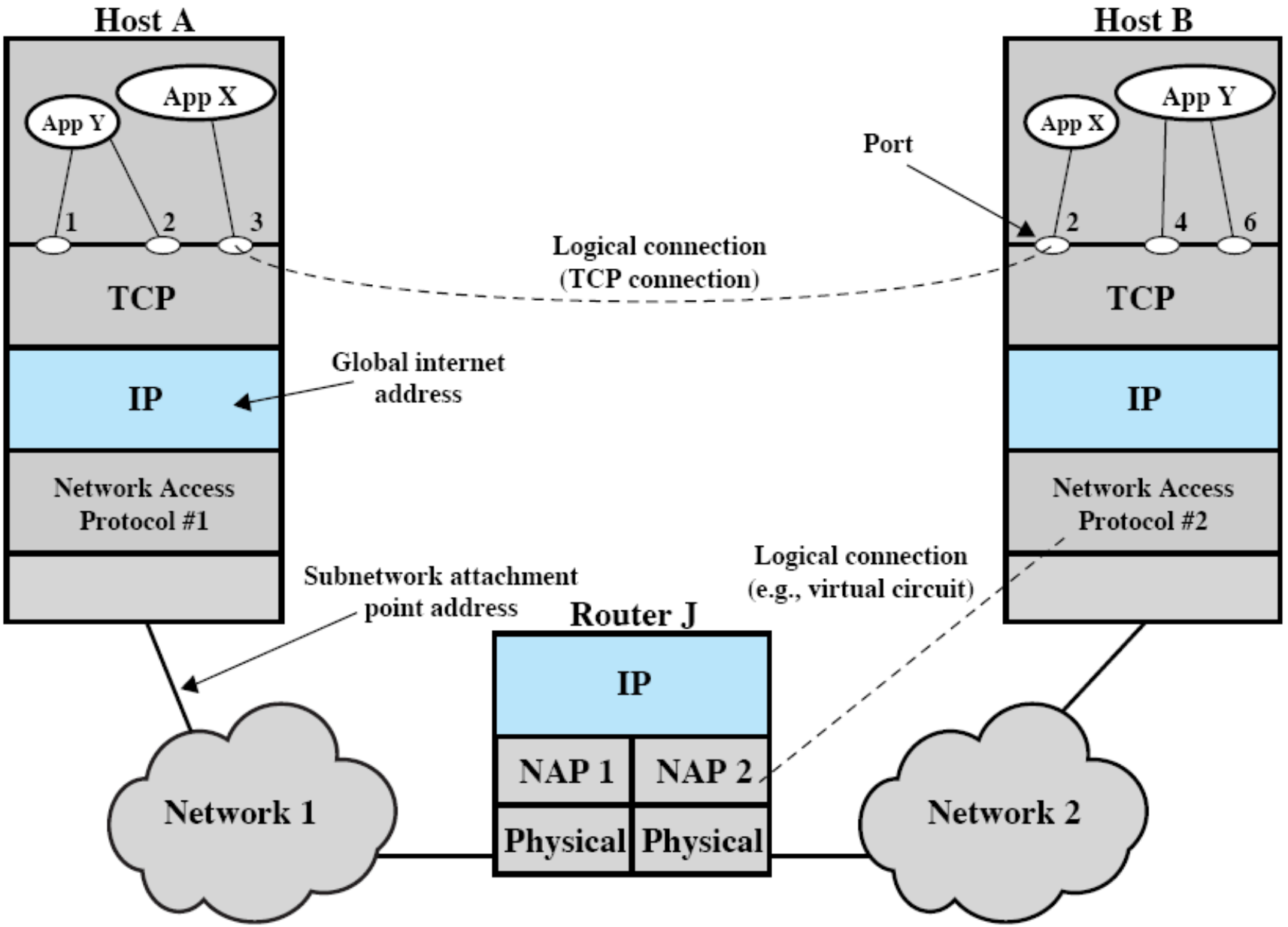


The TCP/IP Reference Model

- TCP/IP is a result of protocol research and development conducted on the experimental packet switched network, ARPANET, funded by the Defense Advanced Research Projects Agency (DARPA), and is generally referred to as the TCP/IP protocol suite.
- This protocol suite consists of a large collection of protocols that have been issued as Internet standards by the Internet Activities Board (IAB).

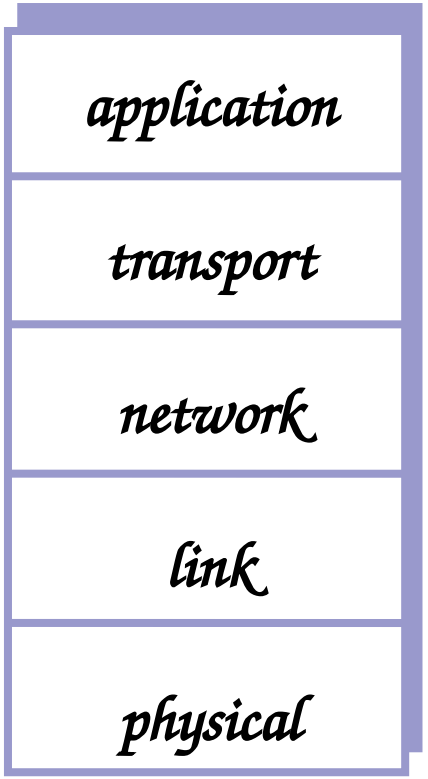
The TCP/IP Reference Model



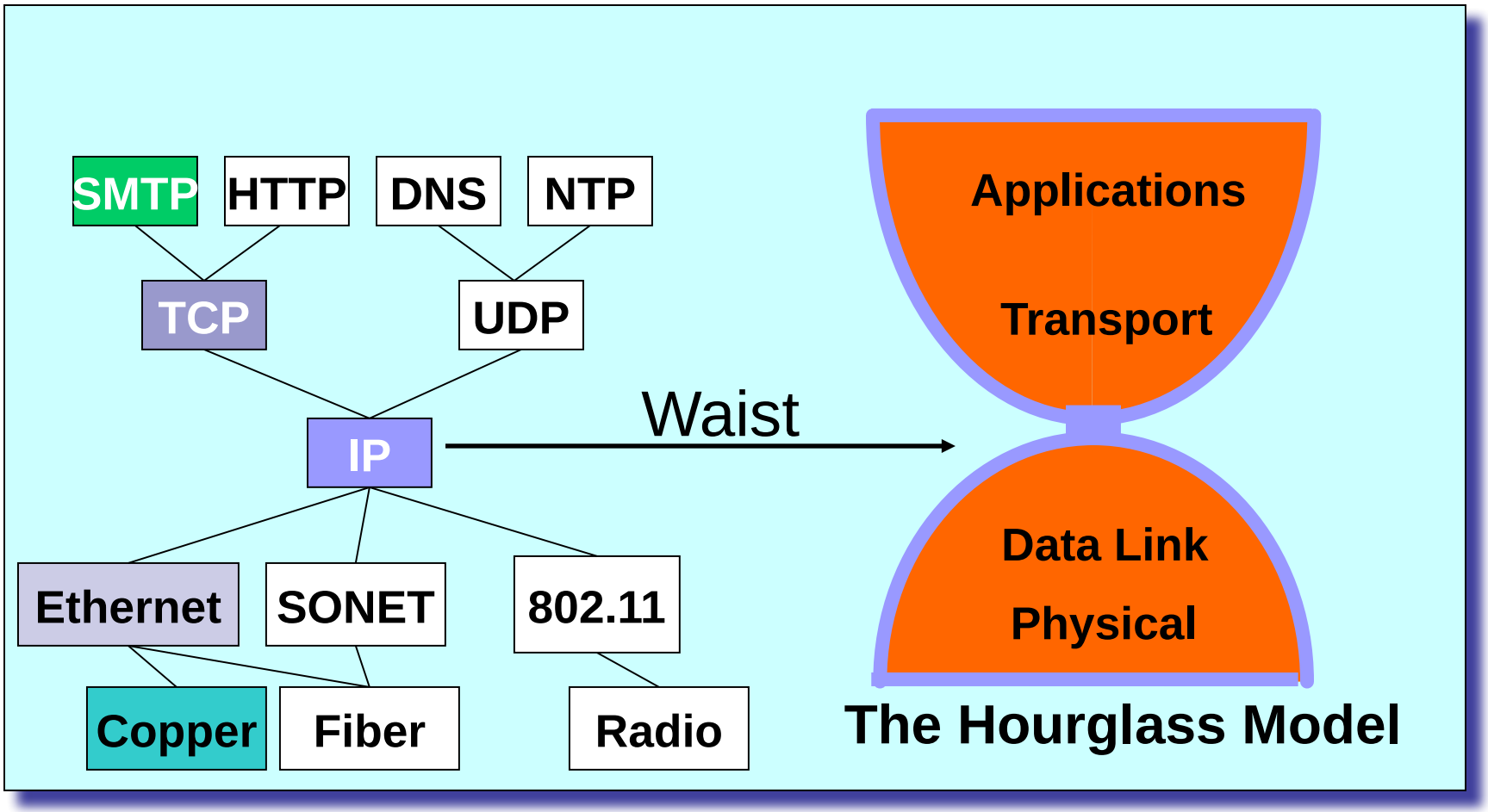

Host A
App Y
App X
1
2
3
TCP
IP
Network Access Protocol #1
Host B
App X
App Y
2
4
6
TCP
IP
Network Access Protocol #2
Port
Logical connection (TCP connection)
Global internet address
Subnetwork attachment point address
Logical connection (e.g., virtual circuit)
Router J
IP
NAP 1
NAP 2
Physical
Physical
Network 1
Network 2

The TCP/IP Reference Model

- **application:** supporting network applications
  - FTP, SMTP, STTP
- **transport:** host-host data transfer
  - TCP, UDP
- **network:** routing of datagrams from source to destination
  - IP, routing protocols
- **link:** data transfer between neighboring network elements
  - PPP, Ethernet
- **physical:** bits “on the wire”

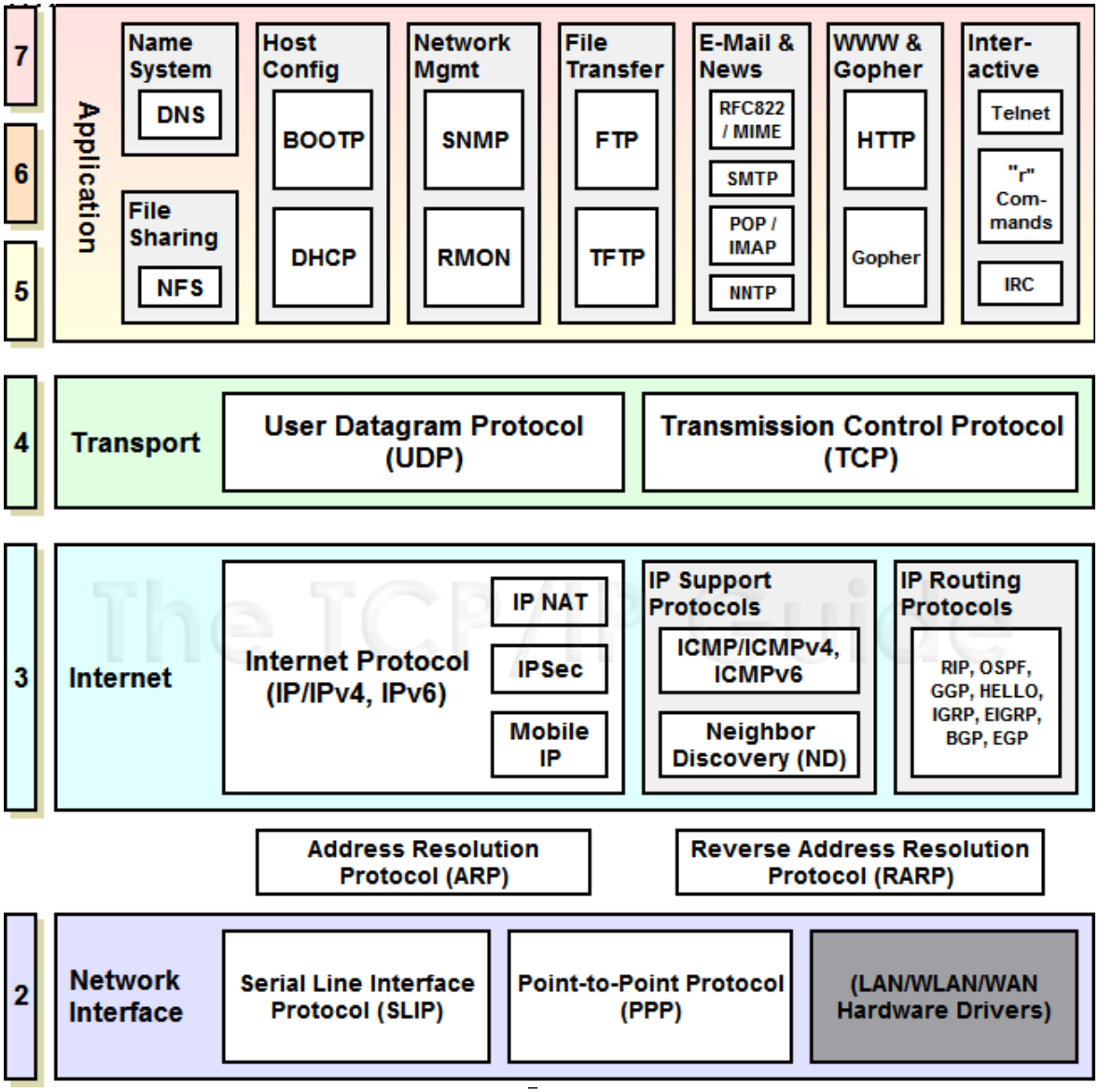


The Internet *Hourglass*



There is just one network-layer protocol, IP.
The "narrow waist" facilitates interoperability.

The TCP/IP Reference Model



The TCP/IP Reference Model

Layer #	Layer Name	Protocol	Protocol Data Unit	Addressing
5	Application	HTTP, SMTP, etc…	Messages	n/a
4	Transport	TCP/UDP	Segments/ Datagrams	Port #s
3	Network or Internet	IP	Packets	IP Address
2	Data Link	Ethernet, Wi-Fi	Frames	MAC Address
1	Physical	10 Base T, 802.11	Bits	n/a

TCP/IP Terminology Reference

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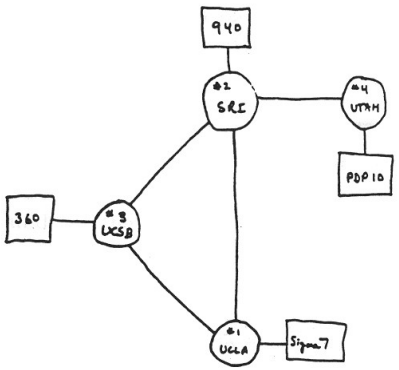
1.9 Internet History

Internet History

1961-1972: Early packet-switching principles

- *1961: Len Kleinrock - queuing theory shows effectiveness of packet-switching*
- *1964: Paul Baran - packet-switching in military networks*
- *1967: ARPAnet conceived by Advanced Research Projects Agency*
- *1969: first ARPAnet node operational*

- *1972:*
  - *ARPAnet demonstrated publicly*
  - *NCP (Network Control Protocol) first host-host protocol*
  - *first e-mail program*
  - *ARPAnet has 15 nodes*



THE ARPA NETWORK

Internet History

1972-1980: Internetworking, new and proprietary nets

- *1970: ALOHAnet satellite network in Hawaii*
- *1973: Metcalfe's PhD thesis proposes Ethernet*
- *1974: Cerf and Kahn - architecture for interconnecting networks*
- *late70's: proprietary architectures: DECnet, SNA, XNA*
- *late 70's: switching fixed length packets (ATM precursor)*
- *1979: ARPAnet has 200 nodes*

Cerf and Kahn's internetworking principles:

- *minimalism, autonomy - no internal changes required to interconnect networks*
- *best effort service model*
- *stateless routers*
- *decentralized control*

define today's Internet architecture

Internet History

1980-1990: new protocols, a proliferation of networks

- 1983: deployment of TCP/IP
- 1982: smtp e-mail protocol defined
- 1983: DNS defined for name-to-IP-address translation
- 1985: ftp protocol defined
- 1988: TCP congestion control

- new national networks: Csnet, BITnet, NSFnet, Minitel
- 100,000 hosts connected to confederation of networks

Internet History

1990's: commercialization, the WWW

- *Early 1990's: ARPAnet decommissioned*
- *1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)*
- *early 1990s: WWW*
  - *hypertext [Bush 1945, Nelson 1960's]*
  - *HTML, http: Berners-Lee*
  - *1994: Mosaic, later Netscape*
  - *late 1990's: commercialization of the web*
- *Late 1990's:*
  - *est. 50 million computers on Internet*
  - *est. 100 million+ users*
  - *backbone links running at 1 Gb/s*
- *2000's:*
  - *backbones at 10 Gb/s*
  - *wireless networks*
  - *broadband home*
  - *IP telephony*

Today's Vision

- **Everything is digital:** voice, video, music, pictures, live events
- **Everything is on-line:** bank statement, medical record, books, airline schedule, weather, highway traffic, toaster, refrigerator …
- **Everyone is connected:** doctor, teacher, broker, mother, son,

Summary

- **What is a Computer Network?**
  - Definition, Objectives, Components, Types
- **How to connect computers together?**
  - Types of Network Connections (Topologies)
- **How to make computers to communicate?**
  - Broadcasting, Switching (Circuit, Packet)

Summary

- How to design and organize a computer network system?
  - Layering, Hierarchical Network Model
- What does network architecture mean?
  - Describe how networks are organized
  - Layered Protocol Architectures
- In a computer network architecture, what is the role and function of each layer?

Summary

Covered a "ton" of materials!

- Network overview.
- What a protocol is.
- Network edge, core, and access network
  - packet-switching vs circuit-switching
- Type & Topology, Internet/ISP structure.
- Performance: loss, delay, utilization, BDP
- Layering and service models
- Architecture: OSI, TCP/IP
- History

Summary

- Architecture of a complex system can be simplified by layering.
- Layer architecture simplifies the network design.
- A set of layers and protocols, together with interfaces between layers, is called Network Architecture.
- Service, Interface, Protocol, SAP, PDU